```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

```
    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from flask_wtf.csrf import CSRFProtect, CSRFError
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
csrf = CSRFProtect()
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # CSRF 防护：WTF_CSRF_ENABLED 只对 FlaskForm 生效，页面上那些手写 POST
    表单
    # （账号增删改、项目 / 图斑删除等）以前是没校验的；同时模板里的 csrf_token()
    # 也需要它才会存在——少了这一行，保存过一次 AHP 权重后 /ahp/ 直接 500。
    csrf.init_app(app)
    # 模板里展示时间统一走 localdt：库里存 UTC，展示成本地时间
    from app.utils.timefmt import to_local
    app.jinja_env.filters["localdt"] = to_local
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
```

```
app.register_blueprint(constraint.bp)
app.register_blueprint(ahp.bp)
app.register_blueprint(indicator.bp)
app.register_blueprint(evaluate.bp)
app.register_blueprint(report.bp)
app.register_blueprint(search.bp)
#  全局登录保护：除登录页与静态资源外，均需登录
from flask import request, redirect, url_for,
send_from_directory, flash
from flask_login import current_user, login_user,
logout_user
PUBLIC_ENDPOINTS = {"auth.login", "auth.register",
"static", "favicon"}
#  浏览器默认会请求根路径下的 /favicon.ico。
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
#  回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"auth.register", "static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    #  账号被停用后立即失效：不然已经登录的会话还能继续用下去
    if not current_user.active:
        logout_user()
        flash("该账号已停用，请联系管理员", "toast-error")
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
#  与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
```

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        ("项目与矿山档案", "/project/", "folder"),
        ("政策与标准知识库", "/standard/", "book"),
        ("限制性指标判定", "/constraint/", "shield"),
        ("微地质单元", "/unit/", "layers"),
        ("指标体系与权重", "/ahp/", "sliders"),
        ("指标数据采集", "/indicator/", "database"),
        ("适宜性定量评价", "/evaluate/", "target"),
        ("报告与结果输出", "/report/", "file"),
    ]
    #  「用户与操作日志」已并入「个人设置」，不再是独立菜单，
    #  因此系统管理组只留一个入口。
    ADMIN_LINKS = [
        ("个人设置", "/profile", "user"),
    ]
    @app.context_processor
    def inject_nav():
        path = request.path or "/"
        def hit(prefix):
            """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
            return path.startswith(prefix.rstrip("/"))
        badges = {}
        if current_user.is_authenticated:
            from app.services import dashboard as dash
            badges = dash.nav_badges()
        def build(links):
            return [{"label": label, "href": href, "icon":
    icon,
                     "active": hit(href), "badge":
    badges.get(href, 0)}
                    for label, href, icon in links]
        groups = [
            {"title": "评价流程", "links": build(FLOW_LINKS)},
            {"title": "系统管理", "links": build(ADMIN_LINKS)},
        ]
        #  面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
        if path == "/":
            page_title = "工作台"
        elif path.startswith("/profile"):
            page_title = "个人设置"
        elif path.startswith(("/users", "/logs")):
            page_title = "账号与日志"
        elif path.startswith("/search"):
            page_title = "搜索结果"
        else:
            page_title = "工作台"
            for label, href, _icon in FLOW_LINKS:
                if hit(href):
                    page_title = label
                    break
        #  层级引导用：项目是归档单位，图斑才是评价对象。
```

```
    #  下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
    from app.models import Project, MinePatch
    projects = Project.query.order_by(Project.id).all()
    return {
        "nav_groups": groups,
        "home_active": path == "/",
        "page_title": page_title,
        "today": date.today().strftime("%Y-%m-%d"),
        "search_q": request.args.get("q", ""),
        "role_labels": ROLE_LABELS,
        "project_count": len(projects),
        "patch_count": MinePatch.query.count(),
        "projects_without_patch": [p for p in projects
                                   if p.patches.count()
== 0],
    }
#  操作审计：统一在响应阶段记录成功的写操作（操作日志）
from app.services import audit
@app.after_request
def audit_writes(response):
    if request.method in ("POST", "PUT", "PATCH",
"DELETE"):
        audit.record(request.endpoint,
response.status_code)
    return response
#  错误页：给出可读提示，而不是默认的英文错误页
from flask import render_template
@app.errorhandler(403)
def forbidden(e):
    return render_template("error.html", code=403,
                           message="没有权限执行该操作"), 403
@app.errorhandler(404)
def not_found(e):
    return render_template("error.html", code=404,
                           message="请求的页面不存在"), 404
@app.errorhandler(413)
def too_large(e):
    """上传超过体积上限时给中文页；默认是英文的 Request Entity Too Large。"""
    limit = app.config.get("MAX_CONTENT_LENGTH") or 0
    return render_template(
        "error.html", code=413,
        message="上传的文件太大（上限 %d MB），请压缩或拆分后重试"
                % (limit // 1024 // 1024)), 413
@app.errorhandler(500)
def server_error(e):
    """未预料的错误也给中文页；生产模式下 Flask 不会带出堆栈。"""
    return render_template("error.html", code=500,
                           message="系统处理时出现异常，本次操作未生效，"
                                   "请返回重试或联系管理员"), 500
@app.errorhandler(CSRFError)
```

```
    def csrf_error(e):
        """令牌过期或来源不符时给一句人话，并把人送回上一页。
        原来校验一失败就是默认英文 400 页，或者（对手写表单）干脆什么都不发生，
        用户只会觉得「点了没反应」。回跳目标只取本站地址，避免开放重定向。
        """
        from flask import flash, redirect, url_for
        flash("页面停留时间过久或提交来源校验未通过，请重新提交一次", "error")
        referrer = request.referrer or ""
        if referrer.startswith(request.host_url):
            return redirect(referrer)
        return redirect(url_for("auth.index"))
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
import secrets
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
def _secret_key():
    """会话签名密钥：优先取环境变量，否则首次运行随机生成并存到本地文件。
    原来是一个写死在代码里的默认值，任何拿到源码的人都能自签会话 cookie。
    单机本地跑风险有限，但改成本机随机值几乎零成本。
    生成的文件已加进 .gitignore，不会进版本库。
    """
    env = os.environ.get("SECRET_KEY")
    if env:
        return env
    path = os.path.join(BASE_DIR, ".secret_key")
    try:
        if os.path.exists(path):
            with open(path, encoding="utf-8") as f:
                saved = f.read().strip()
            if saved:
                return saved
        key = secrets.token_hex(32)
        with open(path, "w", encoding="utf-8") as f:
            f.write(key)
        return key
    except OSError:
        # 目录不可写时退回固定值，保证系统仍能起来（此时会退化为原来的行为）
        return "mine-restoration-dev-key"
class Config:
    SECRET_KEY = _secret_key()
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
```

```
    # 会话 cookie：不交给脚本读取，跨站请求不带
    SESSION_COOKIE_HTTPONLY = True
    SESSION_COOKIE_SAMESITE = "Lax"
    # 模板改动立即生效，不必重启服务。
    # Flask 在非 debug 下默认把模板缓存到进程里，改完模板刷新浏览器还是旧页面，
    # 容易被误当成「改了没生效」。本系统只在本机运行，这点开销可以忽略。
    TEMPLATES_AUTO_RELOAD = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024            # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
from app.utils.timefmt import utcnow    # 替代已弃用的
    datetime.utcnow()
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime, default=utcnow)
    # ---- 个人资料（用户可在「个人设置」里自行维护）----
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    display_name = db.Column(db.String(64), nullable=True)
    # 昵称
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    gender = db.Column(db.String(8), nullable=True)
    # 男 / 女 / 保密
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def is_admin(self):
        return self.role == "admin"
    @property
    def shown_name(self):
        """界面上显示的名字：优先用昵称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字占位。"""
        return (self.shown_name or "?")[0].upper()
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "display_name": self.display_name,
    "gender": self.gender,
                "role": self.role, "active": self.active}
class OperationLog(db.Model):
    """操作日志。"""
    __tablename__ = "operation_logs"
    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"
    ), nullable=True)
    action = db.Column(db.String(128), nullable=False)
    # 操作类型
    target_type = db.Column(db.String(64), nullable=True)
    # 对象类型
    target_id = db.Column(db.Integer, nullable=True)
    # 对象主键
    detail = db.Column(db.Text, nullable=True)
    # 操作详情
    created_at = db.Column(db.DateTime, default=utcnow,
    index=True)
class Project(db.Model):
    """评价项目（项目与矿山档案）。"""
    __tablename__ = "projects"
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(128), nullable=False)
    code = db.Column(db.String(64), nullable=True)
    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
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    created_by = db.Column(db.Integer,                                  401
    db.ForeignKey("users.id"), nullable=True)                           402
    created_at = db.Column(db.DateTime, default=utcnow)                 403
    patches = db.relationship("MinePatch",                              404
    backref="project", lazy="dynamic",                                  405
                                    cascade="all, delete-orphan")       406
class MinePatch(db.Model):                                              407
    """矿山图斑。"""                                                    408
    __tablename__ = "mine_patches"                                      409
    id = db.Column(db.Integer, primary_key=True)                        410
    project_id = db.Column(db.Integer,                                  411
    db.ForeignKey("projects.id"), nullable=False)                       412
    name = db.Column(db.String(128), nullable=False)                    413
    admin_region = db.Column(db.String(128), nullable=True)             414
    # 所在行政区                                                        415
    center_x = db.Column(db.Float, nullable=True)                       416
    # 中心经度 / 横坐标                                                 417
    center_y = db.Column(db.Float, nullable=True)                       418
    # 中心纬度 / 纵坐标                                                 419
    area_ha = db.Column(db.Float, nullable=True)                        420
    # 图斑面积（公顷）                                                  421
    mineral_type = db.Column(db.String(64), nullable=True)              422
    # 开采矿种                                                          423
    mining_method = db.Column(db.String(64), nullable=True)             424
    # 开采方式                                                          425
    stop_years = db.Column(db.Float, nullable=True)                     426
    # 停产年限                                                          427
    responsible_entity_missing = db.Column(db.Boolean,                  428
    default=True)   # 责任主体是否缺失                                 429
    # 遥感底图 / 无人机正射影像挂接                                     430
    image_path = db.Column(db.String(256), nullable=True)               431
    # 相对 static 的路径                                                432
    image_note = db.Column(db.String(256), nullable=True)               433
    # 影像说明（如来源、时相）                                          434
    image_uploaded_at = db.Column(db.DateTime,                          435
    nullable=True)                                                      436
    created_at = db.Column(db.DateTime, default=utcnow)                 437
    units = db.relationship("MicroUnit", backref="patch",               438
    lazy="dynamic",                                                     439
                                    cascade="all, delete-orphan")       440
class ReferenceEcosystem(db.Model):                                     441
    """参照生态系统：标准化基准值，与图斑绑定。"""                     442
    __tablename__ = "reference_ecosystems"                              443
    id = db.Column(db.Integer, primary_key=True)                        444
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat            445
    ches.id"), nullable=False)                                          446
    coverage_benchmark = db.Column(db.Float, default=75.0)              447
    # 植被覆盖度基准 %                                                  448
    shannon_benchmark = db.Column(db.Float, default=0.7)                449
    # Shannon-Wiener 基准                                               450
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    dominant_species = db.Column(db.String(256),                    451
    nullable=True)   # 建群种                                        452
    soil_benchmark = db.Column(db.String(256),                      453
    nullable=True)     # 土壤基准                                   454
class Standard(db.Model):                                           455
    """地方标准 / 政策文件（政策与标准知识库）。"""                    456
    __tablename__ = "standards"                                     457
    id = db.Column(db.Integer, primary_key=True)                    458
    province = db.Column(db.String(32), nullable=False)             459
    # 省份                                                          460
    name = db.Column(db.String(256), nullable=False)                461
    # 名称                                                          462
    doc_number = db.Column(db.String(128), nullable=True)           463
    # 标准编号 / 文号                                               464
    publish_date = db.Column(db.String(32), nullable=True)          465
    implement_date = db.Column(db.String(32), nullable=True)        466
    # 类别：地方标准 / 政策文件                                     467
    category = db.Column(db.String(32), nullable=False,             468
    default="地方标准")                                             469
    summary = db.Column(db.Text, nullable=True)                     470
    # 特点概述                                                      471
    rules = db.relationship("StandardRule",                         472
    backref="standard",                                             473
                                cascade="all, delete-orphan")       474
class StandardRule(db.Model):                                       475
    """结构化规则：将标准条文拆解为可执行规则。"""                    476
    __tablename__ = "standard_rules"                                477
    id = db.Column(db.Integer, primary_key=True)                    478
    standard_id = db.Column(db.Integer,                             479
    db.ForeignKey("standards.id"), nullable=False)                  480
    item = db.Column(db.String(128), nullable=False)                481
    # 判定项                                                        482
    operator = db.Column(db.String(8), nullable=False)              483
    # 比较符：> >= < <= ==                                          484
    threshold = db.Column(db.String(64), nullable=True)             485
    # 阈值                                                          486
    condition_type = db.Column(db.String(32),                       487
    default="必要条件")   # 必要条件 / 充分条件                     488
    region = db.Column(db.String(64), nullable=True)                489
    # 适用分区（宁夏分区差异化）                                    490
    note = db.Column(db.Text, nullable=True)                        491
class ConstraintCheck(db.Model):                                    492
    """限制性判定记录。"""                                          493
    __tablename__ = "constraint_checks"                             494
    id = db.Column(db.Integer, primary_key=True)                    495
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        496
    ches.id"), nullable=False)                                      497
    policy_pass = db.Column(db.Boolean, nullable=True)              498
    # 政策约束                                                      499
    location_pass = db.Column(db.Boolean, nullable=True)            500
```

```
    # 区位约束
    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime, default=utcnow)
class MicroUnit(db.Model):
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
```

```
            return [s for s in (self.enabled_units or                551
        "").split(",") if s]                                         552
class WeightScheme(db.Model):                                        553
    """权重方案：按微地质单元类型分别保存。"""                                554
    __tablename__ = "weight_schemes"                                 555
    id = db.Column(db.Integer, primary_key=True)                     556
    unit_type = db.Column(db.String(32), nullable=False)             557
    # 所属单元类型                                                    558
    name = db.Column(db.String(64), nullable=True)                   559
    # 方案名称                                                        560
    level = db.Column(db.Integer, nullable=False,                    561
    default=1)          # 1=一级指标 2=二级指标                        562
    is_default = db.Column(db.Boolean, default=False)                563
    # 是否默认（系统默认）                                             564
    created_at = db.Column(db.DateTime, default=utcnow)              565
class WeightItem(db.Model):                                          566
    """权重明细：方案内的各项权重。"""                                    567
    __tablename__ = "weight_items"                                   568
    id = db.Column(db.Integer, primary_key=True)                     569
    scheme_id = db.Column(db.Integer,                                570
    db.ForeignKey("weight_schemes.id"), nullable=False)              571
    indicator_code = db.Column(db.String(32),                        572
    nullable=False)         # 指标编码                               573
    weight = db.Column(db.Float, nullable=False)                     574
    # 权重值                                                          575
class ExpertMatrix(db.Model):                                        576
    """专家判断矩阵：多专家打分的原始矩阵。"""                              577
    __tablename__ = "expert_matrices"                                578
    id = db.Column(db.Integer, primary_key=True)                     579
    scheme_id = db.Column(db.Integer,                                580
    db.ForeignKey("weight_schemes.id"), nullable=True)               581
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵                   582
    unit_type = db.Column(db.String(32), nullable=True)              583
    # 微地质单元类型                                                  584
    l1_name = db.Column(db.String(64), nullable=True)                585
    # 所属一级指标（二级矩阵用）                                       586
    expert_name = db.Column(db.String(64), nullable=True)            587
    # 专家姓名                                                        588
    specialty = db.Column(db.String(64), nullable=True)              589
    # 专家专业                                                        590
    level = db.Column(db.Integer, nullable=False,                    591
    default=1)          # 矩阵对应指标层级                             592
    matrix_json = db.Column(db.Text, nullable=False)                 593
    # 判断矩阵（JSON 二维数组）                                       594
    cr = db.Column(db.Float, nullable=True)                          595
    # 该专家矩阵的一致性比例                                          596
    created_at = db.Column(db.DateTime, default=utcnow)              597
class IndicatorData(db.Model):                                       598
    """指标实测数据：按微地质单元组织。"""                                 599
    __tablename__ = "indicator_data"                                 600
```

```
    id = db.Column(db.Integer, primary_key=True)
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_units.id"), nullable=False, unique=True)
    soil_ratio = db.Column(db.Float, nullable=True)
    # 土石比
    thickness = db.Column(db.Float, nullable=True)
    # 土层厚度 cm
    ph = db.Column(db.Float, nullable=True)
    organic_matter = db.Column(db.Float, nullable=True)
    # 有机质 g/kg
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null
    heavy_metals = db.Column(db.Text, nullable=True)
    coverage = db.Column(db.Float, nullable=True)
    # 植被覆盖度 %
    shannon = db.Column(db.Float, nullable=True)
    # Shannon-Wiener 多样性指数
    slope = db.Column(db.Float, nullable=True)
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡/阴阳坡/阴坡
    def heavy_metals_dict(self):
        """把重金属 JSON 解析成字典。
        字段是手工可改的文本，被脏写或导入半个 JSON 时不能让整页 500：
        解析失败按「未检测」处理（空字典），并把原文留在库里供排查。
        """
        if not self.heavy_metals:
            return {}
        try:
            data = json.loads(self.heavy_metals)
        except (ValueError, TypeError):
            return {}
        return data if isinstance(data, dict) else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_patches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    # 计算参数快照：本次用的权重来源、参照生态系统基准、各指标理想取值。
    # 参数现在可以在界面上改，不留快照就无法复现这一次的结果。
    params_json = db.Column(db.Text, nullable=True)
    created_by = db.Column(db.Integer,
```

```
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime, default=utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
# 邮箱格式：够用的轻量校验。
# WTForms 自带的 Email() 需要额外安装 email_validator，
# 为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
```

```
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。
    校验消息用中文写明：登录框带 novalidate，浏览器不会代填提示，
    字段为空时若不自己给消息，用户点了「登录」会以为没反应。
    """
    username = StringField("账户名称",
                           validators=[DataRequired(message=
    "请输入账户名称"),
                                       Length(max=64,

    message="账户名称不能超过 64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired(messag
    e="请输入密码")])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称
```

```
        "),                                                                  751
                                                    Length(min=2, max=64,    752
                                                           message="账号名称   753
        2-64 个字符")])                                                      754
        password = PasswordField("密码",                                     755
                                 validators=[DataRequired("请设置密码        756
        "),                                                                  757
                                                      Length(min=6,          758
        max=64,                                                              759
                                                                             760
        message="密码至少 6 位")])                                           761
        confirm = PasswordField("确认密码",                                  762
                                validators=[DataRequired(),                  763
                                            EqualTo("password",              764
                                                                             765
        message="两次输入的密码不一致")])                                    766
        display_name = StringField("昵称", validators=[Optional(),           767
        Length(max=64)])                                                     768
        gender = SelectField("性别", choices=GENDER_CHOICES,                 769
        validators=[Optional()])                                             770
        email = StringField("邮箱", validators=[Optional(),                  771
        email_format,                                                        772
                                                      Length(max=128)])      773
        phone = StringField("手机号", validators=[Optional(),                774
        Length(max=32)])                                                     775
        submit_reg = SubmitField("注册并登录")                               776
class PasswordForm(FlaskForm):                                               777
        """修改密码：需要先验证当前密码。"""                                 778
        old_password = PasswordField("当前密码",                             779
        validators=[DataRequired()])                                         780
        new_password = PasswordField("新密码",                               781
                                     validators=[DataRequired(),             782
        Length(min=6, max=64,                                                783
                                                                             784
        message="新密码至少 6 位")])                                         785
        confirm = PasswordField("确认新密码",                                786
                                validators=[DataRequired(),                  787
                                                                             788
        EqualTo("new_password",                                              789
                                                                             790
        message="两次输入的新密码不一致")])                                  791
        submit_pwd = SubmitField("修改密码")                                 792
#  ==== app/utils/constants.py ====                                         793
#  -*- coding: utf-8 -*-                                                    794
"""评价指标常量表。                                                         795
全部数值为系统内置值，作为计算的唯一基准：                                  796
- 取值规定：指标标准化理想极值与取值依据                                    797
- 等级划分规定：适宜性等级划分                                              798
- 系统默认权重：按微地质单元分别确定的一级、二级指标权重                    799
- 权重计算方法：平均随机一致性指标 RI                                       800
```

```
"""                                                                    801
#  ----------------------------------------------------------------------  802
    ----------------                                                   803
#  一、微地质单元类型                                                  804
#  ----------------------------------------------------------------------  805
    ----------------                                                   806
#  流程约定："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡    807
#  三种地质微单元类型"                                                 808
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")                          809
#  每个单元类型启用的指标类别（系统约定：岩质边坡无土壤指标，平台无地形指标）  810
#  供数据录入界面与评价引擎判断"该单元是否需要某类指标"               811
UNIT_ENABLED_INDICATORS = {                                            812
    "岩质边坡": ("植被指标", "地形指标"),                  #  不设土壤指标  813
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),                  814
    "平台": ("土壤指标", "植被指标"),                      #  不设地形指标  815
}                                                                      816
#  ----------------------------------------------------------------------  817
    ----------------                                                   818
#  二、指标标准化理想极值（指标取值规定）                              819
#  ----------------------------------------------------------------------  820
    ----------------                                                   821
IDEAL_SOIL_RATIO = 0.8          #  土石比理想值                        822
IDEAL_THICKNESS = 20.0          #  土层厚度理想值（cm）                823
IDEAL_PH_RANGE = (5.0, 9.0)     #  pH 理想区间（适中型指标）           824
IDEAL_ORGANIC = 30.0            #  有机质理想值                        825
    (g/kg)，全国第二次土壤普查养分等级划分标准二等最低值               826
IDEAL_COVERAGE = 75.0           #  植被覆盖度参照生态系统（%）         827
IDEAL_SHANNON = 0.7             #  Shannon-Wiener 多样性指数参照生态系统  828
#  重金属极值（mg/kg），《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值  829
#  取值依据：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800  830
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",              831
    "Zn")                                                              832
HEAVY_METAL_LIMIT = {                                                  833
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,                    834
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,                 835
}                                                                      836
#  污染指标下 8 种重金属"重要性相当，权重平均分配"（权重计算方法）    837
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0                                   838
#  坡度分界，取值规定引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号  839
SLOPE_BREAK = 25.0                                                     840
#  坡向离散赋值，取值依据："阳坡、阴阳坡、阴坡分别赋值 0.2、0.6、1"    841
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}              842
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")                              843
#  pH 适中型标准化的容差边界（区间外线性衰减，避免得分突变）          844
PH_TOL_LOW = 4.0                                                       845
PH_TOL_HIGH = 10.0                                                     846
#  ----------------------------------------------------------------------  847
    ----------------                                                   848
#  三、适宜性等级划分                                                  849
#  ----------------------------------------------------------------------  850
```

```
    ------------------                                                  851
GRADE_LEVELS = (                                                        852
    {"grade": "不适宜", "min": None, "max": 0.3,                        853
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",  854
     "measure": "生态重建、人工辅助"},                                  855
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,                       856
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",  857
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},  858
    {"grade": "适宜", "min": 0.7, "max": None,                          859
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",  860
     "measure": "自然恢复"},                                           861
)                                                                       862
def grade_of(score):                                                    863
    """按适宜性等级划分表判定等级。"""                                  864
    for g in GRADE_LEVELS:                                              865
        lo, hi = g["min"], g["max"]                                     866
        if lo is None and score < hi:                                   867
            return g                                                    868
        if hi is None and score >= lo:                                  869
            return g                                                    870
        if lo is not None and hi is not None and lo <=                  871
    score < hi:                                                         872
            return g                                                    873
    # score 恰好等于 0.7 时落入"适宜"                                   874
    return GRADE_LEVELS[-1]                                             875
# ------------------------------------------------------------------    876
    ------------------                                                  877
# 四、平均随机一致性指标 RI（权重计算方法给出 n=2，3，4，5，其余为常用扩展值）  878
# ------------------------------------------------------------------    879
    ------------------                                                  880
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6:               881
    1.24,                                                               882
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}                        883
# 一致性比例阈值（方法要求：当 CR<0.1 时，判断矩阵满足一致性要求）       884
CR_THRESHOLD = 0.1                                                      885
# ------------------------------------------------------------------    886
    ------------------                                                  887
# 五、默认权重方案（系统内置）                                          888
# ------------------------------------------------------------------    889
    ------------------                                                  890
# 指标取值规定："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，  891
# 分别计算评价指标权重"（方法概述与流程均作"碎石边坡"，                 892
# 系统统一采用"碎石边坡"）。权重由 10 位专家打分后经 AHP 方法确定。      893
# 下表为系统内置的归一化权重，作为系统默认方案。                        894
DEFAULT_WEIGHTS = {                                                     895
    "岩质边坡": {                                                       896
        "l1": {"植被指标": 0.45, "地形指标": 0.55},                     897
        "l2": {                                                         898
            "植被指标": {"植被覆盖度": 0.67, "生物多样性": 0.33},       899
            "地形指标": {"坡度": 0.64, "坡向": 0.36},                   900
```

```
            },
        },
        "碎石边坡":  {
            "l1":  {"土壤指标":  0.44,  "植被指标":  0.31,  "地形指标":  0.25},
            "l2":  {
                "土壤指标":  {"土石比":  0.25,  "土层厚度":  0.25,  "pH":  0.17,
                            "有机质":  0.17,  "污染指标":  0.17},
                "植被指标":  {"植被覆盖度":  0.64,  "生物多样性":  0.36},
                "地形指标":  {"坡度":  0.67,  "坡向":  0.33},
            },
        },
        "平台":  {
            "l1":  {"土壤指标":  0.54,  "植被指标":  0.46},
            "l2":  {
                "土壤指标":  {"土石比":  0.25,  "土层厚度":  0.25,  "pH":  0.17,
                            "有机质":  0.17,  "污染指标":  0.17},
                "植被指标":  {"植被覆盖度":  0.61,  "生物多样性":  0.39},
            },
        },
}
#  ----------------------------------------------------------------------
    ------------------
#  六、限制性指标（指标清单与评价流程）
#  ----------------------------------------------------------------------
    ------------------
#  四项限制性指标：政策、区位约束、土壤污染、地质灾害（地质环境）
CONSTRAINT_ITEMS  =  ("政策",  "区位约束",  "土壤污染",  "地质灾害")
#  ----------------------------------------------------------------------
    ------------------
#  七、账号职务
#  ----------------------------------------------------------------------
    ------------------
#  只分两种：管理员可管理账号、维护基础数据并查看操作日志，操作员用业务功能。
ROLE_LABELS  =  {"admin":  "管理员",  "operator":  "操作员"}
ROLE_CODES  =  ("admin",  "operator")
DEFAULT_ROLE  =  "operator"                #  新注册账号的默认职务
ROLE_ORDER  =  ("admin",  "operator")    #  列表排序用：管理员在前
#  性别（注册与个人资料中选填）
GENDERS  =  ("男",  "女",  "保密")
#  ====  app/utils/schema.py  ====
#  -*-  coding:  utf-8  -*-
"""轻量结构迁移。
SQLite  下  `db.create_all()`  只建缺失的表，不会给已有表补列。
本模块在启动时检查目标列是否存在，缺失则  ALTER  TABLE  补上，
使旧数据库无需重建即可用上新字段。
"""
from  sqlalchemy  import  text
#  （表名，  列名，  列定义）
REQUIRED_COLUMNS  =  [
    #  遥感影像挂接
```

```
        ("mine_patches", "image_path", "VARCHAR(256)"),
        ("mine_patches", "image_note", "VARCHAR(256)"),
        ("mine_patches", "image_uploaded_at", "DATETIME"),
        #  指标维护
        ("indicators", "enabled_units", "VARCHAR(128)"),
        ("indicators", "description", "TEXT"),
        #  多专家矩阵
        ("expert_matrices", "unit_type", "VARCHAR(32)"),
        ("expert_matrices", "l1_name", "VARCHAR(64)"),
        ("expert_matrices", "specialty", "VARCHAR(64)"),
        ("expert_matrices", "cr", "FLOAT"),
        #  用户个人资料（账户页）
        ("users", "display_name", "VARCHAR(64)"),
        ("users", "email", "VARCHAR(128)"),
        ("users", "phone", "VARCHAR(32)"),
        ("users", "gender", "VARCHAR(8)"),
        ("users", "avatar", "VARCHAR(256)"),
        #  评价结果的可追溯信息：计算参数快照与计算人
        ("evaluation_results", "params_json", "TEXT"),
        ("evaluation_results", "created_by", "INTEGER"),
]
#  职务体系由「管理员  /  评价员  /  查看者」三种收敛为「管理员  /  操作员」两种，
#  旧数据里的后两种统一归为操作员。
ROLE_MIGRATION = {"evaluator": "operator", "viewer":
    "operator"}
#  已经废弃的列（表名，  列名）：新库不会再建，旧库启动时删掉
RETIRED_COLUMNS = [
        #  「耕地属性」不再作为限制性指标
        ("constraint_checks", "farmland_pass"),
]
def _add_columns(db):
        added = []
        with db.engine.connect() as conn:
                for table, column, ddl in REQUIRED_COLUMNS:
                        rows = conn.execute(text("PRAGMA
        table_info(%s)" % table)).fetchall()
                        if not rows:
                                continue                                #
        表还不存在，create_all 会建
                        existing = {r[1] for r in rows}
                        if column in existing:
                                continue
                        conn.execute(text(
                                "ALTER TABLE %s ADD COLUMN %s %s" % (table,
        column, ddl)))
                        conn.commit()
                        added.append("%s.%s" % (table, column))
        return added
def _migrate_roles(db):
        """把旧角色名统一为新的两种，返回改动的行数。"""
```

```
    changed = 0                                                    1001
    with db.engine.connect() as conn:                              1002
        rows = conn.execute(text("PRAGMA                           1003
table_info(users)")).fetchall()                                    1004
        if not rows:                                               1005
            return 0                                               1006
        for old, new in ROLE_MIGRATION.items():                    1007
            r = conn.execute(text("UPDATE users SET role =         1008
:new "                                                             1009
                                  "WHERE role = :old"),            1010
                                 {"new": new, "old": old})         1011
            changed += r.rowcount or 0                             1012
        conn.commit()                                              1013
    return changed                                                 1014
def _drop_retired_columns(db):                                     1015
    """删掉已经废弃的列。                                          1016
    「耕地属性」不再作为限制性指标，`constraint_checks.farmland_pass`  1017
    随之废弃：新库由 create_all 直接按新模型建表，不会再有这一列；  1018
    旧库在这里删一次，避免源码与库结构长期对不上。                 1019
    SQLite 3.35 以下不支持 DROP COLUMN，失败就跳过，不影响启动。   1020
    """                                                            1021
    dropped = []                                                   1022
    for table, column in RETIRED_COLUMNS:                          1023
        try:                                                       1024
            with db.engine.connect() as conn:                      1025
                rows = conn.execute(text("PRAGMA                   1026
table_info(%s)" % table)).fetchall()                               1027
                if not rows or column not in {r[1] for r in        1028
rows}:                                                             1029
                    continue                                       1030
                conn.execute(text("ALTER TABLE %s DROP             1031
COLUMN %s" % (table, column)))                                     1032
                conn.commit()                                      1033
                dropped.append("%s.%s" % (table, column))          1034
        except Exception:                                  # 老版本  1035
SQLite 不支持就跳过                                                1036
            continue                                               1037
    return dropped                                                 1038
def ensure_schema(db):                                             1039
    """为已有表补齐缺失列、删掉废弃列，并做必要的数据迁移。返回变更清单。"""  1040
    try:                                                           1041
        changed = _add_columns(db)                                 1042
        changed += _drop_retired_columns(db)                       1043
        n = _migrate_roles(db)                                     1044
        if n:                                                      1045
            changed.append("用户角色归一为管理员 / 操作员 %d 条" % n)  1046
        return changed                                             1047
    except Exception as e:                                  #      1048
迁移失败不应阻断启动                                               1049
        print("[schema] 结构检查跳过：%s" % e)                     1050
```

```
        return []
# ==== app/utils/uploads.py ====
# -*- coding: utf-8 -*-
"""上传文件处理（遥感影像、批量导入）。"""
import os
import re
import uuid
from datetime import datetime
from werkzeug.utils import secure_filename
def _safe_ext(filename):
    return os.path.splitext(filename)[1].lower().lstrip(".")
def ext_allowed(filename, allowed):
    return _safe_ext(filename) in allowed
def save_upload(file_storage, subdir, allowed_ext,
    upload_root):
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。
    """
    if file_storage is None or not file_storage.filename:
        return None, "未选择文件"
    ext = _safe_ext(file_storage.filename)
    if ext not in allowed_ext:
        return None, "不支持的文件类型：.%s" % ext
    target_dir = os.path.join(upload_root, subdir)
    os.makedirs(target_dir, exist_ok=True)
    # 仅保留安全字符，并加时间戳与随机串防重名
    base = secure_filename(os.path.splitext(file_storage.fil
    ename)[0]) or "file"
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or
    "file"
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M
    %S"),
                         uuid.uuid4().hex[:6], ext)
    abs_path = os.path.join(target_dir, name)
    file_storage.save(abs_path)
    rel = os.path.join("uploads", subdir,
    name).replace("\\", "/")
    return rel, None
def remove_upload(rel_path, upload_root):
    """删除已保存的上传文件（相对 static 的路径）。
    路径必须落在上传根目录之内：rel_path 来自数据库，正常都是 save_upload
    生成的，但万一被改脏（手工改库、导入异常），`../` 这种值会删掉上传根目录
    之外的文件。这里用 realpath 做一次围栏校验。
    """
    if not rel_path:
        return
    root = os.path.realpath(upload_root)
    abs_path = os.path.realpath(
        os.path.join(root, rel_path.replace("uploads/", "",
    1)))
```

```
    if abs_path != root and not abs_path.startswith(root +
    os.sep):
        return
    try:
        if os.path.isfile(abs_path):
            os.remove(abs_path)
    except OSError:
        pass
# ==== app/utils/charts.py ====
# -*- coding: utf-8 -*-
"""轻量图表。
用纯 SVG 输出，不依赖任何绘图库或前端脚本：
- 服务端渲染，页面与打印（PDF）中都能稳定显示
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）
- 单色方案，符合界面「一个视图一个强调色」的约定
"""
from html import escape
ACCENT = "#559FCF"
ACCENT_SOFT = "#64BBE0"
INK = "#23262b"
MUTED = "#6b7280"
LINE = "#dfe3e8"
# 科研顶刊配色
GRAYS = ["#6FAE47", "#FCE5D6", "#FFF2CC", "#DAE3F2",
    "#E9C9C9", "#559FCF"]
def _num(v):
    """图表里容忍空值：None / 非数字按 0 处理，避免一条脏数据把页面打成 500。"""
    try:
        return float(v)
    except (TypeError, ValueError):
        return 0.0
def _fmt(v, digits=4):
    try:
        return ("%%.%df" % digits) % float(v)
    except (TypeError, ValueError):
        return str(v)
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,
    value_w=76):
    """水平条形图。items: [(标签, 数值, 显示文本)]"""
    if not items:
        return '<p class="muted">暂无数据</p>'
    n = len(items)
    height = n * (bar_h + gap) + 10
    max_v = max((abs(_num(v)) for _, v, _ in items),
    default=1) or 1
    body_w = width - label_w - value_w
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"
    height="%d" '
             'role="img" aria-label="条形图" '
             'style="max-width:%dpx;font-family:inherit;">'
```

```
    % (width, height, height, width)]
    for i, (label, value, disp) in enumerate(items):
        y = i * (bar_h + gap) + 5
        frac = (abs(_num(value)) / max_v) if max_v else 0
        w = max(2.0, body_w * frac)
        fill = GRAYS[i % len(GRAYS)]
        parts.append(
            '<text x="%d" y="%d" text-anchor="end"
font-size="12" fill="%s">%s</text>'
            % (label_w - 10, y + bar_h * 0.72, INK,
escape(str(label))))
        parts.append(
            '<rect x="%d" y="%d" width="%.1f" height="%d"
rx="3" fill="%s"/>'
            % (label_w, y, w, bar_h, fill))
        parts.append(
            '<text x="%d" y="%d" font-size="12" fill="%s" '
            'style="font-variant-numeric:tabular-nums;">%s</
text>'
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,
escape(str(disp))))
    parts.append('</svg>')
    return "".join(parts)
def donut(items, size=210, thickness=34):
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。
    """
    vals = [(l, _num(v), d) for l, v, d in items if _num(v)
> 0]
    if not vals:
        return '<p class="muted">暂无数据</p>'
    total = sum(v for _, v, _ in vals) or 1
    r = (size - thickness) / 2
    cx = cy = size / 2
    import math
    parts = ['<svg viewBox="0 0 %d %d" width="%d"
height="%d" role="img" '
             'aria-label="权重分布环形图">' % (size, size, size,
size)]
    angle = -math.pi / 2
    for i, (label, v, disp) in enumerate(vals):
        sweep = 2 * math.pi * (v / total)
        a0, a1 = angle, angle + sweep
        x0, y0 = cx + r * math.cos(a0), cy + r *
math.sin(a0)
        x1, y1 = cx + r * math.cos(a1), cy + r *
math.sin(a1)
        large = 1 if sweep > math.pi else 0
        parts.append(
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f
```

```
    %.2f" '                                                              1201
          'fill="none" stroke="%s" stroke-width="%d"/>'                  1202
         % (x0, y0, r, r, large, x1, y1, GRAYS[i %                      1203
    len(GRAYS)], thickness))                                            1204
        angle = a1                                                      1205
    parts.append('<text x="%d" y="%d" text-anchor="middle"              1206
    font-size="12" '                                                    1207
                 'fill="%s">合计</text>' % (cx, cy - 2,                 1208
    MUTED))                                                             1209
    parts.append('<text x="%d" y="%d" text-anchor="middle"              1210
    font-size="15" '                                                    1211
                 'fill="%s" style="font-variant-numeric:tabu            1212
    lar-nums;">%s</text>'                                               1213
                % (cx, cy + 18, INK, _fmt(total, 4)))                   1214
    parts.append('</svg>')                                              1215
    legend = ['<table style="width:auto;margin-top:10px;"><t            1216
    head><tr>'                                                          1217
              '<th scope="col">项目</th><th                             1218
    scope="col">数值</th>'                                              1219
              '<th scope="col">占比</th></tr></thead><tbody>']          1220
    for i, (label, v, disp) in enumerate(vals):                         1221
        legend.append(                                                  1222
            '<tr><td style="text-align:left;">'                         1223
            '<span style="display:inline-block;width:10px;he            1224
    ight:10px;'                                                         1225
            'border-radius:2px;background:%s;margin-right:6p            1226
    x;"></span>%s</td>'                                                 1227
            '<td class="mono">%s</td><td                                1228
    class="mono">%.1f%%</td></tr>'                                      1229
           % (GRAYS[i % len(GRAYS)], escape(str(label)),                1230
    escape(str(disp)),                                                  1231
              100.0 * v / total))                                       1232
    legend.append('</tbody></table>')                                   1233
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p            1234
    x;align-items:center;">'                                            1235
            + "".join(parts) + "".join(legend) + '</div>')              1236
# ==== app/utils/browser.py ====                                        1237
# -*- coding: utf-8 -*-                                                 1238
"""HTML 转 PDF（报告导出）。                                            1239
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、          1240
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，       1241
避免中文路径在 file:// URL 下出问题。                                   1242
"""                                                                     1243
import os                                                               1244
import shutil                                                           1245
import subprocess                                                       1246
import tempfile                                                         1247
import uuid                                                             1248
CANDIDATES = [                                                          1249
    r"C:\Program Files\Google\Chrome\Application\chrome.exe"            1250
```

```
    ,                                                                    1251
    r"C:\Program Files (x86)\Google\Chrome\Application\chrom1252
    e.exe",                                                              1253
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msed1254
    ge.exe",                                                             1255
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe1256
    ",                                                                   1257
]                                                                        1258
#  允许通过环境变量覆盖，便于在其他机器上指定浏览器                                  1259
ENV_KEY = "DSH_BROWSER_PATH"                                             1260
def browser_path():                                                      1261
    """定位可用的浏览器可执行文件，找不到返回 None。"""                              1262
    override = os.environ.get(ENV_KEY)                                   1263
    if override and os.path.isfile(override):                            1264
        return override                                                  1265
    for p in CANDIDATES:                                                 1266
        if os.path.isfile(p):                                            1267
            return p                                                     1268
    for name in ("chrome", "msedge", "chromium"):                        1269
        found = shutil.which(name)                                       1270
        if found:                                                        1271
            return found                                                 1272
    return None                                                          1273
def html_to_pdf(html, timeout=90):                                       1274
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""                1275
    exe = browser_path()                                                 1276
    if exe is None:                                                      1277
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"              1278
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)            1279
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")                          1280
    tag = uuid.uuid4().hex[:8]                                           1281
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)                  1282
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)                    1283
    try:                                                                 1284
        with open(html_path, "w", encoding="utf-8") as f:                1285
            f.write(html)                                                1286
        cmd = [                                                          1287
            exe, "--headless=new", "--disable-gpu",                      1288
    "--no-sandbox",                                                      1289
            "--no-first-run", "--no-default-browser-check",              1290
            "--no-pdf-header-footer",                                    1291
            "--virtual-time-budget=4000",                                1292
            "--print-to-pdf=%s" % pdf_path,                              1293
            "file:///" + html_path.replace("\\", "/"),                   1294
        ]                                                                1295
        p = subprocess.run(cmd, capture_output=True,                     1296
    text=True,                                                           1297
                           timeout=timeout,                              1298
    encoding="utf-8", errors="replace")                                  1299
        if not os.path.isfile(pdf_path) or                               1300
```

```
    os.path.getsize(pdf_path) < 800:
        return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
"").strip()[:200]
    with open(pdf_path, "rb") as f:
        return f.read(), None
    except subprocess.TimeoutExpired:
        return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
        return None, "生成 PDF 失败：%s" % e
    finally:
        shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
按权重计算方法实现：
- 式（2）（3）（4）：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式（5）（6）（7）：一致性检验（λmax、CI、CR）
- 方法要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式（2） 行元素乘积
```

```
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
                        "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
```

```
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按标准化公式实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
```

```
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    取值规定：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（取值规定）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals, limits=None):
    """污染指标：8 种重金属等权（各 0.125）合成。
    方法约定：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    limits : 可选，金属 → 极值；不传则用代码常量（界面改过就传库里的值）。
    """
    table = limits or HEAVY_METAL_LIMIT
    total = 0.0
    for metal, extreme in table.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(table)
def standardize_unit(unit_type, data, weights=None,
    settings=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
    HTS 结构），
                未提供时用默认值。
    settings : app.services.unit_config.UnitSettings，图斑级的基准值
    与理想取值。
                不传则全部按代码常量 —— 历史数据与验证算例走这条路径，结果不变。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
```

```
    """                                                                 1501
    from app.utils.constants import DEFAULT_WEIGHTS                     1502
    from app.services import unit_config as uc                          1503
    if weights is None:                                                 1504
        weights = DEFAULT_WEIGHTS[unit_type]                            1505
    if settings is None:                                                1506
        settings = uc.UnitSettings()                                    1507
    #  理想取值：库里有数字就用库里的，否则用常量                                1508
    def ideal(code, fallback):                                          1509
        return settings.ideal(code, fallback)                           1510
    l1_enabled = weights["l1"].keys()                                   1511
    l2w = weights["l2"]                                                 1512
    result = {}                                                         1513
    #  一级指标是否启用：库里配过就按库里的（界面上的「启用单元」），没配按常量                1514
    def group_on(l1_name, l1_code):                                     1515
        return settings.group_enabled(unit_type, l1_name,               1516
    l1_code) is not False                                               1517
    #  ---- 土壤指标 ----                                               1518
    if "土壤指标" in l1_enabled and group_on("土壤指标", "soil"):           1519
        result["土壤指标"] = {                                            1520
            "土石比": (data.get("soil_ratio"),                          1521
                       std_benefit(data.get("soil_ratio"),              1522
                                   ideal("soil_ratio",                  1523
    IDEAL_SOIL_RATIO)),                                                 1524
                       l2w["土壤指标"]["土石比"]),                          1525
            "土层厚度": (data.get("thickness"),                         1526
                        std_benefit(data.get("thickness"),              1527
                                    ideal("thickness",                  1528
    IDEAL_THICKNESS)),                                                  1529
                        l2w["土壤指标"]["土层厚度"]),                       1530
            "pH": (data.get("ph"),                                      1531
                   std_mid_ph(data.get("ph"),                           1532
    *IDEAL_PH_RANGE),                                                   1533
                   l2w["土壤指标"]["pH"]),                               1534
            "有机质": (data.get("organic_matter"),                      1535
                      std_benefit(data.get("organic_matter")            1536
    ,                                                                   1537
                                  ideal("organic_matter",               1538
    IDEAL_ORGANIC)),                                                    1539
                      l2w["土壤指标"]["有机质"]),                          1540
            "污染指标": (data.get("heavy_metals") or {},                 1541
                        pollution_score(data.get("heavy_meta            1542
    ls") or {},                                                         1543
                                                                        1544
    uc.metal_limits(settings)),                                         1545
                        l2w["土壤指标"]["污染指标"]),                       1546
        }                                                               1547
    #  ---- 植被指标 ----                                               1548
    if "植被指标" in l1_enabled and group_on("植被指标", "veg"):           1549
        result["植被指标"] = {                                            1550
```

```
                "植被覆盖度": (data.get("coverage"),                    1551
                               std_benefit(data.get("coverage"),        1552
                                           ideal("coverage",            1553
    IDEAL_COVERAGE)),                                                   1554
                               l2w["植被指标"]["植被覆盖度"]),          1555
                "生物多样性": (data.get("shannon"),                     1556
                               std_benefit(data.get("shannon"),         1557
                                           ideal("shannon",             1558
    IDEAL_SHANNON)),                                                    1559
                               l2w["植被指标"]["生物多样性"]),          1560
            }                                                           1561
    # ---- 地形指标 ----                                               1562
    if "地形指标" in l1_enabled and group_on("地形指标", "terrain"):    1563
        result["地形指标"] = {                                          1564
            "坡度": (data.get("slope"),                                 1565
                     std_slope(data.get("slope")),                      1566
                     l2w["地形指标"]["坡度"]),                          1567
            "坡向": (data.get("aspect"),                                1568
                     aspect_score(data.get("aspect")),                  1569
                     l2w["地形指标"]["坡向"]),                          1570
        }                                                               1571
    return result                                                       1572
# ==== app/services/weight_service.py ====                            1573
# -*- coding: utf-8 -*-                                               1574
"""权重方案的存取与生效权重解析。                                       1575
系统有两套权重来源：                                                   1576
1. 系统内置的默认权重（app.utils.constants.DEFAULT_WEIGHTS），         1577
   是**权威取值来源**，案例复算与报告结论以此为准；                    1578
2. 通过层次分析法在本系统中实际算出的权重，保存在                      1579
   weight_schemes / weight_items 两张表中，可标记为「生效」。          1580
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：               1581
以系统内置权重为底，用生效方案中已保存的条目覆盖。                     1582
没有任何生效方案时，即完全使用系统内置权重。                           1583
这样既保证默认行为与内置取值一致，又使 AHP 计算结果真正参与评价，      1584
而不是算完即弃。                                                       1585
"""                                                                    1586
import copy                                                            1587
from app.models import db, WeightScheme, WeightItem                    1588
from app.utils.constants import DEFAULT_WEIGHTS                        1589
from app.services.indicator_service import TREE                        1590
# 方案名称：每个微地质单元类型固定一个可生效的方案                      1591
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"                         1592
def _code_maps():                                                      1593
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""    1594
    l1, l2 = {}, {}                                                    1595
    for row in TREE:                                                   1596
        code, name, level, parent = row[0], row[1], row[2],            1597
    row[3]                                                             1598
        if level == 1:                                                 1599
            l1[code] = name                                            1600
```

```
            elif level == 2:                                                1601
                l2[code] = (l1.get(parent, parent), name)                   1602
        return l1, l2                                                       1603
def effective_weights(unit_type):                                           1604
    """返回该单元类型当前生效的权重。                                       1605
    返回 (weights, source_label, scheme)                                    1606
        weights      嵌套权重结构，可直接交给标准化与评价引擎               1607
        source_label 权重来源说明，用于界面与报告标注                       1608
        scheme       生效的权重方案对象，无自定义方案时为 None              1609
    """                                                                     1610
    base = DEFAULT_WEIGHTS.get(unit_type)                                   1611
    if base is None:                                                        1612
        return None, "未定义该微地质单元类型的权重", None                   1613
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ                1614
    e,                                                                      1615
                                                                            1616
    name=ACTIVE_NAME).first()                                               1617
    if scheme is None:                                                      1618
        return copy.deepcopy(base), "系统默认权重", None                    1619
    merged = copy.deepcopy(base)                                            1620
    l1_map, l2_map = _code_maps()                                           1621
    applied = 0                                                             1622
    for item in WeightItem.query.filter_by(scheme_id=scheme.                1623
    id).all():                                                              1624
        code, w = item.indicator_code, item.weight                          1625
        if w is None or w <= 0:                                             1626
            continue                                                        1627
        if code in l1_map and l1_map[code] in merged["l1"]:                 1628
            merged["l1"][l1_map[code]] = w                                  1629
            applied += 1                                                    1630
        elif code in l2_map:                                                1631
            l1_name, l2_name = l2_map[code]                                 1632
            if l1_name in merged["l2"] and l2_name in                       1633
    merged["l2"][l1_name]:                                                  1634
                merged["l2"][l1_name][l2_name] = w                          1635
                applied += 1                                                1636
    if applied == 0:                                                        1637
        return copy.deepcopy(base), "系统默认权重", None                    1638
    return merged, "自定义方案（%d 项覆盖系统内置权重）" % applied, scheme   1639
def save_weights(unit_type, code_weight_pairs, note=None):                  1640
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。      1641
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）          1642
    分次录入而无需一次填完。                                                1643
    """                                                                     1644
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ                1645
    e,                                                                      1646
                                                                            1647
    name=ACTIVE_NAME).first()                                               1648
    if scheme is None:                                                      1649
        scheme = WeightScheme(unit_type=unit_type,                          1650
```

```
    name=ACTIVE_NAME,                                                  1651
                                    level=1, is_default=True)          1652
        db.session.add(scheme)                                         1653
        db.session.flush()                                             1654
    if note:                                                           1655
        scheme.level = scheme.level or 1                               1656
    if not scheme.is_default:                                          1657
        scheme.is_default = True                                       1658
    written = 0                                                        1659
    for code, w in code_weight_pairs:                                  1660
        if w is None or w <= 0:                                        1661
            continue                                                   1662
        item = WeightItem.query.filter_by(scheme_id=scheme.i           1663
    d,                                                                 1664
                                                                       1665
    indicator_code=code).first()                                       1666
        if item is None:                                               1667
            item = WeightItem(scheme_id=scheme.id,                     1668
    indicator_code=code)                                               1669
            db.session.add(item)                                       1670
        item.weight = float(w)                                         1671
        written += 1                                                   1672
    db.session.commit()                                                1673
    return scheme, written                                             1674
def reset_weights(unit_type):                                          1675
    """删除该单元类型的自定义方案，恢复为系统默认权重。"""                1676
    schemes = WeightScheme.query.filter_by(unit_type=unit_ty           1677
    pe,                                                                1678
                                                                       1679
    name=ACTIVE_NAME).all()                                            1680
    n = 0                                                              1681
    for s in schemes:                                                  1682
        WeightItem.query.filter_by(scheme_id=s.id).delete(             1683
            synchronize_session=False)                                 1684
        db.session.delete(s)                                           1685
        n += 1                                                         1686
    db.session.commit()                                                1687
    return n                                                           1688
def scheme_items(unit_type):                                           1689
    """返回该单元类型已保存的自定义权重条目 {指标编码： 权重}。"""          1690
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ           1691
    e,                                                                 1692
                                                                       1693
    name=ACTIVE_NAME).first()                                          1694
    if scheme is None:                                                 1695
        return {}, None                                                1696
    items = {i.indicator_code: i.weight                                1697
             for i in WeightItem.query.filter_by(scheme_id=s           1698
    cheme.id).all()}                                                   1699
    return items, scheme                                               1700
```

```
def summary():
    """各单元类型的权重来源概览，供权重首页展示。"""
    rows = []
    for unit_type in DEFAULT_WEIGHTS:
        weights, label, scheme = effective_weights(unit_type
    )
        rows.append({
            "unit_type": unit_type,
            "label": label,
            "custom": scheme is not None,
        })
    return rows
#  ==== app/services/evaluation_service.py ====
#  -*- coding: utf-8 -*-
"""适宜性评价引擎。
按加权合成公式实现：
- 单元级：F'  =  Σ（二级指标权重  ×  该二级指标下三级指标标准化值）
- 矿山级：F  =  Σ（Si  ×  F'i），Si  为各微地质单元面积占比
- 等级判定按适宜性等级划分表。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None,
    settings=None):
    """计算单个微地质单元的评价因子  F'。
    参数
    ----
    weights   :  该单元类型的权重；不传用系统默认权重
    settings  :  图斑级标准化参数（基准值与理想取值）；不传全部按代码常量
    返回
    ----
    dict：{  l1_scores:  {一级指标:  得分},  unit_score:  F',  details:
    [...]  }
    若某一级指标下存在"不适用"（标准化值为  None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    std_result = std.standardize_unit(unit_type, data,
    weights, settings)
    l1_weights = weights["l1"]
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                #  不适用项：不计入
                details.append({
```

```
                    "l1": l1_name, "l2": l2_name, "raw":
    raw,
                    "std_value": None, "weight": w2,
    "contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
                "contribution": contribution,
            })
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
    #  一级指标权重按实际参与计算的指标归一化。
    #  内置权重本身已归一（和为 1），此处为恒等变换；
    #  若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。
    lw_sum = sum(l1_weights[k] for k in l1_scores if k in
    l1_weights)
    if lw_sum > 0:
        unit_score = sum(l1_weights[k] * l1_scores[k]
                         for k in l1_scores if k in
    l1_weights) / lw_sum
    else:
        unit_score = 0.0
    return {"l1_scores": l1_scores, "unit_score":
    unit_score, "details": details}
def score_mine(units, weights_map=None, settings=None):
    """计算整个矿山的评价因子 F 与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    weights_map : dict，{单元类型: 权重结构}。未提供的单元类型
        使用系统默认权重。权重可来自 AHP 计算并经
        weight_service 保存生效，从而真正参与评价。
    settings : 图斑级标准化参数（参照生态系统基准、指标理想取值），
        由 app.services.unit_config.for_patch() 生成；不传按代码常量。
    返回
    ----
    dict:{ final_score, grade, grade_info, unit_results }
    """
    weights_map = weights_map or {}
    total = 0.0
    unit_results = []
    for u in units:
        r = score_unit(u["unit_type"], u["data"],
                       weights_map.get(u["unit_type"]),
```

```
    settings)                                                    1801
        weighted = u["ratio"] * r["unit_score"]                  1802
        total += weighted                                        1803
        unit_results.append({                                    1804
            "unit_type": u["unit_type"],                         1805
            "ratio": u["ratio"],                                 1806
            "unit_score": r["unit_score"],                       1807
            "weighted": weighted,                                1808
            "l1_scores": r["l1_scores"],                         1809
            "details": r["details"],                             1810
        })                                                       1811
    grade_info = grade_of(total)                                 1812
    return {                                                     1813
        "final_score": total,                                    1814
        "grade": grade_info["grade"],                            1815
        "grade_info": grade_info,                                1816
        "unit_results": unit_results,                            1817
    }                                                            1818
# ==== app/services/report_service.py ====                       1819
# -*- coding: utf-8 -*-                                          1820
"""报告生成：生成 Word 评价报告。                                  1821
格式（规范体例）：                                               1822
- 题目：三号（16pt） 黑体 居中                                   1823
- 一级标题：四号（14pt） 黑体 居中                               1824
- 正文/表格：小四号（12pt） 宋体                                 1825
- 行距：单倍行距                                                 1826
- 全部黑色                                                       1827
"""                                                              1828
import io                                                        1829
import json                                                      1830
from docx import Document                                        1831
from docx.shared import Pt, Cm, RGBColor                         1832
from docx.oxml.ns import qn                                      1833
from docx.enum.text import WD_ALIGN_PARAGRAPH,                   1834
    WD_LINE_SPACING                                              1835
from app.utils.timefmt import to_local, now_utc                  1836
from app.services import workflow as wf                          1837
BLACK = RGBColor(0, 0, 0)                                        1838
def _set_run(run, text=None, size=12, bold=False,                1839
    east_asia="宋体"):                                           1840
    if text is not None:                                         1841
        run.text = text                                          1842
    run.font.name = "Times New Roman"                            1843
    run.font.size = Pt(size)                                     1844
    run.font.bold = bold                                         1845
    run.font.color.rgb = BLACK                                   1846
    run._element.rPr.rFonts.set(qn("w:eastAsia"), east_asia)     1847
    return run                                                   1848
def _set_line_spacing(p):                                        1849
    """单倍行距。"""                                             1850
```

```
    p.paragraph_format.line_spacing_rule =                          1851
    WD_LINE_SPACING.SINGLE                                          1852
def _para(doc, text, size=12, bold=False, align=None,               1853
    space_after=6, east_asia="宋体"):                               1854
    p = doc.add_paragraph()                                         1855
    if align is not None:                                           1856
        p.alignment = align                                         1857
    p.paragraph_format.space_after = Pt(space_after)                1858
    _set_line_spacing(p)                                            1859
    _set_run(p.add_run(), text, size, bold, east_asia)              1860
    return p                                                        1861
def _heading(doc, text, level=1):                                   1862
    """level=0 题目（三号黑体居中）；level=1 一级标题（四号黑体居中）。"""  1863
    p = doc.add_paragraph()                                         1864
    p.alignment = WD_ALIGN_PARAGRAPH.CENTER                         1865
    p.paragraph_format.space_before = Pt(12)                        1866
    p.paragraph_format.space_after = Pt(8)                          1867
    _set_line_spacing(p)                                            1868
    if level == 0:                                                  1869
        size = 16  # 三号                                           1870
    else:                                                           1871
        size = 14  # 四号                                           1872
    _set_run(p.add_run(), text, size=size, bold=True,               1873
    east_asia="黑体")                                               1874
    return p                                                        1875
def _table(doc, headers, rows, widths=None):                        1876
    t = doc.add_table(rows=1, cols=len(headers))                    1877
    t.style = "Table Grid"                                          1878
    hdr = t.rows[0].cells                                           1879
    for i, htext in enumerate(headers):                             1880
        cell_p = hdr[i].paragraphs[0]                               1881
        _set_line_spacing(cell_p)                                   1882
        _set_run(cell_p.add_run(), htext, 12, bold=True)            1883
    for row in rows:                                                1884
        cells = t.add_row().cells                                   1885
        for i, val in enumerate(row):                               1886
            cell_p = cells[i].paragraphs[0]                         1887
            _set_line_spacing(cell_p)                               1888
            _set_run(cell_p.add_run(), str(val), 12)                1889
    return t                                                        1890
def build_report(patch, mine_result, record=None,                   1891
    weak=None, operator=None,                                       1892
                  check=None):                                      1893
    """生成评价报告 Word 文档，返回 BytesIO。                          1894
    check : 标准结论对照（app.services.standard_check.compare 的结果）。1895
            标准不参与计算，只在报告里回答「结论是否符合当地文件要求」。   1896
    """                                                             1897
    doc = Document()                                                1898
    for section in doc.sections:                                    1899
        section.top_margin = Cm(2.5)                                1900
```

```
        section.bottom_margin = Cm(2.5)
        section.left_margin = Cm(2.5)
        section.right_margin = Cm(2.5)
    # 题目：三号黑体居中
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0",
            size=12, align=WD_ALIGN_PARAGRAPH.CENTER)
    _para(doc, "生成时间：%s" % to_local(now_utc(), "%Y-%m-%d
%H：%M：%S"),
            size=12, align=WD_ALIGN_PARAGRAPH.CENTER,
space_after=12)
    # 计算人与参数来源
    if record is not None:
        from app.views.evaluate import param_source_summary
        try:
            params = json.loads(record.params_json) if
record.params_json else {}
        except ValueError:
            params = {}
        meta = [
            ("计算人", operator or "—"),
            ("计算时间", to_local(record.created_at, "%Y-%m-%d
%H：%M：%S")),
            ("评价记录编号", record.id),
            ("计算参数", param_source_summary(params)),
        ]
        _table(doc, ["项目", "内容"], meta)
    # 图斑信息
    _heading(doc, "一、图斑基本信息", 1)
    info = [
        ("图斑名称", patch.name),
        ("所在行政区", patch.admin_region or "—"),
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is
not None else "—"),
        ("开采矿种", patch.mineral_type or "—"),
        ("开采方式", patch.mining_method or "—"),
    ]
    _table(doc, ["项目", "内容"], info)
    # 评价结论
    _heading(doc, "二、评价结论", 1)
    gi = mine_result["grade_info"]
    concl = [
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),
        ("适宜性等级", mine_result["grade"]),
        ("判定依据", gi["meaning"]),
        ("推荐措施", gi["measure"]),
    ]
    _table(doc, ["项目", "结果"], concl)
    # 各单元得分
    _heading(doc, "三、各微地质单元得分", 1)
```

```
    rows = []
    for u in mine_result["unit_results"]:
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"] * 100),
                     "%.4f" % u["unit_score"], "%.4f" % u["weighted"]))
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"],
           rows)
    # 与当地标准 / 政策文件的对照（不参与计算，只回答结论是否符合文件要求）
    if check and check.get("covered"):
        _heading(doc, "四、与地方标准 / 政策文件的对照", 1)
        _para(doc, "图斑所在行政区：%s（对照 %s 的标准与政策文件）。"
                   "对照只把文件里的判定项逐条映射到系统已有数据，"
                   "系统内没有对应数据的定性要求一律标为待人工判断。"
                   % (check.get("region") or "未填写",
                      check.get("province") or ""),
              size=12, space_after=6)
        for s in check.get("standards", []):
            _para(doc, "%s%s" % (s["name"],
                                 ("（%s）" % s["doc_number"])
                                 if s.get("doc_number") else ""),
                  size=12, bold=True, space_after=4)
            _table(doc, ["文件判定项", "条件", "类型", "对照结果", "依据"],
                   [(r["item"], "%s %s" % (r["operator"],
                                           r["threshold"] or ""),
                     r["condition_type"] or "—",
                     r["verdict"], r["evidence"])
                    for r in s["rules"]])
        t = check.get("totals", {})
        _para(doc, "合计：符合 %d 条、不符合 %d 条、待人工判断 %d 条。"
                   % (t.get("符合", 0), t.get("不符合", 0),
                      t.get("待人工判断", 0)),
              size=12)
    # 完整得分明细
    _heading(doc, "五、得分明细", 1)
    for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"],
                                          u["ratio"] * 100),
              size=12, bold=True, space_after=4)
        detail_rows = []
        for d in u["details"]:
            raw = "—" if d["raw"] is None else d["raw"]
            std = "—" if d["std_value"] is None else "%.4f"
                  % d["std_value"]
            w = "%.4f" % d["weight"]
            c = "—" if d["contribution"] is None else
                "%.4f" % d["contribution"]
            detail_rows.append((d["l1"], d["l2"], raw, std,
                                w, c))
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重",
```

```
        "贡献"],                                                                2001
                    detail_rows)                                                2002
            l1_row = []                                                         2003
            for k, v in u["l1_scores"].items():                                 2004
                l1_row.append("%s=%.4f" % (k, v))                               2005
            _para(doc, "一级指标得分：" + ",".join(l1_row), size=12,            2006
        space_after=10)                                                         2007
        # 数据完整度                                                            2008
        undetected = wf.undetected_metals(patch.id)                             2009
        if undetected:                                                          2010
            _heading(doc, "六、数据完整度说明", 1)                              2011
            parts = ["%s：%s" % (t, "、".join(m)) for t, m in                   2012
        undetected]                                                             2013
            _para(doc, "以下重金属项未检测，按方法约定以「未受污染」计入污染指标"  2014
                       "（并非实测达标）：%s。" % "；".join(parts),              2015
        size=12)                                                                2016
        # 短板提示                                                              2017
        if weak:                                                                2018
            _heading(doc, "七、短板提示", 1)                                    2019
            _para(doc, "按「权重 ×（1 − 标准化值）」排序，提升下列指标对总分的拉动最大。",  2020
                    size=12, space_after=4)                                     2021
            _table(doc, ["微地质单元", "指标", "实测值", "标准化值", "提升空间"],  2022
                    [(u, l2, "—" if raw is None else raw, "%.4f"                2023
        % std,                                                                  2024
                      "%.4f" % gain) for u, l2, raw, std, gain                  2025
        in weak])                                                               2026
        # 签字栏                                                               2027
        _para(doc, "", size=12, space_after=18)                                 2028
        _para(doc, "评价人（签字）：_____________________", size=12)            2029
        _para(doc, "日期：_____________________", size=12)                      2030
        buf = io.BytesIO()                                                      2031
        doc.save(buf)                                                           2032
        buf.seek(0)                                                             2033
        return buf                                                              2034
# ==== app/services/indicator_service.py ====                                   2035
# -*- coding: utf-8 -*-                                                         2036
"""指标定义维护。                                                              2037
把内置指标体系写入 indicators 表，使指标名称、单位、                           2038
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。                 2039
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；                  2040
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。         2041
"""                                                                             2042
from app.models import db, Indicator                                            2043
from app.utils.constants import (                                               2044
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,                      2045
    IDEAL_THICKNESS,                                                            2046
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,                              2047
    IDEAL_SHANNON,                                                              2048
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,                              2049
)                                                                               2050
```

```
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "评价指标体系的一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "评价指标体系的一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "评价指标体系的一级指标", False),
    # ---- 二级指标：土壤 ----
    ("soil_ratio", "土石比", 2, "soil", "—",
    str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
     "取值规定：理想值 0.8，根据矿山修复实际情况确定", False),
    ("thickness", "土层厚度", 2, "soil", "cm",
    str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
     "取值规定：理想值 20 cm", False),
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
    "mid", "碎石边坡,平台",
     "取值规定：pH 用适中指标的标准化方法，理想区间 %g~%g" % IDEAL_PH_RANGE,
    False),
    ("organic_matter", "有机质", 2, "soil", "g/kg",
    str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
     "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),
    ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
    "cost", "碎石边坡,平台",
     "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
    # ---- 二级指标：植被 ----
    ("coverage", "植被覆盖度", 2, "veg", "%",
    str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
     "取值规定：参照生态系统取值 75%", False),
    ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
    "benefit", ALL_UNITS,
     "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
    # ---- 二级指标：地形 ----
    ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
    SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
     "取值规定引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25° 为陡坡重要分界",
    False),
    ("aspect", "坡向", 2, "terrain", "—",
    "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
     "取值规定：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),
]
# 污染指标下的三级指标（8 种重金属）
for _m, _limit in HEAVY_METAL_LIMIT.items():
    TREE.append((
        "hm_" + _m, _m, 3, "pollution", "mg/kg",
    str(_limit), "cost",
        "碎石边坡,平台",
```

```
            "《绿化种植土壤》ＣＪ/Ｔ ３４０—２０１６ 土壤重金属含量 ＩＶ 级最高值 %g mg/kg" %
    _limit,
            False,
    ))
def seed_indicators():
    """写入/更新指标定义（幂等）。返回写入条数。"""
    code_map = {
        "benefit": "越大越优", "cost": "越小越优",
        "mid": "适中型", "discrete": "离散赋值",
    }
    n = 0
    for (code, name, level, parent, unit, ideal, itype,
    units, desc, is_con) in TREE:
        ind = Indicator.query.filter_by(code=code).first()
        if ind is None:
            ind = Indicator(code=code)
            db.session.add(ind)
        ind.name = name
        ind.level = level
        ind.parent_code = parent
        ind.unit = unit
        ind.ideal_value = ideal
        ind.ideal_type = itype
        ind.enabled_units = units
        ind.description = desc
        ind.is_constraint = is_con
        ind.is_evaluative = (level in (2, 3))
        n += 1
    db.session.commit()
    return n
def ideal_type_label(code):
    return {"benefit": "越大越优", "cost": "越小越优",
            "mid": "适中型", "discrete": "离散赋值"}.get(code,
    code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg, As, Ni, Cu, Cd, Pb, Cr, Zn,
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
# 表头（顺序即导出模板顺序）
```

```
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
```

```
    for c in rows[0]]                                                    2201
        out = []                                                         2202
        for r in rows[1:]:                                               2203
            if all(c is None or str(c).strip() == "" for c               2204
    in r):                                                               2205
                continue                                                 2206
            out.append({header[i]: (r[i] if i < len(r) else              2207
    None)                                                                2208
                        for i in range(len(header)) if                   2209
    header[i]})                                                          2210
        return out                                                       2211
    raise ValueError("不支持的文件类型：.%s" % ext)                          2212
def _num(value):                                                         2213
    if value is None:                                                    2214
        return None, None                                                2215
    s = str(value).strip()                                               2216
    if s == "":                                                          2217
        return None, None                                                2218
    try:                                                                 2219
        return float(s), None                                            2220
    except ValueError:                                                   2221
        return None, "「%s」不是有效数值" % s                               2222
def parse_upload(filename, stream, unit_type_map):                       2223
    """解析上传文件。                                                      2224
    参数                                                                 2225
    ----                                                                 2226
    unit_type_map : dict，{单元类型: MicroUnit 对象}                       2227
    返回                                                                 2228
    ----                                                                 2229
    (parsed, errors)                                                     2230
      parsed : list[dict]，每项 {unit_type, fields{...},                 2231
    heavy_metals{}}                                                      2232
      errors : list[str]，逐行问题说明                                    2233
    """                                                                  2234
    rows = _read_rows(filename, stream)                                  2235
    parsed, errors = [], []                                              2236
    if not rows:                                                         2237
        return [], ["文件为空或没有数据行"]                                  2238
    # 表头校验                                                           2239
    unknown = [h for h in (rows[0].keys() if rows else [])               2240
    if h not in FIELD_ALIASES]                                           2241
    if unknown:                                                          2242
        errors.append("存在无法识别的列，将被忽略：%s" %                      2243
    "、".join(map(str, unknown)))                                        2244
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头      2245
        rec = {}                                                         2246
        for k, v in raw.items():                                         2247
            key = FIELD_ALIASES.get(str(k).strip())                      2248
            if key:                                                      2249
                rec[key] = v                                             2250
```

```
        unit_type = str(rec.get("unit_type") or "").strip()
        if not unit_type:
            errors.append("第 %d 行：缺少「单元类型」" % idx)
            continue
        if unit_type not in unit_type_map:
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
                          % (idx, unit_type,
"、".join(unit_type_map) or "无"))
            continue
        if unit_type not in UNIT_TYPES:
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"
                          % (idx, unit_type))
            continue
        fields, heavy = {}, {}
        ok = True
        for key, val in rec.items():
            if key in ("unit_type",):
                continue
            if key.startswith("hm_"):
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx,
key[3:], err))
                    ok = False
                elif n is not None:
                    heavy[key[3:]] = n
                continue
            if key == "aspect":
                s = str(val).strip() if val is not None
else ""
                if s == "":
                    fields["aspect"] = None
                elif s in ASPECT_TYPES:
                    fields["aspect"] = s
                else:
                    errors.append("第 %d 行：坡向「%s」无效（应为 %s）"
                                  % (idx, s,
"、".join(ASPECT_TYPES)))
                    ok = False
                continue
            n, err = _num(val)
            if err:
                errors.append("第 %d 行：%s %s" % (idx, key,
err))
                ok = False
            else:
                fields[key] = n
        if ok:
            parsed.append({"unit_type": unit_type,
"fields": fields,
```

```
                                    "heavy_metals": heavy})
    return parsed, errors
# ==== app/services/workflow.py ====
# -*- coding: utf-8 -*-
"""评价流程前置条件（定量评价与报告输出的流程门禁）。
评价流程存在严格的先后关系：
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
流程约定：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                        EvaluationResult)
from app.services import unit_config
#  指标数据非空字段（用于判断某单元是否已录入数据）
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
               "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
        return True
    return bool(ind.heavy_metals_dict())
def _data_of(ind):
    """把 IndicatorData 行转成标准化接口要的 dict。"""
    if ind is None:
        return {}
    data = {f: getattr(ind, f, None) for f in DATA_FIELDS}
    data["heavy_metals"] = ind.heavy_metals_dict()
    return data
def patch_progress(patch_id):
    """检查某图斑的评价流程完成情况。
    返回 dict：
        units       {done, count, ratio_sum, detail}
        constraint  {done, passed, detail}
        data        {done, ready, missing, detail}
        blocking    未满足的前置步骤列表（按流程顺序）
        ready       是否可进入定量评价
    """
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    ratio_sum = sum(u.ratio or 0.0 for u in units)
```

```
    #  ----  步骤一：微地质单元  ----
    units_step = {
        "done": bool(units),
        "count": len(units),
        "ratio_sum": ratio_sum,
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"
                   % (len(units), ratio_sum * 100))
                  if units else "尚未划分微地质单元",
    }
    #  ----  步骤二：限制性条件判定  ----
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id) \
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
        constraint_step = {"done": True, "passed": False,
                           "detail": "限制性条件判定未通过（%s）" %
    (cc.reason or "")}
    #  ----  步骤三：指标数据  ----
    #  判断标准是「该单元类型启用的必填项是否齐全」，而不是「有没有填过任意一个字段」。
    #  早先只要 8 个字段里有一个非空就算齐，于是碎石边坡只填一个坡度也能跑出
    #  F=0.69「基本适宜」；而 score_unit 又会把缺的项剔掉、把权重摊给其余指标，
    #  结论方向还偏乐观。污染指标按方法约定允许「未检测」，不算缺。
    settings = unit_config.for_patch(patch_id)
    missing = []                  #  完全没录数据的单元类型
    incomplete = []               #  录了但不齐的单元类型 → [(单元类型, [缺的字段])]
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        data = _data_of(ind)
        gaps = settings.missing_fields(u.unit_type, data)
        if not _has_data(ind):
            missing.append(u.unit_type)
        elif gaps:
            incomplete.append((u.unit_type, gaps))
    if not units:
        data_step = {"done": False, "ready": False,
    "missing": [],
                     "detail": "尚无可录入数据的微地质单元"}
    elif missing:
        data_step = {"done": False, "ready": False,
    "missing": missing,
                     "detail": "以下微地质单元尚未录入指标数据：%s"
                               % "、".join(missing)}
    elif incomplete:
```

```
        data_step = {"done": False, "ready": False,
                     "missing": [t for t, _ in incomplete],
                     "detail": "以下微地质单元的必填指标未填齐：%s"
                               % "；".join("%s缺 %s" % (t,
"、".join(g))
                                           for t, g in
incomplete)}
else:
        data_step = {"done": True, "ready": True,
"missing": [],
                     "detail": "全部 %d 个微地质单元的指标数据已录入" %
len(units)}
#  占比合计必须落在 １００% 附近才算通过。只挡「偏小」是不够的：
#  库里若出现合计 １．５ 的记录（手工改库、导入异常），F = Σ(Si×F′i)
#  会被放大到 １．５ 倍，所以上下都要挡。
ratio_ok = 0.999 <= ratio_sum <= 1.001 if units else
False
if units and not ratio_ok:
        units_step["detail"] += "（占比合计不等于 １００%，请检查面积录入）"
#  ---- 汇总未满足的前置条件（按评价流程的顺序：限制性判定优先） ----
#  reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
#  缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
blocking = []
if not constraint_step["done"]:
        blocking.append({
                "step": "constraint", "label": "限制性条件判定未执行",
                "reason": "四项限制性条件尚未逐项判定。",
                "fix_url": "/constraint/%d" % patch_id,
"fix_label": "去执行判定",
        })
elif not constraint_step["passed"]:
        blocking.append({
                "step": "constraint", "label": "限制性条件判定未通过",
                "reason": cc.reason or "",
                "fix_url": "/constraint/%d" % patch_id,
"fix_label": "去复核判定结果",
        })
if not units_step["done"]:
        blocking.append({
                "step": "units", "label": "未划分微地质单元",
                "reason": "该图斑下没有任何微地质单元。",
                "fix_url": "/unit/%d" % patch_id, "fix_label":
"去划分",
        })
elif not ratio_ok:
        blocking.append({
                "step": "units", "label": "面积占比合计不为 １００%",
                "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),
                "fix_url": "/unit/%d" % patch_id, "fix_label":
"去核对面积",
```

```
            })                                                                  2451
        if units_step["done"] and not data_step["ready"]:                       2452
            blocking.append({                                                   2453
                "step": "data", "label": "指标数据未录入完整",                  2454
                "reason": data_step["detail"],                                  2455
                "fix_url": "/indicator/", "fix_label": "去录入",                2456
            })                                                                  2457
        return {                                                                2458
            "units": units_step,                                                2459
            "constraint": constraint_step,                                      2460
            "data": data_step,                                                  2461
            "blocking": blocking,                                               2462
            "ready": not blocking,                                              2463
        }                                                                       2464
def undetected_metals(patch_id):                                                2465
    """列出该图斑各单元「未检测」的重金属，供结果页与报告如实标注。            2466
    方法约定允许未检测的重金属按未受污染计入，但报告里必须写明是哪些 ——      2467
    否则读报告的人会以为这些项是实测达标。                                      2468
    只列启用了土壤指标的单元：岩质边坡本来就不评价土壤，列 8 项未检测是噪音。  2469
    """                                                                         2470
    settings = unit_config.for_patch(patch_id)                                  2471
    out = []                                                                    2472
    for u in MicroUnit.query.filter_by(patch_id=patch_id).al                    2473
    l():                                                                        2474
        if settings.group_enabled(u.unit_type, "土壤指标",                      2475
    "soil") is False:                                                           2476
            continue                                                            2477
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi                    2478
    rst()                                                                       2479
        if ind is None:                                                         2480
            continue                                                            2481
        metals = settings.undetected_metals(_data_of(ind))                      2482
        if metals:                                                              2483
            out.append((u.unit_type, metals))                                   2484
    return out                                                                  2485
def latest_result(patch_id):                                                    2486
    """取该图斑最近一次的评价记录。"""                                          2487
    return EvaluationResult.query.filter_by(patch_id=patch_i                    2488
    d) \                                                                        2489
        .order_by(EvaluationResult.id.desc()).first()                           2490
def report_progress(patch_id):                                                  2491
    """报告输出的前置条件：评价流程齐备，且已有评价记录。                      2492
    报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，                  2493
    导出时读取已保存的评价结果，而不是在导出环节重新计算。                      2494
    否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，          2495
    同一份评价出现两个数字，结论无法追溯。                                      2496
    返回 dict：{ready, blocking, result, flow}                                  2497
        result 为最近一次评价记录（EvaluationResult），无记录时为空。          2498
    """                                                                         2499
    flow = patch_progress(patch_id)                                             2500
```

```
    blocking = list(flow["blocking"])
    record = latest_result(patch_id)
    if not blocking and record is None:
        blocking.append({
            "step": "evaluate", "label": "尚未执行适宜性评价",
            "reason": "该图斑还没有评价记录。",
            "fix_url": "/evaluate/%d" % patch_id,
            "fix_label": "去执行评价",
        })
    return {
        "ready": not blocking,
        "blocking": blocking,
        "result": record,
        "flow": flow,
    }
#  ==== app/services/dashboard.py ====
#  -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即评价流程的作业顺序）
#    key          步骤标识
#    no           步骤编号（01~08）
#    title        名称
#    icon         图标名（见 templates/_macros.html 的 ICONS）
#    desc         一句话说明
#    href         入口地址
STEPS = [
    dict(key="project", no="01", title="项目与矿山档案",
    icon="folder",
         href="/project/",
         desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),
    dict(key="standard", no="02", title="政策与标准知识库",
    icon="book",
         href="/standard/",
         desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),
    dict(key="constraint", no="03", title="限制性指标判定",
    icon="shield",
         href="/constraint/",
         desc="政策、区位约束、土壤污染、地质灾害四项一票否决"),
    dict(key="unit", no="04", title="微地质单元", icon="layers",
         href="/unit/",
         desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),
```

```
    dict(key="ahp", no="05", title="指标体系与权重",
    icon="sliders",
        href="/ahp/",
        desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),
    dict(key="indicator", no="06", title="指标数据采集",
    icon="database",
        href="/indicator/",
        desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),
    dict(key="evaluate", no="07", title="适宜性定量评价",
    icon="target",
        href="/evaluate/",
        desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),
    dict(key="report", no="08", title="报告与结果输出",
    icon="file",
        href="/report/",
        desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),
]
def _has_data(ind):
    return wf._has_data(ind)
#  系统的九个功能模块（登录页介绍用，含不在评价流程内的账号与权限管理）
MODULES = [
    ("01", "账号与权限管理", "自助注册与登录注销、管理员与操作员两种职务、写操作自动留痕"),
    ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),
    ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),
    ("04", "限制性指标判定", "政策、区位、污染、地质灾害四项一票否决"),
    ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),
    ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),
    ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),
    ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),
    ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),
]
def snapshot():
    """一次算出工作台需要的全部数据（同一请求内只算一次）。
    返回 dict：
        stats       {projects, patches, unit_types, standards}
        steps       八个步骤，附 available / pending / note
        badges      {key: 待办数量}，仅含有待办的项
        progress    {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
        cached = _compute()
        g._dash_snapshot = cached
    return cached
def _compute():
    projects = Project.query.count()
    patches = MinePatch.query.all()
    patch_ids = [p.id for p in patches]
    units = MicroUnit.query.filter(
```

```
        MicroUnit.patch_id.in_(patch_ids)).all() if
patch_ids else []
unit_by_patch = {}
for u in units:
        unit_by_patch.setdefault(u.patch_id, []).append(u)
unit_ids = [u.id for u in units]
data_unit_ids = set()
if unit_ids:
        for ind in IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).all():
            if _has_data(ind):
                data_unit_ids.add(ind.unit_id)
passed_constraint = set()
if patch_ids:
        for c in ConstraintCheck.query.filter(
                ConstraintCheck.patch_id.in_(patch_ids)).all
():
            if c.passed:
                passed_constraint.add(c.patch_id)
evaluated = set()
if patch_ids:
        for r in EvaluationResult.query.filter(
                EvaluationResult.patch_id.in_(patch_ids)).al
l():
            evaluated.add(r.patch_id)
# ---- 各步骤的待办数量 ----
pending = {
        "constraint": sum(1 for p in patches if p.id not in
passed_constraint),
        "unit": sum(1 for p in patches if not
unit_by_patch.get(p.id)),
        "indicator": 0,
        "evaluate": 0,
}
ready_patches = 0
for p in patches:
        us = unit_by_patch.get(p.id, [])
        if us and any(u.id not in data_unit_ids for u in
us):
            pending["indicator"] += 1
        pr = wf.patch_progress(p.id)
        if pr["ready"]:
            ready_patches += 1
            if p.id not in evaluated:
                pending["evaluate"] += 1
# ---- 步骤可用性：前置数据不具备时置灰 ----
avail = {
        "project": True,
        "standard": True,
        "constraint": bool(patches),
```

```
            "unit": bool(patches),
            "ahp": True,
            "indicator": bool(units),
            "evaluate": ready_patches > 0,
            "report": bool(evaluated),
        }
        notes = {
            "constraint": "需先建立矿山图斑",
            "unit": "需先建立矿山图斑",
            "indicator": "需先划分微地质单元",
            "evaluate": "需前置步骤齐备的图斑",
            "report": "需先执行适宜性评价",
        }
        steps = []
        for s in STEPS:
            steps.append(dict(
                s,
                available=avail[s["key"]],
                pending=pending.get(s["key"], 0),
                note=None if avail[s["key"]] else
        notes.get(s["key"], ""),
            ))
        unit_types = len({u.unit_type for u in units})
        stats = {
            "projects": projects,
            "patches": len(patches),
            "unit_types": unit_types,
            "standards": Standard.query.count(),
        }
        return {
            "stats": stats,
            "steps": steps,
            "badges": {k: v for k, v in pending.items() if v},
            "progress": {
                "patches": len(patches),
                "units": len(units),
                "data_units": len(data_unit_ids),
                "ready": ready_patches,
                "evaluated": len(evaluated),
            },
        }
    def nav_badges():
        """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""
        snap = snapshot()
        return {
            "/constraint/": snap["badges"].get("constraint", 0),
            "/unit/": snap["badges"].get("unit", 0),
            "/indicator/": snap["badges"].get("indicator", 0),
            "/evaluate/": snap["badges"].get("evaluate", 0),
        }
```

```
# ==== app/services/audit.py ====
# -*- coding: utf-8 -*-
"""操作审计。
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
"""
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
# 端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.register": "注册账号",
    "auth.logout": "退出登录",
    "auth.add_user": "新增账号",
    "auth.edit_user": "编辑账号",
    "auth.delete_user": "删除账号",
    "auth.reset_password": "重置用户密码",
    "auth.profile": "修改个人资料",
    "auth.upload_avatar": "更新头像",
    "auth.delete_avatar": "移除头像",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
# 无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    # 个人设置页的两个表单共用一个端点，按提交的动作区分
```

```
    if endpoint == "auth.profile":
        return "修改密码" if request.form.get("action") ==
    "password" \
            else "修改个人资料"
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += "；结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
            target_type=(endpoint or "").split(".")[0] or
    None,
            target_id=extract_target_id(),
            detail=detail,
        ))
        db.session.commit()
    except Exception:
        # 审计失败不应影响业务响应
        db.session.rollback()
# ==== app/views/auth.py ====
# -*- coding: utf-8 -*-
"""账号与权限 —— 登录、注册、个人资料、用户维护、操作日志。
页面组织：
  /login          登录页（含「点我注册」入口）
  /register       自助注册（默认「操作员」）
  /profile        个人设置，分类切换：info / security / users / logs
                  后两个分类仅管理员可见，由本模块统一管控。
"""
from flask import (Blueprint, render_template, redirect,
    url_for, request,
```

```
                    flash, abort, current_app)
from flask_login import login_user, logout_user,
    login_required, current_user
from werkzeug.security import check_password_hash,
    generate_password_hash
from app.forms import (LoginForm, EmptyForm, ProfileForm,
    PasswordForm,
                       RegisterForm, email_ok)
from app.models import db, User, OperationLog
from app.utils import uploads as up
from app.utils.constants import ROLE_LABELS, DEFAULT_ROLE,
    GENDERS
bp = Blueprint("auth", __name__)
#  「个人设置」的四个分类；后两个是管理员专属
PROFILE_TABS = ("info", "security", "users", "logs")
ADMIN_TABS = ("users", "logs")
#  口令最短长度：自助注册与管理员建号保持一致，免得出现「a」这种口令
MIN_PASSWORD_LEN = 6
def require_admin():
    """非管理员直接 403，避免误操作基础数据。"""
    if current_user.role != "admin":
        abort(403)
def _intro_data():
    """登录页下方的系统介绍所用的静态内容。"""
    from app.services import dashboard as dash
    return {"modules": dash.MODULES}
@bp.route("/")
def index():
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        return render_template("auth/home.html",
    dash=dash.snapshot())
    return redirect(url_for("auth.login"))
@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.d
    ata).first()
        if user and check_password_hash(user.password_hash,
    form.password.data):
            if not user.active:
                flash("该账号已停用，请联系管理员", "toast-error")
            else:
                login_user(user, remember=form.remember.data
    )
                #  用 toast 类别：渲染为页面中上方的浮层，淡入停留后淡出。
                #  登录成功只是告知性反馈，不需要用户停下来阅读。
```

```
                flash("登录成功", "toast")                                  2851
                return redirect(url_for("auth.index"))                      2852
            else:                                                           2853
                flash("账户名称或密码错误", "toast-error")                  2854
        #  字段为空之类的校验失败，错误留在表单里逐项显示，不走浮层：      2855
        #  那类消息要对着输入框看，一闪而过反而让人以为「点了没反应」。    2856
        return render_template("auth/login.html", form=form,               2857
        **_intro_data())                                                    2858
@bp.route("/register", methods=["GET", "POST"])                             2859
def register():                                                             2860
        """自助注册。新账号一律为「操作员」，权限只能由管理员事后调整。"""  2861
        if current_user.is_authenticated:                                   2862
            return redirect(url_for("auth.index"))                          2863
        form = RegisterForm()                                               2864
        if form.validate_on_submit():                                       2865
            username = form.username.data.strip()                           2866
            if User.query.filter_by(username=username).first():             2867
                form.username.errors.append("该账户名称已被使用")           2868
            else:                                                           2869
                user = User(username=username,                              2870
                            role=DEFAULT_ROLE,                              2871
                            password_hash=generate_password_hash            2872
        (form.password.data),                                               2873
                            display_name=(form.display_name.data            2874
        or "").strip() or None,                                             2875
                            gender=(form.gender.data or                     2876
        "").strip() or None,                                                2877
                            email=(form.email.data or                       2878
        "").strip() or None,                                                2879
                            phone=(form.phone.data or                       2880
        "").strip() or None)                                                2881
                rel, err = _save_avatar(request.files.get("avata            2882
        r"))                                                                2883
                if err:                                                     2884
                    flash(err, "error")                                     2885
                else:                                                       2886
                    user.avatar = rel                                       2887
                    db.session.add(user)                                    2888
                    db.session.commit()                                     2889
                    login_user(user)                                        2890
                    flash("注册成功，欢迎使用", "toast")                    2891
                    return redirect(url_for("auth.index"))                  2892
        return render_template("auth/register.html", form=form)             2893
@bp.route("/logout")                                                        2894
@login_required                                                             2895
def logout():                                                               2896
        logout_user()                                                       2897
        flash("已退出登录", "ok")                                           2898
        return redirect(url_for("auth.login"))                              2899
# ---------------------------- 个人设置 ----------------------------       2900
```

```
    ----
# 任何登录用户在这里维护自己；管理员额外能管理账号与查看日志。
# 「用户与操作日志」不再是独立菜单，而是本页的管理员分类。
@bp.route("/profile", methods=["GET", "POST"])
@login_required
def profile():
    is_admin = current_user.role == "admin"
    tab = request.values.get("tab") or "info"
    if tab not in PROFILE_TABS or (tab in ADMIN_TABS and
    not is_admin):
        tab = "info"
    # 表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名
    info_form = ProfileForm(prefix="info", obj=current_user)
    pwd_form = PasswordForm(prefix="pwd")
    if request.method == "POST":
        action = request.form.get("action")
        # 校验失败要重新渲染本页，此时必须回到提交表单所在的那个分类，
        # 否则错误提示会出现在另一个分类里，用户根本看不到。
        if action == "password":
            tab = "security"
        elif action == "info":
            tab = "info"
        if action == "info" and info_form.validate_on_submit
    ():
            current_user.display_name =
    (info_form.display_name.data or "").strip() or None
            current_user.gender = (info_form.gender.data or
    "").strip() or None
            current_user.email = (info_form.email.data or
    "").strip() or None
            current_user.phone = (info_form.phone.data or
    "").strip() or None
            db.session.commit()
            flash("个人资料已保存", "ok")
            return redirect(url_for("auth.profile",
    tab="info"))
        if action == "password" and pwd_form.validate_on_sub
    mit():
            if not check_password_hash(current_user.password
    _hash,

    pwd_form.old_password.data):
                pwd_form.old_password.errors.append("当前密码不正确
    ")
            else:
                current_user.password_hash =
    generate_password_hash(
                    pwd_form.new_password.data)
                db.session.commit()
                flash("密码已修改", "ok")
```

```
                return redirect(url_for("auth.profile",
    tab="security"))
    ctx = dict(info_form=info_form, pwd_form=pwd_form,
    avatar_form=EmptyForm(),
                tab=tab, role_labels=ROLE_LABELS,
    genders=GENDERS)
    if tab == "users":
        ctx.update(_users_context())
    elif tab == "logs":
        ctx.update(_logs_context())
    return render_template("auth/profile.html", **ctx)
def _users_context():
    """管理员分类「账号」所需的数据。"""
    return {"users": User.query.order_by(User.id).all(),
            "form": EmptyForm()}
def _logs_context():
    """管理员分类「操作日志」所需的数据。"""
    per_page = 50
    q = OperationLog.query.order_by(OperationLog.id.desc())
    total = q.count()
    pages = max(1, (total + per_page - 1) // per_page)
    # 页码先夹到 [1, pages] 再算 offset：手工把 page 写成天文数字时，
    # (page-1)*per_page 会超出 SQLite 能绑定的整数范围，直接 500。
    try:
        page = int(request.args.get("page", 1))
    except (TypeError, ValueError):
        page = 1
    page = max(1, min(page, pages))
    records = q.offset((page - 1) * per_page).limit(per_page
    ).all()
    users_map = {u.id: u.username for u in User.query.all()}
    return {"records": records, "page": page, "pages":
    pages,
            "total": total, "users_map": users_map}
# ------------------------------ 头像 ------------------------------
--
def _avatar_length_ok(file_storage):
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 *
    1024 * 1024)
    stream = file_storage.stream
    pos = stream.tell()
    stream.seek(0, 2)
    size = stream.tell()
    stream.seek(pos)
    return size <= limit, size
def _save_avatar(file):
    """保存头像文件，返回 (相对路径, 错误信息)。没有文件时返回 (None, None)。"""
    if file is None or not file.filename:
        return None, None
```

```
    ok, size = _avatar_length_ok(file)                                  3001
    if not ok:                                                          3002
        return None, ("头像不能超过 %.0f MB（当前 %.1f MB）"              3003
                      % (current_app.config["MAX_AVATAR_LENG            3004
    TH"] / 1024 / 1024,                                                 3005
                         size / 1024 / 1024))                           3006
    return up.save_upload(file, "avatars",                              3007
                          current_app.config["ALLOWED_AVATAR            3008
    _EXT"],                                                             3009
                          current_app.config["UPLOAD_FOLDER"            3010
    ])                                                                  3011
@bp.route("/profile/avatar", methods=["POST"])                          3012
@login_required                                                         3013
def upload_avatar():                                                    3014
    form = EmptyForm()                                                  3015
    back = url_for("auth.profile", tab=request.form.get("tab            3016
    ") or "info")                                                       3017
    if not form.validate_on_submit():                                   3018
        flash("表单校验失败，请重试", "error")                           3019
        return redirect(back)                                           3020
    rel, err = _save_avatar(request.files.get("avatar"))                3021
    if err:                                                             3022
        flash(err, "error")                                             3023
        return redirect(back)                                           3024
    if rel is None:                                                     3025
        flash("请选择头像图片", "error")                                3026
        return redirect(back)                                           3027
    # 换新头像时删掉旧文件，避免残留                                    3028
    up.remove_upload(current_user.avatar,                               3029
    current_app.config["UPLOAD_FOLDER"])                               3030
    current_user.avatar = rel                                           3031
    db.session.commit()                                                 3032
    flash("头像已更新", "ok")                                           3033
    return redirect(back)                                               3034
@bp.route("/profile/avatar/delete", methods=["POST"])                   3035
@login_required                                                         3036
def delete_avatar():                                                    3037
    form = EmptyForm()                                                  3038
    back = url_for("auth.profile", tab=request.form.get("tab            3039
    ") or "info")                                                       3040
    if form.validate_on_submit():                                       3041
        up.remove_upload(current_user.avatar,                           3042
    current_app.config["UPLOAD_FOLDER"])                               3043
        current_user.avatar = None                                      3044
        db.session.commit()                                             3045
        flash("头像已移除", "ok")                                       3046
    return redirect(back)                                               3047
# ---------------------------- 账号维护（仅管理员）                     3048
    ----------------------------                                        3049
@bp.route("/users")                                                     3050
```

```
@login_required
def users():
    """旧地址保留，统一并入「个人设置 · 账号」。"""
    return redirect(url_for("auth.profile", tab="users"))
@bp.route("/users/add", methods=["GET", "POST"])
@login_required
def add_user():
    require_admin()
    if request.method == "POST":
        username = request.form.get("username", "").strip()
        password = request.form.get("password", "")
        role = request.form.get("role", DEFAULT_ROLE)
        if role not in ROLE_LABELS:
            role = DEFAULT_ROLE
        if not username or not password:
            flash("账户名称和密码不能为空", "error")
            return redirect(url_for("auth.add_user"))
        if len(username) > 64:
            flash("账户名称不能超过 64 个字符", "error")
            return redirect(url_for("auth.add_user"))
        if len(password) < MIN_PASSWORD_LEN:
            flash("密码至少 %d 位（与自助注册保持一致）" % MIN_PASSWORD_LEN,
                  "error")
            return redirect(url_for("auth.add_user"))
        if User.query.filter_by(username=username).first():
            flash("账户名称已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(
            username=username, role=role,
            password_hash=generate_password_hash(password),
            display_name=request.form.get("display_name",
    "").strip() or None,
            gender=request.form.get("gender", "").strip()
    or None,
            email=request.form.get("email", "").strip() or
    None,
            phone=request.form.get("phone", "").strip() or
    None))
        db.session.commit()
        flash("账号已添加", "ok")
        return redirect(url_for("auth.profile",
    tab="users"))
    return render_template("auth/add_user.html",
    role_labels=ROLE_LABELS,
                           genders=GENDERS,
    default_role=DEFAULT_ROLE,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
```

```
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if request.method == "POST":
        role = request.form.get("role", user.role)
        if role not in ROLE_LABELS:
            role = user.role
        active = request.form.get("active") == "1"
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统
        if user.role == "admin" and (role != "admin" or not
active):
            others = User.query.filter(User.role == "admin",
                                       User.id != user.id,

User.active.is_(True)).count()
            if others == 0:
                flash("系统至少需保留一名启用状态的管理员", "error")
                return redirect(url_for("auth.edit_user",
user_id=user.id))
        # 个人资料字段（用户自己也能在「个人设置」里改，管理员在此代改）
        email = request.form.get("email", "").strip()
        if not email_ok(email):
            flash("邮箱格式不正确", "error")
            return redirect(url_for("auth.edit_user",
user_id=user.id))
        user.role = role
        user.active = active
        user.display_name = request.form.get("display_name",
"").strip() or None
        user.gender = request.form.get("gender",
"").strip() or None
        user.email = email or None
        user.phone = request.form.get("phone", "").strip()
or None
        new_password = request.form.get("password",
"").strip()
        if new_password:
            if len(new_password) < MIN_PASSWORD_LEN:
                flash("密码至少 %d 位" % MIN_PASSWORD_LEN,
"error")
                return redirect(url_for("auth.edit_user",
user_id=user.id))
            user.password_hash = generate_password_hash(new_
password)
        db.session.commit()
        flash("账号信息已保存", "ok")
        return redirect(url_for("auth.profile",
tab="users"))
    return render_template("auth/edit_user.html", user=user,
                           role_labels=ROLE_LABELS,
```

```
    genders=GENDERS,                                                    3151
                                    form=EmptyForm())                   3152
@bp.route("/users/<int:user_id>/delete", methods=["POST"])              3153
@login_required                                                         3154
def delete_user(user_id):                                               3155
    require_admin()                                                     3156
    user = db.get_or_404(User, user_id)                                 3157
    if user.id == current_user.id:                                      3158
        flash("不能删除当前登录的账号", "error")                          3159
        return redirect(url_for("auth.profile",                         3160
    tab="users"))                                                       3161
    if user.role == "admin":                                            3162
        others = User.query.filter(User.role == "admin",                3163
                                   User.id != user.id,                  3164
                                   User.active.is_(True)).co            3165
    unt()                                                               3166
        if others == 0:                                                 3167
            flash("系统至少需保留一名启用状态的管理员", "error")          3168
            return redirect(url_for("auth.profile",                     3169
    tab="users"))                                                       3170
    db.session.delete(user)                                             3171
    db.session.commit()                                                 3172
    flash("账号「%s」已删除" % user.username, "ok")                      3173
    return redirect(url_for("auth.profile", tab="users"))               3174
# ------------------------------ 操作日志（仅管理员）                   3175
    ------------------------------                                      3176
@bp.route("/logs")                                                      3177
@login_required                                                         3178
def logs():                                                             3179
    """旧地址保留，统一并入「个人设置 · 操作日志」。"""                   3180
    return redirect(url_for("auth.profile", tab="logs"))                3181
# ==== app/views/project.py ====                                        3182
# -*- coding: utf-8 -*-                                                 3183
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""          3184
from flask import (Blueprint, render_template, request,                 3185
    redirect, url_for,                                                  3186
                      flash, current_app)                               3187
from flask_login import login_required, current_user                    3188
from app.forms import EmptyForm                                         3189
from app.models import db, Project, MinePatch,                          3190
    ReferenceEcosystem                                                  3191
from app.services import cleanup                                        3192
from app.utils.parsing import FieldError, parse_float,                  3193
    parse_text                                                          3194
from app.utils.timefmt import utcnow                                    3195
bp = Blueprint("project", __name__, url_prefix="/project")              3196
@bp.route("/")                                                          3197
@login_required                                                         3198
def index():                                                            3199
    projects = Project.query.order_by(Project.id.desc()).all            3200
```

```
    ()                                                              3201
    return render_template("project/index.html",                    3202
    projects=projects,                                              3203
                                    form=EmptyForm())               3204
@bp.route("/add", methods=["GET", "POST"])                          3205
@login_required                                                     3206
def add():                                                          3207
    form = EmptyForm()                                              3208
    if request.method == "POST" and form.validate_on_submit(        3209
    ):                                                              3210
        try:                                                        3211
            p = Project(                                            3212
                name=parse_text(request.form.get("name"),           3213
    "项目名称",                                                     3214
                                        max_len=128, required=True),3215
                code=parse_text(request.form.get("code"),           3216
    "项目编号", max_len=64),                                        3217
                description=parse_text(request.form.get("des        3218
    cription"), "项目说明",                                         3219
                                            max_len=2000),          3220
                created_by=current_user.id,                         3221
            )                                                       3222
        except FieldError as e:                                     3223
            flash(str(e), "error")                                  3224
            return render_template("project/edit.html",             3225
    form=form, project=None)                                        3226
        db.session.add(p)                                           3227
        db.session.commit()                                         3228
        flash("项目已创建", "ok")                                   3229
        return redirect(url_for("project.index"))                   3230
    return render_template("project/edit.html", form=form,          3231
    project=None)                                                   3232
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])               3233
@login_required                                                     3234
def edit(pid):                                                      3235
    p = db.get_or_404(Project, pid)                                 3236
    form = EmptyForm()                                              3237
    if request.method == "POST" and form.validate_on_submit(        3238
    ):                                                              3239
        try:                                                        3240
            p.name = parse_text(request.form.get("name"),           3241
    "项目名称",                                                     3242
                                    max_len=128, required=True)     3243
            p.code = parse_text(request.form.get("code"),           3244
    "项目编号", max_len=64)                                         3245
            p.description = parse_text(request.form.get("des        3246
    cription"), "项目说明",                                         3247
                                            max_len=2000)           3248
        except FieldError as e:                                     3249
            flash(str(e), "error")                                  3250
```

```
            return render_template("project/edit.html",
    form=form, project=p)
        p.status = request.form.get("status", "进行中")
        db.session.commit()
        flash("项目已保存", "ok")
        return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form,
    project=p)
@bp.route("/<int:pid>/delete", methods=["POST"])
@login_required
def delete(pid):
    p = db.get_or_404(Project, pid)
    # 项目下的图斑由 ORM 级联删除，但图斑下面的指标数据、限制性判定、
    # 评价结果、得分明细、参照生态系统不会跟着走，必须先逐个图斑清理，
    # 否则每删一个项目都在库里留下一批孤儿数据（见 app/services/cleanup.py）。
    cleanup.delete_project(p)
    flash("项目已删除", "ok")
    return redirect(url_for("project.index"))
# ---------------------------- 矿山图斑 ----------------------------
    ----
@bp.route("/<int:pid>/patches")
@login_required
def patches(pid):
    p = db.get_or_404(Project, pid)
    return render_template("project/patches.html",
    project=p,
                           form=EmptyForm())
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])
@login_required
def patch_delete(patch_id):
    """删除矿山图斑，及其微地质单元、指标数据与评价记录。"""
    patch = db.get_or_404(MinePatch, patch_id)
    pid = patch.project_id
    # 清理逻辑统一放在 app/services/cleanup.py：项目删除走的是同一个函数，
    # 免得两条删除路径各写一份、漏掉某张子表（曾经就漏过）。
    deleted, _counts = cleanup.delete_patch(patch.id)
    flash("图斑「%s」及其数据已删除" % (deleted.name if deleted else
    patch.name),
          "ok")
    return redirect(url_for("project.patches", pid=pid))
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])
@login_required
def patch_add(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
```

```
        try:                                                        3301
            area_ha = parse_float(request.form.get("area_ha"        3302
    ), "图斑面积（公顷）",                                          3303
                                  minimum=0)                        3304
            stop_years = parse_float(request.form.get("stop_        3305
    years"), "停产年限",                                            3306
                                    minimum=0)                      3307
            name = parse_text(request.form.get("name"),             3308
    "图斑名称",                                                     3309
                                max_len=128, required=True)         3310
            admin_region = parse_text(request.form.get("admi        3311
    n_region"),                                                     3312
                                    "所在行政区", max_len=128)      3313
            mineral_type = parse_text(request.form.get("mine        3314
    ral_type"),                                                     3315
                                    "开采矿种", max_len=64)         3316
            mining_method = parse_text(request.form.get("min        3317
    ing_method"),                                                   3318
                                     "开采方式", max_len=64)        3319
        except FieldError as e:                                     3320
            flash(str(e), "error")                                  3321
            return redirect(url_for("project.patch_add",            3322
    pid=p.id))                                                      3323
        patch = MinePatch(                                          3324
            project_id=p.id,                                        3325
            name=name,                                              3326
            admin_region=admin_region,                              3327
            area_ha=area_ha,                                        3328
            mineral_type=mineral_type,                              3329
            mining_method=mining_method,                            3330
            stop_years=stop_years,                                  3331
            responsible_entity_missing=bool(request.form.get        3332
    ("responsible_entity_missing")),                                3333
        )                                                           3334
        db.session.add(patch)                                       3335
        db.session.flush()                                          3336
        #  参照生态系统默认取系统内置值                             3337
        db.session.add(ReferenceEcosystem(                          3338
            patch_id=patch.id, coverage_benchmark=75.0,             3339
    shannon_benchmark=0.7))                                         3340
        db.session.commit()                                         3341
        flash("图斑已创建", "ok")                                   3342
        return redirect(url_for("project.patches",                  3343
    pid=p.id))                                                      3344
    return render_template("project/patch_edit.html",               3345
    form=form, project=p, patch=None)                               3346
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",             3347
    "POST"])                                                        3348
@login_required                                                     3349
def patch_edit(patch_id):                                           3350
```

```
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit():
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        # 数值字段统一走 parse_float：填了非数字给中文提示，而不是 500
        try:
            area_ha = parse_float(request.form.get("area_ha"), "图斑面积（公顷）",
                                  minimum=0)
            stop_years = parse_float(request.form.get("stop_years"), "停产年限",
                                     minimum=0)
            cov = parse_float(request.form.get("coverage_benchmark"),
                              "植被覆盖度基准（%）", minimum=0, maximum=100)
            sha = parse_float(request.form.get("shannon_benchmark"),
                              "Shannon 基准", minimum=0)
            name = parse_text(request.form.get("name"), "图斑名称",
                              max_len=128, required=True)
            admin_region = parse_text(request.form.get("admin_region"),
                                      "所在行政区", max_len=128)
            mineral_type = parse_text(request.form.get("mineral_type"),
                                      "开采矿种", max_len=64)
            mining_method = parse_text(request.form.get("mining_method"),
                                       "开采方式", max_len=64)
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("project.patch_edit", patch_id=patch.id))
        patch.name = name
        patch.admin_region = admin_region
        patch.area_ha = area_ha
        patch.mineral_type = mineral_type
        patch.mining_method = mining_method
        patch.stop_years = stop_years
        patch.responsible_entity_missing = bool(request.form.get("responsible_entity_missing"))
        # 参照生态系统
        ref = ReferenceEcosystem.query.filter_by(patch_id=patch.id).first()
        if ref is None:
```

```
                ref = ReferenceEcosystem(patch_id=patch.id)
                db.session.add(ref)
            # 基准值：用户填 0 是合法输入（不能写成 `or 75.0`，那样 0 会被悄悄换成默认值）
            ref.coverage_benchmark = 75.0 if cov is None else
    cov
            ref.shannon_benchmark = 0.7 if sha is None else sha
            ref.dominant_species = fv("dominant_species")
            patch.image_note = fv("image_note")
            db.session.commit()
            flash("图斑已保存", "ok")
            return redirect(url_for("project.patches",
    pid=patch.project_id))
    return render_template("project/patch_edit.html",
    form=form,
                                   project=patch.project,
    patch=patch,
                                   reference=ReferenceEcosystem.quer
    y.filter_by(
                                       patch_id=patch.id).first())
# ------------------------------ 遥感影像挂接
    ------------------------------
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])
@login_required
def upload_image(patch_id):
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""
    from app.utils import uploads as up
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if not form.validate_on_submit():
        flash("表单校验失败，请重试", "error")
        return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
    rel, err = up.save_upload(request.files.get("image"),
    "patches",
                              current_app.config["ALLOWED_IM
    AGE_EXT"],
                              current_app.config["UPLOAD_FOL
    DER"])
    if err:
        flash(err, "error")
        return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
    # 替换旧影像时清理原文件，避免残留
    if patch.image_path:
        up.remove_upload(patch.image_path,
    current_app.config["UPLOAD_FOLDER"])
    patch.image_path = rel
    patch.image_uploaded_at = utcnow()
    note = request.form.get("image_note", "").strip()
    if note:
```

```
            patch.image_note = note                                    3451
        db.session.commit()                                            3452
        flash("遥感影像已挂接", "ok")                                    3453
        return redirect(url_for("project.patch_edit",                  3454
        patch_id=patch.id))                                            3455
@bp.route("/patch/<int:patch_id>/image/delete",                        3456
       methods=["POST"])                                               3457
@login_required                                                        3458
def delete_image(patch_id):                                            3459
        from app.utils import uploads as up                            3460
        patch = db.get_or_404(MinePatch, patch_id)                     3461
        form = EmptyForm()                                             3462
        if form.validate_on_submit():                                  3463
            up.remove_upload(patch.image_path,                         3464
        current_app.config["UPLOAD_FOLDER"])                           3465
            patch.image_path = None                                    3466
            patch.image_uploaded_at = None                             3467
            db.session.commit()                                        3468
            flash("遥感影像已移除", "ok")                                3469
        return redirect(url_for("project.patch_edit",                  3470
        patch_id=patch.id))                                            3471
# ==== app/views/unit.py ====                                          3472
# -*- coding: utf-8 -*-                                                3473
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""                3474
from flask import Blueprint, render_template, request,                 3475
     redirect, url_for, flash                                          3476
from flask_login import login_required                                 3477
from app.forms import EmptyForm                                        3478
from app.models import db, MinePatch, MicroUnit,                       3479
     IndicatorData                                                     3480
from app.utils.constants import UNIT_TYPES                             3481
from app.utils.parsing import FieldError, parse_float                  3482
bp = Blueprint("unit", __name__, url_prefix="/unit")                   3483
def recalc_ratio(patch_id):                                            3484
        """按面积重算该图斑各单元面积占比。"""                                3485
        units = MicroUnit.query.filter_by(patch_id=patch_id).all       3486
        ()                                                             3487
        total = sum(u.area or 0.0 for u in units)                      3488
        if total > 0:                                                  3489
            for u in units:                                            3490
                u.ratio = round((u.area or 0.0) / total, 6)            3491
        return total                                                   3492
@bp.route("/")                                                         3493
@login_required                                                        3494
def list_patches():                                                    3495
        """微地质单元入口：列出各图斑及其单元划分情况。"""                        3496
        patches = MinePatch.query.order_by(MinePatch.id.desc()).       3497
        all()                                                          3498
        rows = []                                                      3499
        for p in patches:                                              3500
```

```
        units = MicroUnit.query.filter_by(patch_id=p.id).all
    ()
        rows.append({
            "patch": p,
            "count": len(units),
            "ratio_sum": sum(u.ratio or 0.0 for u in units),
        })
    return render_template("unit/list.html", rows=rows)
@bp.route("/<int:patch_id>")
@login_required
def index(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
                           unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 评价流程规定将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        try:
            area_value = parse_float(area, "面积（公顷）",
    minimum=0)
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
```

```
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=area_value,
                      description=request.form.get("descript
    ion", "").strip() or None)
        db.session.add(u)
        db.session.flush()
        recalc_ratio(patch.id)
        db.session.commit()
        flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        new_type = request.form.get("unit_type",
    u.unit_type)
        if new_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.edit",
    unit_id=u.id))
        # 改类型时同样不允许与本图斑已有的其他单元重复
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,


    unit_type=new_type).first()
        if dup is not None and dup.id != u.id:
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,
    "error")
            return redirect(url_for("unit.edit",
    unit_id=u.id))
        u.unit_type = new_type
        area = request.form.get("area", "").strip()
        try:
            u.area = parse_float(area, "面积（公顷）", minimum=0)
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("unit.edit",
    unit_id=u.id))
        u.description = request.form.get("description",
    "").strip() or None
        recalc_ratio(u.patch_id)
        db.session.commit()
        flash("微地质单元已保存", "ok")
        return redirect(url_for("unit.index",
    patch_id=u.patch_id))
```

```
    return render_template("unit/edit.html", unit=u,                    3601
    patch=u.patch,                                                      3602
                                  unit_types=UNIT_TYPES, form=form)     3603
@bp.route("/delete/<int:unit_id>", methods=["POST"])                    3604
@login_required                                                         3605
def delete(unit_id):                                                    3606
    u = db.get_or_404(MicroUnit, unit_id)                               3607
    pid = u.patch_id                                                    3608
    name = u.unit_type                                                  3609
    # 一并删除该单元的指标数据，避免残留                                3610
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch            3611
    ronize_session=False)                                               3612
    db.session.delete(u)                                                3613
    db.session.flush()                                                  3614
    recalc_ratio(pid)                                                   3615
    db.session.commit()                                                 3616
    flash("微地质单元「%s」已删除" % name, "ok")                        3617
    return redirect(url_for("unit.index", patch_id=pid))                3618
# ==== app/views/standard.py ====                                       3619
# -*- coding: utf-8 -*-                                                 3620
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""  3621
from flask import Blueprint, render_template, request,                  3622
    redirect, url_for, flash                                            3623
from flask_login import login_required                                  3624
from app.forms import EmptyForm                                         3625
from app.models import db, Standard, StandardRule                       3626
from app.services import standard_check                                 3627
from app.utils.parsing import FieldError, parse_text                    3628
bp = Blueprint("standard", __name__, url_prefix="/standard")            3629
@bp.route("/")                                                          3630
@login_required                                                         3631
def index():                                                            3632
    standards = Standard.query.order_by(Standard.category,              3633
    Standard.province).all()                                            3634
    local = [s for s in standards if s.category == "地方标准"]          3635
    policy = [s for s in standards if s.category != "地方标准"]         3636
    return render_template("standard/index.html",                       3637
    local=local, policy=policy,                                         3638
                                  form=EmptyForm())                     3639
@bp.route("/add", methods=["GET", "POST"])                              3640
@login_required                                                         3641
def add():                                                              3642
    form = EmptyForm()                                                  3643
    if request.method == "POST" and form.validate_on_submit(            3644
    ):                                                                  3645
        try:                                                            3646
            province = parse_text(request.form.get("province            3647
    "), "省份",                                                         3648
                                      max_len=32, required=True)        3649
            name = parse_text(request.form.get("name"), "标准           3650
```

```
    / 文件名称",                                                    3651
                                        max_len=256, required=True)  3652
            doc_number = parse_text(request.form.get("doc_nu        3653
    mber"), "标准编号或文号",                                       3654
                                              max_len=128)          3655
        except FieldError as e:                                     3656
            flash(str(e), "error")                                  3657
            return redirect(url_for("standard.index"))              3658
        s = Standard(                                               3659
            province=province,                                      3660
            name=name,                                              3661
            doc_number=doc_number,                                  3662
            publish_date=request.form.get("publish_date",           3663
    "").strip() or None,                                            3664
            implement_date=request.form.get("implement_date"        3665
    , "").strip() or None,                                          3666
            category=request.form.get("category", "地方标准"),      3667
            summary=request.form.get("summary", "").strip()         3668
    or None,                                                        3669
        )                                                           3670
        if not s.province or not s.name:                            3671
            flash("省份与名称不能为空", "error")                    3672
        else:                                                       3673
            db.session.add(s)                                       3674
            db.session.commit()                                     3675
            flash("标准已添加", "ok")                               3676
            return redirect(url_for("standard.index"))              3677
    return render_template("standard/edit.html", form=form,         3678
    standard=None)                                                  3679
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])               3680
@login_required                                                     3681
def edit(sid):                                                      3682
    s = db.get_or_404(Standard, sid)                                3683
    form = EmptyForm()                                              3684
    if request.method == "POST" and form.validate_on_submit(        3685
    ):                                                              3686
        s.province = request.form.get("province",                   3687
    "").strip()                                                     3688
        s.name = request.form.get("name", "").strip()               3689
        s.doc_number = request.form.get("doc_number",               3690
    "").strip() or None                                             3691
        s.publish_date = request.form.get("publish_date",           3692
    "").strip() or None                                             3693
        s.implement_date = request.form.get("implement_date"        3694
    , "").strip() or None                                           3695
        s.category = request.form.get("category", "地方标准")       3696
        s.summary = request.form.get("summary", "").strip()         3697
    or None                                                         3698
        db.session.commit()                                         3699
        flash("已保存", "ok")                                       3700
```

```
            return redirect(url_for("standard.rules", sid=s.id))
        return render_template("standard/edit.html", form=form,
        standard=s)
@bp.route("/<int:sid>/delete", methods=["POST"])
@login_required
def delete(sid):
        s = db.get_or_404(Standard, sid)
        StandardRule.query.filter_by(standard_id=sid).delete()
        db.session.delete(s)
        db.session.commit()
        flash("已删除", "ok")
        return redirect(url_for("standard.index"))
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])
@login_required
def rules(sid):
        s = db.get_or_404(Standard, sid)
        form = EmptyForm()
        if request.method == "POST" and form.validate_on_submit(
        ):
            r = StandardRule(
                standard_id=s.id,
                item=request.form.get("item", "").strip(),
                operator=request.form.get("operator", ">="),
                threshold=request.form.get("threshold",
        "").strip() or None,
                condition_type=request.form.get("condition_type"
        , "必要条件"),
                region=request.form.get("region", "").strip()
        or None,
                note=request.form.get("note", "").strip() or
        None,
            )
            if not r.item:
                flash("判定项不能为空", "error")
            else:
                db.session.add(r)
                db.session.commit()
                flash("规则已添加", "ok")
            return redirect(url_for("standard.rules", sid=s.id))
        rs = StandardRule.query.filter_by(standard_id=s.id).all(
        )
        return render_template("standard/rules.html",
        form=form, standard=s, rules=rs)
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
        r = db.get_or_404(StandardRule, rid)
        sid = r.standard_id
        db.session.delete(r)
        db.session.commit()
```

```
    flash("规则已删除", "ok")
    return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
@login_required
def compare():
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""
    rules = StandardRule.query.join(Standard).all()
    groups = {}
    for r in rules:
        groups.setdefault(r.item, []).append(r)
    return render_template("standard/compare.html",
    groups=groups)
@bp.route("/check")
@login_required
def check_index():
    """结论对照入口：选择一个图斑，看评价结论是否符合当地文件要求。"""
    from app.models import MinePatch
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    rows = []
    for p in patches:
        found = standard_check.province_of(p.admin_region)
        rows.append({"patch": p, "province": found})
    return render_template("standard/check_index.html",
    rows=rows)
@bp.route("/check/<int:patch_id>")
@login_required
def check(patch_id):
    """某个图斑的评价结论与当地标准 / 政策文件的逐条对照。"""
    result = standard_check.compare(patch_id)
    if result is None:
        flash("未找到该矿山图斑", "error")
        return redirect(url_for("standard.check_index"))
    return render_template("standard/check.html",
    check=result)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""限制性指标判定 —— 一票否决判定。
按指标清单与评价流程，共 4 项：政策、区位约束、土壤污染、地质灾害。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, ConstraintCheck
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
```

```
FIELDS = [                                                              3801
    ("policy_pass", "政策约束"),                                        3802
    ("location_pass", "区位约束"),                                      3803
    ("pollution_pass", "土壤污染"),                                     3804
    ("geo_pass", "地质灾害"),                                           3805
]                                                                       3806
@bp.route("/")                                                          3807
@login_required                                                         3808
def index():                                                            3809
    patches = MinePatch.query.order_by(MinePatch.id.desc()).            3810
    all()                                                               3811
    latest = {}                                                         3812
    for p in patches:                                                   3813
        latest[p.id] = ConstraintCheck.query.filter_by(patch            3814
    _id=p.id) \                                                         3815
            .order_by(ConstraintCheck.id.desc()).first()                3816
    return render_template("constraint/index.html",                     3817
    patches=patches, latest=latest)                                     3818
@bp.route("/<int:patch_id>", methods=["GET", "POST"])                   3819
@login_required                                                         3820
def check(patch_id):                                                    3821
    patch = db.get_or_404(MinePatch, patch_id)                          3822
    form = EmptyForm()                                                  3823
    record = ConstraintCheck.query.filter_by(patch_id=patch_            3824
    id) \                                                               3825
        .order_by(ConstraintCheck.id.desc()).first()                    3826
    if request.method == "POST" and form.validate_on_submit(            3827
    ):                                                                  3828
        if record is None:                                              3829
            record = ConstraintCheck(patch_id=patch.id)                 3830
            db.session.add(record)                                      3831
        for field, _ in FIELDS:                                         3832
            setattr(record, field, request.form.get(field)              3833
    == "1")                                                             3834
        record.passed = all(getattr(record, f) for f, _ in              3835
    FIELDS)                                                             3836
        failed = [name for f, name in FIELDS if not                     3837
    getattr(record, f)]                                                 3838
        record.reason = ("全部通过" if record.passed                    3839
                         else "未通过：" + "、".join(failed))           3840
        db.session.commit()                                             3841
        flash("判定结果已保存", "ok")                                   3842
        return redirect(url_for("constraint.check",                     3843
    patch_id=patch.id))                                                 3844
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）       3845
    field_states = [(f, name, None if record is None else               3846
    getattr(record, f))                                                 3847
                    for f, name in FIELDS]                              3848
    return render_template("constraint/check.html",                     3849
    patch=patch, form=form,                                             3850
```

```
                                    fields=FIELDS,
        field_states=field_states,
                                    record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
import math
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
# 一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
```

```
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 （指标名称列表， 默认权重列表）。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form,
                                   weight_rows=ws.summary())
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form,
                           weight_rows=ws.summary())
@bp.route("/matrix", methods=["GET", "POST"])
```

```
def matrix():
    unit_type = request.values.get("unit_type")
    # level 是手改地址栏就能改的参数，写成 abc 不该把页面打成 500
    try:
        level = int(request.values.get("level", 1))
    except (TypeError, ValueError):
        level = 1
    if level not in (1, 2):
        level = 1
    l1 = normalize_l1(unit_type, level,
    request.values.get("l1"))
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    # 预填默认值：系统默认权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    def compute(upper):
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        bad = ahp.find_most_inconsistent(A)
        return {
            "weights": [round(v, 4) for v in r["weights"]],
            "lambda_max": round(r["lambda_max"], 4),
            "ci": r["ci"],
            "cr": r["cr"],
            "consistent": r["consistent"],
            "matrix": [[round(A[i, j], 4) for j in
    range(n)] for i in range(n)],
            "most_inconsistent": bad,
        }
    result = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        action = request.form.get("action", "calc")
        try:
            upper = [float(request.form.get("m_%d_%d" % (i,
    j), "1"))
                     for i in range(n) for j in range(i +
    1, n)]
            # 0 / 负数 / inf / nan 能过 float()，但会让 n 次方根算出 inf
    或 nan，
            # 保存时直接撞上 NOT NULL 约束（weight 为 nan）→ 500
            if any((not math.isfinite(x)) or x <= 0 for x
    in upper):
                raise ValueError("矩阵元素必须是正数")
            result = compute(upper)
```

```
        except (ValueError, ZeroDivisionError):                    4001
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",  4002
    "error")                                                       4003
            return redirect(url_for("ahp.matrix",                  4004
    unit_type=unit_type,                                           4005
                                          level=level, l1=l1))     4006
        if action == "save":                                       4007
            if not result["consistent"]:                           4008
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"  4009
                      "请调整后再保存" % result["cr"], "error")     4010
            else:                                                  4011
                pairs = to_code_pairs(unit_type, level, l1,        4012
    names,                                                         4013
                                          result["weights"])       4014
                scheme, cnt = ws.save_weights(unit_type,           4015
    pairs)                                                         4016
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"          4017
                      "此后该单元类型的评价将以此权重计算。"                4018
                      % (unit_type, cnt, result["cr"]),            4019
    "ok")                                                          4020
                return redirect(url_for("ahp.matrix",              4021
    unit_type=unit_type,                                           4022
                                            level=level, l1=l1))   4023
    elif request.values.get("calc") == "1":                        4024
        # 按默认（系统默认）矩阵直接试算，便于一键查看权重与一致性          4025
        result = compute([default[i][j] for i in range(n)          4026
    for j in range(i + 1, n)])                                     4027
    _, source_label, custom_scheme = ws.effective_weights(un       4028
    it_type)                                                       4029
    return render_template(                                        4030
        "ahp/matrix.html",                                         4031
        form=form, unit_type=unit_type, level=level, l1=l1,        4032
        names=names, vec=vec, default=default,                     4033
    result=result,                                                 4034
        source_label=source_label, custom_scheme=custom_sche       4035
    me,                                                            4036
    )                                                              4037
@bp.route("/weights/reset", methods=["POST"])                      4038
def weights_reset():                                               4039
    """删除自定义权重方案，恢复为该单元类型的系统默认权重。"""                4040
    unit_type = request.form.get("unit_type")                      4041
    form = EmptyForm()                                             4042
    if form.validate_on_submit():                                  4043
        if unit_type not in DEFAULT_WEIGHTS:                       4044
            flash("无效的微地质单元类型", "error")                    4045
        else:                                                      4046
            n = ws.reset_weights(unit_type)                        4047
            if n:                                                  4048
                flash("已恢复「%s」为系统默认权重" % unit_type, "ok")     4049
            else:                                                  4050
```

```
                flash("「%s」本就是系统默认权重，无需恢复" % unit_type,
    "ok")
    return redirect(url_for("ahp.index"))
# ---------------------------- 指标维护 ----------------------------
    ----
@bp.route("/indicators")
def indicators():
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""
    if Indicator.query.count() == 0:
        ind_svc.seed_indicators()
    all_ind = Indicator.query.order_by(Indicator.level,
    Indicator.id).all()
    by_parent = {}
    for i in all_ind:
        by_parent.setdefault(i.parent_code, []).append(i)
    return render_template("ahp/indicators.html",
    by_parent=by_parent,
                           all_ind=all_ind,
    unit_types=UNIT_TYPES,
                           type_label=ind_svc.ideal_type_lab
    el,
                           form=EmptyForm())
@bp.route("/indicators/<code>/edit", methods=["GET",
    "POST"])
def indicator_edit(code):
    ind = Indicator.query.filter_by(code=code).first_or_404(
    )
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        name = request.form.get("name", "").strip()
        if not name:
            flash("指标名称不能为空", "error")
            return redirect(url_for("ahp.indicator_edit",
    code=code))
        ind.name = name
        ind.unit = request.form.get("unit", "").strip() or
    None
        ind.ideal_value = request.form.get("ideal_value",
    "").strip() or None
        ind.ideal_type = request.form.get("ideal_type") or
    ind.ideal_type
        units = request.form.getlist("enabled_units")
        ind.enabled_units = ",".join(units) if units else
    None
        ind.is_constraint = request.form.get("is_constraint"
    ) == "1"
        ind.description = request.form.get("description",
    "").strip() or None
        db.session.commit()
```

```
            flash("指标「%s」已保存" % ind.name, "ok")                    4101
            return redirect(url_for("ahp.indicators"))                   4102
        return render_template("ahp/indicator_edit.html",                4103
        ind=ind,                                                         4104
                                    unit_types=UNIT_TYPES,               4105
                                    type_label=ind_svc.ideal_type_lab4106
        el, form=form)                                                   4107
@bp.route("/indicators/reset", methods=["POST"])                         4108
def indicators_reset():                                                  4109
    """按系统默认值重置指标定义。"""                                          4110
    form = EmptyForm()                                                   4111
    if form.validate_on_submit():                                        4112
        n = ind_svc.seed_indicators()                                    4113
        flash("已按系统默认值重置 %d 项指标定义" % n, "ok")                      4114
    return redirect(url_for("ahp.indicators"))                           4115
# ---------------------------- 多专家矩阵 ----------------------------4116
    -----                                                                4117
def expert_scope():                                                      4118
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""                          4119
    unit_type = request.values.get("unit_type") or                       4120
    UNIT_TYPES[0]                                                        4121
    if unit_type not in DEFAULT_WEIGHTS:                                 4122
        unit_type = UNIT_TYPES[0]                                        4123
    try:                                                                 4124
        level = int(request.values.get("level", 1))                     4125
    except (TypeError, ValueError):                                      4126
        level = 1                                                        4127
    if level not in (1, 2):                                              4128
        level = 1                                                        4129
    l1 = normalize_l1(unit_type, level,                                  4130
    request.values.get("l1"))                                            4131
    return unit_type, level, l1                                          4132
def expert_query(unit_type, level, l1):                                  4133
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,                4134
    level=level)                                                         4135
    if level == 2:                                                       4136
        q = q.filter_by(l1_name=l1)                                      4137
    else:                                                                4138
        q = q.filter(ExpertMatrix.l1_name.is_(None))                     4139
    return q.order_by(ExpertMatrix.id)                                   4140
def default_scheme(unit_type, level, l1=None):                           4141
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""                        4142
    name = "%s · %s%s" % (unit_type,                                     4143
                          "一级指标" if level == 1 else "二级指标",          4144
                          "(%s)" % l1 if l1 else "")                     4145
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ4146
    e,                                                                   4147
                                          level=level,                   4148
    name=name).first()                                                   4149
    if scheme is None:                                                   4150
```

```
            scheme = WeightScheme(unit_type=unit_type,
        level=level, name=name,
                                  is_default=False)
            db.session.add(scheme)
            db.session.flush()
        return scheme
@bp.route("/experts", methods=["GET", "POST"])
def experts():
        """多专家判断矩阵录入与汇总。"""
        unit_type, level, l1 = expert_scope()
        if unit_type not in DEFAULT_WEIGHTS:
            flash("无效的微地质单元类型", "error")
            return redirect(url_for("ahp.index"))
        names, vec = indicator_list(unit_type, level, l1)
        n = len(names)
        form = EmptyForm()
        if request.method == "POST" and form.validate_on_submit(
        ):
            action = request.form.get("action")
            if action == "add":
                expert_name = request.form.get("expert_name",
        "").strip() or "未具名专家"
                specialty = request.form.get("specialty",
        "").strip()
                try:
                    upper = [float(request.form.get("m_%d_%d" %
        (i, j), "1"))
                             for i in range(n) for j in range(i
        + 1, n)]
                except ValueError:
                    flash("矩阵输入有误，请填入正数", "error")
                    return redirect(url_for("ahp.experts",
        unit_type=unit_type,
                                            level=level, l1=l1))
                # 0 / 负数 / inf / nan 都能过 float()，但会让 n 次方根算出
        inf 或 nan，
                # 于是库里存下 [[1,0],[Infinity,1]] 这种矩阵、CR 还显示
        0.0000。
                if any((not math.isfinite(x)) or x <= 0 for x
        in upper):
                    flash("判断矩阵只能填正数（大于 0 的有限数字）", "error")
                    return redirect(url_for("ahp.experts",
        unit_type=unit_type,
                                            level=level, l1=l1))
                A = ahp.matrix_from_upper(upper)
                r = ahp.weights_and_consistency(A)
                scheme = default_scheme(unit_type, level, l1)
                db.session.add(ExpertMatrix(
                    scheme_id=scheme.id,
                    unit_type=unit_type, l1_name=(l1 if level
```

```
    == 2 else None),                                                    4201
                expert_name=expert_name,                                4202
    specialty=specialty or None, level=level,                           4203
                matrix_json=json.dumps([list(map(float,                 4204
    row)) for row in A]),                                               4205
                cr=round(r["cr"], 6)))                                  4206
            db.session.commit()                                         4207
            flash("已录入专家「%s」%s的判断矩阵（CR=%.4f）" %              4208
    (expert_name, ("(%s)" % specialty) if specialty else                4209
    "", r["cr"]), "ok")                                                 4210
    elif action == "save_agg":                                          4211
            # 把汇总权重落库并生效，使多专家结果真正参与评价计算          4212
            weights = _aggregate(unit_type, level, l1)                  4213
            if weights is None:                                         4214
                flash("暂无可汇总的专家矩阵", "error")                   4215
            elif not weights["consistent"]:                             4216
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"    4217
                      "请先检查或调整专家矩阵" % weights["cr"],           4218
    "error")                                                            4219
            else:                                                       4220
                names_, _ = indicator_list(unit_type,                   4221
    level, l1)                                                          4222
                pairs = to_code_pairs(unit_type, level, l1,             4223
    names_,                                                             4224
                                      weights["weights"])               4225
                _, cnt = ws.save_weights(unit_type, pairs)              4226
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"      4227
                      % (unit_type, cnt, weights["cr"]),                4228
    "ok")                                                               4229
    else:                                                               4230
            # action 缺失或值不认识时以前是静默跳回，用户只觉得「点了没反应」  4231
            flash("未识别的操作，请从页面上的按钮重新提交", "error")       4232
    return redirect(url_for("ahp.experts",                              4233
    unit_type=unit_type,                                                4234
                            level=level, l1=l1))                        4235
    rows = expert_query(unit_type, level, l1).all()                     4236
    default = [[round(vec[i] / vec[j], 4) for j in                      4237
    range(n)] for i in range(n)]                                        4238
    return render_template("ahp/experts.html",                          4239
    unit_type=unit_type, level=level,                                   4240
                           l1=l1, names=names, rows=rows,               4241
    default=default,                                                    4242
                           n=n, form=form)                              4243
@bp.route("/experts/<int:mid>/delete", methods=["POST"])                4244
def expert_delete(mid):                                                 4245
    row = db.get_or_404(ExpertMatrix, mid)                              4246
    unit_type, level, l1 = row.unit_type, row.level,                    4247
    row.l1_name                                                         4248
    form = EmptyForm()                                                  4249
    if form.validate_on_submit():                                       4250
```

```
        db.session.delete(row)                                          4251
        db.session.commit()                                             4252
        flash("已删除专家矩阵", "ok")                                    4253
    return redirect(url_for("ahp.experts",                              4254
    unit_type=unit_type,                                                4255
                                    level=level, l1=l1))                4256
def _aggregate(unit_type, level, l1):                                   4257
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""          4258
    rows = expert_query(unit_type, level, l1).all()                     4259
    mats = []                                                           4260
    for r in rows:                                                      4261
        try:                                                            4262
            mats.append([list(map(float, line))                         4263
                         for line in json.loads(r.matrix_jso4264
    n)])                                                                4265
        except Exception:                                               4266
            continue                                                    4267
    if not mats:                                                        4268
        return None                                                     4269
    agg = ahp.aggregate_matrices(mats)                                  4270
    res = ahp.weights_and_consistency(agg)                              4271
    return {                                                            4272
        "count": len(mats),                                             4273
        "weights": res["weights"],                                      4274
        "lambda_max": res["lambda_max"],                                4275
        "ci": res["ci"], "cr": res["cr"],                               4276
        "consistent": res["consistent"],                                4277
        "most_inconsistent": ahp.find_most_inconsistent(agg)4278
    ,                                                                   4279
        "expert_crs": [r.cr for r in rows if r.cr is not                4280
    None],                                                              4281
    }                                                                   4282
@bp.route("/experts/aggregate")                                         4283
def experts_aggregate():                                                4284
    """按几何平均汇总多专家矩阵，并做一致性检验。"""                          4285
    unit_type, level, l1 = expert_scope()                               4286
    if unit_type not in DEFAULT_WEIGHTS:                                4287
        flash("无效的微地质单元类型", "error")                           4288
        return redirect(url_for("ahp.index"))                           4289
    names, vec = indicator_list(unit_type, level, l1)                   4290
    rows = expert_query(unit_type, level, l1).all()                     4291
    result = _aggregate(unit_type, level, l1)                           4292
    if result is not None:                                              4293
        result["weights"] = [round(v, 4) for v in                       4294
    result["weights"]]                                                  4295
        result["lambda_max"] = round(result["lambda_max"],              4296
    4)                                                                  4297
    return render_template("ahp/experts_aggregate.html",                4298
    unit_type=unit_type,                                                4299
                                level=level, l1=l1, names=names,        4300
```

```
    rows=rows,                                                    4301
                                    result=result, form=EmptyForm())  4302
# ==== app/views/indicator.py ====                                4303
# -*- coding: utf-8 -*-                                           4304
"""指标数据采集与管理 —— 路由与页面逻辑。                          4305
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标        4306
（岩质边坡无土壤指标、平台无地形指标）。                          4307
"""                                                               4308
import io                                                         4309
import json                                                       4310
from flask import (Blueprint, render_template, request,           4311
    redirect, url_for,                                            4312
                      flash, send_file)                           4313
from app.forms import EmptyForm                                   4314
from app.models import db, MinePatch, MicroUnit,                  4315
    IndicatorData                                                 4316
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,       4317
    UNIT_ENABLED_INDICATORS                                       4318
from app.utils.parsing import FieldError, parse_float             4319
from app.services import import_service as imp                    4320
bp = Blueprint("indicator", __name__,                             4321
    url_prefix="/indicator")                                      4322
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)                4323
SOIL_FIELDS = [                                                   4324
    ("soil_ratio", "土石比", ""),                                 4325
    ("thickness", "土层厚度", "cm"),                              4326
    ("ph", "pH", ""),                                             4327
    ("organic_matter", "有机质", "g/kg"),                         4328
]                                                                 4329
VEG_FIELDS = [                                                    4330
    ("coverage", "植被覆盖度", "%"),                              4331
    ("shannon", "Shannon-Wiener 多样性指数", ""),                 4332
]                                                                 4333
TERRAIN_FIELDS = [                                                4334
    ("slope", "坡度", "°"),                                       4335
    ("aspect", "坡向", ""),                                       4336
]                                                                 4337
def unit_field_config(unit_type):                                 4338
    """根据单元类型返回需录入的字段分组。"""                      4339
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())          4340
    groups = []                                                   4341
    if "土壤指标" in enabled:                                     4342
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True  4343
    = 含重金属                                                    4344
    if "植被指标" in enabled:                                     4345
        groups.append(("植被指标", VEG_FIELDS, False))            4346
    if "地形指标" in enabled:                                     4347
        groups.append(("地形指标", TERRAIN_FIELDS, False))        4348
    return groups                                                 4349
@bp.route("/")                                                    4350
```

```
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    return render_template("indicator/index.html", patches=patches)
@bp.route("/<int:unit_id>", methods=["GET", "POST"])
def edit(unit_id):
    unit = db.get_or_404(MicroUnit, unit_id)
    ind = IndicatorData.query.filter_by(unit_id=unit.id).first()
    form = EmptyForm()
    groups = unit_field_config(unit.unit_type)
    if request.method == "POST" and form.validate_on_submit():
        # 各字段的合理范围（覆盖度是百分数、pH 0–14、坡度 0–90，其余非负）。
        # 以前这里直接 float(v)，用户填个「约 3.5」看到的就是英文 500 页。
        LIMITS = {
            "soil_ratio": ("土石比", 0, None),
            "thickness": ("土层厚度（cm）", 0, None),
            "ph": ("pH", 0, 14),
            "organic_matter": ("有机质（g/kg）", 0, None),
            "coverage": ("植被覆盖度（%）", 0, 100),
            "shannon": ("Shannon 指数", 0, None),
            "slope": ("坡度（°）", 0, 90),
        }
        try:
            values = {}
            for name, (label, lo, hi) in LIMITS.items():
                values[name] = parse_float(request.form.get(name), label,
                                           minimum=lo, maximum=hi)
            heavy = {}
            for m in HEAVY_METALS:
                v = parse_float(request.form.get("hm_" + m),
                                "重金属 %s（mg/kg）" % m, minimum=0)
                if v is not None:
                    heavy[m] = v
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("indicator.edit", unit_id=unit.id))
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        for name, value in values.items():
            setattr(ind, name, value)
        ind.heavy_metals = json.dumps(heavy, ensure_ascii=False) if heavy else None
```

```
        ind.aspect = request.form.get("aspect", "").strip() 4401
    or None                                                 4402
        db.session.commit()                                 4403
        flash("指标数据已保存", "ok")                        4404
        return redirect(url_for("indicator.index"))         4405
    # 组装当前值供模板回显                                  4406
    current = {}                                            4407
    if ind:                                                 4408
        current = {                                         4409
            "soil_ratio": ind.soil_ratio, "thickness":      4410
    ind.thickness,                                          4411
            "ph": ind.ph, "organic_matter":                 4412
    ind.organic_matter,                                     4413
            "coverage": ind.coverage, "shannon":            4414
    ind.shannon,                                            4415
            "slope": ind.slope, "aspect": ind.aspect,       4416
        }                                                   4417
        current["heavy"] = ind.heavy_metals_dict()          4418
    return render_template("indicator/edit.html",           4419
                           unit=unit, form=form,            4420
    groups=groups,                                          4421
                           heavy_metals=HEAVY_METALS,       4422
                           aspect_types=ASPECT_TYPES,       4423
    current=current)                                        4424
# ------------------------------ 批量导入 ------------------------------4425
    ----                                                    4426
@bp.route("/import", methods=["GET", "POST"])               4427
def import_data():                                          4428
    """批量导入指标数据（CSV / Excel）。"""                  4429
    patches = MinePatch.query.order_by(MinePatch.id.desc()).4430
    all()                                                   4431
    form = EmptyForm()                                      4432
    report = None                                           4433
    if request.method == "POST" and form.validate_on_submit(4434
    ):                                                      4435
        patch_id = request.form.get("patch_id", type=int)   4436
        patch = db.session.get(MinePatch, patch_id) if      4437
    patch_id else None                                      4438
        if patch is None:                                   4439
            flash("请选择矿山图斑", "error")                 4440
            return redirect(url_for("indicator.import_data")4441
    )                                                       4442
        file = request.files.get("datafile")                4443
        if file is None or not file.filename:               4444
            flash("请选择要导入的文件", "error")             4445
            return redirect(url_for("indicator.import_data")4446
    )                                                       4447
        ext = file.filename.rsplit(".", 1)[-1].lower()      4448
        if ext not in ("csv", "xlsx", "xls"):               4449
            flash("仅支持 CSV 或 Excel（.xlsx/.xls）文件", "error") 4450
```

```
            return redirect(url_for("indicator.import_data")4451
)                                                           4452
        units = MicroUnit.query.filter_by(patch_id=patch.id)4453
.all()                                                      4454
        unit_map = {}                                       4455
        for u in units:                                     4456
            unit_map.setdefault(u.unit_type, u)             4457
        if not unit_map:                                    4458
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")4459
            return redirect(url_for("indicator.import_data")4460
)                                                           4461
        try:                                                4462
            parsed, errors = imp.parse_upload(file.filename,4463
file.stream, unit_map)                                      4464
        except Exception as e:                              4465
            flash("文件解析失败：%s" % e, "error")          4466
            return redirect(url_for("indicator.import_data")4467
)                                                           4468
        saved = 0                                           4469
        for rec in parsed:                                  4470
            unit = unit_map[rec["unit_type"]]               4471
            ind = IndicatorData.query.filter_by(unit_id=unit4472
.id).first()                                                4473
            if ind is None:                                 4474
                ind = IndicatorData(unit_id=unit.id)        4475
                db.session.add(ind)                         4476
            f = rec["fields"]                               4477
            for key in ("soil_ratio", "thickness", "ph",    4478
"organic_matter",                                           4479
                        "coverage", "shannon", "slope",     4480
"aspect"):                                                  4481
                if key in f:                                4482
                    setattr(ind, key, f[key])               4483
            if rec["heavy_metals"]:                         4484
                ind.heavy_metals = json.dumps(rec["heavy_met4485
als"],                                                      4486
                                                            4487
ensure_ascii=False)                                         4488
            saved += 1                                      4489
        db.session.commit()                                 4490
        report = {"patch": patch.name, "saved": saved,      4491
"errors": errors,                                           4492
                  "total": len(parsed) + len(               4493
                      [e for e in errors if e.startswith("第 4494
")])}                                                       4495
        if saved:                                           4496
            flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")4497
        else:                                               4498
            flash("没有可导入的数据行，请检查文件内容", "error")4499
    return render_template("indicator/import.html",         4500
```

```
    patches=patches,                                                    4501
                                form=form, report=report,               4502
    headers=imp.HEADERS)                                                4503
@bp.route("/import/template")                                           4504
def import_template():                                                  4505
    """下载导入模板（CSV，含一行示例）。"""                                4506
    csv_text = imp.build_template_csv()                                 4507
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于            4508
    Excel 正确识别中文                                                    4509
    return send_file(io.BytesIO(data), as_attachment=True,              4510
                     download_name="指标数据导入模板.csv",                4511
                     mimetype="text/csv")                               4512
# ==== app/views/evaluate.py ====                                       4513
# -*- coding: utf-8 -*-                                                 4514
"""适宜性定量评价引擎 —— 路由与页面逻辑。                                  4515
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，                4516
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。                     4517
入口处按评价流程的顺序检查前置条件（见 app/services/workflow.py），       4518
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。                 4519
"""                                                                     4520
import json                                                             4521
from flask import Blueprint, render_template, request,                  4522
    redirect, url_for, flash                                            4523
from flask_login import current_user                                    4524
from app.forms import EmptyForm                                         4525
from app.models import (db, MinePatch, MicroUnit,                       4526
    IndicatorData,                                                      4527
                        EvaluationResult, ScoreDetail)                  4528
from app.services import evaluation_service as ev                       4529
from app.services import workflow as wf                                 4530
from app.services import unit_config                                    4531
from app.services import standard_check                                 4532
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")            4533
def load_patch_input(patch_id):                                         4534
    """从数据库组装评价引擎输入。"""                                       4535
    units = MicroUnit.query.filter_by(patch_id=patch_id).all            4536
    ()                                                                  4537
    result = []                                                         4538
    for u in units:                                                     4539
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi            4540
    rst()                                                               4541
        data = {                                                        4542
            "soil_ratio": ind.soil_ratio if ind else None,              4543
            "thickness": ind.thickness if ind else None,                4544
            "ph": ind.ph if ind else None,                              4545
            "organic_matter": ind.organic_matter if ind                 4546
    else None,                                                          4547
            "heavy_metals": ind.heavy_metals_dict() if ind              4548
    else {},                                                            4549
            "coverage": ind.coverage if ind else None,                  4550
```

```
                "shannon": ind.shannon if ind else None,
                "slope": ind.slope if ind else None,
                "aspect": ind.aspect if ind else None,
            }
            result.append({
                "unit_id": u.id,
                "unit_type": u.unit_type,
                "ratio": u.ratio if u.ratio is not None else
    0.0,
                "data": data,
            })
    return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自 AHP 计算并保存的生效方案；未配置时用系统默认值。
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 JSON。
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
            "unit_score": _round(u["unit_score"]),
            "weighted": _round(u["weighted"]),
            "weight_source": sources.get(u["unit_type"]),
            "l1_scores": {k: _round(v) for k, v in
    u["l1_scores"].items()},
            "details": [{
                "l1": d["l1"], "l2": d["l2"],
                "raw": None if d["raw"] is None else
    str(d["raw"]),
                "std_value": _round(d["std_value"]),
```

```
                    "weight": _round(d["weight"]),                    4601
                    "contribution": _round(d["contribution"]),        4602
                } for d in u["details"]],                             4603
            })                                                        4604
    return out                                                        4605
def pack_params(settings, sources=None):                              4606
    """把本次计算用到的参数整理成可持久化的快照。                          4607
    参数（指标理想取值、图斑参照生态系统基准、权重方案）都能在界面上改，      4608
    所以结果记录必须连「当时用的什么参数」一起留下，否则事后无法复现。        4609
    """                                                               4610
    from app.utils.constants import (IDEAL_SOIL_RATIO,                4611
    IDEAL_THICKNESS,                                                  4612
                                     IDEAL_ORGANIC,                   4613
    IDEAL_COVERAGE,                                                   4614
                                     IDEAL_SHANNON,                   4615
    HEAVY_METAL_LIMIT)                                                4616
    defaults = {                                                      4617
        "soil_ratio": IDEAL_SOIL_RATIO, "thickness":                  4618
    IDEAL_THICKNESS,                                                  4619
        "organic_matter": IDEAL_ORGANIC, "coverage":                  4620
    IDEAL_COVERAGE,                                                   4621
        "shannon": IDEAL_SHANNON,                                     4622
    }                                                                 4623
    for metal, limit in HEAVY_METAL_LIMIT.items():                    4624
        defaults["hm_" + metal] = limit                               4625
    used = {}                                                         4626
    for code, fallback in sorted(defaults.items()):                   4627
        value = settings.params.get(code) if settings else            4628
    None                                                              4629
        if value is None:                                             4630
            used[code] = {"value": fallback, "default":               4631
    fallback,                                                         4632
                          "source": "系统默认取值"}                    4633
        elif abs(float(value) - float(fallback)) < 1e-9:              4634
            #  库里存着这一项，但取值与系统默认值一致 —— 不能标成「界面配置」，  4635
            #  否则每份报告都会声称用了自定义参数，反而看不出真正的偏离。      4636
            used[code] = {"value": value, "default":                  4637
    fallback,                                                         4638
                          "source": "系统默认取值"}                    4639
        else:                                                         4640
            used[code] = {"value": value, "default":                  4641
    fallback,                                                         4642
                          "source": "界面配置（系统默认值 %.4g）" %     4643
    fallback}                                                         4644
    return {"ideals": used, "weight_sources": dict(sources            4645
    or {})}                                                           4646
def save_evaluation(patch_id, units):                                 4647
    """执行评价并把结果与逐层明细落库，返回评价记录。                      4648
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用              4649
    系统默认权重；来源一并记录。结果 JSON 保留各单元得分、                  4650
```

```
    一级指标得分与逐条二级指标明细，结果页与报告据此复现
    记录下来的那次评价，不再重新推算。
    """
    weights, sources = load_weights(units)
    #  标准化参数取自该图斑的参照生态系统与指标定义（界面上可改），
    #  没配过的项自动回落到代码常量，历史结果与验证算例不受影响。
    settings = unit_config.for_patch(patch_id)
    res = ev.score_mine(units, weights, settings)
    saved = EvaluationResult(
        patch_id=patch_id,
        final_score=round(res["final_score"], 4),
        grade=res["grade"],
        measures=res["grade_info"]["measure"],
        unit_scores=json.dumps(pack_units(res["unit_results"
    ], sources),
                               ensure_ascii=False),
        #  参数快照 + 计算人：参数现在可以在界面上改，不留快照就复现不了这次结果
        params_json=json.dumps(pack_params(settings,
    sources),
                               ensure_ascii=False),
        created_by=current_user.id if getattr(current_user,
    "is_authenticated", False)
        else None,
    )
    db.session.add(saved)
    db.session.flush()        #  先取得 result.id，再写明细
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}
    for u in res["unit_results"]:
        for d in u["details"]:
            db.session.add(ScoreDetail(
                result_id=saved.id,
                unit_id=unit_ids.get(u["unit_type"]),
                l1_name=d["l1"], l2_name=d["l2"],
                raw_value=None if d["raw"] is None else
    str(d["raw"]),
                std_value=d["std_value"],
    weight=d["weight"],
                contribution=d["contribution"]))
    db.session.commit()
    return saved
def weak_points(unit_scores, top=3):
    """从已保存的结果里挑出「提升空间最大」的指标。
    提升空间 = 权重 ×（1 − 标准化值）：标准化值越低、权重越大，抬它越划算。
    只读已保存的记录，不重新计算，保证与报告口径一致。
    """
    rows = []
    for u in unit_scores or []:
        for d in u.get("details") or []:
            std = d.get("std_value")
            weight = d.get("weight")
```

```
            if std is None or weight is None:
                continue                # 不适用项（未参与计算）不进短板
            rows.append({
                "unit_type": u.get("unit_type"),
                "l2": d.get("l2"),
                "raw": d.get("raw"),
                "std_value": std,
                "gain": weight * (1.0 - std),
            })
    rows.sort(key=lambda x: x["gain"], reverse=True)
    return [x for x in rows if x["gain"] > 1e-9][:top]
def user_label(user_id):
    """把计算人 id 换成「昵称（账户名称）」。"""
    from app.models import User
    if not user_id:
        return None
    u = db.session.get(User, user_id)
    if u is None:
        return "已删除的账号"
    return u.shown_name
def param_source_summary(params):
    """一句话说清参数从哪来：有没有偏离系统默认值。
    多数情况下全部等于系统默认值，报告应直接说明「未偏离」；
    只有真被改过时才点名，便于审阅者一眼看到差异。
    """
    ideals = (params or {}).get("ideals") or {}
    custom = sorted(k for k, v in ideals.items()
                    if v.get("source",
    "").startswith("界面配置"))
    if not custom:
        return "全部采用系统默认取值，未被改动"
    return "以下 %d 项被改为界面配置值：%s" % (len(custom),
    "、".join(custom))
def block_reasons(progress):
    """把未满足的前置条件转成可读的拦截原因。"""
    return ["%s：%s" % (b["label"], b["reason"]) for b in
    progress["blocking"]]
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    progress = {}
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patc
    h_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
```

```
                                latest=latest, progress=progress)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
    if request.method == "POST" and form.validate_on_submit(
    ):
        if not progress["ready"]:
            for reason in block_reasons(progress):
                flash("无法执行评价 —— " + reason, "error")
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        saved = save_evaluation(patch.id, units)
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    # 展示将采用的权重来源，便于在执行前确认
    units_now = load_patch_input(patch_id)
    _, sources = load_weights(units_now) if units_now else
    ({}, {})
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                           progress=progress,
    weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    params = json.loads(r.params_json) if r.params_json
    else {}
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                           unit_scores=unit_scores,
                           undetected=wf.undetected_metals(p
    atch.id)
                           if patch else [],
                           weak=weak_points(unit_scores),
                           params=params.get("ideals") or
    {},
                           param_sources=param_source_summar
    y(params),
                           check=standard_check.compare(patc
    h.id) if patch else None,
                           operator=user_label(r.created_by)
    )
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
```

```
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
import json
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
from app.services import report_service as rs
from app.services import standard_check
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.utils.constants import grade_of
from app.utils.timefmt import now_utc
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查
    结果，由调用方拦截并提示先去执行评价。
    """
    patch = db.get_or_404(MinePatch, patch_id)
    prog = wf.report_progress(patch_id)
    if not prog["ready"]:
        return patch, None, None, None, prog
    rec = prog["result"]
    result = {
        "final_score": rec.final_score,
        "grade": rec.grade,
        "grade_info": grade_of(rec.final_score),
        "unit_results": json.loads(rec.unit_scores) if
    rec.unit_scores else [],
    }
    reference = ReferenceEcosystem.query.filter_by(patch_id=
    patch_id).first()
    return patch, reference, result, rec, prog
def _deny(progress, patch_id, endpoint="report.detail"):
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""
    for b in progress["blocking"]:
        flash("暂不能生成报告 —— %s：%s" % (b["label"],
    b["reason"]), "error")
    if progress["blocking"]:
        flash("请先完成上述步骤后再生成报告。", "error")
    return redirect(url_for("report.index"))
def _weak_points(result, top=3):
    """提升空间最大的指标：权重 ×（1 − 标准化值）。
    与结果页用同一套口径，便于向使用方解释「先改哪几项」。
    """
    rows = []
```

```
    for u in result.get("unit_results") or []:
        for d in u.get("details") or []:
            std, w = d.get("std_value"), d.get("weight")
            if std is None or w is None:
                continue
            rows.append((u["unit_type"], d["l2"],
    d.get("raw"), std,
                         w * (1.0 - std)))
    rows.sort(key=lambda x: x[4], reverse=True)
    return [r for r in rows if r[4] > 1e-9][:top]
def _operator(record):
    """报告里的计算人：昵称（账户名称）。"""
    from app.models import User
    uid = getattr(record, "created_by", None)
    if not uid:
        return None
    u = db.session.get(User, uid)
    return u.shown_name if u else "已删除的账号"
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
    chart_units = charts.hbars(units)
    #  各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
            acc[k] = acc.get(k, 0.0) + v
            cnt[k] = cnt.get(k, 0) + 1
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
    items.sort(key=lambda x: -x[1])
    chart_weight = charts.donut(items)
    return chart_units, chart_weight
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    progress = {p.id: wf.report_progress(p.id) for p in
    patches}
    return render_template("report/index.html",
    patches=patches,
                           progress=progress)
@bp.route("/<int:patch_id>")
def detail(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
```

```
    chart_units, chart_weight = _charts(result)                    4901
    return render_template("report/detail.html",                   4902
    patch=patch, result=result,                                    4903
                                record=record,                     4904
    reference=reference,                                           4905
                                chart_units=chart_units,           4906
    chart_weight=chart_weight,                                     4907
                                check=standard_check.compare(patc4908
    h.id),                                                         4909
                                undetected=wf.undetected_metals(p4910
    atch.id))                                                      4911
@bp.route("/<int:patch_id>/word")                                  4912
def word(patch_id):                                                4913
    patch, reference, result, record, prog =                       4914
    _prepare(patch_id)                                             4915
    if result is None:                                             4916
        return _deny(prog, patch_id)                               4917
    buf = rs.build_report(patch, result, record=record,            4918
                          weak=_weak_points(result),               4919
                          operator=_operator(record),              4920
                          check=standard_check.compare(patch4921
    .id))                                                          4922
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name          4923
    return send_file(buf, as_attachment=True,                      4924
    download_name=filename,                                        4925
                     mimetype="application/vnd.openxmlformat4926
    s-officedocument."                                             4927
                              "wordprocessingml.document")         4928
@bp.route("/<int:patch_id>/pdf")                                   4929
def pdf(patch_id):                                                 4930
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""          4931
    patch, reference, result, record, prog =                       4932
    _prepare(patch_id)                                             4933
    if result is None:                                             4934
        return _deny(prog, patch_id)                               4935
    chart_units, chart_weight = _charts(result)                    4936
    html = render_template(                                        4937
        "report/print.html", patch=patch,                          4938
    reference=reference, result=result,                            4939
        chart_units=chart_units, chart_weight=chart_weight,        4940
        check=standard_check.compare(patch.id),                    4941
        generated_at=now_utc())                                    4942
    data, err = browser.html_to_pdf(html)                          4943
    if err:                                                        4944
        flash("PDF 导出失败：%s" % err, "error")                     4945
        return redirect(url_for("report.detail",                   4946
    patch_id=patch_id))                                            4947
    return send_file(io.BytesIO(data), as_attachment=True,         4948
                     download_name="自然恢复适宜性评价报告_%s.pdf" %    4949
    patch.name,                                                    4950
```

```
                                mimetype="application/pdf")
# ==== app/views/search.py ====
# -*- coding: utf-8 -*-
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。
"""
from sqlalchemy import or_
from flask import Blueprint, render_template, request
from app.models import db, Project, MinePatch, Standard
bp = Blueprint("search", __name__)
LIMIT = 50              # 每类结果上限，避免一次渲染过多
@bp.route("/search")
def index():
    q = (request.args.get("q") or "").strip()
    projects, patches, standards = [], [], []
    if q:
        like = "%" + q + "%"
        projects = Project.query.filter(or_(
            Project.name.like(like),
            Project.code.like(like),
            Project.description.like(like),
        )).order_by(Project.id.desc()).limit(LIMIT).all()
        patches = MinePatch.query.filter(or_(
            MinePatch.name.like(like),
            MinePatch.admin_region.like(like),
            MinePatch.mineral_type.like(like),
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()
        standards = Standard.query.filter(or_(
            Standard.name.like(like),
            Standard.doc_number.like(like),
            Standard.province.like(like),
            Standard.summary.like(like),
        )).order_by(Standard.id).limit(LIMIT).all()
    total = len(projects) + len(patches) + len(standards)
    return render_template("search.html", q=q,
    projects=projects,
                                   patches=patches,
    standards=standards, total=total)
# ==== app/templates/base.html ====
{% import "_macros.html" as ui %}<!DOCTYPE html>
<html lang="zh-CN">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
    initial-scale=1.0">
  <link rel="icon" href="{{ url_for('static',
    filename='favicon.ico') }}">
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock
    %}</title>
```

```
<style>
  /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */
  :root {
    --accent: #2c6b5a;                  /* 主色：青绿，千里江山石绿偏青 */
    --accent-hover: #1e5548;
    --accent-soft: #e6f0ec;
    --side-1: #1a4a4a;                  /* 侧边栏：深青墨 */
    --side-2: #0f3338;
    --danger: #a0392a;                  /* 朱砂红 */
    --danger-soft: #fbeae6;
    --ok: #2c6b5a;
    --ok-soft: #e6f0ec;
    --todo: #c08a3e;                    /* 赭石色待办 */
    --ink: #2a2520;                     /* 深褐墨色 */
    --ink-soft: #6b5d4f;                /* 褐灰 */
    --line: #e0d8c8;                    /* 绢本边线 */
    --surface: #fdfaf2;                 /* 暖白绢本 */
    --canvas: #f5efe0;                  /* 绢本米黄背景 */
    --radius: 10px;
    --shadow: 0 2px 8px rgba(42, 37, 32, .06);
    --header-h: 60px;
    --nav-w: 226px;
  }
  * { box-sizing: border-box; }
  html, body { margin: 0; padding: 0; }
  body {
    font-family: "Microsoft YaHei", "PingFang SC", "Noto
  Serif SC", "SimSun", serif;
    font-size: 16.5px;
    line-height: 1.6;
    color: var(--ink);
    background: var(--canvas);
  }
  /* ---------- 跳到主要内容（键盘可达性） ---------- */
  .skip-link {
    position: absolute;
    left: 12px;
    top: -60px;
    z-index: 30;
    padding: 8px 14px;
    background: var(--accent);
    color: #fff;
    border-radius: var(--radius);
    text-decoration: none;
    transition: top .15s ease-out;
  }
  .skip-link:focus { top: 12px; }
  /* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ----------
     固定为一屏高，页面滚动交给右侧内容区，
     这样左侧导航栏（含底部用户区）不会随长页面一起被拉到下面去。 */
```

```
.app-shell {
  display: flex;
  align-items: stretch;
  height: 100vh;
  height: 100dvh;
  overflow: hidden;
}
/* ---------- 侧边栏 ---------- */
.app-side {
  flex: 0 0 var(--nav-w);
  width: var(--nav-w);
  display: flex;
  flex-direction: column;
  background: linear-gradient(180deg, var(--side-1) 0%, var(--side-2) 100%);
  color: #fff;
}
.app-brand {
  display: flex;
  align-items: center;
  gap: 11px;
  padding: 17px 16px 15px;
  border-bottom: 1px solid rgba(255, 255, 255, .10);
}
.app-brand__mark {
  flex: none;
  display: grid;
  place-items: center;
  width: 36px;
  height: 36px;
  border-radius: 10px;
  background: rgba(255, 255, 255, .15);
}
.app-brand__name {
  font-size: 16px;
  font-weight: 600;
  line-height: 1.35;
  letter-spacing: .2px;
}
.app-brand__sub {
  font-size: 11.5px;
  line-height: 1.35;
  color: rgba(255, 255, 255, .58);
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y: auto; }
.app-nav__group {
  margin: 12px 0 5px;
  padding: 0 10px;
  font-size: 11.5px;
```

```
    letter-spacing: .05em;
    color: rgba(255, 255, 255, .48);
  }
  .app-nav ul { margin: 0; padding: 0; list-style: none; }
  .app-nav a {
    display: flex;
    align-items: center;
    gap: 10px;
    padding: 9px 11px;
    margin-bottom: 2px;
    border-radius: 8px;
    color: rgba(255, 255, 255, .80);
    text-decoration: none;
    transition: background-color .16s ease, color .16s
  ease;
  }
  .app-nav a:hover { background: rgba(255, 255, 255,
  .10); color: #fff; }
  .app-nav a:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
  }
  .app-nav a[aria-current="page"] {
    background: rgba(255, 255, 255, .17);
    color: #fff;
    font-weight: 600;
  }
  .app-nav .ico { flex: none; opacity: .92; }
  .app-nav__badge {
    margin-left: auto;
    min-width: 20px;
    padding: 0 6px;
    font-size: 11px;
    font-weight: 600;
    line-height: 18px;
    text-align: center;
    border-radius: 9px;
    background: var(--todo);
    color: #fff;
    font-variant-numeric: tabular-nums;
  }
  .app-user {
    display: flex;
    align-items: center;
    gap: 8px;
    padding: 10px 14px;
    border-top: 1px solid rgba(255, 255, 255, .10);
  }
  .app-user__me {
    display: flex;
```

```
    flex: 1;
    min-width: 0;
    align-items: center;
    gap: 10px;
    padding: 4px 6px;
    border-radius: 8px;
    color: inherit;
    text-decoration: none;
    transition: background-color .16s ease;
}
.app-user__me:hover { background: rgba(255, 255, 255, .10); }
.app-user__me:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
}
.app-user__avatar {
    flex: none;
    display: grid;
    place-items: center;
    width: 32px;
    height: 32px;
    border-radius: 50%;
    background: rgba(255, 255, 255, .16);
    font-size: 15.5px;
    font-weight: 600;
    object-fit: cover;
}
.app-user__name {
    display: block;
    font-size: 15.5px;
    line-height: 1.3;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}
.app-user__state {
    display: flex;
    align-items: center;
    gap: 5px;
    font-size: 11.5px;
    line-height: 1.3;
    color: rgba(255, 255, 255, .60);
}
.app-user__dot {
    width: 6px;
    height: 6px;
    border-radius: 50%;
    background: #6ed49a;
}
```

```
.app-user__out {
  flex: none;
  padding: 6px;
  border-radius: 6px;
  color: rgba(255, 255, 255, .68);
  text-decoration: none;
}
.app-user__out:hover { color: #fff; background:
rgba(255, 255, 255, .12); }
/* ---------- 右侧内容区 ----------
   整个骨架固定一屏高，滚动只发生在内容区内部：
   左侧导航栏（含底部用户区）始终留在原位，右侧页面自己滚。 */
.app-content {
  flex: 1;
  min-width: 0;
  display: flex;
  flex-direction: column;
  overflow-y: auto;
  overflow-x: hidden;
  /* 滚动条槽位始终占位。滚动条是出现在这个滚动容器内部的，
     内容高的页面（工作台、标准库、操作日志）会因此把顶栏挤窄约 10px，
     内容刚好放得下的页面（限制性判定、项目档案…）则不会 ——
     结果就是「顶栏在不同页面不一样宽」。预留槽位后各页一致。 */
  scrollbar-gutter: stable;
  /* 细滚动条：默认半透明，悬停时加深，不抢视线 */
  scrollbar-width: thin;
  scrollbar-color: rgba(31, 107, 69, .28) transparent;
}
.app-content::-webkit-scrollbar { width: 12px; }
.app-content::-webkit-scrollbar-track { background:
transparent; }
.app-content::-webkit-scrollbar-thumb {
  background: rgba(31, 107, 69, .26);
  border: 3px solid transparent;
  background-clip: content-box;
  border-radius: 999px;
}
.app-content::-webkit-scrollbar-thumb:hover {
  background: rgba(31, 107, 69, .46);
  background-clip: content-box;
}
/* ---------- 顶栏：面包屑 + 搜索 ----------
   粘在内容区顶部，滚动时面包屑与搜索框始终可用 */
.app-header {
  position: sticky;
  top: 0;
  z-index: 20;
  display: flex;
  align-items: center;
  gap: 14px;
```

```
  height: var(--header-h);
  padding: 0 22px;
  background: var(--surface);
  border-bottom: 1px solid var(--line);
}
.crumb {
  display: flex;
  align-items: center;
  gap: 7px;
  font-size: 15.5px;
  color: var(--ink-soft);
  white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
  display: flex;
  align-items: center;
  gap: 8px;
  width: 268px;
  padding: 7px 12px;
  background: var(--canvas);
  border: 1px solid var(--line);
  border-radius: 9px;
  color: var(--ink-soft);
  transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
  flex: 1;
  min-width: 0;
  border: none;
  outline: none;
  background: transparent;
  font-family: inherit;
  font-size: 15.5px;
  color: var(--ink);
}
.search-box button {
  display: grid;
  place-items: center;
  padding: 0;
  border: none;
  background: transparent;
```

```
    color: inherit;                                                    5301
    cursor: pointer;                                                   5302
}                                                                      5303
.icon-btn {                                                            5304
    display: grid;                                                     5305
    place-items: center;                                               5306
    width: 34px;                                                       5307
    height: 34px;                                                      5308
    border-radius: 9px;                                                5309
    border: 1px solid transparent;                                     5310
    color: var(--ink-soft);                                            5311
    text-decoration: none;                                             5312
    transition: background-color .16s ease, color .16s                 5313
ease,                                                                  5314
                border-color .16s ease;                                5315
}                                                                      5316
.icon-btn:hover {                                                      5317
    color: var(--accent);                                              5318
    background: var(--accent-soft);                                    5319
    border-color: rgba(31, 107, 69, .18);                              5320
}                                                                      5321
.app-main {                                                            5322
    flex: 1;                                                           5323
    min-width: 0;                                                      5324
    padding: 22px 26px 26px;                                           5325
}                                                                      5326
.container { max-width: none; width: 100%; }                           5327
/* ---------- 页脚 ---------- */                                       5328
.app-foot {                                                            5329
    display: flex;                                                     5330
    align-items: center;                                               5331
    justify-content: space-between;                                    5332
    gap: 14px;                                                         5333
    padding: 0 26px 18px;                                              5334
    font-size: 14.5px;                                                 5335
    color: var(--ink-soft);                                            5336
}                                                                      5337
/* ---------- 未登录时的裸页面（登录页） ---------- */                  5338
.login-shell {                                                         5339
    display: flex;                                                     5340
    flex-direction: column;                                            5341
    min-height: 100dvh;                                                5342
}                                                                      5343
/* ---------- 卡片 ---------- */                                       5344
.card {                                                                5345
    background: var(--surface);                                        5346
    border: 1px solid var(--line);                                     5347
    border-radius: var(--radius);                                      5348
    box-shadow: var(--shadow);                                         5349
    padding: 22px 24px;                                                5350
```

```
    margin-bottom: 18px;                                                5351
}                                                                       5352
.card > h2 {                                                            5353
    margin: 0 0 14px;                                                   5354
    font-size: 18.5px;                                                  5355
    font-weight: 600;                                                   5356
    color: var(--accent);                                               5357
    text-wrap: balance;                                                 5358
}                                                                       5359
.card > h3 {                                                            5360
    margin: 20px 0 10px;                                                5361
    font-size: 16.5px;                                                  5362
    font-weight: 600;                                                   5363
    color: var(--accent);                                               5364
}                                                                       5365
/* ---------- 表格 ---------- */                                        5366
table {                                                                 5367
    width: 100%;                                                        5368
    border-collapse: collapse;                                          5369
    margin: 10px 0 16px;                                                5370
    font-variant-numeric: tabular-nums;                                 5371
}                                                                       5372
caption {                                                               5373
    text-align: left;                                                   5374
    color: var(--ink-soft);                                             5375
    font-size: 14.5px;                                                  5376
    padding-bottom: 6px;                                                5377
}                                                                       5378
th, td {                                                                5379
    border: 1px solid var(--line);                                      5380
    padding: 8px 12px;                                                  5381
    text-align: center;                                                 5382
    vertical-align: middle;                                             5383
}                                                                       5384
th {                                                                    5385
    background: var(--accent-soft);                                     5386
    font-weight: 600;                                                   5387
    white-space: nowrap;                                                5388
}                                                                       5389
tbody tr:hover { background: #fafbfc; }                                 5390
/* ---------- 表单控件 ---------- */                                    5391
label { display: inline-block; }                                        5392
.field { margin-bottom: 12px; }                                         5393
.field__label {                                                         5394
    display: block;                                                     5395
    margin-bottom: 4px;                                                 5396
    color: var(--ink-soft);                                             5397
    font-size: 15.5px;                                                  5398
}                                                                       5399
input[type="text"], input[type="number"],                               5400
```

```
input[type="password"],
select, textarea {
  padding: 6px 9px;
  font-family: inherit;
  font-size: 16.5px;
  color: var(--ink);
  background: #fff;
  border: 1px solid #c6ccd4;
  border-radius: 4px;
  max-width: 100%;
}
input[type="number"] {
  width: 104px;
  text-align: center;
  font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 1px;
  border-color: var(--accent);
}
.checkbox {
  display: inline-flex;
  align-items: center;
  gap: 7px;
  cursor: pointer;
  user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 8px 18px;
  font-family: inherit;
  font-size: 16.5px;
  line-height: 1.4;
  color: #fff;
  background: rgba(31, 107, 69, .80);
  border: 1px solid rgba(31, 107, 69, .30);
  border-radius: 8px;
  cursor: pointer;
  text-decoration: none;
```

```
    transition: background-color .16s ease, border-color .16s ease,
                color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.btn:hover {
    background: var(--accent);
    border-color: var(--accent);
    box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow: none; }
.btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.btn--ghost {
    color: var(--accent);
    background: rgba(31, 107, 69, .06);
    border-color: rgba(31, 107, 69, .22);
}
.btn--ghost:hover {
    color: #fff;
    background: var(--accent);
    border-color: var(--accent);
}
.btn--danger {
    color: var(--danger);
    background: rgba(179, 38, 30, .07);
    border-color: rgba(179, 38, 30, .26);
}
.btn--danger:hover {
    color: #fff;
    background: var(--danger);
    border-color: var(--danger);
    box-shadow: 0 3px 12px rgba(179, 38, 30, .20);
}
.btn--sm { padding: 5px 13px; font-size: 15.5px; border-radius: 7px; }
.btn:disabled { opacity: .45; cursor: not-allowed; }
.btn:disabled:hover {
    background: rgba(31, 107, 69, .80);
    border-color: rgba(31, 107, 69, .30);
    box-shadow: none;
}
.btn--ghost:disabled:hover { background: rgba(31, 107, 69, .06); color: var(--accent); }
/* ---------- 表格操作列的行内小按钮 ---------- */
    取代原来「文字 | 文字 | 文字」的写法：
    圆角矩形 + 半透明底色，悬停时变实色。 */
.row-actions {
    display: inline-flex;
```

```
    flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */
    align-items: center;
    gap: 6px;
    white-space: nowrap;
}
.row-actions form { display: inline-flex; margin: 0; }
.link-btn {
    display: inline-flex;
    align-items: center;
    padding: 4px 11px;
    font-family: inherit;
    font-size: 15.5px;
    line-height: 1.5;
    color: var(--accent);
    background: rgba(31, 107, 69, .07);
    border: 1px solid rgba(31, 107, 69, .20);
    border-radius: 7px;
    cursor: pointer;
    text-decoration: none;
    white-space: nowrap;
    transition: background-color .16s ease, border-color
.16s ease,
                color .16s ease, box-shadow .16s ease,
transform .16s ease;
}
.link-btn:hover {
    color: #fff;
    background: var(--accent);
    border-color: var(--accent);
    box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
}
.link-btn:active { transform: translateY(1px);
box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid
var(--accent); outline-offset: 2px; }
.link-btn--danger {
    color: var(--danger);
    background: rgba(179, 38, 30, .07);
    border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
    color: #fff;
    background: var(--danger);
    border-color: var(--danger);
    box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items:
center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
```

```
.flash {
  padding: 10px 14px;
  margin-bottom: 14px;
  border-radius: var(--radius);
  border-left: 4px solid;
}
.flash.error {
  color: var(--danger);
  background: var(--danger-soft);
  border-left-color: var(--danger);
}
.flash.ok {
  color: var(--ok);
  background: var(--ok-soft);
  border-left-color: var(--ok);
}
/* ---------- 浮层提示（登录成功、各类操作完成、登录失败） ---------- */
   外层 toast-stack 固定在页面中上方，内层每条各自淡入 → 停留 → 淡出，
   多条提示会依次排开而不重叠。
   底色是浅绿半透明：既有一层底色把文字托住，又能透出后面的内容，
   不用高饱和色块，提示不该比页面内容更抢眼。
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
  position: fixed;
  top: 16px;
  left: 50%;
  z-index: 100;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 6px;
  transform: translateX(-50%);
  pointer-events: none;
}
.toast-center {
  max-width: min(620px, 82vw);
  padding: 8px 22px;
  border-radius: 999px;
  /* 浅绿、半透明：不遮住背后的内容，也比纯透明更托得住文字 */
  background: rgba(223, 241, 231, .76);
  border: 1px solid rgba(31, 107, 69, .14);
  color: #2f4a3c;
  font-size: 18.5px;
  font-weight: 500;
  line-height: 1.65;
  letter-spacing: .03em;
  text-align: center;
  text-wrap: pretty;
  backdrop-filter: blur(6px) saturate(1.1);
  -webkit-backdrop-filter: blur(6px) saturate(1.1);
```

```
  box-shadow: 0 4px 16px rgba(23, 32, 27, .07);
  animation: toast-fade 2.6s ease-in-out forwards;
}
/* 失败类提示：同一套观感，换成浅红底，一眼能区分成败 */
.toast-center--error {
  background: rgba(252, 233, 230, .78);
  border-color: rgba(163, 43, 34, .16);
  color: #9c2b23;
}
@keyframes toast-fade {
  0%         { opacity: 0; transform: translateY(-6px); }
  14%, 74%   { opacity: 1; transform: none; }
  100%       { opacity: 0; transform: none; }
}
/* 系统设置里要求「减少动态效果」时，去掉位移，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
  .toast-center { animation: toast-fade-plain 2.6s linear forwards; }
  @keyframes toast-fade-plain {
    0%         { opacity: 0; }
    14%, 74%   { opacity: 1; }
    100%       { opacity: 0; }
  }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 14px; }
.mono { font-family: Consolas, "Courier New", monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap: 14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
  display: inline-block;
  padding: 1px 9px;
  font-size: 14px;
  border-radius: 10px;
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background: var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background: var(--danger-soft); border-color: #eec4c0; }
```

```
.tag--muted { color: var(--ink-soft); background:
#f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */
.body--login {
  background:
    radial-gradient(1100px 640px at 50% 8%,
                    #edf6f1 0%, rgba(237, 246, 241, 0)
72%),
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);
}
.body--login .app-main { padding: 0; }
.body--login .container { max-width: none; width: 100%;
}
/* 仅供读屏软件的隐藏文本 */
.visually-hidden {
  position: absolute !important;
  width: 1px;
  height: 1px;
  margin: -1px;
  padding: 0;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap;
  border: 0;
}
/* ---------- 工作台：页首横幅 ---------- */
.hero {
  position: relative;
  overflow: hidden;
  min-height: 260px;
  padding: 26px 28px;
  margin-bottom: 16px;
  border-radius: 12px;
  /* 左侧压深保证文字可读，右侧透出远山插画；
     背景按底部对齐，避免远山被居中裁掉 */
  background:
    linear-gradient(100deg, rgba(15, 40, 45, .90) 0%,
                            rgba(20, 55, 58, .55) 45%,
                            rgba(20, 55, 58, .12) 100%),
    url("{{ url_for('static', filename='hero_landscape.j
pg') }}")
    center center / cover no-repeat;
  color: #fff;
}
.hero__body { max-width: 720px; }
.hero__title {
  margin: 0 0 10px;
  font-size: 23.5px;
```

```
    font-weight: 700;
    line-height: 1.4;
    letter-spacing: .3px;
}
.hero__desc {
    margin: 0;
    max-width: 640px;
    font-size: 14px;
    line-height: 1.85;
    color: rgba(255, 255, 255, .86);
}
.hero__actions {
    display: flex;
    flex-wrap: wrap;
    gap: 10px;
    margin-top: 18px;
}
.hero-btn {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    padding: 7px 15px;
    font-size: 15.5px;
    border-radius: 8px;
    text-decoration: none;
    transition: background-color .16s ease, border-color .16s ease,
                color .16s ease;
}
.hero-btn--solid {
    color: #14432f;
    background: rgba(255, 255, 255, .92);
    border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
    color: #fff;
    background: rgba(255, 255, 255, .14);
    border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255, 255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff; outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
.stat-cards {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(190px, 1fr));
    gap: 12px;
```

```
  margin-bottom: 22px;
}
.stat-card {
  display: block;
  padding: 15px 17px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s ease;
}
.stat-card:hover {
  border-color: rgba(31, 107, 69, .45);
  box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
  display: block;
  font-size: 14px;
  color: var(--ink-soft);
}
.stat-card__value {
  display: block;
  margin: 3px 0 2px;
  font-size: 28.5px;
  font-weight: 700;
  line-height: 1.25;
  color: var(--accent);
  font-variant-numeric: tabular-nums;
}
.stat-card__unit {
  margin-left: 3px;
  font-size: 15.5px;
  font-weight: 500;
  color: var(--ink-soft);
}
.stat-card__note {
  display: block;
  font-size: 14.5px;
  color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
```

```
  margin-bottom: 12px;
}
.section__title { margin: 0; font-size: 18.5px; }
.section__desc { margin: 0; font-size: 14px; color:
var(--ink-soft); }
/* ---------- 工作台：流程步骤卡 ---------- */
.step-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill,
minmax(238px, 1fr));
  gap: 12px;
}
.step-card {
  position: relative;
  display: block;
  padding: 16px 16px 15px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s
ease,
              transform .16s ease;
}
.step-card:hover {
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 5px 16px rgba(19, 60, 43, .10);
  transform: translateY(-1px);
}
.step-card:focus-visible { outline: 2px solid
var(--accent); outline-offset: 2px; }
.step-card__no {
  position: absolute;
  top: 13px;
  right: 15px;
  font-size: 19.5px;
  font-weight: 700;
  color: #d7e3db;
  font-variant-numeric: tabular-nums;
}
.step-card__icon {
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  margin-bottom: 10px;
  border-radius: 9px;
  background: var(--accent-soft);
```

```
    color: var(--accent);
}
.step-card__title {
    display: block;
    margin-bottom: 5px;
    font-size: 16.5px;
    font-weight: 600;
    color: var(--accent);
}
.step-card__desc {
    display: block;
    font-size: 17px;
    line-height: 1.65;
    color: var(--ink-soft);
    text-wrap: pretty;
}
.step-card__badge {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #fdf1e2;
    color: #9a5a12;
    border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
    background: #fafbfa;
    border-color: #eceff0;
    box-shadow: none;
    cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
    background: #f1f3f2;
    color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #f1f3f2;
    color: #8b948f;
}
/* ---------- 结果摘要 ---------- */
```

```
.stat-row {
  display: flex;
  flex-wrap: wrap;
  gap: 28px;
  margin: 4px 0 6px;
}
.stat__label { font-size: 14px; color: var(--ink-soft);
}
.stat__value { font-size: 22.5px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
margin: 4px 0 18px; }
.step {
  display: flex; align-items: center; gap: 7px;
  padding: 8px 12px; font-size: 15.5px;
  background: #fff; border: 1px solid var(--line);
  border-radius: var(--radius);
}
.step__no {
  display: inline-flex; align-items: center;
justify-content: center;
  width: 20px; height: 20px; border-radius: 50%;
  background: #eef1f5; color: var(--ink-soft);
  font-size: 14.5px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 14px; color: var(--ink-soft);
}
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块（中性底色，不用整块警示色压页面） ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-left: 3px solid var(--todo);
  border-radius: var(--radius);
```

```
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
/* ---------- 前置条件表（与页面主表同款，只把待处理的行标成浅红） ---------- */
.prereq-table { margin: 0 0 18px; }
.prereq-table > caption { padding-bottom: 7px; }
.prereq-row > td { background: var(--danger-soft); }
.prereq-row:hover > td { background: #f6dcd7; }
.prereq-flag { color: var(--danger); font-weight: 600; }
.prereq-row .muted { color: #9a6b62; }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
  border: 1px solid var(--line);
  border-radius: var(--radius);
  background: var(--surface);
  padding: 14px 16px;
  margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin: 0 0 12px;
  padding-bottom: 9px;
  border-bottom: 1px solid var(--line);
}
.patch-panel__name {
  margin: 0;
  font-size: 17.5px;
  font-weight: 600;
}
.patch-panel__meta {
  font-size: 14px;
  color: var(--ink-soft);
}
.patch-panel__meta a { color: var(--accent); }
.patch-panel__tail { margin-left: auto; }
.patch-panel .notice { margin: 0; }
.patch-panel table { margin: 0; }
/* ---------- 窄屏适配 ---------- */
@media (max-width: 960px) {
  /* 上下布局时取消「一屏高 + 内容区内部滚动」，
     否则导航会占掉半屏、内容被挤在剩下的小格子里滚动 */
  .app-shell {
    flex-direction: column;
    height: auto;
    min-height: 100vh;
    min-height: 100dvh;
    overflow: visible;
```

```
        }                                                                    6001
        .app-content { overflow: visible; }                                  6002
        .app-header { position: static; }                                    6003
        .app-side { width: auto; flex: none; }                               6004
        .app-brand { padding: 12px 14px; }                                   6005
        .app-nav { padding: 4px 10px 10px; }                                 6006
        .app-nav ul { display: flex; flex-wrap: wrap; gap:                   6007
    4px; }                                                                   6008
        .app-nav a { margin-bottom: 0; }                                     6009
        .app-nav__group { margin: 8px 0 4px; }                               6010
        .app-user { padding: 10px 14px; }                                    6011
        .app-header { padding: 0 14px; gap: 10px; }                          6012
        /* 搜索框保持与宽屏一致的宽度：原来这里缩到 180px，                  6013
           窗口一窄搜索框就变窄，同一台机器不同窗口大小看着像两个尺寸 */     6014
        .search-box { width: 268px; }                                        6015
        .app-main { padding: 16px 14px 20px; }                               6016
        .app-foot { padding: 0 14px 14px; }                                  6017
    }                                                                        6018
    @media (max-width: 640px) {                                              6019
        .crumb { display: none; }                                            6020
        .search-box { width: auto; flex: 1; }                                6021
    }                                                                        6022
  </style>                                                                   6023
</head>                                                                      6024
<body class="{% if not current_user.is_authenticated                         6025
    %}body--login{% endif %}">                                               6026
  <a class="skip-link" href="#main">跳到主要内容</a>                         6027
  {# 取一次即可（同一请求内重复调用返回相同内容）。                          6028
     类别为 toast / ok / toast-error 的渲染成中上方浮层，出现后自动淡出；   6029
     其余仍是页面顶部的提示条 ——                                            6030
     需要对照着改表单的错误，留在页面上比一闪而过更有用。 #}                6031
  {% set flash_msgs = get_flashed_messages(with_categories=t                 6032
    rue) %}                                                                  6033
  <div class="toast-stack">                                                  6034
    {% for category, message in flash_msgs if category in                    6035
    ("toast", "ok", "toast-error") %}                                        6036
    <div class="toast-center{% if category == 'toast-error'                  6037
    %} toast-center--error{% endif %}"                                       6038
         role="status" aria-live="polite">{{ message                         6039
    }}</div>                                                                 6040
    {% endfor %}                                                             6041
  </div>                                                                     6042
  {% if current_user.is_authenticated %}                                     6043
  <div class="app-shell">                                                    6044
    <aside class="app-side">                                                 6045
      <div class="app-brand">                                                6046
        <span class="app-brand__mark">{{                                     6047
    ui.icon("mountain", 21) }}</span>                                        6048
        <span>                                                               6049
          <span class="app-brand__name">矿山自然恢复评价</span><br>          6050
```

```
        <span class="app-brand__sub">适宜性评价系统 V1.0</span>
      </span>
    </div>
    <nav class="app-nav" aria-label="功能导航">
      <ul>
        <li>
          <a href="/"{% if home_active %}
aria-current="page"{% endif %}>
            {{ ui.icon("home") }}<span>工作台</span>
          </a>
        </li>
      </ul>
      {% for g in nav_groups %}
      <p class="app-nav__group">{{ g.title }}</p>
      <ul>
        {% for item in g.links %}
        <li>
          <a href="{{ item.href }}"{% if item.active %}
aria-current="page"{% endif %}>
            {{ ui.icon(item.icon) }}<span>{{ item.label
}}</span>
            {% if item.badge %}<span
class="app-nav__badge">{{ item.badge }}</span>{% endif
%}
          </a>
        </li>
        {% endfor %}
      </ul>
      {% endfor %}
    </nav>
    <div class="app-user">
      <a class="app-user__me" href="/profile"
title="个人设置">
        {% if current_user.avatar %}
        <img class="app-user__avatar"
             src="{{ url_for('static',
filename=current_user.avatar) }}" alt="">
        {% else %}
        <span class="app-user__avatar"
aria-hidden="true">{{ current_user.initial }}</span>
        {% endif %}
        <span style="min-width:0;">
          <span class="app-user__name">{{
current_user.shown_name }}</span>
          <span class="app-user__state">
            <span class="app-user__dot"
aria-hidden="true"></span>
            {{ role_labels.get(current_user.role,
current_user.role) }}
          </span>
```

```
            </span>                                                    6101
          </a>                                                         6102
          <a class="app-user__out" href="/logout" title="退出登录"     6103
             aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>     6104
        </div>                                                         6105
      </aside>                                                         6106
      <div class="app-content">                                        6107
        <header class="app-header">                                    6108
          <nav class="crumb" aria-label="当前位置">                    6109
            <a href="/">首页</a>                                       6110
            {{ ui.icon("chevron", 13) }}                               6111
            <strong>{{ page_title }}</strong>                          6112
          </nav>                                                       6113
          <span class="app-header__spacer"></span>                     6114
          <form class="search-box" method="get"                        6115
      action="/search" role="search">                                  6116
            {{ ui.icon("search", 16) }}                                6117
            <label for="q" class="visually-hidden">搜索项目、图斑、标准<  6118
      /label>                                                          6119
            <input id="q" type="search" name="q"                       6120
      autocomplete="off"                                               6121
                   placeholder="搜索项目、图斑、标准" value="{{          6122
      search_q }}">                                                    6123
            <button type="submit" aria-label="搜索">{{                 6124
      ui.icon("chevron", 15) }}</button>                               6125
          </form>                                                      6126
          {% if current_user.is_admin %}                               6127
          <a class="icon-btn" href="/profile?tab=logs"                 6128
      title="操作日志"                                                 6129
             aria-label="操作日志">{{ ui.icon("bell", 18) }}</a>       6130
          {% endif %}                                                  6131
        </header>                                                      6132
  {% else %}                                                           6133
  <div class="login-shell">                                            6134
  {% endif %}                                                          6135
      <main id="main" class="app-main">                                6136
        <div class="container">                                        6137
          {% for category, message in flash_msgs if category           6138
      not in ("toast", "ok", "toast-error") %}                         6139
          <p class="flash {{ category }}" role="status">{{             6140
      message }}</p>                                                   6141
          {% endfor %}                                                 6142
          {% block content %}{% endblock %}                            6143
        </div>                                                         6144
      </main>                                                          6145
  {% if current_user.is_authenticated %}                               6146
        <footer class="app-foot">                                      6147
          <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０</span>    6148
          <span>数据截止 {{ today }}</span>                            6149
        </footer>                                                      6150
```

```
    </div>
  </div>
  {% else %}
  </div>
  {% endif %}
  {% block scripts %}{% endblock %}
</body>
</html>
# ==== app/templates/_macros.html ====
{# 共用宏：图标、流程状态面板 #}
{# ============================================================
   ================== 
   图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）
   不引图标字体或外部图片，避免额外请求与资源依赖；
   stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。
   ============================================================
   ================== #}
{% set ICONS = {
  "dot": '<circle cx="12" cy="12" r="3"/>',
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5
    17h5"/>',
  "database": '<ellipse cx="12" cy="5.5" rx="8"
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2"/>',
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"
    cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',
  "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0
    2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14
    2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',
  "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5
    20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6
    3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path
    d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
  "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5
    21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4
    4.4V21"/>',
  "mail": '<rect x="3" y="5" width="18" height="14"
```

```
      rx="2.5"/><path d="m3.8 7 8.2 5.8L20.2 7"/>',
  "phone": '<path d="M6.4 3h2.8l1.5 4.4-2 1.4a12.4 12.4 0 0
      0 6.5 6.5l1.4-2L21 14.8v2.8a2.4 2.4 0 0 1-2.6 2.4A16.6
      16.6 0 0 1 3.5 5.6 2.4 2.4 0 0 1 5.9 3z"/>',
  "lock": '<rect x="4" y="10.4" width="16" height="10.6"
      rx="2.6"/><path d="M8 10.4V7.6a4 4 0 0 1 8 0v2.8"/>',
  "briefcase": '<rect x="3" y="7.4" width="18"
      height="13.1" rx="2.6"/><path d="M8.4 7.4V6A2 2 0 0 1
      10.4 4h3.2a2 2 0 0 1 2 2v1.4"/><path d="M3 12.6h18"/>',
  "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
      20.5-4-4"/>',
  "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
      6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
      1-3.6 0"/>',
  "plus": '<path d="M12 5v14M5 12h14"/>',
  "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
      4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
      21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
  "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
      2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
      7.5"/><path d="M5.5 12H15"/>',
  "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
  "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
      10v4.2"/><path d="M12 17.4h.01"/>',
  "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
      20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
     fill="none" stroke="currentColor" stroke-width="1.7"
     stroke-linecap="round" stroke-linejoin="round"
     aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按评价流程顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   流程约定：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
      {% elif progress.constraint.done %}未通过
      {% else %}待完成{% endif %}
```

```
      </span>                                                                  6251
    </div>                                                                     6252
    <div class="step {% if progress.units.done %}step--done{%                 6253
      else %}step--todo{% endif %}">                                           6254
      <span class="step__no">2</span>                                          6255
      <span class="step__name">划分微地质单元</span>                              6256
      <span class="step__state">                                               6257
        {% if progress.units.done %}已完成{% else %}待完成{% endif             6258
      %}                                                                       6259
      </span>                                                                  6260
    </div>                                                                     6261
    <div class="step {% if progress.data.ready %}step--done{%                 6262
      else %}step--todo{% endif %}">                                           6263
      <span class="step__no">3</span>                                          6264
      <span class="step__name">录入指标数据</span>                               6265
      <span class="step__state">                                               6266
        {% if progress.data.ready %}已完成{% else %}待完成{% endif             6267
      %}                                                                       6268
      </span>                                                                  6269
    </div>                                                                     6270
    <div class="step step--next">                                              6271
      <span class="step__no">4</span>                                          6272
      <span class="step__name">执行适宜性评价</span>                             6273
      <span class="step__state">当前步骤</span>                                  6274
    </div>                                                                     6275
</div>                                                                         6276
{% endmacro %}                                                                 6277
{# 前置条件未满足时的提示：做成与页面主表同款的表格，只把待处理的那几行标成浅红，                  6278
   说明写在 caption 里，缺什么、去哪补都在行内。 #}                              6279
{% macro workflow_blocking(progress) %}                                        6280
{% if progress.blocking %}                                                     6281
<table class="prereq-table">                                                   6282
  <caption>本步骤的前置条件尚未满足，按评价流程的顺序先补上：</caption>                  6283
  <thead>                                                                      6284
    <tr>                                                                       6285
      <th scope="col">前置步骤</th>                                             6286
      <th scope="col">说明</th>                                                6287
      <th scope="col">操作</th>                                                6288
    </tr>                                                                      6289
  </thead>                                                                     6290
  <tbody>                                                                      6291
    {% for b in progress.blocking %}                                           6292
    <tr class="prereq-row">                                                    6293
      <td>{{ b.label }}</td>                                                   6294
      <td class="muted">{{ b.reason }}</td>                                    6295
      <td><a class="btn btn--sm" href="{{ b.fix_url }}">{{                     6296
    b.fix_label }}</a></td>                                                    6297
    </tr>                                                                      6298
    {% endfor %}                                                               6299
  </tbody>                                                                     6300
```

```
</table>
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label }}</span>
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
   只在流程里**第一个**按图斑操作的页面（限制性指标判定）列一次，后面各页
   不再重复；建好图斑才会出现在下面的表里，但离「能进入本页流程」还差别的步骤，
   所以措辞只说到「出现在这里」为止。 #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
<table class="prereq-table">
  <caption>以下项目下尚未建立矿山图斑，本页按图斑组织数据，建好图斑后才会出现在这里：</caption>
  <thead>
    <tr>
      <th scope="col">所属项目</th>
      <th scope="col">矿山图斑</th>
      <th scope="col">状态</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for p in projects_without_patch %}
    <tr class="prereq-row">
      <td>项目「{{ p.name }}」</td>
      <td class="muted">未建立</td>
      <td><span class="prereq-flag">缺少矿山图斑</span></td>
      <td>
        <a class="btn btn--sm" href="/project/{{ p.id }}/patch/add">
          {{ icon("plus", 14) }}去新建图斑
        </a>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endif %}
{% endmacro %}
# ==== app/templates/_login_badge.svg ====
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg" viewBox="0 0 240 240" role="img"
```

```
    aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">                                    6351
  <title>系统徽标</title>                                                                  6352
  <g class="lp-ring" style="--k: 2.7">                                                     6353
    <circle cx="162.00" cy="120.00" r="2.60"                                               6354
    fill="#c4f5ab"/>                                                                       6355
    <circle cx="157.84" cy="138.22" r="2.97"                                               6356
    fill="#b8f5ab"/>                                                                       6357
    <circle cx="146.19" cy="152.84" r="3.06"                                               6358
    fill="#acf5ab"/>                                                                       6359
    <circle cx="129.35" cy="160.95" r="2.80"                                               6360
    fill="#abf5b7"/>                                                                       6361
    <circle cx="110.65" cy="160.95" r="2.40"                                               6362
    fill="#abf5c2"/>                                                                       6363
    <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce"/>                               6364
    <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da"/>                               6365
    <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6"/>                               6366
    <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da"/>                               6367
    <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce"/>                                6368
    <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2"/>                               6369
    <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7"/>                               6370
    <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab"/>                               6371
    <circle cx="157.84" cy="101.78" r="2.23"                                               6372
    fill="#b8f5ab"/>                                                                       6373
  </g>                                                                                     6374
  <g class="lp-ring" style="--k: 2.15">                                                    6375
    <circle cx="177.29" cy="129.07" r="3.92"                                               6376
    fill="#a4e28d"/>                                                                       6377
    <circle cx="171.68" cy="146.33" r="4.01"                                               6378
    fill="#9be28d"/>                                                                       6379
    <circle cx="161.01" cy="161.01" r="3.87"                                               6380
    fill="#91e28d"/>                                                                       6381
    <circle cx="146.33" cy="171.68" r="3.55"                                               6382
    fill="#8de292"/>                                                                       6383
    <circle cx="129.07" cy="177.29" r="3.17"                                               6384
    fill="#8de29c"/>                                                                       6385
    <circle cx="110.93" cy="177.29" r="2.88"                                               6386
    fill="#8de2a6"/>                                                                       6387
    <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af"/>                               6388
    <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9"/>                               6389
    <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2"/>                               6390
    <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc"/>                               6391
    <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc"/>                               6392
    <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2"/>                                6393
    <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9"/>                                6394
    <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af"/>                                6395
    <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6"/>                               6396
    <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c"/>                               6397
    <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292"/>                               6398
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d"/>                               6399
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d"/>                               6400
```

```
    <circle cx="177.29" cy="110.93" r="3.63"                    6401
    fill="#a4e28d"/>                                            6402
  </g>                                                          6403
  <g class="lp-ring" style="--k: 1.85">                         6404
    <circle cx="194.00" cy="120.00" r="4.94"                    6405
    fill="#91cf72"/>                                            6406
    <circle cx="191.85" cy="137.71" r="4.92"                    6407
    fill="#88cf72"/>                                            6408
    <circle cx="185.52" cy="154.39" r="4.73"                    6409
    fill="#80cf72"/>                                            6410
    <circle cx="175.39" cy="169.07" r="4.42"                    6411
    fill="#78cf72"/>                                            6412
    <circle cx="162.04" cy="180.90" r="4.06"                    6413
    fill="#72cf73"/>                                            6414
    <circle cx="146.24" cy="189.19" r="3.73"                    6415
    fill="#72cf7b"/>                                            6416
    <circle cx="128.92" cy="193.46" r="3.51"                    6417
    fill="#72cf83"/>                                            6418
    <circle cx="111.08" cy="193.46" r="3.45"                    6419
    fill="#72cf8b"/>                                            6420
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93"/>    6421
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c"/>    6422
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4"/>    6423
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac"/>    6424
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4"/>    6425
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc"/>    6426
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4"/>    6427
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac"/>     6428
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4"/>     6429
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c"/>     6430
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93"/>     6431
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b"/>    6432
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83"/>    6433
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b"/>    6434
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73"/>    6435
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72"/>    6436
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72"/>    6437
    <circle cx="191.85" cy="102.29" r="4.80"                    6438
    fill="#88cf72"/>                                            6439
  </g>                                                          6440
  <g class="lp-ring" style="--k: 1.62">                         6441
    <circle cx="209.57" cy="128.82" r="5.68"                    6442
    fill="#76bb59"/>                                            6443
    <circle cx="206.12" cy="146.13" r="5.40"                    6444
    fill="#6fbb59"/>                                            6445
    <circle cx="199.37" cy="162.43" r="5.06"                    6446
    fill="#68bb59"/>                                            6447
    <circle cx="189.57" cy="177.10" r="4.71"                    6448
    fill="#61bb59"/>                                            6449
    <circle cx="177.10" cy="189.57" r="4.41"                    6450
```

```
    fill="#5abb59" />                                                    6451
    <circle cx="162.43" cy="199.37" r="4.19"                             6452
    fill="#59bb5f" />                                                    6453
    <circle cx="146.13" cy="206.12" r="4.10"                             6454
    fill="#59bb66" />                                                    6455
    <circle cx="128.82" cy="209.57" r="4.15"                             6456
    fill="#59bb6d" />                                                    6457
    <circle cx="111.18" cy="209.57" r="4.32"                             6458
    fill="#59bb73" />                                                    6459
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a" />            6460
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81" />            6461
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88" />            6462
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f" />            6463
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96" />            6464
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d" />            6465
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4" />            6466
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4" />            6467
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d" />             6468
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96" />             6469
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f" />             6470
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88" />             6471
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81" />             6472
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a" />             6473
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73" />            6474
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d" />            6475
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66" />            6476
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f" />            6477
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59" />            6478
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59" />            6479
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59" />            6480
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59" />            6481
    <circle cx="209.57" cy="111.18" r="5.85"                             6482
    fill="#76bb59" />                                                    6483
  </g>                                                                   6484
  <g class="lp-ring" style="--k: 1.46">                                  6485
    <circle cx="226.00" cy="120.00" r="5.94"                             6486
    fill="#65a843" />                                                    6487
    <circle cx="224.55" cy="137.45" r="5.62"                             6488
    fill="#5fa843" />                                                    6489
    <circle cx="220.26" cy="154.42" r="5.29"                             6490
    fill="#59a843" />                                                    6491
    <circle cx="213.22" cy="170.45" r="5.00"                             6492
    fill="#53a843" />                                                    6493
    <circle cx="203.65" cy="185.11" r="4.77"                             6494
    fill="#4da843" />                                                    6495
    <circle cx="191.79" cy="197.99" r="4.63"                             6496
    fill="#47a843" />                                                    6497
    <circle cx="177.98" cy="208.74" r="4.59"                             6498
    fill="#43a846" />                                                    6499
    <circle cx="162.58" cy="217.07" r="4.67"                             6500
```

```
      fill="#43a84c"/>                                                    6501
      <circle cx="146.02" cy="222.76" r="4.85"                            6502
      fill="#43a852"/>                                                    6503
      <circle cx="128.75" cy="225.64" r="5.10"                            6504
      fill="#43a858"/>                                                    6505
      <circle cx="111.25" cy="225.64" r="5.42"                            6506
      fill="#43a85e"/>                                                    6507
      <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864"/>            6508
      <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a"/>            6509
      <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870"/>            6510
      <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876"/>            6511
      <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c"/>            6512
      <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882"/>            6513
      <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888"/>            6514
      <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e"/>            6515
      <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894"/>            6516
      <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e"/>            6517
      <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888"/>             6518
      <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882"/>             6519
      <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c"/>             6520
      <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876"/>             6521
      <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870"/>             6522
      <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a"/>             6523
      <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864"/>             6524
      <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e"/>            6525
      <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858"/>            6526
      <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852"/>            6527
      <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c"/>            6528
      <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846"/>            6529
      <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843"/>            6530
      <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843"/>            6531
      <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843"/>            6532
      <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843"/>            6533
      <circle cx="224.55" cy="102.55" r="6.23"                            6534
      fill="#5fa843"/>                                                    6535
    </g>                                                                  6536
    <g class="lp-badge__mark">                                            6537
      <circle cx="120" cy="120" r="27" fill="#14432f"/>                   6538
      <g transform="translate(120 120) scale(1.5)                         6539
      translate(-12 -12)" fill="none" stroke="#ffffff"                    6540
      stroke-width="1.9" stroke-linecap="round"                           6541
      stroke-linejoin="round">                                            6542
        <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>                       6543
        <circle cx="17" cy="6" r="2.2"/>                                  6544
      </g>                                                                6545
    </g>                                                                  6546
</svg>                                                                    6547
# ==== app/templates/error.html ====                                      6548
{% extends "base.html" %}                                                 6549
{% block title %}{{ code }} · {{ message }}{% endblock %}                 6550
```

```
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>{{ code }} · {{ message }}</h2>
  <p class="muted">
    请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
  </p>
  <p class="actions" style="margin-top:16px;">
    <a class="btn" href="/">返回首页</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 登录页（仿 Apple 账户页） ==================
     纯白底、居中、大量留白；色彩只用在徽标与按钮上。
     首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
     随滚动逐条弹入、往回滚时弹回。 */
  .lp {
    color: #1d1d1f;
    background:
      linear-gradient(180deg, rgba(245, 239, 224, .88) 0%,
                              rgba(245, 239, 224, .92)
      100%),
      url("../static/hero_qinglv.jpg")
      center top / cover no-repeat fixed;
  }
  /* ---------- 首屏 ---------- */
  .lp-hero {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
    position: relative;
    background:
      linear-gradient(180deg, rgba(15, 40, 45, .72) 0%,
                              rgba(20, 55, 58, .50) 50%,
                              rgba(15, 40, 45, .76) 100%),
      url("../static/hero_qinglv.jpg")
      center center / cover no-repeat;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
     --p 为 0~1 的滚动进度，由页面脚本写入；
     每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
```

```
.lp-badge {
  width: 212px;
  height: 212px;
  margin-bottom: 36px;
  --p: 0;
  animation: lp-in .7s ease both;
}
.lp-badge__svg {
  display: block;
  width: 100%;
  height: 100%;
  overflow: visible;                    /* 散开时圆点不被画布裁掉 */
  opacity: calc(1 - var(--p) * .94);
  filter: blur(calc(var(--p) * 2.6px));
}
.lp-ring {
  transform-box: fill-box;
  transform-origin: center;
  transform: scale(calc(1 + (var(--k) - 1) * var(--p)));
}
.lp-badge__mark {
  transform-box: fill-box;
  transform-origin: center;
  transform: scale(calc(1 + .18 * var(--p)));
}
.lp-title {
  margin: 0 0 18px;
  font-size: clamp(28px, 4.2vw, 42px);
  font-weight: 700;
  line-height: 1.3;
  letter-spacing: -.4px;
  color: #fff;
  text-shadow: 0 2px 12px rgba(0, 0, 0, .3);
  animation: lp-in .7s ease .06s both;
}
.lp-cta {
  padding: 13px 52px;
  font-family: inherit;
  font-size: 17px;
  font-weight: 500;
  color: #fff;
  background: var(--accent);
  border: none;
  border-radius: 980px;
  cursor: pointer;
  animation: lp-in .7s ease .18s both;
  transition: background-color .2s ease, box-shadow .2s
  ease,
              transform .2s ease;
}
```

```
.lp-cta:hover {                                                    6651
  background: var(--accent-hover);                                 6652
  box-shadow: 0 10px 26px rgba(44, 107, 90, .4);                   6653
}                                                                  6654
.lp-cta:active { transform: translateY(1px); box-shadow:           6655
  none; }                                                          6656
.lp-cta:focus-visible { outline: 3px solid rgba(31, 107,           6657
  69, .40); outline-offset: 3px; }                                 6658
.lp-more {                                                         6659
  margin-top: 38px;                                                6660
  font-size: 13px;                                                 6661
  color: rgba(255, 255, 255, .7);                                  6662
  text-decoration: none;                                           6663
  animation: lp-in .7s ease .24s both;                             6664
}                                                                  6665
.lp-more:hover { color: #fff; }                                    6666
@keyframes lp-in {                                                 6667
  from { opacity: 0; transform: translateY(12px); }                6668
  to   { opacity: 1; transform: none; }                            6669
}                                                                  6670
/* ---------- 滚动进场 ----------                                  6671
   进入视口时从屏外飞入，离开视口时飞回，可反复。                  6672
   交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */ 6673
.lp-reveal {                                                       6674
  opacity: 0;                                                      6675
  transition: opacity .7s ease,                                    6676
              transform 1.05s cubic-bezier(.18, 1.04, .3,          6677
  1),                                                              6678
              border-color .62s ease, background-color             6679
  .62s ease;                                                       6680
  transition-delay: 0s;                                            6681
  will-change: opacity, transform;                                 6682
}                                                                  6683
/* 起始位置整体在屏外：横向按视口宽度取位移，纵向按视口高度取位移，  6684
   再带一点缩小，落位时按贝塞尔曲线的过冲轻微回弹。 */              6685
.lp-reveal[data-dir="left"]  { transform:                          6686
  translate3d(-56vw, 5vh, 0) scale(.9); }                          6687
.lp-reveal[data-dir="right"] { transform:                          6688
  translate3d(56vw, 5vh, 0) scale(.9); }                           6689
.lp-reveal[data-dir="up"]    { transform: translate3d(0,           6690
  24vh, 0) scale(.94); }                                           6691
.lp-reveal.is-in {                                                 6692
  opacity: 1;                                                      6693
  transform: none;                                                 6694
  transition-delay: calc(var(--i, 0) * 110ms);                     6695
}                                                                  6696
/* 标题颜色由淡到深，强化「浮现」的层次感 */                       6697
.lp-reveal h2,                                                     6698
.lp-reveal h3,                                                     6699
.lp-reveal .lp-mod__name { color: #c2d2c8; transition:             6700
```

```
    color .7s ease; }
.lp-reveal.is-in h3,
.lp-reveal.is-in .lp-mod__name { color: var(--accent); }
.lp-reveal.is-in h2 { color: #1d1d1f; }
/* ---------- 系统介绍 ---------- */
.lp-intro {
  max-width: none;
  margin: 0;
  padding: 56px 24px 72px;
  background: transparent;
  /* 卡片从屏外飞入，横向超出的部分裁掉，免得出现横向滚动条；
     用 clip 而不是 hidden，纵向滚动仍由页面本身负责 */
  overflow-x: clip;
}
.lp-intro > * { max-width: 1040px; margin-left: auto;
  margin-right: auto; }
.lp-intro__title {
  margin: 0 auto 26px;
  font-size: clamp(22px, 2.6vw, 30px);
  text-align: center;
  letter-spacing: -.3px;
}
/* 三张特点卡与九个模块卡都按三等分排版，并整体居中 */
.lp-points {
  display: grid;
  grid-template-columns: repeat(3, minmax(0, 1fr));
  gap: 16px;
  margin-bottom: 46px;
}
.lp-point {
  padding: 30px 26px;
  background: rgba(253, 250, 242, .82);
  backdrop-filter: blur(12px);
  -webkit-backdrop-filter: blur(12px);
  border: 1px solid rgba(224, 216, 200, .6);
  border-radius: 22px;
  box-shadow: 0 24px 60px rgba(42, 37, 32, .10), 0 8px
  20px rgba(42, 37, 32, .05);
  transition: transform .25s ease, box-shadow .25s ease;
}
.lp-point:hover {
  transform: translateY(-4px);
  box-shadow: 0 32px 72px rgba(42, 37, 32, .14), 0 12px
  28px rgba(42, 37, 32, .07);
}
.lp-point__icon {
  display: grid;
  place-items: center;
  width: 40px;
  height: 40px;
```

```
    margin-bottom: 16px;
    border-radius: 12px;
    background: rgba(44, 107, 90, .1);
    color: var(--accent);
  }
  .lp-point h3 { margin: 0 0 8px; font-size: 16px; }
  .lp-point p {
    margin: 0;
    font-size: 13.5px;
    line-height: 1.85;
    color: #6e6e73;
    text-wrap: pretty;
  }
  .lp-sub-title {
    margin: 0 auto 20px;
    font-size: 17px;
    text-align: center;
  }
  .lp-mods {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: 12px;
    margin-bottom: 46px;
    align-items: start;
  }
  /* 模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明 */
  .lp-mod {
    position: relative;
    padding: 17px 18px 17px 48px;
    background: rgba(253, 250, 242, .75);
    backdrop-filter: blur(8px);
    -webkit-backdrop-filter: blur(8px);
    border: 1px solid rgba(224, 216, 200, .6);
    border-radius: 14px;
    outline: none;
    transition: border-color .3s ease, box-shadow .3s ease;
  }
  .lp-mod:hover,
  .lp-mod:focus-within,
  .lp-mod:focus {
    border-color: rgba(44, 107, 90, .5);
    box-shadow: 0 8px 22px rgba(42, 37, 32, .08);
  }
  .lp-mod__no {
    position: absolute;
    left: 17px;
    top: 17px;
    font-size: 13px;
    font-weight: 700;
    color: #c8d6cd;
```

```
    font-variant-numeric: tabular-nums;
  }
  .lp-mod__name {
    display: block;
    font-size: 14px;
    font-weight: 600;
  }
  /* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */
  .lp-mod__body {
    display: grid;
    grid-template-rows: 0fr;
    transition: grid-template-rows .34s ease, margin-top
    .34s ease;
  }
  .lp-mod:hover .lp-mod__body,
  .lp-mod:focus-within .lp-mod__body {
    grid-template-rows: 1fr;
    margin-top: 7px;
  }
  .lp-mod__desc {
    overflow: hidden;
    margin: 0;
    font-size: 12.5px;
    line-height: 1.7;
    color: #6e6e73;
    opacity: 0;
    transition: opacity .3s ease;
    text-wrap: pretty;
  }
  .lp-mod:hover .lp-mod__desc,
  .lp-mod:focus-within .lp-mod__desc { opacity: 1; }
  /* 触摸设备没有悬停，直接把说明摊开 */
  @media (hover: none) {
    .lp-mod__body { grid-template-rows: 1fr; margin-top:
    7px; }
    .lp-mod__desc { opacity: 1; }
  }
  .lp-foot {
    /* margin 简写会把 .lp-intro > * 里的左右 auto 一起清掉，块就贴到左边
       （文字自己居中，看着却整体偏左）—— 与标题那两处是同一个坑 */
    margin: 0 auto;
    padding-top: 26px;
    border-top: 1px solid #e6e6e8;
    text-align: center;
    font-size: 12.5px;
    color: #86868b;
  }
  /* ---------- 登录面板：点击后渐变浮现 ---------- */
  .lp-auth {
    position: fixed;
```

```
    inset: 0;                                                          6851
    z-index: 60;                                                       6852
    display: grid;                                                     6853
    place-items: center;                                               6854
    padding: 24px;                                                     6855
    opacity: 0;                                                        6856
    visibility: hidden;                                                6857
    transition: opacity .42s ease, visibility 0s linear                6858
    .42s;                                                              6859
}                                                                      6860
.lp-auth.is-open {                                                     6861
    opacity: 1;                                                        6862
    visibility: visible;                                               6863
    transition: opacity .42s ease, visibility 0s;                      6864
}                                                                      6865
/* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */                                  6866
.lp-auth__scrim {                                                      6867
    position: absolute;                                                6868
    inset: 0;                                                          6869
    background:                                                        6870
      radial-gradient(1000px 640px at 50% 0%,                          6871
                      rgba(31, 107, 69, .22), rgba(31, 107,            6872
    69, 0) 70%),                                                       6873
      linear-gradient(180deg, rgba(255, 255, 255, .84),                6874
                              rgba(246, 250, 247, .95));               6875
    backdrop-filter: blur(16px) saturate(1.25);                        6876
    -webkit-backdrop-filter: blur(16px) saturate(1.25);                6877
}                                                                      6878
.lp-auth__panel {                                                      6879
    position: relative;                                                6880
    width: 100%;                                                       6881
    max-width: 396px;                                                  6882
    padding: 36px 36px 26px;                                           6883
    background: #fff;                                                  6884
    border-radius: 20px;                                               6885
    box-shadow: 0 28px 80px rgba(14, 46, 32, .18);                     6886
    opacity: 0;                                                        6887
    transform: translateY(18px) scale(.975);                           6888
    transition: transform .55s cubic-bezier(.16, .84, .34,             6889
    1) .05s,                                                           6890
                opacity .55s ease .05s;                                6891
}                                                                      6892
.lp-auth.is-open .lp-auth__panel { opacity: 1; transform:              6893
    none; }                                                            6894
.lp-auth__close {                                                      6895
    position: absolute;                                                6896
    top: 14px;                                                         6897
    right: 14px;                                                       6898
    display: grid;                                                     6899
    place-items: center;                                               6900
```

```
    width: 30px;
    height: 30px;
    color: #86868b;
    background: transparent;
    border: none;
    border-radius: 50%;
    cursor: pointer;
    transition: background-color .18s ease, color .18s ease;
}
.lp-auth__close:hover { background: #f5f5f7; color:
    #1d1d1f; }
.lp-auth__title {
    margin: 0 0 6px;
    text-align: center;
    font-size: 24px;
    font-weight: 700;
    letter-spacing: -.2px;
}
.lp-auth__sub {
    margin: 0 0 26px;
    text-align: center;
    font-size: 13px;
    color: #86868b;
}
/* 字段校验错误：贴着输入框显示，不做成会消失的浮层 ——
    这类消息要对着输入框看，一闪而过会让人以为「点了没反应」 */
.lp-field__error {
    margin: 7px 0 0;
    font-size: 12.5px;
    color: #9c2b23;
}
.lp-field input[aria-invalid="true"] {
    background: #fdeeec;
    border-color: rgba(156, 43, 35, .35);
}
.lp-field { margin-bottom: 15px; }
.lp-field label {
    display: block;
    margin-bottom: 7px;
    font-size: 13px;
    color: #6e6e73;
}
.lp-field input {
    width: 100%;
    padding: 13px 16px;
    font-family: inherit;
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */
    color: #1d1d1f;
    background: #f5f5f7;
    border: 1px solid transparent;
```

```
    border-radius: 12px;                                                    6951
    outline: none;                                                          6952
    transition: background-color .18s ease, border-color                    6953
    .18s ease,                                                              6954
                box-shadow .18s ease;                                       6955
  }                                                                         6956
  .lp-field input::placeholder { color: #a1a1a6; }                          6957
  .lp-field input:focus {                                                   6958
    background: #fff;                                                       6959
    border-color: rgba(31, 107, 69, .55);                                   6960
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                           6961
  }                                                                         6962
  .lp-remember {                                                            6963
    display: flex;                                                          6964
    align-items: center;                                                    6965
    gap: 8px;                                                               6966
    margin: 4px 0 20px;                                                     6967
    font-size: 13px;                                                        6968
    color: #6e6e73;                                                         6969
    cursor: pointer;                                                        6970
  }                                                                         6971
  .lp-remember input {                                                      6972
    width: 16px;                                                            6973
    height: 16px;                                                           6974
    margin: 0;                                                              6975
    accent-color: var(--accent);                                            6976
    cursor: pointer;                                                        6977
  }                                                                         6978
  .lp-submit {                                                              6979
    width: 100%;                                                            6980
    padding: 13px 20px;                                                     6981
    font-family: inherit;                                                   6982
    font-size: 16px;                                                        6983
    font-weight: 500;                                                       6984
    color: #fff;                                                            6985
    background: var(--accent);                                              6986
    border: none;                                                           6987
    border-radius: 980px;                                                   6988
    cursor: pointer;                                                        6989
    transition: background-color .2s ease, box-shadow .2s                   6990
    ease,                                                                   6991
                transform .2s ease;                                         6992
  }                                                                         6993
  .lp-submit:hover {                                                        6994
    background: var(--accent-hover);                                        6995
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                         6996
  }                                                                         6997
  .lp-submit:active { transform: translateY(1px);                           6998
    box-shadow: none; }                                                     6999
  .lp-submit:focus-visible { outline: 3px solid rgba(31,                    7000
```

```
      107, 69, .40); outline-offset: 3px; }
    .lp-auth__hint {
      margin: 18px 0 0;
      text-align: center;
      font-size: 12.5px;
      color: #a1a1a6;
    }
    .lp-auth__reg {
      color: var(--accent);
      font-weight: 500;
      text-decoration: none;
      border-bottom: 1px solid rgba(31, 107, 69, .35);
      transition: border-color .18s ease;
    }
    .lp-auth__reg:hover { border-bottom-color: var(--accent);
      }
    /* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */
    @media (prefers-reduced-motion: reduce) {
      .lp-badge, .lp-title, .lp-cta, .lp-more { animation:
      none; }
      .lp-badge__svg { opacity: 1; filter: none; }
      .lp-ring, .lp-badge__mark { transform: none; }
      .lp-reveal { opacity: 1; transform: none !important; }
      .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name
      { color: inherit; }
      .lp-reveal.is-in h2 { color: #1d1d1f; }
      .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name {
      color: var(--accent); }
      .lp-auth, .lp-auth__panel { transition: opacity .2s
      linear; }
      .lp-auth__panel { transform: none; }
    }
    /* 窄屏退成两列、一列，避免卡片被挤变形 */
    @media (max-width: 900px) {
      .lp-points, .lp-mods { grid-template-columns: repeat(2,
      minmax(0, 1fr)); }
    }
    @media (max-width: 620px) {
      .lp-points, .lp-mods { grid-template-columns: minmax(0,
      1fr); }
    }
    @media (max-width: 560px) {
      .lp-hero { padding: 36px 18px 48px; min-height: auto; }
      .lp-badge { width: 160px; height: 160px; margin-bottom:
      26px; }
      .lp-auth__panel { padding: 30px 24px 22px; }
    }
</style>
<div class="lp">
    {# ---------------- 首屏：徽标 + 标题 + 说明 + 登录入口
```

```
    ----------------- #}
<section class="lp-hero">
    <div class="lp-badge" id="lp-badge" aria-hidden="true">
      {% include "_login_badge.svg" %}
    </div>
    <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>
    <button type="button" class="lp-cta" id="lp-open"
            aria-expanded="false" aria-controls="lp-auth">登录
    </button>
    <a class="lp-more" href="#intro">
      向下滑动，了解系统 <span aria-hidden="true">↓</span>
    </a>
</section>
{# ----------------- 系统介绍：随滚动逐条弹入 ----------------- #}
<section class="lp-intro" id="intro">
    <h2 class="lp-intro__title lp-reveal"
    data-dir="up">一套把评价流程固定下来的工具</h2>
    <div class="lp-points">
      {% set pdirs = ['left', 'up', 'right'] %}
      {% set points = [
        ('sliders', '方法可溯',
         '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回
    看。'),
        ('shield', '流程可控',
         '政策、区位约束、土壤污染、地质灾害四项限制性条件一票否决；未通过即终止评价，直接给出生态重建或人工辅
    助的结论。'),
        ('file', '结果可复算',
         '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')
      ] %}
      {% for ic, title, text in points %}
      <div class="lp-point lp-reveal" data-dir="{{
    pdirs[loop.index0 % 3] }}"
           style="--i: {{ loop.index0 }}">
        <span class="lp-point__icon">{{ icon(ic, 20)
    }}</span>
        <h3>{{ title }}</h3>
        <p>{{ text }}</p>
      </div>
      {% endfor %}
    </div>
    <h3 class="lp-sub-title lp-reveal"
    data-dir="up">九个功能模块</h3>
    <div class="lp-mods">
      {% set mdirs = ['left', 'up', 'right'] %}
      {% for no, name, desc in modules %}
      <div class="lp-mod lp-reveal" data-dir="{{
    mdirs[loop.index0 % 3] }}"
           style="--i: {{ loop.index0 % 3 }}" tabindex="0">
        <span class="lp-mod__no">{{ no }}</span>
        <span class="lp-mod__name">{{ name }}</span>
```

```
          <div class="lp-mod__body"><p class="lp-mod__desc">{{
    desc }}</p></div>
        </div>
        {% endfor %}
    </div>
    <p class="lp-foot lp-reveal" data-dir="up">
        历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0 ·
        运行于浏览器，数据保存在本机 ＳＱＬｉｔｅ 数据库
    </p>
</section>
{# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}
<div class="lp-auth" id="lp-auth" aria-hidden="true">
    <div class="lp-auth__scrim" data-lp-close></div>
    <div class="lp-auth__panel" role="dialog"
    aria-modal="true"
          aria-labelledby="lp-auth-title">
        <button type="button" class="lp-auth__close"
    data-lp-close
                aria-label="关闭登录框">
          <svg width="16" height="16" viewBox="0 0 24 24"
    fill="none"
              stroke="currentColor" stroke-width="2"
    stroke-linecap="round"
              aria-hidden="true"><path d="M6 6l12 12M18 6L6
    18"/></svg>
        </button>
        <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>
        <p class="lp-auth__sub">使用系统账号登录</p>
        <form method="post" novalidate>
          {{ form.hidden_tag() }}
          <div class="lp-field">
            <label for="{{ form.username.id }}">账户名称</label>
            {{ form.username(id=form.username.id,
    autocomplete="username",
                             placeholder="请输入账户名称",
                             **({'aria-invalid': 'true'} if
    form.username.errors else {})) }}
            {% for e in form.username.errors %}
            <p class="lp-field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <div class="lp-field">
            <label for="{{ form.password.id }}">密码</label>
            {{ form.password(id=form.password.id,
    autocomplete="current-password",
                             placeholder="请输入密码",
                             **({'aria-invalid': 'true'} if
    form.password.errors else {})) }}
            {% for e in form.password.errors %}
            <p class="lp-field__error">{{ e }}</p>
```

```
            {% endfor %}                                                    7151
          </div>                                                            7152
          <label class="lp-remember" for="{{ form.remember.id               7153
    }}">                                                                    7154
            {{ form.remember(id=form.remember.id) }}                        7155
            <span>记住我（下次自动登录，一个月内有效）</span>                  7156
          </label>                                                          7157
          {{ form.submit(class_="lp-submit") }}                             7158
        </form>                                                             7159
        <p class="lp-auth__hint">                                           7160
          还没有账号？<a class="lp-auth__reg"                               7161
                         href="{{ url_for('auth.register')                  7162
    }}">点我注册</a>                                                        7163
        </p>                                                                7164
        <p class="lp-auth__hint" style="margin-top:10px;">                  7165
          演示账号：admin 密码：admin123                                    7166
        </p>                                                                7167
      </div>                                                                7168
    </div>                                                                  7169
</div>                                                                      7170
{% endblock %}                                                              7171
{% block scripts %}                                                         7172
<script>                                                                    7173
(function () {                                                              7174
  var reduce = window.matchMedia &&                                         7175
    window.matchMedia('(prefers-reduced-motion:                             7176
    reduce)').matches;                                                      7177
  /* ---------- 登录面板 ---------- */                                       7178
  var btn = document.getElementById('lp-open');                             7179
  var auth = document.getElementById('lp-auth');                            7180
  var lastFocus = null;                                                     7181
  /* 清掉上一次提交留下的字段错误（红框 + 输入框下方的提示）。               7182
     登录失败时页面是直接渲染的，错误留在 DOM 里；如果只是把面板关掉再打开，  7183
     那些红框还在，看起来就像「刚点开就报错」。 */                          7184
  function clearFieldErrors(scope) {                                        7185
    var root = scope || auth;                                               7186
    Array.prototype.forEach.call(                                           7187
      root.querySelectorAll('.lp-field__error'),                            7188
      function (el) { el.remove(); });                                      7189
    Array.prototype.forEach.call(                                           7190
      root.querySelectorAll('[aria-invalid="true"]'),                       7191
      function (el) { el.removeAttribute('aria-invalid'); });               7192
    });                                                                     7193
  }                                                                         7194
  function openPanel(keepErrors) {                                          7195
    // keepErrors：由「登录失败自动展开」触发时保留错误提示；                7196
    // 用户自己点「登录」重新打开时则清掉，给一个干净的开始。                7197
    if (!keepErrors) clearFieldErrors();                                    7198
    lastFocus = document.activeElement;                                     7199
    auth.classList.add('is-open');                                          7200
```

```
    auth.setAttribute('aria-hidden', 'false');                    7201
    btn.setAttribute('aria-expanded', 'true');                    7202
    document.body.style.overflow = 'hidden';                      7203
    // 记到地址栏，直接访问 /login#login 也能展开登录框                  7204
    if (location.hash !== '#login') {                             7205
      history.replaceState(null, '', location.pathname +          7206
    '#login');                                                    7207
    }                                                             7208
    var first = auth.querySelector('input');                      7209
    if (first) window.setTimeout(function () {                    7210
    first.focus(); }, 300);                                       7211
  }                                                               7212
  function closePanel() {                                         7213
    auth.classList.remove('is-open');                             7214
    auth.setAttribute('aria-hidden', 'true');                     7215
    btn.setAttribute('aria-expanded', 'false');                   7216
    document.body.style.overflow = '';                            7217
    if (location.hash === '#login') {                             7218
      history.replaceState(null, '', location.pathname);          7219
    }                                                             7220
    if (lastFocus && lastFocus.focus) lastFocus.focus();          7221
  }                                                               7222
  if (btn && auth) {                                              7223
    // 注意别把事件对象当成参数传进去（那样会被当成 keepErrors=true）    7224
    btn.addEventListener('click', function () {                   7225
    openPanel(false); });                                         7226
    Array.prototype.forEach.call(                                 7227
      auth.querySelectorAll('[data-lp-close]'),                   7228
      function (el) { el.addEventListener('click',                7229
    closePanel); }                                                7230
    );                                                            7231
    document.addEventListener('keydown', function (e) {           7232
      if (e.key === 'Escape' && auth.classList.contains('is-      7233
    open')) closePanel();                                         7234
    });                                                           7235
    // 登录失败回到本页、或直接带 #login 访问时，直接展开登录框。          7236
    // 失败可能落在两处：字段校验错误在表单里，凭据错误在浮层里。          7237
    // 这条路径保留错误提示（keepErrors），用户才知道刚才哪儿没填。        7238
    if (location.hash === '#login'                                7239
        || document.querySelector('.lp-field__error')             7240
        || document.querySelector('.toast-center')) {             7241
      openPanel(true);                                            7242
    }                                                             7243
    // 一开始输入就把这一项的红色状态撤掉：错误已经不被犯了，红框留着只会让人困惑  7244
    Array.prototype.forEach.call(                                 7245
      auth.querySelectorAll('input'),                             7246
      function (input) {                                          7247
        input.addEventListener('input', function () {             7248
          var field = input.closest('.lp-field');                 7249
          if (field) clearFieldErrors(field);                     7250
```

```
        });
      });
}
/* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- 
   原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，
   内容始终停在 opacity:0。改为直接按元素与视口的相对位置判断，
   与徽标用同一次滚动回调，行为可预期、也便于排查。
   判断「进没进视野」用的是**布局位置**，不是 getBoundingClientRect()：
   进场动画本身就是位移（左右两列横向 56vw、中列自下方 24vh 升起），
   用视觉位置判断时，中列的盒子一直比自己的位置低一截，
   于是它总要再往下滚一段才触发，看起来比左右两列慢半拍。
   transform 不影响 offsetTop，所以这里沿 offsetParent 把布局位置累加出来，
   只在加载时算一次；滚动过程中不再测量，也就没有读写交替带来的重排。 */
var items = document.querySelectorAll('.lp-reveal');
var badge = document.getElementById('lp-badge');
var ticking = false;
var slots = [];
function layoutTop(el) {
  var top = 0;
  var node = el;
  while (node) {
    top += node.offsetTop || 0;
    node = node.offsetParent;
  }
  return top;
}
if (!reduce) {
  for (var k = 0; k < items.length; k++) {
    slots.push({
      el: items[k],
      top: layoutTop(items[k]),
      height: items[k].offsetHeight
    });
  }
}
function applyScroll() {
  ticking = false;
  var vh = window.innerHeight || 800;
  var y = window.scrollY || window.pageYOffset || 0;
  /* 徽标：滚动前约三分之一屏内完成散开淡出 */
  if (badge) {
    var range = vh * 0.34;
    var p = range > 0 ? y / range : 0;
    badge.style.setProperty('--p', Math.min(1,
  Math.max(0, p)).toFixed(3));
  }
  if (reduce || !slots.length) return;
  for (var j = 0; j < slots.length; j++) {
    var s = slots[j];
    var viewTop = s.top - y;
```

```
        //  上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内
        var shown = viewTop < vh * 0.88 && viewTop + s.height
    > vh * 0.06;
        if (shown) s.el.classList.add('is-in');
        else s.el.classList.remove('is-in');
      }
    }
    function onScroll() {
      if (!ticking) {
        ticking = true;
        window.requestAnimationFrame(applyScroll);
      }
    }
    //  减少动态效果时不做滚动特效，内容直接显示
    if (reduce) {
      Array.prototype.forEach.call(items, function (el) {
        el.classList.add('is-in');
      });
      if (badge) badge.style.setProperty('--p', '0');
    } else {
      window.addEventListener('scroll', onScroll, { passive:
      true });
      window.addEventListener('resize', onScroll);
      applyScroll();
    }
})();
</script>
{% endblock %}
#  ==== app/templates/auth/register.html ====
{% extends "base.html" %}
{% block title %}注册账号 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
    /*  ================== 注册页 ==================
        与登录页同一套观感：浅绿渐变底 + 白色卡片。
        必填只有账户名称与密码，其余收在「选填」分隔线以下，
        这样首次注册不会被一堆输入框挡住。  */
    .rg {
      min-height: 100dvh;
      display: grid;
      place-items: center;
      padding: 40px 20px 56px;
      background:
        radial-gradient(900px 520px at 50% 0%,
                        rgba(31, 107, 69, .14), rgba(31, 107,
      69, 0) 72%),
        #f6faf7;
    }
    .rg-card {
      width: 100%;
```

```
    max-width: 470px;                                                    7351
    padding: 38px 40px 30px;                                             7352
    background: #fff;                                                    7353
    border-radius: 20px;                                                 7354
    box-shadow: 0 26px 70px rgba(14, 46, 32, .13);                       7355
    animation: rg-in .5s cubic-bezier(.16, .84, .34, 1)                  7356
    both;                                                                7357
}                                                                        7358
@keyframes rg-in {                                                       7359
    from { opacity: 0; transform: translateY(16px)                       7360
    scale(.98); }                                                        7361
    to   { opacity: 1; transform: none; }                                7362
}                                                                        7363
.rg-title {                                                              7364
    margin: 0 0 6px;                                                     7365
    text-align: center;                                                  7366
    font-size: 25px;                                                     7367
    font-weight: 700;                                                    7368
    letter-spacing: -.2px;                                               7369
}                                                                        7370
.rg-sub {                                                                7371
    margin: 0 0 28px;                                                    7372
    text-align: center;                                                  7373
    font-size: 13px;                                                     7374
    color: #86868b;                                                      7375
}                                                                        7376
.rg-field { margin-bottom: 15px; }                                       7377
.rg-field > label {                                                      7378
    display: block;                                                      7379
    margin-bottom: 7px;                                                  7380
    font-size: 13px;                                                     7381
    color: #6e6e73;                                                      7382
}                                                                        7383
.rg-field > label .rg-opt {                                              7384
    margin-left: 5px;                                                    7385
    font-size: 12px;                                                     7386
    color: #a1a1a6;                                                      7387
}                                                                        7388
.rg-field input[type="text"],                                            7389
.rg-field input[type="password"] {                                       7390
    width: 100%;                                                         7391
    padding: 12px 15px;                                                  7392
    font-family: inherit;                                                7393
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */    7394
    color: #1d1d1f;                                                      7395
    background: #f5f5f7;                                                 7396
    border: 1px solid transparent;                                       7397
    border-radius: 12px;                                                 7398
    outline: none;                                                       7399
    transition: background-color .18s ease, border-color                 7400
```

```
    .18s ease,                                                    7401
                box-shadow .18s ease;                             7402
  }                                                               7403
  .rg-field input::placeholder { color: #a1a1a6; }                7404
  .rg-field input:focus {                                         7405
    background: #fff;                                             7406
    border-color: rgba(31, 107, 69, .55);                         7407
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                 7408
  }                                                               7409
  .rg-field input[aria-invalid="true"] {                          7410
    border-color: rgba(180, 60, 50, .55);                         7411
    background: #fff;                                             7412
  }                                                               7413
  .rg-field select {                                              7414
    width: 100%;                                                  7415
    padding: 12px 15px;                                           7416
    font-family: inherit;                                         7417
    font-size: 15px;                                              7418
    color: #1d1d1f;                                               7419
    background: #f5f5f7;                                          7420
    border: 1px solid transparent;                                7421
    border-radius: 12px;                                          7422
    outline: none;                                                7423
  }                                                               7424
  .rg-field select:focus {                                        7425
    background: #fff;                                             7426
    border-color: rgba(31, 107, 69, .55);                         7427
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                 7428
  }                                                               7429
  .rg-err {                                                       7430
    margin: 6px 0 0;                                              7431
    font-size: 12.5px;                                            7432
    color: var(--danger);                                         7433
  }                                                               7434
  /* 「选填」分隔线 */                                             7435
  .rg-split {                                                     7436
    display: flex;                                                7437
    align-items: center;                                          7438
    gap: 12px;                                                    7439
    margin: 24px 0 18px;                                          7440
    font-size: 12.5px;                                            7441
    color: #a1a1a6;                                               7442
  }                                                               7443
  .rg-split::before,                                              7444
  .rg-split::after {                                              7445
    content: "";                                                  7446
    flex: 1;                                                      7447
    height: 1px;                                                  7448
    background: #ececee;                                          7449
  }                                                               7450
```

```
.rg-pair {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 0 14px;
}
@media (max-width: 480px) { .rg-pair {
  grid-template-columns: 1fr; } }
/* 头像：圆形预览 + 选择文件 */
.rg-avatar {
  display: flex;
  align-items: center;
  gap: 14px;
  margin-bottom: 15px;
}
.rg-avatar__face {
  display: grid;
  place-items: center;
  flex: none;
  width: 54px;
  height: 54px;
  border-radius: 50%;
  background: #eef2f0;
  color: #9aa8a1;
  font-size: 12px;
  overflow: hidden;
  background-size: cover;
  background-position: center;
}
.rg-avatar__pick { min-width: 0; }
.rg-avatar__pick input[type="file"] {
  max-width: 100%;
  font-family: inherit;
  font-size: 12.5px;
  color: #6e6e73;
}
.rg-avatar__tip {
  margin: 4px 0 0;
  font-size: 12px;
  color: #a1a1a6;
}
.rg-submit {
  width: 100%;
  margin-top: 24px;
  padding: 13px 20px;
  font-family: inherit;
  font-size: 16px;
  font-weight: 500;
  color: #fff;
  background: var(--accent);
  border: none;
```

```
      border-radius: 980px;                                          7501
      cursor: pointer;                                               7502
      transition: background-color .2s ease, box-shadow .2s          7503
      ease,                                                          7504
                  transform .2s ease;                                7505
    }                                                                7506
    .rg-submit:hover {                                               7507
      background: var(--accent-hover);                               7508
      box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                7509
    }                                                                7510
    .rg-submit:active { transform: translateY(1px);                  7511
      box-shadow: none; }                                            7512
    .rg-submit:focus-visible { outline: 3px solid rgba(31,           7513
      107, 69, .40); outline-offset: 3px; }                          7514
    .rg-foot {                                                       7515
      margin: 20px 0 0;                                              7516
      text-align: center;                                            7517
      font-size: 13px;                                               7518
      color: #86868b;                                                7519
    }                                                                7520
    .rg-foot a { color: var(--accent); text-decoration: none;        7521
      }                                                              7522
    .rg-foot a:hover { text-decoration: underline; }                 7523
    @media (prefers-reduced-motion: reduce) {                        7524
      .rg-card { animation: none; }                                  7525
    }                                                                7526
</style>                                                             7527
<div class="rg">                                                     7528
  <div class="rg-card">                                              7529
    <h1 class="rg-title">创建账号</h1>                                7530
    <p class="rg-sub">新账号为「操作员」，职务由管理员调整</p>          7531
    <form method="post" enctype="multipart/form-data"                7532
    novalidate>                                                      7533
      {{ form.hidden_tag() }}                                        7534
      <div class="rg-field">                                         7535
        <label for="{{ form.username.id }}">账户名称</label>          7536
        {{ form.username(id=form.username.id,                        7537
    autocomplete="username",                                         7538
                         placeholder="2–64 个字符",                  7539
                         **({'aria-invalid': 'true'} if              7540
    form.username.errors else {})) }}                                7541
        {% for e in form.username.errors %}<p                        7542
    class="rg-err">{{ e }}</p>{% endfor %}                           7543
      </div>                                                         7544
      <div class="rg-field">                                         7545
        <label for="{{ form.password.id }}">密码</label>             7546
        {{ form.password(id=form.password.id,                        7547
    autocomplete="new-password",                                     7548
                         placeholder="至少 6 位",                    7549
                         **({'aria-invalid': 'true'} if              7550
```

```
form.password.errors else {})) }}                                    7551
    {% for e in form.password.errors %}<p                            7552
class="rg-err">{{ e }}</p>{% endfor %}                               7553
  </div>                                                             7554
  <div class="rg-field">                                             7555
    <label for="{{ form.confirm.id }}">确认密码</label>              7556
    {{ form.confirm(id=form.confirm.id,                              7557
autocomplete="new-password",                                         7558
                    placeholder="再次输入密码",                      7559
                    **({'aria-invalid': 'true'} if                   7560
form.confirm.errors else {})) }}                                     7561
    {% for e in form.confirm.errors %}<p                             7562
class="rg-err">{{ e }}</p>{% endfor %}                               7563
  </div>                                                             7564
  <p class="rg-split">以下均为选填</p>                               7565
  <div class="rg-avatar">                                            7566
    <span class="rg-avatar__face" id="rg-face"                       7567
aria-hidden="true">头像</span>                                       7568
    <div class="rg-avatar__pick">                                    7569
      <label class="visually-hidden"                                 7570
for="rg-avatar">选择头像图片</label>                                 7571
      <input id="rg-avatar" type="file" name="avatar"                7572
             accept=".png,.jpg,.jpeg,.gif,.webp">                    7573
      <p class="rg-avatar__tip">PNG / JPG / GIF /                    7574
WebP，不超过 4 MB</p>                                                7575
    </div>                                                           7576
  </div>                                                             7577
  <div class="rg-pair">                                              7578
    <div class="rg-field">                                           7579
      <label for="{{ form.display_name.id }}">昵称</label>           7580
      {{ form.display_name(id=form.display_name.id,                  7581
                           placeholder="留空则显示账户名称") }}      7582
    </div>                                                           7583
    <div class="rg-field">                                           7584
      <label for="{{ form.gender.id }}">性别</label>                 7585
      {{ form.gender(id=form.gender.id) }}                           7586
    </div>                                                           7587
  </div>                                                             7588
  <div class="rg-field">                                             7589
    <label for="{{ form.email.id }}">邮箱</label>                    7590
    {{ form.email(id=form.email.id,                                  7591
autocomplete="email",                                                7592
                  placeholder="用于联系，可留空",                    7593
                  **({'aria-invalid': 'true'} if                     7594
form.email.errors else {})) }}                                       7595
    {% for e in form.email.errors %}<p                               7596
class="rg-err">{{ e }}</p>{% endfor %}                               7597
  </div>                                                             7598
  <div class="rg-field" style="margin-bottom:0;">                    7599
    <label for="{{ form.phone.id }}">手机号</label>                  7600
```

```
            {{ form.phone(id=form.phone.id, autocomplete="tel",
      placeholder="可留空") }}
        </div>
        {{ form.submit_reg(class_="rg-submit") }}
      </form>
      <p class="rg-foot">已有账号？<a href="{{
      url_for('auth.login') }}">返回登录</a></p>
    </div>
</div>
{% endblock %}
{% block scripts %}
<script>
//  选好头像后立刻在左侧圆框里预览，省得传上去才知道选错了
(function () {
  var input = document.getElementById('rg-avatar');
  var face = document.getElementById('rg-face');
  if (!input || !face || !window.FileReader) return;
  input.addEventListener('change', function () {
    var file = input.files && input.files[0];
    if (!file) return;
    var reader = new FileReader();
    reader.onload = function (e) {
      face.textContent = '';
      face.style.backgroundImage = 'url("' +
    e.target.result + '")';
    };
    reader.readAsDataURL(file);
  });
})();
</script>
{% endblock %}
# ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
{# ---------- 页首横幅 ---------- #}
<section class="hero">
  <div class="hero__body">
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>
    <div class="hero__actions">
      <a class="hero-btn hero-btn--ghost"
    href="/indicator/import">
        {{ icon("upload", 16) }}批量导入
      </a>
      <a class="hero-btn hero-btn--solid"
    href="/project/add">
        {{ icon("plus", 16) }}新建评价项目
      </a>
    </div>
```

```
  </div>
</section>
{# ---------- 数据概览 ---------- #}
<section class="stat-cards">
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">评价项目</span>
    <span class="stat-card__value">{{ dash.stats.projects
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {%- if projects_without_patch -%}
        其中 {{ projects_without_patch|length }} 个项目尚未建立图斑
      {%- elif dash.progress.patches -%}
        已建立 {{ dash.progress.patches }} 个矿山图斑
      {%- else -%}
        尚未建立矿山图斑
      {%- endif -%}
    </span>
  </a>
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">矿山图斑</span>
    <span class="stat-card__value">{{ dash.stats.patches
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {% if dash.progress.ready %}其中 {{ dash.progress.ready
    }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}
    </span>
  </a>
  <a class="stat-card" href="/unit/">
    <span class="stat-card__label">微地质单元</span>
    <span class="stat-card__value">{{ dash.stats.unit_types
    }}<span class="stat-card__unit">类</span></span>
    <span class="stat-card__note">
      {% if dash.progress.units %}共划分 {{
    dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}
    </span>
  </a>
  <a class="stat-card" href="/standard/">
    <span class="stat-card__label">标准与政策</span>
    <span class="stat-card__value">{{ dash.stats.standards
    }}<span class="stat-card__unit">项</span></span>
    <span class="stat-card__note">地方标准与政策文件</span>
  </a>
</section>
{# ---------- 评价流程 ---------- #}
<section class="section">
  <div class="section__head">
    <h2 class="section__title">评价流程</h2>
  </div>
  <div class="step-grid">
    {% for s in dash.steps %}
```

```
      {% if s.available %}                                                    7701
      <a class="step-card" href="{{ s.href }}">                               7702
        <span class="step-card__no">{{ s.no }}</span>                         7703
        <span class="step-card__icon">{{ icon(s.icon, 18)                     7704
      }}</span>                                                               7705
        <span class="step-card__title">{{ s.title }}</span>                   7706
        <span class="step-card__desc">{{ s.desc }}</span>                     7707
        {% if s.pending %}                                                    7708
        <span class="step-card__badge">待处理 {{ s.pending                    7709
      }}</span>                                                               7710
        {% endif %}                                                           7711
      </a>                                                                    7712
      {% else %}                                                              7713
      <div class="step-card step-card--locked"                                7714
      aria-disabled="true">                                                   7715
        <span class="step-card__no">{{ s.no }}</span>                         7716
        <span class="step-card__icon">{{ icon(s.icon, 18)                     7717
      }}</span>                                                               7718
        <span class="step-card__title">{{ s.title }}</span>                   7719
        <span class="step-card__desc">{{ s.desc }}</span>                     7720
        <span class="step-card__lock">{{ s.note }}</span>                     7721
      </div>                                                                  7722
      {% endif %}                                                             7723
      {% endfor %}                                                            7724
    </div>                                                                    7725
</section>                                                                    7726
{# ---------- 系统管理 ---------- #}                                          7727
{# 个人设置、账号与操作日志的入口都在左侧导航栏，工作台不再重复列出 #}                  7728
{% endblock %}                                                                7729
# ==== app/templates/auth/add_user.html ====                                  7730
{% extends "base.html" %}                                                     7731
{% block title %}新增账号{% endblock %}                                         7732
{% block content %}                                                           7733
<div class="card" style="max-width:560px;">                                   7734
  <h2>新增账号</h2>                                                           7735
  <form method="post">                                                        7736
    {{ form.hidden_tag() }}                                                   7737
    <div class="field">                                                       7738
      <label class="field__label" for="username">账户名称</label              7739
    >                                                                         7740
      <input id="username" type="text" name="username"                        7741
    required                                                                  7742
            maxlength="64" autocomplete="off"                                 7743
    style="width:280px;">                                                     7744
    </div>                                                                    7745
    <div class="field">                                                       7746
      <label class="field__label" for="password">密码</label>                 7747
      <input id="password" type="password" name="password"                    7748
    required                                                                  7749
            autocomplete="new-password"                                       7750
```

```
      style="width:280px;">                                                    7751
      </div>                                                                   7752
      <div class="field">                                                      7753
        <label class="field__label" for="role">职务</label>                    7754
        <select id="role" name="role">                                         7755
          {% for code, label in role_labels.items() %}                         7756
          <option value="{{ code }}" {% if code ==                             7757
      default_role %}selected{% endif %}>{{ label }}</option>                  7758
          {% endfor %}                                                         7759
        </select>                                                              7760
        <p class="muted" style="margin:6px 0 0;">                              7761
          管理员可管理系统、维护基础数据并查看操作日志；操作员可使用全部业务功能。                    7762
        </p>                                                                   7763
      </div>                                                                   7764
      <hr style="margin:22px 0;border:none;border-top:1px                      7765
      solid var(--line);">                                                     7766
      <h3 style="margin:0 0 12px;font-size:15px;">个人资料（选填）</h3         7767
      >                                                                        7768
      <div class="field">                                                      7769
        <label class="field__label" for="display_name">昵称</lab               7770
      el>                                                                      7771
        <input id="display_name" type="text"                                   7772
      name="display_name" maxlength="64"                                       7773
             placeholder="留空则显示账户名称">                                   7774
      </div>                                                                   7775
      <div class="field">                                                      7776
        <label class="field__label" for="gender">性别</label>                  7777
        <select id="gender" name="gender">                                     7778
          <option value="">不填</option>                                       7779
          {% for g in genders %}<option value="{{ g }}">{{ g                   7780
      }}</option>{% endfor %}                                                  7781
        </select>                                                              7782
      </div>                                                                   7783
      <div class="field">                                                      7784
        <label class="field__label" for="email">邮箱</label>                   7785
        <input id="email" type="text" name="email"                             7786
      maxlength="128">                                                         7787
      </div>                                                                   7788
      <div class="field">                                                      7789
        <label class="field__label" for="phone">手机号</label>                 7790
        <input id="phone" type="text" name="phone"                             7791
      maxlength="32">                                                          7792
      </div>                                                                   7793
      <p class="actions" style="margin-top:18px;">                             7794
        <button type="submit" class="btn">保存</button>                        7795
        <a class="btn btn--ghost" href="{{                                     7796
      url_for('auth.profile', tab='users') }}">返回</a>                        7797
      </p>                                                                     7798
    </form>                                                                    7799
</div>                                                                         7800
```

```
{% endblock %}                                                    7801
# ==== app/templates/auth/edit_user.html ====                     7802
{% extends "base.html" %}                                         7803
{% block title %}编辑账号 · {{ user.username }}{% endblock %}     7804
{% block content %}                                               7805
<div class="card" style="max-width:560px;">                       7806
  <h2>编辑账号 · {{ user.username }}</h2>                         7807
  <h3 style="margin:0 0 12px;font-size:15px;">个人资料</h3>       7808
  <p class="muted" style="margin:0 0 14px;font-size:12.5px;"      7809
    >                                                             7810
    以下内容用户也可在「个人设置」中自行修改，此处为管理员代改。 7811
  </p>                                                            7812
  <form method="post">                                            7813
    {{ form.hidden_tag() }}                                       7814
    <div class="field">                                           7815
      <label class="field__label" for="display_name">昵称</lab    7816
    el>                                                           7817
      <input id="display_name" type="text"                        7818
    name="display_name" maxlength="64"                            7819
             value="{{ user.display_name or '' }}"                7820
             placeholder="留空则显示账户名称">                    7821
    </div>                                                        7822
    <div class="field">                                           7823
      <label class="field__label" for="gender">性别</label>       7824
      <select id="gender" name="gender">                          7825
        <option value="">不填</option>                            7826
        {% for g in genders %}                                    7827
        <option value="{{ g }}" {% if user.gender == g            7828
    %}selected{% endif %}>{{ g }}</option>                        7829
        {% endfor %}                                              7830
      </select>                                                   7831
    </div>                                                        7832
    <div class="field">                                           7833
      <label class="field__label" for="email">邮箱</label>        7834
      <input id="email" type="text" name="email"                  7835
    maxlength="128"                                               7836
             value="{{ user.email or '' }}">                      7837
    </div>                                                        7838
    <div class="field">                                           7839
      <label class="field__label" for="phone">手机号</label>      7840
      <input id="phone" type="text" name="phone"                  7841
    maxlength="32"                                                7842
             value="{{ user.phone or '' }}">                      7843
    </div>                                                        7844
    <hr style="margin:22px 0;border:none;border-top:1px           7845
    solid var(--line);">                                          7846
    <h3 style="margin:0 0 12px;font-size:15px;">职务与权限</h3>   7847
    <p class="muted" style="margin:0 0                            7848
    14px;font-size:12.5px;">                                      7849
      职务只能由管理员设置：管理员可管理系统、维护基础数据并查看操作日志； 7850
```

```
        操作员可使用全部业务功能。                                                    7851
      </p>                                                                          7852
      <div class="field">                                                           7853
        <label class="field__label" for="role">职务</label>                         7854
        <select id="role" name="role">                                              7855
          {% for code, label in role_labels.items() %}                              7856
          <option value="{{ code }}" {% if user.role == code                        7857
      %}selected{% endif %}>{{ label }}</option>                                    7858
          {% endfor %}                                                              7859
        </select>                                                                   7860
      </div>                                                                        7861
      <div class="field">                                                           7862
        <label class="checkbox" for="active">                                       7863
          <input id="active" type="checkbox" name="active"                          7864
      value="1"                                                                     7865
                 {% if user.active %}checked{% endif %}>                            7866
          <span>启用该账号</span>                                                   7867
        </label>                                                                    7868
        <p class="muted" style="margin:6px 0 0;">                                   7869
          停用后该账号无法登录，但其历史操作记录会保留。                           7870
          系统至少需保留一名启用状态的管理员。                                     7871
        </p>                                                                        7872
      </div>                                                                        7873
      <div class="field">                                                           7874
        <label class="field__label" for="password">重置密码（留空则不修改7875
      ）</label>                                                                    7876
        <input id="password" type="password" name="password"                        7877
               autocomplete="new-password"                                          7878
      style="width:280px;"                                                          7879
               placeholder="留空表示保持原密码">                                    7880
      </div>                                                                        7881
      <p class="actions" style="margin-top:18px;">                                  7882
        <button type="submit" class="btn">保存</button>                             7883
        <a class="btn btn--ghost" href="{{                                          7884
      url_for('auth.profile', tab='users') }}">返回</a>                             7885
      </p>                                                                          7886
    </form>                                                                         7887
</div>                                                                              7888
{% endblock %}                                                                      7889
# ==== app/templates/auth/profile.html ====                                         7890
{% extends "base.html" %}                                                           7891
{% from "_macros.html" import icon %}                                               7892
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}    7893
{% block content %}                                                                 7894
<style>                                                                             7895
  /* ================== 个人设置（仿 Apple 账户页） ==================             7896
     左栏是身份与分类，右栏是该分类下的功能卡。                                    7897
     每张卡默认只显示当前值，点开才展开对应的修改表单 ——                          7898
     不点就不会误改，页面也不会一上来堆满输入框。 */                               7899
  .acct {                                                                           7900
```

```
    display: grid;                                                    7901
    grid-template-columns: 244px 1fr;                                 7902
    gap: 40px;                                                        7903
    align-items: start;                                               7904
  }                                                                   7905
  @media (max-width: 900px) {                                         7906
    .acct { grid-template-columns: 1fr; gap: 24px; }                  7907
  }                                                                   7908
  /* ---------- 左栏 ---------- */                                    7909
  .acct-rail__avatar {                                                7910
    display: grid;                                                    7911
    place-items: center;                                              7912
    width: 88px;                                                      7913
    height: 88px;                                                     7914
    border-radius: 50%;                                               7915
    background: #e9edeb;                                              7916
    color: #97a09b;                                                   7917
    font-size: 32px;                                                  7918
    font-weight: 500;                                                 7919
    object-fit: cover;                                                7920
  }                                                                   7921
  .acct-rail__name {                                                  7922
    margin: 16px 0 2px;                                               7923
    font-size: 16px;                                                  7924
    font-weight: 600;                                                 7925
  }                                                                   7926
  .acct-rail__sub {                                                   7927
    margin: 0 0 24px;                                                 7928
    font-size: 13px;                                                  7929
    color: #86868b;                                                   7930
    word-break: break-all;                                            7931
  }                                                                   7932
  .acct-tabs {                                                        7933
    display: flex;                                                    7934
    flex-direction: column;                                           7935
    gap: 2px;                                                         7936
    margin-bottom: 24px;                                              7937
  }                                                                   7938
  .acct-tabs a {                                                      7939
    padding: 7px 12px;                                                7940
    border: 1px solid transparent;                                    7941
    border-radius: 7px;                                               7942
    font-size: 15px;                                                  7943
    color: #1d1d1f;                                                   7944
    text-decoration: none;                                            7945
    transition: background-color .16s ease, border-color              7946
    .16s ease;                                                        7947
  }                                                                   7948
  .acct-tabs a:hover { background: #f5f5f7; }                         7949
  .acct-tabs a.is-active {                                            7950
```

```
    font-weight: 600;
    border-color: #bcc9c1;
    background: #fff;
}
/* 更换头像：折叠在左栏，点开才出现文件选择 */
.acct-avatar-edit > summary {
    display: inline-block;
    padding: 5px 12px;
    font-size: 13px;
    color: var(--accent);
    background: var(--accent-soft);
    border-radius: 7px;
    cursor: pointer;
    list-style: none;
}
.acct-avatar-edit > summary::-webkit-details-marker {
    display: none; }
.acct-avatar-edit > summary:hover { background: #dcebe2; }
.acct-avatar-edit__body { margin-top: 12px; }
.acct-avatar-edit__body .btn { width: 100%;
    margin-bottom: 8px; }
.acct-avatar-edit__file {
    margin: 0 0 10px;
    font-size: 12px;
    color: var(--ink-soft);
    word-break: break-all;
}
/* ---------- 右栏 ---------- */
.acct-title {
    margin: 0 0 8px;
    font-size: 26px;
    font-weight: 700;
    letter-spacing: -.3px;
}
.acct-lead {
    margin: 0 0 26px;
    font-size: 14px;
    color: #6e6e73;
}
/* 固定两列，与 Apple 账户页的 2×2 一致；
   卡片也宽一些，邮箱之类的长值不至于被拆成两行 */
.acct-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 16px;
    align-items: start;
}
@media (max-width: 760px) {
    .acct-grid { grid-template-columns: 1fr; }
}
```

```
/* 功能卡：details/summary 实现，点开才展开表单 */
.acct-card {
  background: #fff;
  border: 1px solid #e6e6e8;
  border-radius: 14px;
  transition: border-color .2s ease, box-shadow .2s ease;
}
.acct-card[open] {
  border-color: rgba(31, 107, 69, .45);
  box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
}
.acct-card > summary {
  position: relative;
  display: block;
  padding: 18px 56px 18px 20px;
  border-radius: 14px;
  cursor: pointer;
  list-style: none;
}
.acct-card > summary::-webkit-details-marker { display:
  none; }
.acct-card > summary:hover { background: #fafbfa; }
.acct-card > summary:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 2px;
}
.acct-card__title {
  display: block;
  font-size: 15px;
  font-weight: 600;
}
.acct-card__value {
  display: block;
  margin-top: 3px;
  font-size: 14px;
  color: #6e6e73;
  overflow-wrap: anywhere;
}
.acct-card__icon {
  position: absolute;
  top: 17px;
  right: 17px;
  display: grid;
  place-items: center;
  width: 27px;
  height: 27px;
  border-radius: 8px;
  background: var(--accent-soft);
  color: var(--accent);
}
```

```
  .acct-card__body {                                                    8051
    padding: 4px 20px 20px;                                             8052
    border-top: 1px solid #f0f1f2;                                      8053
    margin-top: 2px;                                                    8054
  }                                                                     8055
  .acct-card__body .field:first-child { margin-top: 16px; }             8056
  .acct-card__body .btn { width: 100%; }                                8057
  /* 只读信息列表 */                                                    8058
  .acct-kv {                                                            8059
    display: grid;                                                      8060
    grid-template-columns: 88px 1fr;                                    8061
    gap: 7px 12px;                                                      8062
    margin: 16px 0 0;                                                   8063
    font-size: 13.5px;                                                  8064
  }                                                                     8065
  .acct-kv dt { color: var(--ink-soft); }                               8066
  .acct-kv dd { margin: 0; }                                            8067
  .field__error {                                                       8068
    margin: 5px 0 0;                                                    8069
    font-size: 12.5px;                                                  8070
    color: var(--danger);                                               8071
  }                                                                     8072
  /* 管理员分类（账号 / 操作日志）里的表格面板 */                       8073
  .acct-panel {                                                         8074
    background: #fff;                                                   8075
    border: 1px solid #e6e6e8;                                          8076
    border-radius: 14px;                                                8077
    overflow: hidden;                                                   8078
  }                                                                     8079
  .acct-panel__head {                                                   8080
    display: flex;                                                      8081
    align-items: center;                                                8082
    justify-content: space-between;                                     8083
    gap: 12px;                                                          8084
    padding: 13px 20px;                                                 8085
    border-bottom: 1px solid #f0f1f2;                                   8086
  }                                                                     8087
  .acct-panel__count {                                                  8088
    margin: 0;                                                          8089
    font-size: 13.5px;                                                  8090
    color: #6e6e73;                                                     8091
  }                                                                     8092
  .acct-panel table { margin: 0; }                                      8093
  .acct-panel th,                                                       8094
  .acct-panel td { border: 0; border-bottom: 1px solid                  8095
    #f0f1f2; }                                                          8096
  .acct-panel thead th { background: #f7f9f8; }                         8097
  .acct-panel tbody tr:last-child td { border-bottom: 0; }              8098
  .acct-panel tbody tr:hover { background: #fafbfa; }                   8099
</style>                                                                8100
```

```
<div class="acct">                                                    8101
  {# ---------------- 左栏：身份 + 分类 ---------------- #}              8102
  <aside class="acct-rail">                                           8103
    {% if current_user.avatar %}                                      8104
    <img class="acct-rail__avatar"                                    8105
         src="{{ url_for('static', filename=current_user.ava8106
    tar) }}"                                                          8107
         alt="{{ current_user.shown_name }} 的头像">                   8108
    {% else %}                                                        8109
    <span class="acct-rail__avatar" aria-hidden="true">{{             8110
    current_user.initial }}</span>                                    8111
    {% endif %}                                                       8112
    <p class="acct-rail__name">{{ current_user.shown_name             8113
    }}</p>                                                            8114
    <p class="acct-rail__sub">                                        8115
      {{ role_labels.get(current_user.role,                           8116
    current_user.role) }}                                             8117
      {% if current_user.email %} · {{ current_user.email             8118
    }}{% endif %}                                                     8119
    </p>                                                              8120
    <nav class="acct-tabs" aria-label="个人设置分类">                  8121
      <a href="{{ url_for('auth.profile', tab='info') }}"             8122
         {% if tab == 'info' %}class="is-active"                      8123
    aria-current="page"{% endif %}>个人信息</a>                       8124
      <a href="{{ url_for('auth.profile', tab='security')             8125
    }}"                                                               8126
         {% if tab == 'security' %}class="is-active"                  8127
    aria-current="page"{% endif %}>登录与安全性</a>                   8128
      {% if current_user.is_admin %}                                  8129
      <a href="{{ url_for('auth.profile', tab='users') }}"            8130
         {% if tab == 'users' %}class="is-active"                     8131
    aria-current="page"{% endif %}>账号</a>                           8132
      <a href="{{ url_for('auth.profile', tab='logs') }}"             8133
         {% if tab == 'logs' %}class="is-active"                      8134
    aria-current="page"{% endif %}>操作日志</a>                       8135
      {% endif %}                                                     8136
    </nav>                                                            8137
    <details class="acct-avatar-edit">                                8138
      <summary>更换头像</summary>                                     8139
      <div class="acct-avatar-edit__body">                            8140
        <form method="post" action="/profile/avatar"                  8141
              enctype="multipart/form-data">                          8142
          {{ avatar_form.hidden_tag() }}                              8143
          <input type="hidden" name="tab" value="{{ tab }}">          8144
          <label class="visually-hidden"                              8145
    for="avatar-file">选择头像图片</label>                            8146
          <input id="avatar-file" type="file" name="avatar"           8147
                 accept=".png,.jpg,.jpeg,.gif,.webp">                 8148
          <p class="acct-avatar-edit__file">PNG / JPG / GIF           8149
    / WebP，不超过 4 MB</p>                                           8150
```

```
          <button type="submit" class="btn                    8151
    btn--sm">上传头像</button>                                8152
        </form>                                               8153
        {% if current_user.avatar %}                          8154
        <form method="post" action="/profile/avatar/delete"   8155
              onsubmit="return confirm('确定移除当前头像吗？');">  8156
          {{ avatar_form.hidden_tag() }}                      8157
          <input type="hidden" name="tab" value="{{ tab }}">  8158
          <button type="submit" class="btn btn--sm            8159
    btn--ghost">移除头像</button>                             8160
        </form>                                               8161
        {% endif %}                                           8162
      </div>                                                  8163
    </details>                                                8164
  </aside>                                                    8165
  {# ---------------- 右栏：功能卡 ---------------- #}          8166
  <div class="acct-main">                                     8167
    {% if tab == 'security' %}                                8168
    <h1 class="acct-title">登录与安全性</h1>                    8169
    <p class="acct-lead">管理与登录账号、账号安全有关的设置。</p>  8170
    <div class="acct-grid">                                   8171
      <details class="acct-card"                              8172
               {% if pwd_form.old_password.errors or          8173
    pwd_form.new_password.errors or pwd_form.confirm.errors   8174
    %}open{% endif %}>                                        8175
        <summary>                                             8176
          <span class="acct-card__title">密码</span>          8177
          <span class="acct-card__value">修改登录密码</span>  8178
          <span class="acct-card__icon">{{ icon("lock", 15)   8179
    }}</span>                                                 8180
        </summary>                                            8181
        <div class="acct-card__body">                         8182
          <form method="post">                                8183
            {{ pwd_form.hidden_tag() }}                       8184
            <input type="hidden" name="action"                8185
    value="password">                                         8186
            <input type="hidden" name="tab"                   8187
    value="security">                                         8188
            <div class="field">                               8189
              <label class="field__label" for="{{             8190
    pwd_form.old_password.id }}">当前密码</label>             8191
              {{ pwd_form.old_password(id=pwd_form.old_passw  8192
    ord.id,                                                   8193
                                                              8194
    autocomplete="current-password") }}                       8195
              {% for e in pwd_form.old_password.errors %}     8196
              <p class="field__error">{{ e }}</p>             8197
              {% endfor %}                                    8198
            </div>                                            8199
            <div class="field">                               8200
```

```
            <label class="field__label" for="{{                8201
pwd_form.new_password.id }}">新密码</label>                    8202
            {{ pwd_form.new_password(id=pwd_form.new_passw8203
ord.id,                                                        8204
                                                               8205
autocomplete="new-password") }}                                8206
            {% for e in pwd_form.new_password.errors %}        8207
            <p class="field__error">{{ e }}</p>                8208
            {% endfor %}                                       8209
          </div>                                               8210
          <div class="field">                                  8211
            <label class="field__label" for="{{                8212
pwd_form.confirm.id }}">确认新密码</label>                     8213
            {{ pwd_form.confirm(id=pwd_form.confirm.id,        8214
                                autocomplete="new-password8215
") }}                                                          8216
            {% for e in pwd_form.confirm.errors %}             8217
            <p class="field__error">{{ e }}</p>                8218
            {% endfor %}                                       8219
          </div>                                               8220
          <p class="actions">{{ pwd_form.submit_pwd(class_8221
="btn") }}</p>                                                 8222
        </form>                                                8223
      </div>                                                   8224
    </details>                                                 8225
    <details class="acct-card">                                8226
      <summary>                                                8227
        <span class="acct-card__title">登录账号</span>          8228
        <span class="acct-card__value">{{                      8229
current_user.username }}</span>                                8230
        <span class="acct-card__icon">{{ icon("user", 15)      8231
}}</span>                                                      8232
      </summary>                                               8233
      <div class="acct-card__body">                            8234
        <dl class="acct-kv">                                   8235
          <dt>登录账号</dt><dd class="mono">{{                 8236
current_user.username }}</dd>                                  8237
          <dt>职务</dt>                                        8238
          <dd>{{ role_labels.get(current_user.role,            8239
current_user.role) }}</dd>                                     8240
          <dt>状态</dt>                                        8241
          <dd>                                                 8242
            {% if current_user.active %}                       8243
            <span class="tag tag--ok">启用</span>              8244
            {% else %}                                         8245
            <span class="tag tag--warn">停用</span>            8246
            {% endif %}                                        8247
          </dd>                                                8248
          <dt>创建时间</dt>                                    8249
          <dd class="mono">{{ current_user.created_at |        8250
```

```
localdt }}</dd>
        </dl>
      </div>
    </details>
</div>
{% elif tab == 'users' %}
<h1 class="acct-title">账号</h1>
<p class="acct-lead">管理本系统的账号、职务与启用状态。新注册的账号默认为「操作员」。</p
>
{% include "auth/_users.html" %}
{% elif tab == 'logs' %}
<h1 class="acct-title">操作日志</h1>
<p class="acct-lead">记录各账号在本系统中的关键操作。</p>
{% include "auth/_logs.html" %}
{% else %}
<h1 class="acct-title">个人信息</h1>
<p class="acct-lead">管理你的昵称、性别以及可以联系到你的邮箱与手机号。</p>
<form method="post">
  {{ info_form.hidden_tag() }}
  <input type="hidden" name="action" value="info">
  <input type="hidden" name="tab" value="info">
  <div class="acct-grid">
    <details class="acct-card" {% if
info_form.display_name.errors %}open{% endif %}>
      <summary>
        <span class="acct-card__title">昵称</span>
        <span class="acct-card__value">
          {{ current_user.display_name or "未设置" }}
        </span>
        <span class="acct-card__icon">{{ icon("user",
15) }}</span>
      </summary>
      <div class="acct-card__body">
        <div class="field">
          <label class="field__label" for="{{
info_form.display_name.id }}">
            昵称
          </label>
          {{ info_form.display_name(id=info_form.display
_name.id,

placeholder="留空则显示账户名称") }}
          {% for e in info_form.display_name.errors %}
          <p class="field__error">{{ e }}</p>
          {% endfor %}
        </div>
        <p class="actions">{{ info_form.submit_info(clas
s_="btn") }}</p>
      </div>
    </details>
```

```
      <details class="acct-card" {% if                          8301
info_form.gender.errors %}open{% endif %}>                      8302
        <summary>                                               8303
          <span class="acct-card__title">性别</span>            8304
          <span class="acct-card__value">{{                     8305
current_user.gender or "未设置" }}</span>                       8306
          <span class="acct-card__icon">{{ icon("user",         8307
15) }}</span>                                                   8308
        </summary>                                              8309
        <div class="acct-card__body">                           8310
          <div class="field">                                   8311
            <label class="field__label" for="{{                 8312
info_form.gender.id }}">性别</label>                            8313
            {{ info_form.gender(id=info_form.gender.id) }}      8314
            {% for e in info_form.gender.errors %}              8315
            <p class="field__error">{{ e }}</p>                 8316
            {% endfor %}                                        8317
          </div>                                                8318
          <p class="actions">{{ info_form.submit_info(clas      8319
s_="btn") }}</p>                                                8320
        </div>                                                  8321
      </details>                                                8322
      <details class="acct-card" {% if                          8323
info_form.email.errors %}open{% endif %}>                       8324
        <summary>                                               8325
          <span class="acct-card__title">邮箱</span>            8326
          <span class="acct-card__value">{{                     8327
current_user.email or "未设置" }}</span>                        8328
          <span class="acct-card__icon">{{ icon("mail",         8329
15) }}</span>                                                   8330
        </summary>                                              8331
        <div class="acct-card__body">                           8332
          <div class="field">                                   8333
            <label class="field__label" for="{{                 8334
info_form.email.id }}">邮箱</label>                             8335
            {{ info_form.email(id=info_form.email.id) }}        8336
            {% for e in info_form.email.errors %}               8337
            <p class="field__error">{{ e }}</p>                 8338
            {% endfor %}                                        8339
          </div>                                                8340
          <p class="actions">{{ info_form.submit_info(clas      8341
s_="btn") }}</p>                                                8342
        </div>                                                  8343
      </details>                                                8344
      <details class="acct-card" {% if                          8345
info_form.phone.errors %}open{% endif %}>                       8346
        <summary>                                               8347
          <span class="acct-card__title">手机号</span>          8348
          <span class="acct-card__value">{{                     8349
current_user.phone or "未设置" }}</span>                        8350
```

```
                <span class="acct-card__icon">{{ icon("phone",
    15) }}</span>
              </summary>
              <div class="acct-card__body">
                <div class="field">
                  <label class="field__label" for="{{
    info_form.phone.id }}">手机号</label>
                  {{ info_form.phone(id=info_form.phone.id) }}
                  {% for e in info_form.phone.errors %}
                  <p class="field__error">{{ e }}</p>
                  {% endfor %}
                </div>
                <p class="actions">{{ info_form.submit_info(clas
    s_="btn") }}</p>
              </div>
            </details>
          </div>
        </form>
        {% endif %}
      </div>
    </div>
    {% endblock %}
    {% block scripts %}
    <script>
    // 同时只展开一张卡，避免点开多张后页面被撑得很长
    (function () {
      var cards = document.querySelectorAll('.acct-card');
      Array.prototype.forEach.call(cards, function (card) {
        card.addEventListener('toggle', function () {
          if (!card.open) return;
          Array.prototype.forEach.call(cards, function (other) {
            if (other !== card && other.open) other.open =
    false;
          });
        });
      });
    })();
    </script>
    {% endblock %}
    # ==== app/templates/auth/_users.html ====
    {# 管理员的「账号」分类。由 auth/profile.html 内联引入，不单独成页。 #}
    <div class="acct-panel">
      <div class="acct-panel__head">
        <p class="acct-panel__count">共 {{ users|length }}
        个账号</p>
        <a class="btn btn--sm" href="{{ url_for('auth.add_user')
        }}">新增账号</a>
      </div>
      <table>
        <thead>
```

```
    <tr>                                                                    8401
      <th scope="col">ID</th>                                               8402
      <th scope="col">账户名称</th>                                         8403
      <th scope="col">昵称</th>                                             8404
      <th scope="col">职务</th>                                             8405
      <th scope="col">状态</th>                                             8406
      <th scope="col">操作</th>                                             8407
    </tr>                                                                   8408
  </thead>                                                                  8409
  <tbody>                                                                   8410
    {% for u in users %}                                                    8411
    <tr>                                                                    8412
      <td class="num">{{ u.id }}</td>                                       8413
      <td class="mono">                                                     8414
        {{ u.username }}                                                    8415
        {% if u.id == current_user.id %}<span                               8416
class="muted">（当前登录）</span>{% endif %}                                8417
      </td>                                                                 8418
      <td style="text-align:left;">                                         8419
        {% if u.display_name %}{{ u.display_name }}                         8420
        {% else %}<span class="muted">—</span>{% endif %}                   8421
      </td>                                                                 8422
      <td>                                                                  8423
        <span class="tag {{ 'tag--ok' if u.role ==                          8424
'admin' else 'tag--muted' }}">                                              8425
          {{ role_labels.get(u.role, u.role) }}                             8426
        </span>                                                             8427
      </td>                                                                 8428
      <td>                                                                  8429
        {% if u.active %}<span class="tag                                   8430
tag--ok">启用</span>                                                        8431
        {% else %}<span class="tag tag--muted">停用</span>{%                8432
endif %}                                                                    8433
      </td>                                                                 8434
      <td class="nowrap">                                                   8435
        <span class="row-actions">                                          8436
          <a class="link-btn" href="{{                                      8437
url_for('auth.edit_user', user_id=u.id) }}">编辑</a>                        8438
          <form method="post"                                               8439
                action="{{ url_for('auth.delete_user',                      8440
user_id=u.id) }}"                                                           8441
                onsubmit="return confirm('确定删除账号「{{                  8442
u.username }}」吗？\n该操作无法恢复。');">                                  8443
            {{ form.hidden_tag() }}                                         8444
            <button type="submit" class="link-btn                           8445
link-btn--danger"                                                           8446
                    {% if u.id == current_user.id                           8447
%}disabled                                                                  8448
                    title="不能删除当前登录的账号"{% endif                  8449
%}>删除</button>                                                            8450
```

```
                </form>                                                    8451
              </span>                                                      8452
            </td>                                                          8453
          </tr>                                                            8454
          {% endfor %}                                                     8455
        </tbody>                                                           8456
      </table>                                                             8457
</div>                                                                     8458
#  ==== app/templates/auth/_logs.html ====                                 8459
{# 管理员的「操作日志」分类。由 auth/profile.html 内联引入，不单独成页。 #}  8460
<div class="acct-panel">                                                   8461
  <div class="acct-panel__head">                                           8462
    <p class="acct-panel__count">                                          8463
      共 {{ total }} 条记录，当前第 {{ page }} / {{ pages }} 页            8464
    </p>                                                                   8465
    {% if pages > 1 %}                                                     8466
    <nav class="row-actions" aria-label="日志分页">                        8467
      {% if page > 1 %}                                                    8468
      <a class="link-btn"                                                  8469
         href="{{ url_for('auth.profile', tab='logs',                      8470
    page=page - 1) }}">上一页</a>                                          8471
      {% endif %}                                                          8472
      {% if page < pages %}                                                8473
      <a class="link-btn"                                                  8474
         href="{{ url_for('auth.profile', tab='logs',                      8475
    page=page + 1) }}">下一页</a>                                          8476
      {% endif %}                                                          8477
    </nav>                                                                 8478
    {% endif %}                                                            8479
  </div>                                                                   8480
  {% if not records %}                                                     8481
  <p class="muted" style="padding:18px 20px;">暂无操作记录。</p>           8482
  {% else %}                                                               8483
  <table>                                                                  8484
    <thead>                                                                8485
      <tr>                                                                 8486
        <th scope="col">时间</th>                                          8487
        <th scope="col">操作人</th>                                        8488
        <th scope="col">操作</th>                                          8489
        <th scope="col">对象</th>                                          8490
        <th scope="col">说明</th>                                          8491
      </tr>                                                                8492
    </thead>                                                               8493
    <tbody>                                                                8494
      {% for r in records %}                                               8495
      <tr>                                                                 8496
        <td class="mono nowrap">{{ r.created_at |                          8497
    localdt("%Y-%m-%d %H:%M:%S") }}</td>                                   8498
        <td>{{ users_map.get(r.user_id, "—") }}</td>                       8499
        <td style="text-align:left;">{{ r.action }}</td>                   8500
```

```
            <td>                                                                    8501
              {% if r.target_type %}{{ r.target_type }}{% if                        8502
        r.target_id %}#{{ r.target_id }}{% endif %}                                 8503
              {% else %}—{% endif %}                                                8504
            </td>                                                                   8505
            <td style="text-align:left;">{{ r.detail or "—"                         8506
        }}</td>                                                                     8507
          </tr>                                                                     8508
          {% endfor %}                                                              8509
        </tbody>                                                                    8510
      </table>                                                                      8511
      {% endif %}                                                                   8512
</div>                                                                              8513
#  ==== app/templates/project/index.html ====                                       8514
{% extends "base.html" %}                                                           8515
{% block title %}项目与矿山档案{% endblock %}                                        8516
{% block content %}                                                                 8517
<div class="card">                                                                  8518
  <h2>项目与矿山档案</h2>                                                           8519
  <p class="actions" style="margin-bottom:12px;">                                   8520
    <a class="btn btn--sm" href="/project/add">新建项目</a>                         8521
  </p>                                                                              8522
  {% if not projects %}                                                             8523
  <p class="muted">                                                                 8524
    暂无项目。点击上方「新建项目」开始建立评价项目，                               8525
    或运行 <span class="mono">python seed_demo.py</span>                            8526
    载入示例案例。                                                                  8527
  </p>                                                                              8528
  {% else %}                                                                        8529
  <table>                                                                           8530
    <caption class="muted">共 {{ projects|length }}                                 8531
    个项目</caption>                                                                8532
    <thead>                                                                         8533
      <tr>                                                                          8534
        <th scope="col">项目名称</th>                                               8535
        <th scope="col">编号</th>                                                   8536
        <th scope="col">状态</th>                                                   8537
        <th scope="col">图斑数</th>                                                 8538
        <th scope="col">操作</th>                                                   8539
      </tr>                                                                         8540
    </thead>                                                                        8541
    <tbody>                                                                         8542
      {% for p in projects %}                                                       8543
      <tr>                                                                          8544
        <td style="text-align:left;">{{ p.name }}</td>                              8545
        <td class="mono">{{ p.code or "—" }}</td>                                   8546
        <td><span class="tag tag--muted">{{ p.status                                8547
    }}</span></td>                                                                  8548
        <td class="num">                                                            8549
          {% if p.patches.count() %}                                                8550
```

```
                {{ p.patches.count() }}                                    8551
              {% else %}                                                   8552
                <span class="tag tag--warn">0</span>                       8553
              {% endif %}                                                  8554
            </td>                                                          8555
            <td class="nowrap">                                            8556
              <span class="row-actions">                                   8557
                <a class="link-btn" href="/project/{{ p.id                 8558
    }}/patches">{% if p.patches.count() %}矿山图斑{% else %}+              8559
    新建图斑{% endif %}</a>                                                8560
                <a class="link-btn" href="/project/{{ p.id                 8561
    }}/edit">编辑</a>                                                      8562
                <form method="post" action="/project/{{ p.id               8563
    }}/delete"                                                             8564
                      onsubmit="return confirm('确定删除项目「{{           8565
    p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">  8566
                  {{ form.hidden_tag() }}                                  8567
                  <button type="submit" class="link-btn                    8568
    link-btn--danger">删除</button>                                        8569
                </form>                                                    8570
              </span>                                                      8571
            </td>                                                          8572
          </tr>                                                            8573
          {% endfor %}                                                     8574
        </tbody>                                                           8575
      </table>                                                             8576
      {% endif %}                                                          8577
</div>                                                                     8578
{% endblock %}                                                             8579
# ==== app/templates/project/edit.html ====                                8580
{% extends "base.html" %}                                                  8581
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%             8582
    endblock %}                                                            8583
{% block content %}                                                        8584
<div class="card" style="max-width:640px;">                                8585
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>                     8586
  <form method="post">                                                     8587
    {{ form.hidden_tag() }}                                                8588
    <div class="field">                                                    8589
      <label class="field__label" for="name">项目名称</label>              8590
      <input id="name" type="text" name="name"                             8591
    style="width:100%;" required                                           8592
             value="{{ project.name if project else '' }}">                8593
    </div>                                                                 8594
    <div class="field">                                                    8595
      <label class="field__label" for="code">项目编号</label>              8596
      <input id="code" type="text" name="code"                             8597
    style="width:240px;"                                                   8598
             value="{{ project.code or '' if project else                  8599
    '' }}">                                                                8600
```

```
      </div>                                                                    8601
      {% if project %}                                                          8602
      <div class="field">                                                       8603
        <label class="field__label" for="status">状态</label>                   8604
        <select id="status" name="status">                                      8605
          {% for s in ["进行中", "已完成", "已归档"] %}                          8606
          <option value="{{ s }}" {% if project.status == s                     8607
      %}selected{% endif %}>{{ s }}</option>                                    8608
          {% endfor %}                                                          8609
        </select>                                                               8610
      </div>                                                                    8611
      {% endif %}                                                               8612
      <div class="field">                                                       8613
        <label class="field__label" for="description">项目说明</la              8614
      bel>                                                                      8615
        <textarea id="description" name="description">{{                        8616
      project.description or '' if project else ''                              8617
      }}</textarea>                                                             8618
      </div>                                                                    8619
      <p class="actions" style="margin-top:18px;">                              8620
        <button type="submit" class="btn">保存</button>                         8621
        <a class="btn btn--ghost" href="/project/">返回</a>                     8622
      </p>                                                                      8623
    </form>                                                                     8624
</div>                                                                          8625
{% endblock %}                                                                  8626
# ==== app/templates/project/patches.html ====                                  8627
{% extends "base.html" %}                                                       8628
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}                    8629
{% block content %}                                                             8630
<div class="card">                                                              8631
  <h2>矿山图斑 · {{ project.name }}</h2>                                        8632
  <p class="actions" style="margin-bottom:12px;">                               8633
    <a class="btn btn--sm" href="/project/{{ project.id                         8634
    }}/patch/add">新建图斑</a>                                                  8635
    <a class="btn btn--sm btn--ghost"                                           8636
    href="/project/">返回项目列表</a>                                           8637
  </p>                                                                          8638
  {% if not project.patches.count() %}                                          8639
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>  8640
  {% else %}                                                                    8641
  <table>                                                                       8642
    <caption class="muted">共 {{ project.patches.count() }}                     8643
    个图斑</caption>                                                            8644
    <thead>                                                                     8645
      <tr>                                                                      8646
        <th scope="col">图斑名称</th>                                           8647
        <th scope="col">行政区</th>                                             8648
        <th scope="col">面积（公顷）</th>                                       8649
        <th scope="col">开采矿种</th>                                           8650
```

```
            <th scope="col">微地质单元</th>                                    8651
            <th scope="col">操作</th>                                          8652
          </tr>                                                                8653
        </thead>                                                               8654
        <tbody>                                                                8655
          {% for p in project.patches %}                                       8656
          <tr>                                                                 8657
            <td style="text-align:left;">{{ p.name }}</td>                     8658
            <td>{{ p.admin_region or "—" }}</td>                               8659
            <td class="num">{{ p.area_ha if p.area_ha is not                   8660
        none else "—" }}</td>                                                  8661
            <td>{{ p.mineral_type or "—" }}</td>                               8662
            <td class="num">{{ p.units.count() }}</td>                         8663
            <td class="nowrap">                                                8664
              <span class="row-actions">                                       8665
                <a class="link-btn" href="/unit/{{ p.id                        8666
        }}">微地质单元</a>                                                     8667
                <a class="link-btn" href="/constraint/{{ p.id                  8668
        }}">限制性判定</a>                                                     8669
                <a class="link-btn" href="/project/patch/{{                    8670
        p.id }}/edit">编辑</a>                                                 8671
                <form method="post" action="/project/patch/{{                  8672
        p.id }}/delete"                                                        8673
                      onsubmit="return confirm('确定删除图斑「{{               8674
        p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">  8675
                  {{ form.hidden_tag() }}                                      8676
                  <button type="submit" class="link-btn                        8677
        link-btn--danger">删除</button>                                        8678
                </form>                                                        8679
              </span>                                                          8680
            </td>                                                              8681
          </tr>                                                                8682
          {% endfor %}                                                         8683
        </tbody>                                                               8684
      </table>                                                                 8685
      {% endif %}                                                              8686
    </div>                                                                     8687
    {% endblock %}                                                             8688
    # ==== app/templates/project/patch_edit.html ====                          8689
    {% extends "base.html" %}                                                  8690
    {% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%               8691
        endblock %}                                                            8692
    {% block content %}                                                        8693
    <div class="card" style="max-width:720px;">                                8694
      <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name          8695
        }}</h2>                                                                8696
      <form method="post">                                                     8697
        {{ form.hidden_tag() }}                                                8698
        <div class="field">                                                    8699
          <label class="field__label" for="patch-name">图斑名称</lab           8700
```

```
el>                                                                    8701
  <input id="patch-name" type="text" name="name"                       8702
style="width:100%;" required                                           8703
         value="{{ patch.name if patch else '' }}">                    8704
</div>                                                                 8705
<div class="field">                                                    8706
  <label class="field__label" for="admin-region">所在行政区</        8707
label>                                                                 8708
  <input id="admin-region" type="text"                                 8709
name="admin_region" style="width:100%;"                                8710
         value="{{ patch.admin_region or '' if patch                   8711
else '' }}">                                                           8712
</div>                                                                 8713
<div class="field">                                                    8714
  <label class="field__label" for="area-ha">图斑面积（公顷）</la8715
bel>                                                                   8716
  <input id="area-ha" type="number" step="any"                         8717
name="area_ha"                                                         8718
         value="{{ patch.area_ha if patch and                          8719
patch.area_ha is not none else '' }}">                                 8720
</div>                                                                 8721
<div class="field">                                                    8722
  <label class="field__label" for="mineral-type">开采矿种</l8723
abel>                                                                  8724
  <input id="mineral-type" type="text"                                 8725
name="mineral_type" style="width:240px;"                               8726
         value="{{ patch.mineral_type or '' if patch                   8727
else '' }}">                                                           8728
</div>                                                                 8729
<div class="field">                                                    8730
  <label class="field__label" for="mining-method">开采方式</        8731
label>                                                                 8732
  <input id="mining-method" type="text"                                8733
name="mining_method" style="width:240px;"                              8734
         value="{{ patch.mining_method or '' if patch                  8735
else '' }}">                                                           8736
</div>                                                                 8737
<div class="field">                                                    8738
  <label class="field__label" for="stop-years">停产年限</lab8739
el>                                                                    8740
  <input id="stop-years" type="number" step="any"                      8741
name="stop_years"                                                      8742
         value="{{ patch.stop_years if patch and                       8743
patch.stop_years is not none else '' }}">                              8744
</div>                                                                 8745
<div class="field">                                                    8746
  <label class="checkbox" for="responsible-missing">                   8747
    <input id="responsible-missing" type="checkbox"                    8748
name="responsible_entity_missing" value="1"                            8749
           {% if not patch or patch.responsible_entity_m               8750
```

```
      issing %}checked{% endif %}>                                          8751
          <span>责任主体缺失</span>                                          8752
        </label>                                                            8753
      </div>                                                                8754
      <h3>参照生态系统（指标标准化基准）</h3>                                  8755
      <p class="muted">                                                     8756
        系统默认取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。          8757
        该基准与图斑绑定管理，可按不同参照生态系统分别设定。                  8758
      </p>                                                                  8759
      <div class="field">                                                   8760
        <label class="field__label" for="coverage-benchmark">植8761
      被覆盖度基准（%）</label>                                              8762
        <input id="coverage-benchmark" type="number"                        8763
      step="any" name="coverage_benchmark"                                  8764
              value="{{ reference.coverage_benchmark if                     8765
      reference else 75 }}">                                                8766
      </div>                                                                8767
      <div class="field">                                                   8768
        <label class="field__label" for="shannon-benchmark">Sh8769
      annon-Wiener 多样性指数基准</label>                                    8770
        <input id="shannon-benchmark" type="number"                         8771
      step="any" name="shannon_benchmark"                                   8772
              value="{{ reference.shannon_benchmark if                      8773
      reference else 0.7 }}">                                               8774
      </div>                                                                8775
      <div class="field">                                                   8776
        <label class="field__label" for="dominant-species">建群种8777
      </label>                                                              8778
        <input id="dominant-species" type="text"                            8779
      name="dominant_species" style="width:100%;"                           8780
              value="{{ reference.dominant_species or '' if                8781
      reference else '' }}">                                                8782
      </div>                                                                8783
      <p class="actions" style="margin-top:18px;">                          8784
        <button type="submit" class="btn">保存</button>                     8785
        <a class="btn btn--ghost" href="/project/{{                         8786
      project.id }}/patches">返回</a>                                       8787
      </p>                                                                  8788
    </form>                                                                 8789
</div>                                                                      8790
{% if patch %}                                                              8791
<div class="card" style="max-width:720px;">                                 8792
    <h2>遥感影像挂接</h2>                                                   8793
    <p class="muted">                                                       8794
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。  8795
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。           8796
    </p>                                                                    8797
    {% if patch.image_path %}                                               8798
    <figure style="margin:0 0 16px;">                                       8799
      <img src="{{ url_for('static', filename=patch.image_path 8800
```

```
      ) }}"                                                                    8801
            alt="图斑「{{ patch.name }}」已挂接的遥感影像"                      8802
            style="max-width:100%;border:1px solid                             8803
      var(--line);border-radius:6px;">                                         8804
      <figcaption class="muted" style="margin-top:6px;">                       8805
        {{ patch.image_note or "遥感影像" }}                                   8806
        {% if patch.image_uploaded_at %}                                       8807
        · 上传于 {{ patch.image_uploaded_at | localdt }}                       8808
        {% endif %}                                                            8809
      </figcaption>                                                            8810
    </figure>                                                                  8811
    <form method="post" action="/project/patch/{{ patch.id                     8812
      }}/image/delete"                                                         8813
          onsubmit="return confirm('确定移除该遥感影像吗？');">                8814
      {{ form.hidden_tag() }}                                                  8815
      <button type="submit" class="btn btn--sm                                 8816
      btn--danger">移除影像</button>                                           8817
    </form>                                                                    8818
    <hr style="margin:16px 0;border:none;border-top:1px solid                  8819
      var(--line);">                                                           8820
    {% endif %}                                                                8821
    <form method="post" action="/project/patch/{{ patch.id                     8822
      }}/image"                                                                8823
          enctype="multipart/form-data">                                       8824
      {{ form.hidden_tag() }}                                                  8825
      <div class="field">                                                      8826
        <label class="field__label" for="image-file">                          8827
          {{ "替换影像文件" if patch.image_path else "选择影像文件" }}         8828
        </label>                                                               8829
        <input id="image-file" type="file" name="image"                        8830
               accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t                 8831
      iff" required>                                                           8832
      </div>                                                                   8833
      <div class="field">                                                      8834
        <label class="field__label" for="image-note">影像说明（如来源、       8835
      时相）</label>                                                           8836
        <input id="image-note" type="text" name="image_note"                   8837
      style="width:100%;"                                                      8838
               placeholder="如：2025 年 6 月无人机正射影像"                    8839
               value="{{ patch.image_note or '' }}">                           8840
      </div>                                                                   8841
      <p class="actions">                                                      8842
        <button type="submit" class="btn">{{ "替换影像" if                    8843
      patch.image_path else "上传并挂接" }}</button>                           8844
      </p>                                                                     8845
    </form>                                                                    8846
</div>                                                                         8847
{% endif %}                                                                    8848
{% endblock %}                                                                 8849
# ==== app/templates/unit/list.html ====                                       8850
```

```
{% extends "base.html" %}
{% block title %}微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元</h2>
  {% if not rows %}
  <p class="muted">
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。
  </p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">已划分单元</th>
        <th scope="col">占比合计</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.patch.project.name if r.patch.project else
"—" }}</td>
        <td style="text-align:left;">{{ r.patch.name }}</td>
        <td class="num">{{ r.count }} 个</td>
        <td class="num">
          {% if r.count %}
            {{ "%.2f%%"|format(r.ratio_sum * 100) }}
            {% if r.ratio_sum < 0.999 or r.ratio_sum >
1.001 %}
              <span class="warn">（不等于 100%）</span>
            {% endif %}
          {% else %}—{% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/{{ r.patch.id
}}">划分与维护</a>
            <a class="link-btn" href="/indicator/">录入指标数据</a
>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
</div>
```

```
{% endblock %}
# ==== app/templates/unit/index.html ====
{% extends "base.html" %}
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元 · {{ patch.name }}</h2>
  {% if units %}
  <table>
    <caption class="muted">
      共 {{ units|length }} 个单元；
      面积合计 <span class="num">{{ "%.4f"|format(total_area)
    }}</span>；
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *
    100) }}</span>
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >
    1.001) %}
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>
      {% endif %}
    </caption>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积</th>
        <th scope="col">面积占比</th>
        <th scope="col">说明</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ u.area if u.area is not none
    else "—" }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ u.description or
    "—" }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/edit/{{ u.id
    }}">编辑</a>
            <form method="post" action="/unit/delete/{{
    u.id }}"
                  onsubmit="return confirm('确定删除微地质单元「{{
    u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
```

```
                    <button type="submit" class="link-btn
    link-btn--danger">删除</button>
                  </form>
                </span>
              </td>
            </tr>
            {% endfor %}
          </tbody>
        </table>
        {% else %}
        <p class="muted">尚未划分微地质单元。请在下方添加。</p>
        {% endif %}
        <h3>添加微地质单元</h3>
        <form method="post" action="/unit/{{ patch.id }}/add">
          {{ form.hidden_tag() }}
          <div class="actions">
            <label class="field__label" for="add-unit-type"
          style="margin:0;">类型</label>
            <select id="add-unit-type" name="unit_type">
              {% for t in unit_types %}<option value="{{ t }}">{{
          t }}</option>{% endfor %}
            </select>
            <label class="field__label" for="add-unit-area"
          style="margin:0;">面积</label>
            <input id="add-unit-area" type="number" step="any"
          name="area">
            <label class="field__label" for="add-unit-desc"
          style="margin:0;">说明</label>
            <input id="add-unit-desc" type="text"
          name="description" style="width:220px;">
            <button type="submit" class="btn btn--sm">添加</button>
          </div>
        </form>
        <p class="actions" style="margin-top:20px;">
          <a class="btn btn--sm btn--ghost" href="/project/{{
          patch.project_id }}/patches">返回图斑列表</a>
          <a class="btn btn--sm btn--ghost"
          href="/indicator/">去录入指标数据</a>
        </p>
</div>
{% endblock %}
# ==== app/templates/unit/edit.html ====
{% extends "base.html" %}
{% block title %}编辑微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>编辑微地质单元 · {{ patch.name }}</h2>
  <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
  <form method="post">
    {{ form.hidden_tag() }}
```

```
      <div class="field">                                                    9001
        <label class="field__label" for="unit-type">微地质单元类型</l 9002
      abel>                                                                  9003
        <select id="unit-type" name="unit_type">                             9004
          {% for t in unit_types %}                                          9005
          <option value="{{ t }}" {% if unit.unit_type == t     9006
      %}selected{% endif %}>{{ t }}</option>                                 9007
          {% endfor %}                                                       9008
        </select>                                                            9009
      </div>                                                                 9010
      <div class="field">                                                    9011
        <label class="field__label" for="unit-area">面积</label>9012
        <input id="unit-area" type="number" step="any"                       9013
      name="area"                                                            9014
                 value="{{ unit.area if unit.area is not none     9015
      else '' }}">                                                           9016
      </div>                                                                 9017
      <div class="field">                                                    9018
        <label class="field__label" for="unit-desc">说明</label>9019
        <input id="unit-desc" type="text" name="description"   9020
      style="width:420px;"                                                   9021
                 value="{{ unit.description or '' }}">                       9022
      </div>                                                                 9023
      <p class="actions" style="margin-top:16px;">                           9024
        <button type="submit" class="btn">保存</button>                      9025
        <a class="btn btn--ghost" href="/unit/{{                             9026
      unit.patch_id }}">返回</a>                                             9027
      </p>                                                                   9028
    </form>                                                                  9029
</div>                                                                       9030
{% endblock %}                                                               9031
#  ==== app/templates/standard/index.html ====                               9032
{% extends "base.html" %}                                                    9033
{% block title %}政策与标准知识库{% endblock %}                               9034
{% block content %}                                                          9035
<div class="card">                                                           9036
    <h2>政策与标准知识库</h2>  <p class="actions"                            9037
      style="margin-bottom:14px;">                                           9038
      <a class="btn btn--sm" href="/standard/add">添加标准</a>               9039
      <a class="btn btn--sm btn--ghost"                                      9040
      href="/standard/compare">省际口径比对</a>                              9041
      <a class="btn btn--sm btn--ghost"                                      9042
      href="/standard/check">结论对照</a>                                    9043
    </p>                                                                     9044
    <h3>地方标准</h3>                                                       9045
    {% if local %}                                                           9046
    <table>                                                                  9047
      <thead>                                                                9048
        <tr>                                                                 9049
          <th scope="col">省份</th>                                          9050
```

```
          <th scope="col">名称</th>
          <th scope="col">编号</th>
          <th scope="col">规则数</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for s in local %}
        <tr>
          <td>{{ s.province }}</td>
          <td style="text-align:left;">{{ s.name }}</td>
          <td class="mono">{{ s.doc_number or "—" }}</td>
          <td class="num">{{ s.rules|length }}</td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/standard/{{ s.id
      }}/rules">规则</a>
              <a class="link-btn" href="/standard/{{ s.id
      }}/edit">编辑</a>
              <form method="post" action="/standard/{{ s.id
      }}/delete"
                    onsubmit="return confirm('确定删除「{{
      s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length
      }} 条规则将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
      link-btn--danger">删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% else %}
    <p class="muted">暂无地方标准。运行 <span class="mono">python
      seed_demo.py</span> 可载入内置的 １２ 项标准与政策文件。</p>
    {% endif %}
    <h3>政策文件</h3>
    {% if policy %}
    <table>
      <thead>
        <tr>
          <th scope="col">省份</th>
          <th scope="col">名称</th>
          <th scope="col">文号</th>
          <th scope="col">规则数</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
```

```
      <tbody>                                                                  9101
        {% for s in policy %}                                                  9102
        <tr>                                                                   9103
          <td>{{ s.province }}</td>                                            9104
          <td style="text-align:left;">{{ s.name }}</td>                       9105
          <td class="mono">{{ s.doc_number or "—" }}</td>                      9106
          <td class="num">{{ s.rules|length }}</td>                            9107
          <td class="nowrap">                                                  9108
            <span class="row-actions">                                         9109
              <a class="link-btn" href="/standard/{{ s.id                      9110
      }}/rules">规则</a>                                                       9111
              <a class="link-btn" href="/standard/{{ s.id                      9112
      }}/edit">编辑</a>                                                        9113
              <form method="post" action="/standard/{{ s.id                    9114
      }}/delete"                                                               9115
                    onsubmit="return confirm('确定删除「{{                     9116
      s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length           9117
      }} 条规则将一并删除，且无法恢复。');">                                   9118
                {{ form.hidden_tag() }}                                        9119
                <button type="submit" class="link-btn                          9120
      link-btn--danger">删除</button>                                          9121
              </form>                                                          9122
            </span>                                                            9123
          </td>                                                                9124
        </tr>                                                                  9125
        {% endfor %}                                                           9126
      </tbody>                                                                 9127
    </table>                                                                   9128
    {% else %}                                                                 9129
    <p class="muted">暂无政策文件。</p>                                        9130
    {% endif %}                                                                9131
</div>                                                                         9132
{% endblock %}                                                                 9133
# ==== app/templates/standard/edit.html ====                                   9134
{% extends "base.html" %}                                                      9135
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%                9136
    endblock %}                                                                9137
{% block content %}                                                            9138
<div class="card" style="max-width:800px;">                                    9139
    <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                      9140
    <form method="post">                                                       9141
      {{ form.hidden_tag() }}                                                  9142
      <div class="field">                                                      9143
        <label class="field__label" for="province">省份</label>                9144
        <input id="province" type="text" name="province"                       9145
      style="width:160px;" required                                            9146
              value="{{ standard.province if standard else                     9147
      '' }}">                                                                  9148
      </div>                                                                   9149
      <div class="field">                                                      9150
```

```
    <label class="field__label" for="std-name">名称</label>
    <input id="std-name" type="text" name="name"
style="width:100%;" required
           value="{{ standard.name if standard else ''
}}">
</div>
<div class="field">
    <label class="field__label" for="doc-number">标准编号 /
文号</label>
    <input id="doc-number" type="text" name="doc_number"
style="width:320px;"
           value="{{ standard.doc_number or '' if
standard else '' }}">
</div>
<div class="field">
    <label class="field__label" for="category">类别</label>
    <select id="category" name="category">
      {% for c in ["地方标准", "政策文件"] %}
      <option value="{{ c }}" {% if standard and
standard.category == c %}selected{% endif %}>{{ c
}}</option>
      {% endfor %}
    </select>
</div>
<div class="field">
    <label class="field__label" for="publish-date">发布时间</l
abel>
    <input id="publish-date" type="text"
name="publish_date" style="width:200px;"
           placeholder="如 2023-03-30"
           value="{{ standard.publish_date or '' if
standard else '' }}">
</div>
<div class="field">
    <label class="field__label" for="implement-date">实施时间<
/label>
    <input id="implement-date" type="text"
name="implement_date" style="width:200px;"
           placeholder="如 2023-04-30（未明确可留空）"
           value="{{ standard.implement_date or '' if
standard else '' }}">
</div>
<div class="field">
    <label class="field__label" for="summary">特点 /
判定标准</label>
    <textarea id="summary" name="summary"
style="min-height:96px;">{{ standard.summary or '' if
standard else '' }}</textarea>
</div>
<p class="actions" style="margin-top:18px;">
```

```
        <button type="submit" class="btn">保存</button>
        {% if standard %}
        <form method="post" action="/standard/{{ standard.id
    }}/delete" style="display:inline;"
              onsubmit="return confirm('确定删除「{{
    standard.province }} {{ standard.name
    }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn
    btn--danger">删除该标准</button>
        </form>
        {% endif %}
        <a class="btn btn--ghost" href="/standard/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/standard/rules.html ====
{% extends "base.html" %}
{% block title %}结构化规则 · {{ standard.province }}{% endblock
    %}
{% block content %}
<div class="card">
    <h2>结构化规则 · {{ standard.province }} {{ standard.name
      }}</h2>
    {% if rules %}
    <table>
      <caption class="muted">共 {{ rules|length }}
      条规则</caption>
      <thead>
        <tr>
          <th scope="col">判定项</th>
          <th scope="col">条件</th>
          <th scope="col">条件类型</th>
          <th scope="col">适用分区</th>
          <th scope="col">备注</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for r in rules %}
        <tr>
          <td style="text-align:left;">{{ r.item }}</td>
          <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
          <td>{{ r.condition_type }}</td>
          <td>{{ r.region or "—" }}</td>
          <td style="text-align:left;">{{ r.note or "—"
    }}</td>
          <td>
```

```
          <form method="post" action="/standard/rule/{{ r.id }}/delete"
                onsubmit="return confirm('确定删除规则「{{ r.item }} {{ r.operator }} {{ r.threshold or '' }}」吗？');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}
  <h3>添加规则</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="actions">
      <div>
        <label class="field__label" for="r-item">判定项</label>
        <input id="r-item" type="text" name="item" placeholder="如：植被覆盖度" required>
      </div>
      <div>
        <label class="field__label" for="r-op">比较符</label>
        <select id="r-op" name="operator">
          {% for o in [">=", ">", "<=", "<", "=="] %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
        </select>
      </div>
      <div>
        <label class="field__label" for="r-thr">阈值</label>
        <input id="r-thr" type="text" name="threshold" placeholder="如：50" style="width:110px;">
      </div>
      <div>
        <label class="field__label" for="r-cond">条件类型</label>
        <select id="r-cond" name="condition_type">
          <option value="必要条件">必要条件</option>
          <option value="充分条件">充分条件</option>
        </select>
      </div>
      <div>
        <label class="field__label" for="r-region">适用分区</label>
        <input id="r-region" type="text" name="region" placeholder="如宁夏分区" style="width:130px;">
```

```
        </div>                                                                  9301
        <div>                                                                   9302
          <label class="field__label" for="r-note">备注</label>                 9303
          <input id="r-note" type="text" name="note"                            9304
    style="width:220px;">                                                       9305
        </div>                                                                  9306
        <div>                                                                   9307
          <button type="submit" class="btn">添加规则</button>                   9308
        </div>                                                                  9309
      </div>                                                                    9310
    </form>                                                                     9311
    <p class="actions" style="margin-top:20px;">                                9312
      <a class="btn btn--sm btn--ghost"                                         9313
      href="/standard/">返回标准列表</a>                                        9314
      <a class="btn btn--sm btn--ghost"                                         9315
      href="/standard/compare">省际口径比对</a>                                 9316
    </p>                                                                        9317
</div>                                                                          9318
{% endblock %}                                                                  9319
# ==== app/templates/standard/compare.html ====                                 9320
{% extends "base.html" %}                                                       9321
{% block title %}省际口径比对{% endblock %}                                     9322
{% block content %}                                                             9323
<div class="card">                                                              9324
  <h2>省际口径比对</h2>                                                        9325
  {% if not groups %}                                                           9326
  <p class="muted">                                                             9327
    暂无规则数据。请先在「标准知识库」中为标准录入规则，                       9328
    或运行 <span class="mono">python seed_demo.py</span>                        9329
    载入内置标准库数据。                                                        9330
  </p>                                                                          9331
  {% else %}                                                                    9332
  {% for item, rs in groups.items() %}                                          9333
  <h3>{{ item }}</h3>                                                           9334
  <table>                                                                       9335
    <thead>                                                                     9336
      <tr>                                                                      9337
        <th scope="col">省份</th>                                               9338
        <th scope="col">标准 / 文件</th>                                        9339
        <th scope="col">条件</th>                                               9340
        <th scope="col">条件类型</th>                                           9341
        <th scope="col">备注</th>                                               9342
      </tr>                                                                     9343
    </thead>                                                                    9344
    <tbody>                                                                     9345
      {% for r in rs %}                                                         9346
      <tr>                                                                      9347
        <td>{{ r.standard.province }}</td>                                      9348
        <td style="text-align:left;">{{ r.standard.name                         9349
    }}</td>                                                                     9350
```

```
            <td class="mono nowrap">{{ r.operator }} {{
      r.threshold or "" }}</td>
            <td>{{ r.condition_type }}</td>
            <td style="text-align:left;">{{ r.note or "—"
      }}</td>
          </tr>
          {% endfor %}
      </tbody>
    </table>
    {% endfor %}
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
    <h2>限制性指标判定</h2>
    {{ patch_guide(projects_without_patch) }}
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">所属项目</th>
          <th scope="col">矿山图斑</th>
          <th scope="col">判定结果</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in patches %}
        {% set r = latest.get(p.id) %}
        <tr>
          <td>{{ p.project.name if p.project else "—" }}</td>
          <td style="text-align:left;">{{ p.name }}</td>
          <td style="text-align:left;">
            {% if r %}
              {% if r.passed %}
              <span class="tag tag--ok">通过</span>
              {% else %}
              <span class="tag tag--warn">不通过</span>
              {% endif %}
              <span class="muted">{{ r.reason }}</span>
```

```
            {% else %}                                                    9401
            <span class="tag tag--muted">未判定</span>                    9402
            {% endif %}                                                   9403
          </td>                                                           9404
          <td class="nowrap">                                             9405
            <span class="row-actions">                                    9406
              <a class="link-btn" href="/constraint/{{ p.id               9407
    }}">执行判定</a>                                                      9408
            </span>                                                       9409
          </td>                                                           9410
        </tr>                                                             9411
        {% endfor %}                                                      9412
      </tbody>                                                            9413
    </table>                                                              9414
    {% endif %}                                                           9415
</div>                                                                    9416
{% endblock %}                                                            9417
# ==== app/templates/constraint/check.html ====                           9418
{% extends "base.html" %}                                                 9419
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}          9420
{% block content %}                                                       9421
<div class="card" style="max-width:760px;">                               9422
  <h2>限制性条件判定 · {{ patch.name }}</h2>                              9423
  <form method="post">                                                    9424
    {{ form.hidden_tag() }}                                               9425
    {% for field, name, val in field_states %}                            9426
    <fieldset style="border:1px solid                                     9427
    var(--line);border-radius:6px;padding:10px                            9428
    14px;margin:0 0 10px;">                                               9429
      <legend style="padding:0 6px;font-weight:600;">{{                   9430
    name }}</legend>                                                      9431
      <div class="actions">                                               9432
        <label class="checkbox">                                          9433
          <input type="radio" name="{{ field }}" value="1"                9434
                 {% if val is none or val %}checked{% endif               9435
    %}>                                                                   9436
          <span>通过</span>                                               9437
        </label>                                                          9438
        <label class="checkbox">                                          9439
          <input type="radio" name="{{ field }}" value="0"                9440
                 {% if val is not none and not val                        9441
    %}checked{% endif %}>                                                 9442
          <span>不通过</span>                                             9443
        </label>                                                          9444
      </div>                                                              9445
    </fieldset>                                                           9446
    {% endfor %}                                                          9447
    <p class="actions" style="margin-top:16px;">                          9448
      <button type="submit" class="btn">保存判定</button>                 9449
      <a class="btn btn--ghost" href="/constraint/">返回</a>              9450
```

```
      </p>                                                                    9451
    </form>                                                                   9452
    {% if record %}                                                           9453
    <div style="margin-top:18px;padding:14px;border-radius:6px 9454
      ;border-left:4px solid;                                                 9455
                  {% if record.passed %}background:var(--ok-soft 9456
      );border-left-color:var(--ok);                                          9457
                  {% else %}background:var(--danger-soft);border 9458
      -left-color:var(--danger);{% endif %}">                                 9459
      <p style="margin:0 0 4px;font-weight:600;">                             9460
        判定结论：                                                            9461
        {% if record.passed %}                                                9462
        <span class="ok">通过，可进入定量评价</span>                          9463
        {% else %}                                                            9464
        <span class="warn">不通过 —— 建议生态重建或人工辅助</span>            9465
        {% endif %}                                                           9466
      </p>                                                                    9467
      <p class="muted" style="margin:0;">{{ record.reason                     9468
      }}</p>                                                                  9469
    </div>                                                                    9470
    {% endif %}                                                               9471
</div>                                                                        9472
{% endblock %}                                                                9473
# ==== app/templates/ahp/index.html ====                                      9474
{% extends "base.html" %}                                                     9475
{% block title %}AHP 权重计算{% endblock %}                                   9476
{% block content %}                                                           9477
<div class="card" style="max-width:720px;">                                   9478
    <h2>AHP 权重计算 · 选择指标层级</h2>                                     9479
    <form method="post">                                                      9480
      {{ form.hidden_tag() }}                                                 9481
      <div class="field">                                                     9482
        <label class="field__label" for="{{ form.unit_type.id                 9483
      }}">微地质单元类型</label>                                            9484
        {{ form.unit_type(id=form.unit_type.id) }}                            9485
      </div>                                                                  9486
      <div class="field">                                                     9487
        <label class="field__label" for="{{ form.level.id                     9488
      }}">指标层级</label>                                                  9489
        {{ form.level(id=form.level.id) }}                                    9490
      </div>                                                                  9491
      <div class="field">                                                     9492
        <label class="field__label" for="{{ form.l1.id }}">                   9493
          一级指标（层级选「二级指标」时需选择）                              9494
        </label>                                                              9495
        {{ form.l1(id=form.l1.id) }}                                          9496
      </div>                                                                  9497
      <p class="actions" style="margin-top:16px;">                            9498
        {{ form.submit(class_="btn") }}                                       9499
        <a class="btn btn--ghost" href="/ahp/indicators">指标定义维 9500
```

```
      护</a>                                                
```

```
  </table>
</div>
{% endblock %}
# ==== app/templates/ahp/matrix.html ====
{% extends "base.html" %}
{% block title %}判断矩阵录入{% endblock %}
{% block content %}
<div class="card">
  <h2>
    判断矩阵录入
    · {{ unit_type }}
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%
    endif %}
  </h2>
  <p class="muted">
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
    矩阵已按系统默认权重预填。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <table style="width:auto;">
      <thead>
        <tr>
          <th></th>
          {% for name in names %}<th scope="col">{{ name
    }}</th>{% endfor %}
        </tr>
      </thead>
      <tbody>
        {% for i in range(names|length) %}
        <tr>
          <th scope="row">{{ names[i] }}</th>
          {% for j in range(names|length) %}
            {% if i == j %}
            <td class="muted num">1</td>
            {% elif i < j %}
            <td>
              <label class="muted" style="position:absolute;
    left:-9999px;"
                     for="m-{{ i }}-{{ j }}">{{ names[i] }}
    相对 {{ names[j] }} 的重要性</label>
              <input id="m-{{ i }}-{{ j }}" type="number"
    step="0.01" min="0.01"
                     name="m_{{ i }}_{{ j }}" value="{{
    default[i][j] }}">
            </td>
            {% else %}
            <td class="muted num">{{ default[i][j] }}</td>
            {% endif %}
```

```
            {% endfor %}                                                    9601
          </tr>                                                             9602
          {% endfor %}                                                      9603
        </tbody>                                                            9604
    </table>                                                                9605
    <p class="actions">                                                     9606
      <button type="submit" class="btn" name="action"                       9607
    value="calc">计算权重</button>                                          9608
      {% if result and result.consistent %}                                 9609
      <button type="submit" class="btn" name="action"                       9610
    value="save">保存并生效</button>                                        9611
      {% endif %}                                                           9612
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>                   9613
    </p>                                                                    9614
</form>                                                                     9615
<p class="muted" style="margin-top:10px;">                                  9616
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label                9617
    }}</strong>。                                                           9618
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；    9619
    如需回到系统默认值，可在 <a href="/ahp/">权重首页</a> 恢复默认。       9620
</p>                                                                        9621
{% if result %}                                                             9622
<hr style="margin:22px 0;border:none;border-top:1px solid                   9623
    var(--line);">                                                          9624
<h3>计算结果</h3>                                                           9625
<table style="width:auto;">                                                 9626
    <thead>                                                                 9627
      <tr>                                                                  9628
        <th scope="col">指标</th>                                           9629
        {% for name in names %}<th scope="col">{{ name                      9630
    }}</th>{% endfor %}                                                     9631
      </tr>                                                                 9632
    </thead>                                                                9633
    <tbody>                                                                 9634
      <tr>                                                                  9635
        <th scope="row">权重</th>                                           9636
        {% for w in result.weights %}<td class="mono">{{ w                  9637
    }}</td>{% endfor %}                                                     9638
      </tr>                                                                 9639
    </tbody>                                                                9640
</table>                                                                    9641
<div class="stat-row">                                                      9642
    <div>                                                                   9643
      <div class="stat__label">最大特征值 λmax</div>                        9644
      <div class="stat__value mono">{{ result.lambda_max                    9645
    }}</div>                                                                9646
    </div>                                                                  9647
    <div>                                                                   9648
      <div class="stat__label">一致性指标 CI</div>                          9649
      <div class="stat__value mono">{{                                      9650
```

```
      "%.4f"|format(result.ci) }}</div>                                  9651
      </div>                                                             9652
      <div>                                                              9653
        <div class="stat__label">一致性比例 CR</div>                      9654
        <div class="stat__value mono">{{                                 9655
      "%.4f"|format(result.cr) }}</div>                                  9656
      </div>                                                             9657
      <div>                                                              9658
        <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>     9659
        <div class="stat__value">                                        9660
          {% if result.consistent %}<span class="ok">通过</span>         9661
          {% else %}<span class="warn">不通过</span>{% endif %}          9662
        </div>                                                           9663
      </div>                                                             9664
    </div>                                                               9665
    {% if not result.consistent and result.most_inconsistent             9666
      %}                                                                 9667
    <p style="margin-top:14px;padding:12px;border-radius:6px;b           9668
      ackground:var(--danger-soft);                                      9669
                 border-left:4px solid var(--danger);">                  9670
      <strong class="warn">建议优先调整：</strong>                       9671
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                  9672
      result.most_inconsistent.j + 1 }} 列的比较                         9673
      （当前 <span class="mono num">{{ result.most_inconsistent.v        9674
      alue|round(3) }}</span>，                                          9675
      理想约 <span class="mono num">{{ result.most_inconsistent.e        9676
      xpected|round(3) }}</span>）                                       9677
    </p>                                                                 9678
    {% endif %}                                                          9679
    {% endif %}                                                          9680
</div>                                                                   9681
{% endblock %}                                                           9682
# ==== app/templates/ahp/indicators.html ====                            9683
{% extends "base.html" %}                                                9684
{% block title %}指标定义维护{% endblock %}                              9685
{% block content %}                                                      9686
<div class="card">                                                       9687
    <h2>指标体系维护</h2>                                                9688
    <p class="actions" style="margin-bottom:14px;">                      9689
      <a class="btn btn--sm btn--ghost"                                  9690
      href="/ahp/">返回权重计算</a>                                      9691
      <a class="btn btn--sm btn--ghost"                                  9692
      href="/ahp/experts">多专家矩阵</a>                                 9693
      <form method="post" action="/ahp/indicators/reset"                 9694
      style="display:inline;"                                            9695
            onsubmit="return confirm('将按系统默认值重置全部指标定义，自定义修改会丢失 9696
      。确定继续？');">                                                  9697
        {{ form.hidden_tag() }}                                          9698
        <button type="submit" class="btn btn--sm                         9699
      btn--danger">按系统默认值重置</button>                             9700
```

```
    </form>
</p>
{% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,
    "三级指标")] %}
{% set level_items = all_ind|selectattr("level",
    "equalto", level)|list %}
{% if level_items %}
<h3>{{ title }}（{{ level_items|length }} 项）</h3>
<table>
    <thead>
      <tr>
        <th scope="col">指标</th>
        <th scope="col">上级指标</th>
        <th scope="col">单位</th>
        <th scope="col">标准化类型</th>
        <th scope="col">理想取值</th>
        <th scope="col">启用单元</th>
        <th scope="col">限制性</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for i in level_items %}
      <tr>
        <td style="text-align:left;">
          <span class="mono muted">{{ i.code
    }}</span><br>{{ i.name }}
        </td>
        <td>{{ (all_ind|selectattr("code", "equalto",
    i.parent_code)|list|first).name if i.parent_code else
    "—" }}</td>
        <td>{{ i.unit or "—" }}</td>
        <td>{{ type_label(i.ideal_type) }}</td>
        <td class="mono">{{ i.ideal_value or "—" }}</td>
        <td style="text-align:left;">{{ i.enabled_units or
    "—" }}</td>
        <td>
          {% if i.is_constraint %}<span class="tag
    tag--warn">是</span>
          {% else %}<span class="tag tag--muted">否</span>{%
    endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/ahp/indicators/{{
    i.code }}/edit">编辑</a>
          </span>
        </td>
      </tr>
      {% endfor %}
```

```
      </tbody>                                                                  9751
    </table>                                                                    9752
    {% endif %}                                                                 9753
    {% endfor %}                                                                9754
</div>                                                                          9755
{% endblock %}                                                                  9756
# ==== app/templates/ahp/indicator_edit.html ====                               9757
{% extends "base.html" %}                                                       9758
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}                        9759
{% block content %}                                                             9760
<div class="card" style="max-width:680px;">                                     9761
  <h2>编辑指标 · {{ ind.name }}</h2>                                            9762
  <p class="muted">                                                             9763
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{                  9764
    ind.level }} 级，                                                           9765
    由系统指标体系确定，不可修改。                                              9766
  </p>                                                                          9767
  <form method="post">                                                          9768
    {{ form.hidden_tag() }}                                                     9769
    <div class="field">                                                         9770
      <label class="field__label" for="name">指标名称</label>                   9771
      <input id="name" type="text" name="name"                                  9772
    style="width:100%;" required                                                9773
             value="{{ ind.name }}">                                            9774
    </div>                                                                      9775
    <div class="field">                                                         9776
      <label class="field__label" for="unit">计量单位</label>                   9777
      <input id="unit" type="text" name="unit"                                  9778
    style="width:180px;"                                                        9779
             value="{{ ind.unit or '' }}">                                      9780
    </div>                                                                      9781
    <div class="field">                                                         9782
      <label class="field__label" for="ideal-type">标准化类型</la9783
    bel>                                                                        9784
      <select id="ideal-type" name="ideal_type">                                9785
        {% for code, label in [("benefit","越大越优"),("cost","越9786
    小越优"),("mid","适中型"),("discrete","离散赋值")] %}                       9787
        <option value="{{ code }}" {% if ind.ideal_type ==                      9788
    code %}selected{% endif %}>{{ label }}</option>                             9789
        {% endfor %}                                                            9790
      </select>                                                                 9791
    </div>                                                                      9792
    <div class="field">                                                         9793
      <label class="field__label" for="ideal-value">理想取值</la9794
    bel>                                                                        9795
      <input id="ideal-value" type="text"                                       9796
    name="ideal_value" style="width:100%;"                                      9797
             value="{{ ind.ideal_value or '' }}">                              9798
      <p class="muted" style="margin:4px 0 0;">                                 9799
        如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。                      9800
```

```
    </p>
  </div>
  <fieldset style="border:1px solid var(--line);border-radius:6px;padding:10px 14px;margin:0 0 12px;">
    <legend style="padding:0 6px;font-weight:600;">启用该指标的微地质单元</legend>
    {% for t in unit_types %}
    <label class="checkbox" style="margin-right:18px;">
      <input type="checkbox" name="enabled_units" value="{{ t }}"
             {% if t in ind.units_list() %}checked{% endif %}>
      <span>{{ t }}</span>
    </label>
    {% endfor %}
    <p class="muted" style="margin:8px 0 0;">
      系统设定：岩质边坡不启用土壤指标，平台不启用地形指标。
    </p>
  </fieldset>
  <div class="field">
    <label class="checkbox" for="is-constraint">
      <input id="is-constraint" type="checkbox" name="is_constraint" value="1"
             {% if ind.is_constraint %}checked{% endif %}>
      <span>同时作为限制性指标（参与一票否决判定）</span>
    </label>
  </div>
  <div class="field">
    <label class="field__label" for="description">取值依据说明</label>
    <textarea id="description" name="description" style="min-height:88px;">{{ ind.description or '' }}</textarea>
  </div>
  <p class="actions" style="margin-top:16px;">
    <button type="submit" class="btn">保存</button>
    <a class="btn btn--ghost" href="/ahp/indicators">返回指标列表</a>
  </p>
</form>
</div>
{% endblock %}
# ==== app/templates/ahp/experts.html ====
{% extends "base.html" %}
{% block title %}多专家判断矩阵{% endblock %}
{% block content %}
<div class="card">
  <h2>多专家判断矩阵 · 录入与汇总</h2>
  <form method="get" class="actions" style="margin-bottom:14
```

```
    px;">                                                                    9851
    <div>                                                                    9852
      <label class="field__label" for="unit-type">微地质单元类型</l 9853
    abel>                                                                    9854
      <select id="unit-type" name="unit_type"                                9855
    onchange="this.form.submit()">                                           9856
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}                      9857
        <option value="{{ t }}" {% if unit_type == t                         9858
    %}selected{% endif %}>{{ t }}</option>                                   9859
        {% endfor %}                                                         9860
      </select>                                                              9861
    </div>                                                                   9862
    <div>                                                                    9863
      <label class="field__label" for="level">指标层级</label>               9864
      <select id="level" name="level"                                        9865
    onchange="this.form.submit()">                                           9866
        <option value="1" {% if level == 1 %}selected{%                      9867
    endif %}>一级指标</option>                                               9868
        <option value="2" {% if level == 2 %}selected{%                      9869
    endif %}>二级指标</option>                                               9870
      </select>                                                              9871
    </div>                                                                   9872
  </form>                                                                    9873
  <p class="muted">                                                          9874
    当前范围：<strong>{{ unit_type }}</strong> ·                             9875
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif       9876
    %}。                                                                     9877
    若选二级指标，请先到                                                     9878
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。                    9879
  </p>                                                                       9880
  {% if rows %}                                                              9881
  <table>                                                                    9882
    <caption class="muted">已录入 {{ rows|length }}                          9883
    位专家的矩阵</caption>                                                   9884
    <thead>                                                                  9885
      <tr>                                                                   9886
        <th scope="col">专家</th>                                            9887
        <th scope="col">专业</th>                                            9888
        <th scope="col">矩阵规模</th>                                        9889
        <th scope="col">该矩阵 CR</th>                                       9890
        <th scope="col">一致性</th>                                          9891
        <th scope="col">录入时间</th>                                        9892
        <th scope="col">操作</th>                                            9893
      </tr>                                                                  9894
    </thead>                                                                 9895
    <tbody>                                                                  9896
      {% for r in rows %}                                                    9897
      <tr>                                                                   9898
        <td>{{ r.expert_name }}</td>                                         9899
        <td>{{ r.specialty or "—" }}</td>                                    9900
```

```
        <td class="num">{{ n }} × {{ n }}</td>
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is not none else "—" }}</td>
        <td>
          {% if r.cr is not none and r.cr < 0.1 %}<span class="tag tag--ok">通过</span>
          {% else %}<span class="tag tag--warn">不通过</span>{% endif %}
        </td>
        <td class="mono nowrap">{{ r.created_at | localdt }}</td>
        <td>
          <form method="post" action="/ahp/experts/{{ r.id }}/delete"
                onsubmit="return confirm('确定删除专家「{{ r.expert_name }}」的矩阵吗？');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  <p class="actions" style="margin-bottom:10px;">
    <a class="btn" href="/ahp/experts/aggregate?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">
      汇总计算（几何平均）
    </a>
  </p>
  {% else %}
  <p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>
  {% endif %}
  <h3>录入一位专家的判断矩阵</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <input type="hidden" name="action" value="add">
    <input type="hidden" name="unit_type" value="{{ unit_type }}">
    <input type="hidden" name="level" value="{{ level }}">
    {% if l1 %}<input type="hidden" name="l1" value="{{ l1 }}">{% endif %}
    <div style="display:flex; gap:16px;">
      <div class="field">
        <label class="field__label" for="expert-name">专家姓名</label>
        <input id="expert-name" type="text" name="expert_name"
```

```
                style="width:180px;" placeholder="如：张工">
    </div>
    <div class="field">
      <label class="field__label" for="expert-specialty">专家专业（选填）</label>
      <input id="expert-specialty" type="text" name="specialty"
                style="width:220px;" placeholder="如：地质/生态/土壤">
    </div>
  </div>
  <table style="width:auto;">
    <thead>
      <tr>
        <th></th>
        {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      {% for i in range(n) %}
      <tr>
        <th scope="row">{{ names[i] }}</th>
        {% for j in range(n) %}
          {% if i == j %}
          <td class="muted num">1</td>
          {% elif i < j %}
          <td>
            <label class="muted" style="position:absolute;left:-9999px;"
                   for="e-{{ i }}-{{ j }}">{{ names[i] }} 相对 {{ names[j] }} 的重要性</label>
            <input id="e-{{ i }}-{{ j }}" type="number" step="0.01" min="0.01"
                   name="m_{{ i }}_{{ j }}" value="{{ default[i][j] }}">
          </td>
          {% else %}
          <td class="muted num">{{ default[i][j] }}</td>
          {% endif %}
        {% endfor %}
      </tr>
      {% endfor %}
    </tbody>
  </table>
  <p class="actions" style="margin-top:12px;">
    <button type="submit" class="btn">录入该专家矩阵</button>
    <a class="btn btn--ghost" href="/ahp/indicators">指标维护</a>
    <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
```

```
      </p>                                                                    10001
    </form>                                                                   10002
</div>                                                                        10003
{% endblock %}                                                                10004
# ==== app/templates/ahp/experts_aggregate.html ====                          10005
{% extends "base.html" %}                                                     10006
{% block title %}多专家矩阵汇总{% endblock %}                                 10007
{% block content %}                                                           10008
<div class="card">                                                            10009
    <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{%     10010
      else %}二级指标（{{ l1 }}）{% endif %}</h2>                             10011
    {% if not result %}                                                       10012
    <p class="muted">                                                         10013
      该范围内尚未录入专家矩阵。                                              10014
      <a href="/ahp/experts?unit_type={{ unit_type                            10015
      }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif                     10016
      %}">去录入</a>                                                          10017
    </p>                                                                      10018
    {% else %}                                                                10019
    <div class="stat-row">                                                    10020
      <div>                                                                   10021
        <div class="stat__label">参与汇总的专家</div>                         10022
        <div class="stat__value">{{ result.count }} 位</div>                  10023
      </div>                                                                  10024
      <div>                                                                   10025
        <div class="stat__label">λmax</div>                                   10026
        <div class="stat__value mono">{{ result.lambda_max                    10027
      }}</div>                                                                10028
      </div>                                                                  10029
      <div>                                                                   10030
        <div class="stat__label">CI</div>                                     10031
        <div class="stat__value mono">{{                                      10032
      "%.4f"|format(result.ci) }}</div>                                       10033
      </div>                                                                  10034
      <div>                                                                   10035
        <div class="stat__label">CR（要求 &lt; 0.1）</div>                    10036
        <div class="stat__value mono">{{                                      10037
      "%.4f"|format(result.cr) }}</div>                                       10038
      </div>                                                                  10039
      <div>                                                                   10040
        <div class="stat__label">一致性检验</div>                             10041
        <div class="stat__value">                                             10042
          {% if result.consistent %}<span class="ok">通过</span>              10043
          {% else %}<span class="warn">不通过</span>{% endif %}               10044
        </div>                                                                10045
      </div>                                                                  10046
    </div>                                                                    10047
    <h3>汇总后权重</h3>                                                       10048
    <table style="width:auto;">                                               10049
      <thead>                                                                 10050
```

```
      <tr>
        <th scope="col">指标</th>
        {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      <tr>
        <th scope="row">权重</th>
        {% for w in result.weights %}<td class="mono">{{ w }}</td>{% endfor %}
      </tr>
    </tbody>
  </table>
  <h3>各专家矩阵的一致性</h3>
  <table style="width:auto;">
    <thead>
      <tr><th scope="col">专家</th><th scope="col">该矩阵CR</th><th scope="col">判定</th></tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.expert_name }}</td>
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is not none else "—" }}</td>
        <td>
          {% if r.cr is not none and r.cr < 0.1 %}<span class="ok">通过</span>
          {% else %}<span class="warn">不通过</span>{% endif %}
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% if not result.consistent and result.most_inconsistent %}
  <p style="padding:12px;border-radius:6px;background:var(--danger-soft);
            border-left:4px solid var(--danger);">
    <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{ result.most_inconsistent.j + 1 }} 列的比较
    （当前 <span class="mono">{{ result.most_inconsistent.value|round(3) }}</span>，
    理想约 <span class="mono">{{ result.most_inconsistent.expected|round(3) }}</span>）。
    该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。
  </p>
  {% endif %}
```

```
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <form method="post"
          action="/ahp/experts?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"
          style="display:inline;"
          onsubmit="return confirm('保存后，「{{ unit_type }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">
      {{ form.hidden_tag() }}
      <input type="hidden" name="action" value="save_agg">
      <button type="submit" class="btn" {% if not result.consistent %}disabled{% endif %}>
        保存为生效权重
      </button>
    </form>
    <a class="btn btn--ghost" href="/ahp/experts?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">返回录入</a>
    <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/indicator/index.html ====
{% extends "base.html" %}
{% block title %}指标数据采集{% endblock %}
{% block content %}
<div class="card">
  <h2>指标数据采集与管理</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/indicator/import">批量导入（CSV / Excel）</a>
  </p>
  {% if not patches %}
  <p class="muted">
    暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，
    或运行 <span class="mono">python seed_demo.py</span>
    载入示例案例。
  </p>
  {% else %}
  {% for p in patches %}
  <div class="patch-panel">
    <div class="patch-panel__head">
      <h3 class="patch-panel__name">{{ p.name }}</h3>
      <span class="patch-panel__meta">
        所属项目：{{ p.project.name if p.project else "—" }}
      </span>
      <span class="patch-panel__meta">
        微地质单元：{{ p.units.count() }} 个
      </span>
      {% if p.units.count() %}
```

```
    <span class="patch-panel__meta patch-panel__tail">
      <a class="link-btn" href="/indicator/import">批量导入本图斑数据</a>
    </span>
    {% endif %}
</div>
{% if not p.units.count() %}
<div class="notice">
    <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
    <p style="margin:0;">
      指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
      请先完成划分，再回到本页录入。
    </p>
    <p style="margin:10px 0 0;">
      <a class="btn btn--sm" href="/unit/{{ p.id }}">去划分微地质单元</a>
    </p>
</div>
{% else %}
<table>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">数据状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in p.units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{ "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td>
          {% if u.indicator_data %}
          <span class="tag tag--ok">已录入</span>
          {% else %}
          <span class="tag tag--muted">未录入</span>
          {% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/indicator/{{ u.id }}">录入 / 编辑</a>
          </span>
        </td>
      </tr>
```

```
            {% endfor %}                                                    10201
          </tbody>                                                          10202
        </table>                                                            10203
        {% endif %}                                                         10204
      </div>                                                                10205
      {% endfor %}                                                          10206
      {% endif %}                                                           10207
</div>                                                                      10208
{% endblock %}                                                              10209
# ==== app/templates/indicator/edit.html ====                               10210
{% extends "base.html" %}                                                   10211
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}          10212
{% block content %}                                                         10213
<div class="card" style="max-width:820px;">                                 10214
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>                              10215
  <p class="muted">                                                         10216
    除标注「可选」的项外，本页字段均为必填：填齐后评价才能执行，缺项会在评价页逐条列出。  10217
    <br>标注「未检测」的重金属可以留空，按方法约定以「未受污染」计入污染指标，          10218
    评价结果与导出报告里会如实写明是哪几项未检测 —— 它不等于实测达标。              10219
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。          10220
  </p>                                                                      10221
  <form method="post">                                                      10222
    {{ form.hidden_tag() }}                                                 10223
    {% for group_name, fields, has_hm in groups %}                          10224
    <h3>{{ group_name }}</h3>                                               10225
    {% for fname, cname, unit_ in fields %}                                 10226
    <div class="field">                                                     10227
      <label class="field__label" for="f-{{ fname }}">                      10228
        {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}                 10229
      </label>                                                              10230
      {% if fname == "aspect" %}                                            10231
      <select id="f-{{ fname }}" name="aspect">                             10232
        <option value="">— 不适用 —</option>                                10233
        {% for a in aspect_types %}                                         10234
        <option value="{{ a }}" {% if current.get("aspect") 10235
    == a %}selected{% endif %}>{{ a }}</option>                             10236
        {% endfor %}                                                        10237
      </select>                                                             10238
      {% else %}                                                            10239
      <input id="f-{{ fname }}" type="number" step="any"                    10240
    name="{{ fname }}"                                                      10241
             value="{{ current.get(fname) if                                10242
    current.get(fname) is not none else '' }}">                             10243
      {% endif %}                                                           10244
    </div>                                                                  10245
    {% endfor %}                                                            10246
    {% endfor %}                                                            10247
    {% for group_name, fields, has_hm in groups %}                          10248
      {% if has_hm %}                                                       10249
      <h3>重金属含量（mg/kg）<span class="muted">（可选，未检测可留空）</span>10250
```

```
      </h3>                                                                  10251
        <p class="muted">未检测的金属可留空：按方法约定以「未受污染」计入污染指标；      10252
          评价结果页与导出报告中会写明未检测的是哪几项，避免被误读成实测达标。</p>      10253
        <table style="width:auto;">                                          10254
          <thead>                                                            10255
            <tr>                                                             10256
              {% for m in heavy_metals %}<th scope="col">{{ m              10257
      }}</th>{% endfor %}                                                    10258
            </tr>                                                            10259
          </thead>                                                           10260
          <tbody>                                                            10261
            <tr>                                                             10262
              {% for m in heavy_metals %}                                    10263
              <td>                                                           10264
                <label class="muted" style="position:absolute;             10265
      left:-9999px;"                                                         10266
                       for="hm-{{ m }}">{{ m }} 含量</label>                 10267
                <input id="hm-{{ m }}" type="number"                         10268
      step="any" name="hm_{{ m }}"                                           10269
                       value="{{ current.get('heavy',                        10270
      {}).get(m) if current.get('heavy', {}).get(m) is not                   10271
      none else '' }}">                                                      10272
              </td>                                                          10273
              {% endfor %}                                                   10274
            </tr>                                                            10275
          </tbody>                                                           10276
        </table>                                                             10277
        {% endif %}                                                          10278
      {% endfor %}                                                           10279
      <p class="actions" style="margin-top:18px;">                           10280
        <button type="submit" class="btn">保存</button>                      10281
        <a class="btn btn--ghost" href="/indicator/">返回</a>                10282
      </p>                                                                   10283
    </form>                                                                  10284
</div>                                                                       10285
{% endblock %}                                                               10286
# ==== app/templates/indicator/import.html ====                              10287
{% extends "base.html" %}                                                    10288
{% block title %}批量导入指标数据{% endblock %}                               10289
{% block content %}                                                          10290
<div class="card" style="max-width:860px;">                                  10291
  <h2>批量导入指标数据</h2>                                                  10292
  <p class="muted">                                                          10293
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，            10294
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。                    10295
    先下载模板按格式填写，可减少出错。                                      10296
  </p>                                                                       10297
  <p class="actions" style="margin-bottom:14px;">                            10298
    <a class="btn btn--sm btn--ghost"                                        10299
    href="/indicator/import/template">下载导入模板（CSV）</a>                10300
```

```
      <a class="btn btn--sm btn--ghost"                                  10301
      href="/indicator/">返回指标数据</a>                                 10302
    </p>                                                                 10303
    <form method="post" enctype="multipart/form-data">                   10304
      {{ form.hidden_tag() }}                                            10305
      <div class="field">                                                10306
        <label class="field__label" for="patch-id">目标矿山图斑</lab     10307
      el>                                                                10308
        <select id="patch-id" name="patch_id" required>                  10309
          <option value="">— 请选择 —</option>                           10310
          {% for p in patches %}                                         10311
          <option value="{{ p.id }}">{{ p.name }}（{{                   10312
      p.units.count() }} 个单元）</option>                               10313
          {% endfor %}                                                   10314
        </select>                                                        10315
      </div>                                                             10316
      <div class="field">                                                10317
        <label class="field__label" for="datafile">数据文件</label      10318
      >                                                                  10319
        <input id="datafile" type="file" name="datafile"                 10320
               accept=".csv,.xlsx,.xls" required>                        10321
      </div>                                                             10322
      <p class="actions" style="margin-top:16px;">                       10323
        <button type="submit" class="btn">开始导入</button>              10324
      </p>                                                               10325
    </form>                                                              10326
    <h3>模板列说明</h3>                                                  10327
    <p class="muted">                                                    10328
      共 {{ headers|length }} 列。留空表示该指标「不适用」。             10329
      坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可       10330
      （按方法约定视为未受污染）。                                       10331
    </p>                                                                 10332
    <p class="mono muted" style="line-height:1.9;">{{                   10333
      headers|join(" , ") }}</p>                                         10334
    <h3>导入规则</h3>                                                    10335
    <ul class="muted" style="padding-left:20px;line-height:1.9          10336
      ;">                                                                10337
      <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>   10338
      <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>         10339
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>   10340
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非   10341
      <span class="mono">10cm</span>）。</li>                            10342
    </ul>                                                                10343
</div>                                                                   10344
{% if report %}                                                          10345
<div class="card" style="max-width:860px;">                              10346
    <h2>导入结果</h2>                                                    10347
    <div class="stat-row">                                               10348
      <div>                                                              10349
        <div class="stat__label">目标图斑</div>                          10350
```

```
        <div style="font-size:15px;">{{ report.patch }}</div>
      </div>
      <div>
        <div class="stat__label">成功导入</div>
        <div class="stat__value ok">{{ report.saved }}
      个单元</div>
      </div>
      <div>
        <div class="stat__label">问题行</div>
        <div class="stat__value {% if report.errors %}warn{%
      endif %}">{{ report.errors|length }}</div>
      </div>
    </div>
    {% if report.errors %}
    <h3>未导入的行</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
    </ul>
    {% else %}
    <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
    </p>
  </div>
  {% endif %}
  {% endblock %}
  # ==== app/templates/evaluate/index.html ====
  {% extends "base.html" %}
  {% block title %}适宜性定量评价{% endblock %}
  {% block content %}
  <div class="card">
    <h2>适宜性定量评价</h2>
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">所属项目</th>
          <th scope="col">矿山图斑</th>
          <th scope="col">前置条件</th>
          <th scope="col">最近评价结果</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in patches %}
        {% set pr = progress.get(p.id) %}
```

```
        <tr>                                                                10401
          <td>{{ p.project.name if p.project else "—" }}</td>                10402
          <td style="text-align:left;">{{ p.name }}</td>                    10403
          <td style="text-align:left;">                                     10404
            {% if pr and pr.ready %}                                        10405
            <span class="tag tag--ok">已具备</span>                          10406
            {% elif pr %}                                                   10407
            <span class="tag tag--warn">缺：{{                              10408
      pr.blocking[0].label }}</span>                                        10409
            <span class="muted">{{ pr.blocking[0].reason                    10410
      }}</span>                                                             10411
            {% endif %}                                                     10412
          </td>                                                             10413
          <td>                                                              10414
            {% if latest.get(p.id) %}                                       10415
            F = <span class="mono num">{{                                   10416
      latest[p.id].final_score }}</span>,                                   10417
            <span class="tag tag--ok">{{ latest[p.id].grade                 10418
      }}</span>                                                             10419
            {% else %}                                                      10420
            <span class="tag tag--muted">未评价</span>                       10421
            {% endif %}                                                     10422
          </td>                                                             10423
          <td class="nowrap">                                               10424
            <span class="row-actions">                                      10425
              <a class="link-btn" href="/evaluate/{{ p.id                   10426
      }}">执行评价</a>                                                      10427
              {% if latest.get(p.id) %}                                     10428
              <a class="link-btn" href="/evaluate/result/{{                 10429
      latest[p.id].id }}">查看结果</a>                                      10430
              {% endif %}                                                   10431
            </span>                                                         10432
          </td>                                                             10433
        </tr>                                                               10434
        {% endfor %}                                                        10435
      </tbody>                                                              10436
    </table>                                                                10437
    {% endif %}                                                             10438
</div>                                                                      10439
{% endblock %}                                                              10440
# ==== app/templates/evaluate/run.html ====                                 10441
{% extends "base.html" %}                                                   10442
{% import "_macros.html" as ui %}                                           10443
{% block title %}执行评价 · {{ patch.name }}{% endblock %}                  10444
{% block content %}                                                         10445
<div class="card" style="max-width:880px;">                                 10446
  <h2>执行适宜性评价 · {{ patch.name }}</h2>                                10447
  {{ ui.workflow_steps(progress) }}                                         10448
  {{ ui.workflow_blocking(progress) }}                                      10449
  {% if progress.ready %}                                                   10450
```

```
  <form method="post">
    {{ form.hidden_tag() }}
    <p class="actions">
      <button type="submit" class="btn">开始评价</button>
      <a class="btn btn--ghost" href="/evaluate/">返回</a>
    </p>
  </form>
  {% else %}
  <p class="actions">
    <button type="button" class="btn" disabled>开始评价</button>
    <a class="btn btn--ghost" href="/evaluate/">返回</a>
  </p>
  {% endif %}
</div>
<div class="card" style="max-width:880px;">
  <h2>当前流程状态</h2>
  <table>
    <thead>
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th
    scope="col">说明</th></tr>
    </thead>
    <tbody>
      <tr>
        <td>限制性条件判定</td>
        <td>
          {% if progress.constraint.passed %}<span
    class="tag tag--ok">已通过</span>
          {% elif progress.constraint.done %}<span
    class="tag tag--warn">未通过</span>
          {% else %}<span class="tag tag--muted">待完成</span>{
    % endif %}
        </td>
        <td style="text-align:left;">{{
    progress.constraint.detail }}</td>
      </tr>
      <tr>
        <td>划分微地质单元</td>
        <td>
          {% if progress.units.done %}<span class="tag
    tag--ok">已完成</span>
          {% else %}<span class="tag tag--muted">待完成</span>{
    % endif %}
        </td>
        <td style="text-align:left;">{{
    progress.units.detail }}</td>
      </tr>
      <tr>
        <td>录入指标数据</td>
        <td>
          {% if progress.data.ready %}<span class="tag
```

```
      tag--ok">已完成</span>                                                    10501
            {% else %}<span class="tag tag--muted">待完成</span>{%             10502
      % endif %}                                                               10503
          </td>                                                                10504
          <td style="text-align:left;">{{                                     10505
      progress.data.detail }}</td>                                             10506
        </tr>                                                                  10507
      </tbody>                                                                 10508
    </table>                                                                   10509
</div>                                                                         10510
{% if weight_sources %}                                                        10511
<div class="card" style="max-width:880px;">                                    10512
    <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">            10513
      <thead>                                                                  10514
        <tr><th scope="col">微地质单元</th><th                                 10515
      scope="col">权重来源</th></tr>                                           10516
      </thead>                                                                 10517
      <tbody>                                                                  10518
        {% for ut, label in weight_sources.items() %}                          10519
        <tr>                                                                   10520
          <td style="text-align:left;">{{ ut }}</td>                          10521
          <td style="text-align:left;">                                        10522
            {% if "自定义" in label %}                                         10523
            <span class="tag tag--warn">{{ label }}</span>                     10524
            {% else %}                                                         10525
            <span class="tag tag--muted">{{ label }}</span>                    10526
            {% endif %}                                                        10527
          </td>                                                                10528
        </tr>                                                                  10529
        {% endfor %}                                                           10530
      </tbody>                                                                 10531
    </table>                                                                   10532
</div>                                                                         10533
{% endif %}                                                                    10534
{% endblock %}                                                                 10535
# ==== app/templates/evaluate/result.html ====                                 10536
{% extends "base.html" %}                                                      10537
{% block title %}评价结果{% endblock %}                                        10538
{% block content %}                                                            10539
<div class="card">                                                             10540
    <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>                10541
    <div class="stat-row">                                                     10542
      <div>                                                                    10543
        <div class="stat__label">最终评价因子 F</div>                          10544
        <div class="stat__value mono">{{ r.final_score                         10545
      }}</div>                                                                 10546
      </div>                                                                   10547
      <div>                                                                    10548
        <div class="stat__label">适宜性等级</div>                              10549
        <div class="stat__value ok">{{ r.grade }}</div>                        10550
```

```
    </div>                                                                  10551
    <div style="flex:1;min-width:240px;">                                   10552
      <div class="stat__label">推荐措施</div>                               10553
      <div style="font-size:15px;">{{ r.measures }}</div>                   10554
    </div>                                                                  10555
  </div>                                                                    10556
  {% if unit_scores %}                                                      10557
  <h3>各微地质单元得分</h3>                                                 10558
  <table style="max-width:860px;">                                          10559
    <thead>                                                                 10560
      <tr>                                                                  10561
        <th scope="col">微地质单元</th>                                     10562
        <th scope="col">面积占比</th>                                       10563
        <th scope="col">单元评价因子 F′</th>                                10564
        <th scope="col">加权贡献</th>                                       10565
        <th scope="col">权重来源</th>                                       10566
      </tr>                                                                 10567
    </thead>                                                                10568
    <tbody>                                                                 10569
      {% for u in unit_scores %}                                            10570
      <tr>                                                                  10571
        <td>{{ u.unit_type }}</td>                                          10572
        <td class="num">{{ "%.2f%%"|format(u.ratio * 100)                   10573
  }}</td>                                                                   10574
        <td class="mono">{{ u.unit_score }}</td>                            10575
        <td class="mono">{{ "%.4f"|format(u.ratio *                         10576
  u.unit_score) }}</td>                                                     10577
        <td class="muted">{{ u.weight_source or "系统默认权重"              10578
  }}</td>                                                                   10579
      </tr>                                                                 10580
      {% endfor %}                                                          10581
    </tbody>                                                                10582
  </table>                                                                  10583
  {% endif %}                                                               10584
  {% if weak %}                                                             10585
  <h3>短板提示</h3>                                                         10586
  <p class="muted" style="margin-top:-4px;">                                10587
    按「权重 ×（1 − 标准化值）」排序，提升这些指标对总分的拉动最大：       10588
  </p>                                                                      10589
  <table style="max-width:760px;">                                          10590
    <thead>                                                                 10591
      <tr>                                                                  10592
        <th scope="col">微地质单元</th>                                     10593
        <th scope="col">指标</th>                                           10594
        <th scope="col">实测值</th>                                         10595
        <th scope="col">标准化值</th>                                       10596
        <th scope="col">提升空间</th>                                       10597
      </tr>                                                                 10598
    </thead>                                                                10599
    <tbody>                                                                 10600
```

```
        {% for w in weak %}                                                  10601
        <tr>                                                                 10602
          <td>{{ w.unit_type }}</td>                                         10603
          <td>{{ w.l2 }}</td>                                                10604
          <td class="mono">{{ w.raw if w.raw is not none else              10605
    "—" }}</td>                                                              10606
          <td class="mono">{{ "%.4f"|format(w.std_value)                     10607
    }}</td>                                                                  10608
          <td class="mono">{{ "%.4f"|format(w.gain) }}</td>                  10609
        </tr>                                                                10610
        {% endfor %}                                                         10611
      </tbody>                                                               10612
    </table>                                                                 10613
    {% endif %}                                                              10614
    {% if check and check.covered %}                                         10615
    {% include "standard/_check_block.html" %}                               10616
    {% endif %}                                                              10617
    <p class="muted" style="margin-top:14px;">                               10618
      计算人：{{ operator or "—" }} ·                                        10619
      计算时间：{{ r.created_at | localdt("%Y-%m-%d %H:%M:%S") }}            10620
      参数来源：{{ param_sources }}                                          10621
    </p>                                                                     10622
    {% if params %}                                                          10623
    <details style="margin-top:8px;">                                        10624
      <summary class="muted">本次计算使用的参数快照</summary>                10625
      <table style="max-width:520px;">                                       10626
        <thead>                                                              10627
          <tr><th scope="col">参数</th><th                                   10628
      scope="col">取值</th><th scope="col">来源</th></tr>                    10629
        </thead>                                                             10630
        <tbody>                                                              10631
          {% for code, item in params.items() %}                             10632
          <tr>                                                               10633
            <td class="mono">{{ code }}</td>                                 10634
            <td class="mono">{{ item.value }}</td>                           10635
            <td class="muted">{{ item.source }}</td>                         10636
          </tr>                                                              10637
          {% endfor %}                                                       10638
        </tbody>                                                             10639
      </table>                                                               10640
    </details>                                                               10641
    {% endif %}                                                              10642
    <p class="actions" style="margin-top:18px;">                             10643
      <a class="btn btn--ghost" href="/report/{{ r.patch_id                  10644
      }}">查看得分明细</a>                                                   10645
      <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>           10646
    </p>                                                                     10647
</div>                                                                       10648
{% endblock %}                                                               10649
# ==== app/templates/report/index.html ====                                  10650
```

```
{% extends "base.html" %}
{% block title %}报告输出{% endblock %}
{% block content %}
<div class="card">
  <h2>报告与结果输出</h2>
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">可生成报告</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set pr = progress.get(p.id) %}
      <tr>
        <td>{{ p.project.name if p.project else "—" }}</td>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if pr and pr.ready %}
          <span class="tag tag--ok">可以</span>
          {% elif pr %}
          <span class="tag tag--warn">暂不可</span>
          <span class="muted">{{ pr.blocking[0].reason
    }}</span>
          {% endif %}
        </td>
        <td class="nowrap">
          {% if pr and pr.ready %}
          <span class="row-actions">
            <a class="link-btn" href="/report/{{ p.id
    }}">查看得分明细</a>
            <a class="link-btn" href="/report/{{ p.id
    }}/pdf">导出 PDF</a>
            <a class="link-btn" href="/report/{{ p.id
    }}/word">导出 Word</a>
          </span>
          {% else %}
          <span class="muted">—</span>
          {% endif %}
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
```

```
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/report/detail.html ====
{% extends "base.html" %}
{% block title %}得分明细 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>得分明细 · {{ patch.name }}</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">最终评价因子 F</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.final_score) }}</div>
    </div>
    <div>
      <div class="stat__label">适宜性等级</div>
      <div class="stat__value ok">{{ result.grade }}</div>
    </div>
    <div style="flex:1;min-width:240px;">
      <div class="stat__label">推荐措施</div>
      <div style="font-size:15px;">{{
    result.grade_info.measure }}</div>
    </div>
  </div>
  <p class="muted" style="margin-top:12px;">
    本页与导出报告均取自
    <strong>{% if record %}{{ record.created_at |
    localdt("%Y-%m-%d %H:%M:%S") }}{% endif %}
    记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else
    "—" }}），
    不会因事后修改指标数据而变化。
  </p>
  {% if undetected %}
  <p class="muted" style="margin-top:8px;">
    数据完整度说明：以下重金属项<strong>未检测</strong>，按方法约定以「未受污染」计入污染指标，
    并非实测达标 ——
    {% for unit_type, metals in undetected %}
      {{ unit_type }}：{{ metals|join("、") }}{% if not
    loop.last %}；{% endif %}
    {% endfor %}。
  </p>
  {% endif %}
  {% if check and check.covered %}
  {% include "standard/_check_block.html" %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF
    报告</a>
    <a class="btn btn--ghost" href="/report/{{ patch.id
```

```
      }}/word">导出 Word 报告</a>                                    10751
      <a class="btn btn--ghost" href="/report/">返回</a>             10752
   </p>                                                              10753
</div>                                                               10754
<div class="card">                                                   10755
   <h2>结果图表</h2>                                                 10756
   <h3>各微地质单元得分对比</h3>                                     10757
   <div>{{ chart_units|safe }}</div>                                 10758
   <h3>各一级指标平均得分</h3>                                       10759
   <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>  10760
   <div>{{ chart_weight|safe }}</div>                                10761
</div>                                                               10762
{% for u in result.unit_results %}                                   10763
<div class="card">                                                   10764
   <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *         10765
      100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>      10766
   {% if u.weight_source %}                                          10767
   <p class="muted" style="margin-top:-6px;">权重来源：{{            10768
      u.weight_source }}</p>                                         10769
   {% endif %}                                                       10770
   <table>                                                           10771
      <thead>                                                        10772
         <tr>                                                        10773
            <th scope="col">一级指标</th>                            10774
            <th scope="col">二级指标</th>                            10775
            <th scope="col">实测值</th>                              10776
            <th scope="col">标准化值</th>                            10777
            <th scope="col">权重</th>                                10778
            <th scope="col">贡献</th>                                10779
         </tr>                                                       10780
      </thead>                                                       10781
      <tbody>                                                        10782
         {% for d in u.details %}                                    10783
         <tr>                                                        10784
            <td>{{ d.l1 }}</td>                                      10785
            <td style="text-align:left;">{{ d.l2 }}</td>             10786
            <td class="mono" style="text-align:left;">{{ "—" if      10787
      d.raw is none else d.raw }}</td>                               10788
            <td class="mono">{{ "—" if d.std_value is none else      10789
      "%.4f"|format(d.std_value) }}</td>                             10790
            <td class="mono">{{ "%.4f"|format(d.weight) }}</td>      10791
            <td class="mono">{{ "—" if d.contribution is none        10792
      else "%.4f"|format(d.contribution) }}</td>                     10793
         </tr>                                                       10794
         {% endfor %}                                                10795
      </tbody>                                                       10796
   </table>                                                          10797
   <p class="muted" style="margin:0;">                               10798
      一级指标得分：                                                 10799
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span           10800
```

```
      class="num">{{ "%.4f"|format(v) }}</span>{% if not
      loop.last %};{% endif %}{% endfor %}
    </p>
</div>
{% endfor %}
{% endblock %}
# ==== app/templates/report/print.html ====
<!DOCTYPE html>
<html lang="zh-CN">
<head>
<meta charset="UTF-8">
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>
<style>
  @page { size: A4; margin: 25mm 25mm; }
  * { box-sizing: border-box; }
  body {
    margin: 0; color: #000; background: #fff;
    font-family: "SimSun", "宋体", serif;
    font-size: 12pt; line-height: 1.0;
  }
  h1 {
    font-family: "SimHei", "黑体", sans-serif;
    font-size: 16pt; text-align: center;
    font-weight: bold; color: #000;
    margin: 0 0 6px;
  }
  h2 {
    font-family: "SimHei", "黑体", sans-serif;
    font-size: 14pt; text-align: center;
    font-weight: bold; color: #000;
    margin: 16px 0 8px;
    border-bottom: none; padding-bottom: 0;
  }
  h3 {
    font-family: "SimSun", "宋体", serif;
    font-size: 12pt; color: #000; font-weight: bold;
    margin: 12px 0 6px;
  }
  .sub { text-align: center; color: #000; font-size: 12pt;
    margin-bottom: 14px; }
  table { width: 100%; border-collapse: collapse; margin:
    8px 0 14px;
          font-size: 12pt; font-variant-numeric:
    tabular-nums; }
  th, td { border: 1px solid #000; padding: 3px 6px;
    text-align: center; }
  th { background: #fff; font-weight: bold; }
  .left { text-align: left; }
  .mono { font-family: "SimSun", "宋体", serif; }
  .kv { width: auto; }
```

```
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; font-size:
    12pt; }
  .note { color: #000; font-size: 12pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 12pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} {{ patch.admin_region }}{%
    endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０
    生成时间 {{ generated_at | localdt }}
  </p>
  <h2>一、图斑基本信息</h2>
  <table class="kv">
    <tr><th>图斑名称</td><td>{{ patch.name }}</td></tr>
    <tr><th>所在行政区</td><td>{{ patch.admin_region or "—"
    }}</td></tr>
    <tr><th>图斑面积（公顷）</td><td>{{ patch.area_ha if
    patch.area_ha is not none else "—" }}</td></tr>
    <tr><th>开采矿种</td><td>{{ patch.mineral_type or "—"
    }}</td></tr>
    <tr><th>开采方式</td><td>{{ patch.mining_method or "—"
    }}</td></tr>
    <tr><th>责任主体缺失</td><td>{{ "是" if patch.responsible_entit
    y_missing else "否" }}</td></tr>
    {% if reference %}
    <tr><th>参照生态系统</td>
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%，
            多样性指数基准 {{ reference.shannon_benchmark
    }}</td></tr>
    {% endif %}
  </table>
  <h2>二、评价结论</h2>
  <table class="kv">
    <tr><th>最终评价因子 F</td><td class="mono"><strong>{{
    "%.4f"|format(result.final_score) }}</strong></td></tr>
    <tr><th>适宜性等级</td><td><strong>{{ result.grade
    }}</strong></td></tr>
    <tr><th>判定依据</td><td>{{ result.grade_info.meaning
    }}</td></tr>
    <tr><th>推荐措施</td><td>{{ result.grade_info.measure
    }}</td></tr>
```

```
</table>
{% if check and check.covered %}
<h2>三、与地方标准 / 政策文件的对照</h2>
<p class="note">
  图斑所在行政区：{{ check.region or "未填写" }}（对照 {{
  check.province }} 的标准与政策文件）。
  对照只把文件里的判定项逐条映射到系统已有数据，系统内没有对应数据的定性要求一律标为待人工判断。
  合计：符合 {{ check.totals["符合"] }} 条、不符合 {{
  check.totals["不符合"] }} 条、待人工判断 {{ check.totals["待人工判断"]
  }} 条。
</p>
{% for s in check.standards %}
<h3>{{ s.name }}{% if s.doc_number %}（{{ s.doc_number
  }}）{% endif %}</h3>
<table>
  <thead>
    <tr><th>文件判定项</th><th>条件</th><th>类型</th><th>对照结果</th><
  th>依据</th></tr>
  </thead>
  <tbody>
    {% for r in s.rules %}
    <tr>
      <td class="left">{{ r.item }}</td>
      <td class="mono">{{ r.operator }} {{ r.threshold or
  "" }}</td>
      <td>{{ r.condition_type or "—" }}</td>
      <td>{{ r.verdict }}</td>
      <td class="left">{{ r.evidence }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endfor %}
{% endif %}
<h2>四、各微地质单元得分</h2>
<table>
  <thead>
    <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子
  F'</th><th>加权贡献</th></tr>
  </thead>
  <tbody>
    {% for u in result.unit_results %}
    <tr>
      <td>{{ u.unit_type }}</td>
      <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
  }}</td>
      <td class="mono">{{ "%.4f"|format(u.unit_score)
  }}</td>
      <td class="mono">{{ "%.4f"|format(u.weighted)
  }}</td>
```

```
      </tr>
      {% endfor %}
    </tbody>
</table>
<div class="chart">{{ chart_units|safe }}</div>
<h2>五、一级指标权重分布</h2>
<div class="chart">{{ chart_weight|safe }}</div>
<h2>六、得分明细</h2>
{% for u in result.unit_results %}
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *
  100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
<table>
  <thead>
    <tr>
      <th>一级指标</th><th>二级指标</th><th>实测值</th>
      <th>标准化值</th><th>权重</th><th>贡献</th>
    </tr>
  </thead>
  <tbody>
    {% for d in u.details %}
    <tr>
      <td>{{ d.l1 }}</td>
      <td class="left">{{ d.l2 }}</td>
      <td class="mono left">{{ "—" if d.raw is none else
  d.raw }}</td>
      <td class="mono">{{ "—" if d.std_value is none else
  "%.4f"|format(d.std_value) }}</td>
      <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
      <td class="mono">{{ "—" if d.contribution is none
  else "%.4f"|format(d.contribution) }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
<p class="note">
  一级指标得分：
  {% for k, v in u.l1_scores.items() %}{{ k }} = {{
  "%.4f"|format(v) }}{% if not loop.last %}；{% endif %}{%
  endfor %}
</p>
{% endfor %}
<h2>七、评价依据说明</h2>
<ol class="note">
  <li>指标标准化按标准化公式执行，按指标性质分为越大越优、越小越优、
    适中型与离散赋值四类。</li>
  <li>单元评价因子按加权合成公式由二级指标权重加权求得；
    整矿评价因子由各微地质单元按面积占比加权合成。</li>
  <li>适宜性等级按等级划分表判定：F&lt;0.3 不适宜、0.3≤F≤0.7 基本适宜、F≥0.7
  适宜。</li>
  <li>指标权重取自系统内置权重，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>
```

```
    </ol>                                                                    11001
    <p class="sig">评价人（签字）：_____</p>                                  11002
    <p class="sig">日期：_____</p>                                            11003
</body>                                                                      11004
</html>                                                                      11005
# ==== app/templates/search.html ====                                        11006
{% extends "base.html" %}                                                    11007
{% from "_macros.html" import icon %}                                        11008
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}      11009
{% block content %}                                                          11010
<div class="card">                                                           11011
  <h2>搜索结果</h2>                                                          11012
  {% if not q %}                                                             11013
  <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区/矿种、                11014
    标准名称与文号。</p>                                                     11015
  {% else %}                                                                 11016
  <p class="muted">                                                          11017
    关键词「<strong>{{ q }}</strong>」共找到                                 11018
    <span class="num">{{ total }}</span> 条结果。                            11019
  </p>                                                                       11020
  {% if not total %}                                                         11021
  <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>          11022
  {% endif %}                                                                11023
  {% endif %}                                                                11024
</div>                                                                       11025
{% if projects %}                                                            11026
<div class="card">                                                           11027
  <h2>评价项目（{{ projects|length }}）</h2>                                 11028
  <table>                                                                    11029
    <thead>                                                                  11030
      <tr>                                                                   11031
        <th scope="col">项目名称</th>                                        11032
        <th scope="col">编号</th>                                            11033
        <th scope="col">状态</th>                                            11034
        <th scope="col">图斑数</th>                                          11035
        <th scope="col">操作</th>                                            11036
      </tr>                                                                  11037
    </thead>                                                                 11038
    <tbody>                                                                  11039
      {% for p in projects %}                                                11040
      <tr>                                                                   11041
        <td style="text-align:left;">{{ p.name }}</td>                       11042
        <td class="mono">{{ p.code or "—" }}</td>                            11043
        <td><span class="tag tag--muted">{{ p.status                         11044
    }}</span></td>                                                           11045
        <td class="num">{{ p.patches.count() }}</td>                         11046
        <td class="nowrap">                                                  11047
          <span class="row-actions">                                         11048
            <a class="link-btn" href="/project/{{ p.id                       11049
    }}/patches">矿山图斑</a>                                                 11050
```

```
                <a class="link-btn" href="/project/{{ p.id
    }}/edit">编辑</a>
                </span>
            </td>
        </tr>
        {% endfor %}
    </tbody>
  </table>
</div>
{% endif %}
{% if patches %}
<div class="card">
  <h2>矿山图斑（{{ patches|length }}）</h2>
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">图斑名称</th>
        <th scope="col">行政区</th>
        <th scope="col">矿种</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for m in patches %}
      <tr>
        <td>{{ m.project.name if m.project else "—" }}</td>
        <td style="text-align:left;">{{ m.name }}</td>
        <td>{{ m.admin_region or "—" }}</td>
        <td>{{ m.mineral_type or "—" }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/constraint/{{ m.id
    }}">限制性判定</a>
            <a class="link-btn" href="/unit/{{ m.id
    }}">微地质单元</a>
            <a class="link-btn" href="/evaluate/{{ m.id
    }}">执行评价</a>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
</div>
{% endif %}
{% if standards %}
<div class="card">
  <h2>标准与政策文件（{{ standards|length }}）</h2>
  <table>
```

```
      <thead>                                                                  11101
        <tr>                                                                   11102
          <th scope="col">类别</th>                                            11103
          <th scope="col">省份</th>                                            11104
          <th scope="col">名称</th>                                            11105
          <th scope="col">文号</th>                                            11106
          <th scope="col">操作</th>                                            11107
        </tr>                                                                  11108
      </thead>                                                                 11109
      <tbody>                                                                  11110
        {% for s in standards %}                                               11111
        <tr>                                                                   11112
          <td>{{ s.category }}</td>                                            11113
          <td>{{ s.province }}</td>                                            11114
          <td style="text-align:left;">{{ s.name }}</td>                       11115
          <td class="mono">{{ s.doc_number or "—" }}</td>                      11116
          <td class="nowrap">                                                  11117
            <span class="row-actions">                                         11118
              <a class="link-btn" href="/standard/{{ s.id                      11119
      }}/rules">规则</a>                                                       11120
            </span>                                                            11121
          </td>                                                                11122
        </tr>                                                                  11123
        {% endfor %}                                                           11124
      </tbody>                                                                 11125
    </table>                                                                   11126
</div>                                                                         11127
{% endif %}                                                                    11128
{% endblock %}                                                                 11129
#  ==== seed_demo.py ====                                                      11130
#  -*- coding: utf-8 -*-                                                       11131
"""内置玉田县示例案例：把示例数据写入数据库。                                  11132
用于功能验证与操作演示。数据逐字取自内置示例数据。                            11133
运行：python seed_demo.py                  幂等，只补写缺失的演示数据，        11134
                                               不会删除用户自建的项目与图斑    11135
      python seed_demo.py --reset          连同标准库与指标定义一并重建        11136
"""                                                                            11137
import json                                                                    11138
import sys                                                                     11139
sys.stdout.reconfigure(encoding="utf-8")                                       11140
from app import create_app                                                     11141
from app.models import (db, User, Project, MinePatch,                          11142
    ReferenceEcosystem,                                                        11143
                          MicroUnit, IndicatorData, Standard,                  11144
    StandardRule,                                                              11145
                          ConstraintCheck, EvaluationResult,                   11146
    ScoreDetail)                                                               11147
app = create_app()                                                             11148
#  内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」              11149
DEMO_CODE = "DEMO-001"                                                         11150
```

```
#  内置标准库：１２  项地方标准与政策文件
STANDARDS = [
    dict(category="地方标准", province="辽宁",
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="辽宁",
    name="废弃矿山自然恢复评估规程",
        doc_number="DB21/T 4421—2026",
    publish_date="2026-05-07",
        implement_date="2026-06-07",
        summary="按水热条件分成湿润区、半湿润区、半干旱区，分区确定植被覆盖度要求；"
                "评估考虑地质灾害、水土污染、立地条件、土地损毁程度、植被发育情况、"
                "植被群落六个方面",
        #  植被覆盖度按水热条件分区给阈值，分区写进判定项名称，便于对照时看清适用分区
        rules=[("植被覆盖度（湿润区）", ">=", "70", "必要条件", "湿润区"),
               ("植被覆盖度（半湿润区）", ">=", "60", "必要条件", "半湿润区"),
               ("植被覆盖度（半干旱区）", ">=", "50", "必要条件", "半干旱区"),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="内蒙古",
    name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
    publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="江西",
    name="废弃矿山自然恢复技术规范",
        doc_number="DB36/T 2221—2026",
    publish_date="2026-02-02",
        implement_date="2026-08-01",
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),
```

```
                ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),
    dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",
        doc_number=None, publish_date="2026-06-18",
    implement_date=None,
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="政策文件", province="安徽",
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",
        doc_number="皖自然资修函[2020]34号",
    publish_date="2020-03-03", implement_date=None,
        summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",
        rules=[("责任主体", "==", "政府", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="山东",
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",
        doc_number="鲁自然资字[2022]104号",
    publish_date="2022-08-02", implement_date=None,
        summary="条件与安徽省相同，另增加自然恢复认定标准",
        rules=[("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="湖南",
        name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",
        doc_number="湘自资办发[2023]58号",
    publish_date="2023-07-20", implement_date=None,
        summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度
    >75%、连续3年正向增长",
        rules=[("植被覆盖度", ">", "75", "必要条件", "%"),
               ("植被覆盖率趋势", "==", "连续3年正向增长", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("符合国土空间规划", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="新疆",
        name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",
        doc_number="新自然资源办函〔2024〕219号",
    publish_date="2024-12-20", implement_date=None,
        summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
        rules=[("植被覆盖度", ">=", "周边30%", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="广西",
        name="广西壮族自治区历史遗留矿山自然恢复技术指南",
        doc_number="桂自然资办〔2024〕250号",
    publish_date="2024-12-30", implement_date=None,
        summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
        rules=[("无地质灾害", "==", "是", "必要条件", None),
               ("植被自然恢复趋势", "==", "明显", "必要条件", None),
               ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="西藏",
```

```
            name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
            doc_number="藏自然资发〔2025〕19号",
        publish_date="2025-09-24", implement_date=None,
            summary="含修复成本高、生态修复效益不显著等适用条件",
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    # 三类微地质单元的面积占比与各项实测值，与最新版评价方法表 5 一致：
    # 平台 26.3%、碎石边坡 69.5%、岩质边坡 4.2%；
    # 单元面积按占比 × 图斑面积（5.6 公顷）换算，保证面积与占比互相对得上。
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.042, "area":
    0.2352,
            "data": {"coverage": 68.88, "shannon": 0.51,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.695, "area":
    3.892,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
                     "slope": 28.75, "aspect": "阳坡"},
        },
        {
            "unit_type": "平台", "ratio": 0.263, "area":
    1.4728,
            "data": {"soil_ratio": 0.56, "thickness": 30,
    "ph": 8.72,
                     "organic_matter": 10.92,
                     "heavy_metals": {"Hg": 0.02, "As":
    3.13, "Ni": 12.33,
                                      "Cu": 11.14, "Cd":
    0.09, "Pb": 9.98},
                     "coverage": 58.42, "shannon": 0.72},
        },
    ],
}
```

```
def clean_case(reset=False):
    """清理数据。
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，
    以便把演示环境恢复到初始状态。
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目
    会在每次启动时被删除。
    """
    from app.models import EvaluationResult, ScoreDetail,
    ConstraintCheck
    demo_projects = Project.query.filter_by(code=DEMO_CODE)
    all()
    demo_ids = [p.id for p in demo_projects]
    patch_ids = []
    if demo_ids:
        patch_ids = [p.id for p in MinePatch.query.filter(
            MinePatch.project_id.in_(demo_ids)).all()]
    if patch_ids:
        unit_ids = [u.id for u in MicroUnit.query.filter(
            MicroUnit.patch_id.in_(patch_ids)).all()]
        if unit_ids:
            IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).delete(
                synchronize_session=False)
        result_ids = [r.id for r in EvaluationResult.query.f
    ilter(
            EvaluationResult.patch_id.in_(patch_ids)).all()]
        if result_ids:
            ScoreDetail.query.filter(
                ScoreDetail.result_id.in_(result_ids)).delet
    e(
                synchronize_session=False)
        EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).delete
    (
            synchronize_session=False)
        ConstraintCheck.query.filter(
            ConstraintCheck.patch_id.in_(patch_ids)).delete(
            synchronize_session=False)
        ReferenceEcosystem.query.filter(
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele
    te(
            synchronize_session=False)
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_
    ids)).delete(
            synchronize_session=False)
        MinePatch.query.filter(MinePatch.id.in_(patch_ids))
    delete(
            synchronize_session=False)
    if demo_ids:
```

```
        Project.query.filter(Project.id.in_(demo_ids)).delet
    e(
            synchronize_session=False)
    if reset:
        StandardRule.query.delete()
        Standard.query.delete()
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，
    #  避免长期运行后 score_details 表堆积无效数据。
    _purge_orphan_details()
    db.session.commit()
    return len(demo_ids)
def demo_patch():
    """取内置演示案例的图斑。
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。
    没有演示案例时才退回第一个图斑。
    """
    proj = Project.query.filter_by(code=DEMO_CODE).first()
    if proj is not None:
        m = MinePatch.query.filter_by(project_id=proj.id).fi
    rst()
        if m is not None:
            return m
    return MinePatch.query.first()
def _purge_orphan_details():
    """清理库里的孤儿行（指向已删除父记录的子表数据）。
    原先只清 score_details，实测还不够：删图斑若绕过 cleanup 的级联，
    indicator_data、constraint_checks 等同样会留孤儿（自检夹具中断时尤其常见），
    孤儿的 unit_id 还会占住自增号，导致新数据插不进去。
    判定规则统一在 app/services/cleanup.py 的 ORPHAN_RULES。
    """
    from app.services import cleanup
    cleanup.purge_orphans(apply=True)
def seed(reset=False):
    with app.app_context():
        db.create_all()          #  全新环境下先建表，保证本脚本可独立运行
        _purge_orphan_details()
        admin = User.query.filter_by(username="admin").first
    ()
        if not admin:
            from werkzeug.security import
    generate_password_hash
            admin = User(username="admin",
                         password_hash=generate_password_has
    h("admin123"),
                         role="admin")
            db.session.add(admin)
            db.session.commit()
        #  幂等：演示案例已存在就不重复写入，更不清空用户自己的项目
        existing = Project.query.filter_by(code=DEMO_CODE).fi
```

```
irst()                                                                  11401
    if existing and not reset:                                          11402
        print("演示案例已存在，跳过写入（用户自建项目不受影响）")                  11403
        _report()                                                       11404
        _seed_reference_data(reset=False)                               11405
        return                                                          11406
    clean_case(reset=reset)                                             11407
    project = Project(name=CASE["name"], code=DEMO_CODE,                11408
                      description="内置示例案例",                          11409
created_by=admin.id)                                                    11410
    db.session.add(project)                                             11411
    db.session.flush()                                                  11412
    patch = MinePatch(project_id=project.id,                            11413
name=CASE["patch_name"],                                                11414
                      admin_region="河北省唐山市玉田县",                    11415
                      area_ha=CASE["area_ha"],                          11416
                      mineral_type=CASE["mineral_type"],                11417
                      mining_method="露天开采",                           11418
stop_years=None,                                                        11419
                      responsible_entity_missing=True)                  11420
    db.session.add(patch)                                               11421
    db.session.flush()                                                  11422
    ref = ReferenceEcosystem(patch_id=patch.id,                         11423
                             coverage_benchmark=75.0,                   11424
shannon_benchmark=0.7)                                                  11425
    db.session.add(ref)                                                 11426
    for u in CASE["units"]:                                             11427
        unit = MicroUnit(patch_id=patch.id,                             11428
unit_type=u["unit_type"],                                               11429
                         ratio=u["ratio"],                              11430
area=u.get("area"))                                                     11431
        db.session.add(unit)                                            11432
        db.session.flush()                                              11433
        d = u["data"]                                                   11434
        ind = IndicatorData(                                            11435
            unit_id=unit.id,                                            11436
            soil_ratio=d.get("soil_ratio"),                             11437
            thickness=d.get("thickness"),                               11438
            ph=d.get("ph"),                                             11439
            organic_matter=d.get("organic_matter"),                     11440
            heavy_metals=json.dumps(d["heavy_metals"],                  11441
ensure_ascii=False)                                                     11442
                              if d.get("heavy_metals") else             11443
None,                                                                   11444
            coverage=d.get("coverage"),                                 11445
            shannon=d.get("shannon"),                                   11446
            slope=d.get("slope"),                                       11447
            aspect=d.get("aspect"),                                     11448
        )                                                               11449
        db.session.add(ind)                                             11450
```

```
            #  限制性条件判定记录：示例案例属「无地质灾害隐患点」、
            #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。
            db.session.add(ConstraintCheck(
                patch_id=patch.id,
                policy_pass=True, location_pass=True,
        pollution_pass=True,
                geo_pass=True,
                passed=True, reason="全部通过"))
            db.session.commit()
            print("已写入案例：", CASE["name"])
            print("  图斑：", CASE["patch_name"])
            print("  微地质单元：平台 26.3% / 碎石边坡 69.5% / 岩质边坡 4.2%")
            print("  管理员账号：admin / admin123")
            _seed_reference_data(reset=reset)
            _report()
def _seed_reference_data(reset=False):
    """写入标准库与指标定义。
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护
    的标准条目与指标定义。
    """
    if reset or Standard.query.count() == 0:
        seed_standards()
    from app.services.indicator_service import
    seed_indicators
    n = seed_indicators()
    print("指标定义：%d 项" % n)
def _report():
    total_p = Project.query.count()
    total_patch = MinePatch.query.count()
    print()
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))
    if total_p > total_patch:
        print("  提示：项目是归档单位，评价针对的是图斑。")
        print("        新项目需先建立图斑，才会出现在后续各模块中。")
def seed_standards():
    """写入内置标准库的 12 项标准与政策文件及其结构化规则。"""
    StandardRule.query.delete()
    Standard.query.delete()
    db.session.commit()
    for s in STANDARDS:
        std = Standard(category=s["category"],
    province=s["province"],
                       name=s["name"],
    doc_number=s["doc_number"],
                       publish_date=s["publish_date"],
                       implement_date=s["implement_date"],
                       summary=s["summary"])
        db.session.add(std)
        db.session.flush()
        for item, op, thr, ctype, note in s["rules"]:
```

```
                db.session.add(StandardRule(standard_id=std.id,
    item=item,
                                                operator=op,
    threshold=thr,

    condition_type=ctype, note=note))
    db.session.commit()
    print("已写入标准/政策文件：%d 项，规则 %d 条"
          % (len(STANDARDS), sum(len(s["rules"]) for s in
    STANDARDS)))
if __name__ == "__main__":
    # --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。
    # 无论是否 --reset，都不会删除用户自建的项目与图斑。
    seed(reset="--reset" in sys.argv[1:])
# ==== 测试/test_ahp.py ====
# -*- coding: utf-8 -*-
"""地基自检：验证 AHP 引擎与应用可启动。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import numpy as np
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import ahp_service as ahp
LINE = "=" * 66
# ------------------------------------------------------------
    ------------------
# 1) 完全一致矩阵 → 权重应精确还原，CR=0
# ------------------------------------------------------------
    ------------------
print(LINE)
print("测试 1  完全一致判断矩阵的权重还原与一致性")
print(LINE)
# 由权重构造完全一致矩阵：a_ij = w_i / w_j
w_true = np.array([0.44, 0.31, 0.25])
A = np.outer(w_true, 1.0 / w_true)
r = ahp.weights_and_consistency(A)
print("  判断矩阵 A：")
for row in A:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print()
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in
    r["weights"]))
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))
print("  λmax = %.6f   CI = %.3e   CR = %.3e   一致: %s"
      % (r["lambda_max"], r["ci"], r["cr"],
```

```
    r["consistent"]))                                                    11551
assert np.allclose(r["weights"], w_true, atol=1e-6),                     11552
    "权重未精确还原"                                                      11553
assert r["consistent"], "完全一致矩阵应通过检验"                          11554
# ------------------------------------------------------------------     11555
    ----------------                                                     11556
# 2） 上三角输入 → 完整矩阵 → 一致性定位                                  11557
# ------------------------------------------------------------------     11558
    ----------------                                                     11559
print()                                                                  11560
print(LINE)                                                              11561
print("测试 2  上三角录入与最不一致项定位")                               11562
print(LINE)                                                              11563
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）       11564
M = ahp.matrix_from_upper([2, 5, 3])                                     11565
r2 = ahp.weights_and_consistency(M)                                      11566
bad = ahp.find_most_inconsistent(M)                                      11567
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")                           11568
for row in M:                                                            11569
    print("    " + "  ".join(f"{v:8.4f}" for v in row))                  11570
print("  权重：", "  ".join(f"{v:.4f}" for v in r2["weights"]),          11571
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))            11572
print("  最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"              11573
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],                       11574
    bad["expected"]))                                                    11575
# ------------------------------------------------------------------     11576
    ----------------                                                     11577
# 3） 多专家聚合（几何平均）                                             11578
# ------------------------------------------------------------------     11579
    ----------------                                                     11580
print()                                                                  11581
print(LINE)                                                              11582
print("测试 3  多专家判断矩阵聚合（几何平均）")                           11583
print(LINE)                                                              11584
m1 = ahp.matrix_from_upper([2, 4, 3])                                    11585
m2 = ahp.matrix_from_upper([3, 5, 2])                                    11586
agg = ahp.aggregate_matrices([m1, m2])                                   11587
print("  专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")                   11588
print("  聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"                       11589
      % (agg[0, 1], agg[0, 2], agg[1, 2]))                               11590
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"            11591
# ------------------------------------------------------------------     11592
    ----------------                                                     11593
# 4） 论文表 3 权重可还原性                                              11594
# ------------------------------------------------------------------     11595
    ----------------                                                     11596
print()                                                                  11597
print(LINE)                                                              11598
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")            11599
print(LINE)                                                              11600
```

```
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f    CR = %.3e    一致: %s" %
    (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ------------------------------------------------------------------
    ----------------
# 5) 应用可启动 + 建表
# ------------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表:" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检:用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录,先把仓库根目录加入模块搜索路径,
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------
    ----------------
# 论文第 5 节案例数据(表 5:占比与各单元实测值逐字抄录)
# ------------------------------------------------------------------
    ----------------
# 三类单元面积占比:平台 26.3%、碎石边坡 69.5%、岩质边坡 4.2%
# 注:岩质边坡无土壤指标、平台无地形指标,对应"不适用"。
case = [
    {
```

```
            "unit_type": "岩质边坡",
            "ratio": 0.042,
            "data": {
                "coverage": 68.88, "shannon": 0.51,
                "slope": 63.33, "aspect": "阴阳坡",
            },
        },
        {
            "unit_type": "碎石边坡",
            "ratio": 0.695,
            "data": {
                "soil_ratio": 0.34, "thickness": 10, "ph":
        9.24, "organic_matter": 4.89,
                "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
        12.60,
                                 "Cu": 11.70, "Cd": 0.04, "Pb":
        6.50},
                "coverage": 87.92, "shannon": 0.60,
                "slope": 28.75, "aspect": "阳坡",
            },
        },
        {
            "unit_type": "平台",
            "ratio": 0.263,
            "data": {
                "soil_ratio": 0.56, "thickness": 30, "ph":
        8.72, "organic_matter": 10.92,
                "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
        12.33,
                                 "Cu": 11.14, "Cd": 0.09, "Pb":
        9.98},
                "coverage": 58.42, "shannon": 0.72,
            },
        },
]
# ------------------------------------------------------------------
    ------------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
```

```
            print("      %s 得分 = %.4f" % (l1, score))                    11701
        print("      单元评价因子 F' = %.4f" % r["unit_score"])            11702
print()                                                                   11703
print(LINE)                                                               11704
print("面积加权合成（式(9)）")                                              11705
print(LINE)                                                               11706
res = ev.score_mine(case)                                                 11707
total = res["final_score"]                                                11708
print()                                                                   11709
for ur in res["unit_results"]:                                            11710
    print("  %s  %.2f%% × %.4f = %.4f"                                    11711
          % (ur["unit_type"], ur["ratio"] * 100,                          11712
    ur["unit_score"], ur["weighted"]))                                    11713
print("  " + "-" * 60)                                                   11714
print("  最终评价因子 F = %.4f" % total)                                   11715
print()                                                                   11716
print("  适宜性等级：%s" % res["grade"])                                    11717
print("  判定依据  ：%s" % res["grade_info"]["meaning"])                    11718
print("  推荐措施  ：%s" % res["grade_info"]["measure"])                    11719
print()                                                                   11720
print(LINE)                                                               11721
print("结论")                                                             11722
print(LINE)                                                               11723
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")                       11724
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")11725
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"    11726
      % (total, res["grade"]))                                            11727
print("  - 论文表 5 载 0.77、正文载 0.76，而按表 5 单元得分×占比加权为 0.7614；")11728
print("    系统差异集中在坡度、坡向的标准化口径（standardization.std_slope   11729
    可配置）")                                                            11730
print(LINE)                                                               11731
#  ==== 测试/test_web_ahp.py ====                                         11732
#  -*- coding: utf-8 -*-                                                  11733
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""                          11734
import sys                                                                11735
sys.stdout.reconfigure(encoding="utf-8")                                  11736
import os                                                                 11737
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                  11738
#  使 `from app import ...` 与 app/app.db 的定位不受影响。                 11739
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__11740
    e__)))                                                                11741
if ROOT not in sys.path:                                                  11742
    sys.path.insert(0, ROOT)                                              11743
from app import create_app                                                11744
LINE = "=" * 66                                                           11745
#  测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）             11746
class TestConfig:                                                         11747
    SECRET_KEY = "test"                                                   11748
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"                        11749
    SQLALCHEMY_TRACK_MODIFICATIONS = False                                11750
```

```
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
```

```
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
```

```
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
      r.headers.get("Location"))
assert r.status_code == 302 and "login" in
      r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
      "password": "admin123"},
                  follow_redirects=True)
body = r.get_data(as_text=True)
#  工作台:页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含账户 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码:", r.status_code)
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
#  新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
      "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True
      ))
assert "测试项目A" in r.get_data(as_text=True)
#  清理测试数据,避免污染演示库
with app.app_context():
      from app.models import db, Project
      from app.services import cleanup
      for p in Project.query.filter_by(name="测试项目A").all():
            cleanup.delete_project(p)              #  级联清子表,避免留孤儿
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  微地质单元(面积占比自动计算)")
print(LINE)
from app.models import MinePatch
with app.app_context():
      patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
      irst()
```

```
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元:", all(t in body for t in ("岩质边坡",
    "碎石边坡", "平台")))
print("  含占比合计:", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5  标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准:", all(p in body for p in ("辽宁", "湖南",
    "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组:", "植被覆盖度" in body)
print("  含多省份并列:", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6  限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  含四项条件:", all(x in body for x in ("政策约束", "区位约束",
    "土壤污染", "地质灾害")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
    "地质灾害"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  全通过后显示「通过，可进入定量评价」:", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
```

```
#  恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过：", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
#  ==== 测试/test_web_evaluate.py ====
#  -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
#  与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
#  登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
```

```
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡").first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.id.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
```

```
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
    print("         逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
```

```
print(LINE)
r = client.get("/report/")
print("   状态码：", r.status_code)
body = r.get_data(as_text=True)
print("   含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2   得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("   状态码：", r.status_code)
print("   含「得分明细」：", "得分明细" in body)
print("   含「最终评价因子 F」：", "最终评价因子 F" in body)
print("   含二级指标明细（植被覆盖度 / 坡度 / 坡向）：", all(n in body for n in
    ("植被覆盖度", "坡度", "坡向")))
print("   含权重 / 标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3   Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("   状态码：", r.status_code)
print("   Content-Type:", r.headers.get("Content-Type",
    "")[:60])
data = r.data
print("   文件大小：", len(data), "字节")
print("   是否为有效 docx（PK 头）：", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("   含报告标题：", "自然恢复适宜性评价报告" in text)
print("   含评价结论章节：", "评价结论" in text)
print("   含得分明细章节：", "得分明细" in text)
print("   段落数：", len(d.paragraphs), " 表格数：", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import io
import os
```

```
import re
import sys
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timedelta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
                    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
              in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
      else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
      "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
```

```
      "[不符合]"))                                                    12201
assert rt, "未下发 remember_token"                                    12202
exp = re.search(r"Expires=([^;]+)", rt[0])                            12203
print("   %s 该 cookie 带过期时间：%s"                                 12204
        % ("[符合]" if exp else "[不符合]", exp.group(1) if exp       12205
      else "无（会话 cookie）"))                                      12206
assert exp, "remember_token 无过期时间，关浏览器即失效"               12207
#  3） 不勾选时不应下发 remember_token                                 12208
c2 = app.test_client()                                                12209
r2 = c2.post("/login", data={"username": "admin",                     12210
      "password": "admin123"},                                        12211
                 follow_redirects=False)                              12212
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if                 12213
      h.startswith("remember_token=")]                                12214
print("   %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2    12215
      else "[不符合]"))                                               12216
assert not rt2                                                        12217
#  4） 携带 remember_token 的新会话应自动登录                          12218
c3 = app.test_client()                                                12219
c3.set_cookie("remember_token",                                       12220
                  rt[0].split(";")[0].split("=", 1)[1],               12221
      domain="localhost")                                             12222
r3 = c3.get("/", follow_redirects=False)                              12223
print("   %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"     12224
        % ("[符合]" if r3.status_code == 200 else "[不符合]",         12225
      r3.status_code))                                                12226
assert r3.status_code == 200                                          12227
#  ------------------------------------------------------------------12228
      ------------------                                              12229
print()                                                               12230
print(LINE)                                                           12231
print("二、删除功能")                                                 12232
print(LINE)                                                           12233
client.post("/login", data={"username": "admin",                      12234
      "password": "admin123"})                                        12235
#  项目：列表页须有删除按钮                                           12236
html = client.get("/project/").get_data(as_text=True)                 12237
has_del = "/delete" in html and "btn--danger" in html                 12238
print("   %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))12239
assert has_del, "项目列表缺少删除按钮"                                12240
print("   %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else 12241
      "[不符合]"))                                                    12242
assert "confirm(" in html                                             12243
#  图斑列表：地址用的是项目主键，不是图斑主键                         12244
with app.app_context():                                               12245
      from app.models import MinePatch, Project                       12246
      import seed_demo                                                12247
      patch = seed_demo.demo_patch()                                  12248
      patch_id = patch.id                                             12249
      project_id = patch.project_id                                   12250
```

```
html = client.get("/project/%d/patches" %
    project_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
    html and "btn--danger" in html else "[不符合]"))
assert "/project/patch/" in html and "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
    ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("六、顶栏宽度在各页面保持一致")
print(LINE)
# 滚动条出现在 .app-content 内部：内容高的页面会因此把顶栏挤窄约 10px，
# 于是「顶栏在不同页面不一样宽」。scrollbar-gutter: stable 预留槽位后各页一致。
base_src = io.open(os.path.join(ROOT, "app", "templates",
    "base.html"),
                   encoding="utf-8").read()
has_gutter = "scrollbar-gutter: stable" in base_src
print("  %s 内容区预留滚动条槽位（顶栏宽度不再随页面变化）"
      % ("[符合]" if has_gutter else "[不符合]"))
assert has_gutter, "缺少 scrollbar-gutter:
    stable，顶栏宽度会因滚动条而不一致"
```

```
# 搜索框宽度不随窗口大小变：原来 960px 断点把它缩到 180px，同一台机器换窗口
# 大小看着就像两个尺寸。统一保持 268px（640px 以下才改为铺满，属手机版式）。
no_shrink = "width: 180px" not in base_src
print("  %s 搜索框宽度在窄屏断点下不再缩小"
      % ("[符合]" if no_shrink else "[不符合]"))
assert no_shrink, "搜索框在 960px 断点被缩成 180px，宽度会随窗口变化"
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
```

```
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/profile?tab=logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 账号编辑 / 删除（职务只有管理员与操作员两种）
with app.app_context():
    u = User(username="tmp_user", role="operator",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "账号编辑页可访问")
# 保护校验要趁系统里只有一名管理员时做：
# 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
with app.app_context():
```

```
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id,
       data={"role": "operator", "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/edit" % uid, data={"role": "admin",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "admin",
    "职务已修改并保存")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(db.session.get(User, uid) is None, "账号删除生效")
# 清理日志核对用的项目
with app.app_context():
    from app.services import cleanup
    for p in Project.query.filter_by(name="日志核对项目").all():
        cleanup.delete_project(p)              # 级联清子表，避免留孤儿
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
```

```
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
       follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ----------------------------------------------------------------
    ----------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按系统默认值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
```

```
    "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
     body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
     else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
     0.01,
    "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
     vals[0] if vals else "")
# ------------------------------------------------------------
     ------------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
     "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
     ,坡向\n"
     "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
     87.92,0.60,28.75,阳坡\n"
     "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
     8.42,0.72,,,\n"
     "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
     f-8")), "d.csv")},
           content_type="multipart/form-data",
     follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
     u = MicroUnit.query.filter_by(patch_id=patch_id,
     unit_type="碎石边坡").first()
     ind = IndicatorData.query.filter_by(unit_id=u.id).first(
     )
     ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
     abs(ind.coverage - 87.92) < 1e-6,
         "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
```

```
        follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------
        ------------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
        "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
        "0.4000")])
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")
#  报告是评价结论的呈现，需先真实执行一次评价才能打开
with app.app_context():
        from app.views.evaluate import load_patch_input,
        save_evaluation
        save_evaluation(patch_id, load_patch_input(patch_id))
r = c.get("/report/%d" % patch_id)
body = r.get_data(as_text=True)
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %
        body.count("<svg"))
exe = browser.browser_path()
if exe is None:
        print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
        r = c.get("/report/%d/pdf" % patch_id)
        data = r.data
        ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
        len(data))
        d = fitz.open(stream=data, filetype="pdf")
        txt = "".join(d[i].get_text() for i in
        range(d.page_count))
        ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
        ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
        ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
        d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
#  ==== 测试/test_workflow.py ====
#  -*- coding: utf-8 -*-
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
```

```
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于  测试/  子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit, IndicatorData,
                        ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password": "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]", label, extra))
    if not flag:
        _fails.append(label)
def purge_fixtures():
    """开跑前先清掉上一次可能残留的自检项目。
    自检项目按 code 前缀命名（WF-）；如果上一次跑到一半被打断（报错、被中断），
    清理函数没执行到，残留的项目就会出现在工作台与项目列表里 ——
    生成软著截图时这些测试数据会被拍进去。
    """
    with app.app_context():
        from app.services import cleanup as cleanup_svc
        leftovers = Project.query.filter(Project.code.like("WF-%")).all()
        for p in leftovers:
            cleanup_svc.delete_project(p)  # 级联清掉图斑下面的子表，避免留孤儿
        return len(leftovers)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
```

```
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    """删掉这套自检项目。
    原先只手工删了得分明细 / 评价记录 / 判定 / 单元 / 指标数据，图斑下面
    其余子表（参照生态系统等）会留孤儿；改用 cleanup 服务统一级联。
    """
    with app.app_context():
        from app.services import cleanup as cleanup_svc
        if db.session.get(MinePatch, pid):
            cleanup_svc.delete_patch(pid)
        pr = Project.query.filter_by(name="门禁-" +
    tag).first()
        if pr:
            cleanup_svc.delete_project(pr)
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
```

```
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ------------------------------------------------------------
    ----------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
print("    （开跑前清掉 %d 个上一次残留的自检项目）" % purge_fixtures())
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
```

```
ok(bool(blocked) and "未通过" in blocked[0]["label"],
    "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
     "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
    "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
     passed=True)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
    "结果数 %d" % n)
with app.app_context():
     pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
    "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
    "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 １００%")
pid = new_patch("5"); add_constraint(pid, passed=True)
with app.app_context():
     u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
     ratio=0.5)
     db.session.add(u); db.session.commit()
     db.session.add(IndicatorData(unit_id=u.id,
     soil_ratio=0.3, thickness=10,
                                  ph=8.0,
     organic_matter=5.0, coverage=80.0,
                                  shannon=0.6, slope=20.0,
     aspect="阳坡"))
     db.session.commit()
r = post_eval(pid)
ok(results_of(pid) == 0, "占比不足 １００% → 不产生评价结果")
cleanup(pid, "5")
print()
print(LINE)
print("二、前置条件齐备时才放行")
print(LINE)
pid = new_patch("OK"); add_units(pid); add_constraint(pid,
     passed=True); add_data(pid)
with app.app_context():
     pr = wf.patch_progress(pid)
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")
r = post_eval(pid)
n = results_of(pid)
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)
```

```
print("   重定向到:", r.headers.get("Location"))
print()
print(LINE)
print("三、报告不得先于评价产生")
print(LINE)
pid2 = new_patch("R")     # 什么都不做
r = get_report(pid2)
ok(r.status_code == 302 and "/report/" ==
     (r.headers.get("Location") or ""),
    "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
     rr = c.get("/report/%d%s" % (pid2, suffix),
     follow_redirects=False)
     ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
     %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
# 报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
# 也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
     add_data(pid3)
with app.app_context():
     pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
     "evaluate",
    "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
    "直接访问 /report/<id> 被退回，且不产生任何评价结果",
    "HTTP %d" % r.status_code)
# 真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
     pr = wf.report_progress(pid3)
     rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
     rr.status_code)
ok(("%.4f" % rec.final_score) in body,
    "报告数值取自评价记录，而非重新计算",
    "F=%.4f" % rec.final_score)
```

```
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
# 回归保护：seed_demo.clean_case() 曾执行 Project.query.delete()，
# 导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
# 演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    from app.services import cleanup as cleanup_svc
    if db.session.get(MinePatch, keep_patch_id):
        cleanup_svc.delete_patch(keep_patch_id)        # 级联清子表
    p = db.session.get(Project, keep_pid)
    if p:
        cleanup_svc.delete_project(p)
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
# 论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
# 允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
```

```
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
            follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
            follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
        follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first()
    ).id
r = c.post("/unit/edit/%d" % uid,
            data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
```

```
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fi
    st()
```

```
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
            pollution_pass=True, geo_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        return patch.id
def purge_fixtures():
    """开跑前清掉上一次可能残留的自检项目（code 前缀 WEIGHT-）。
    上一次跑到一半被打断时清理函数执行不到，残留项目会出现在工作台与
    项目列表里，生成软著截图时会被拍进去。
    """
    with app.app_context():
        from app.models import Project, ReferenceEcosystem
        leftovers = Project.query.filter(Project.code.like("
    WEIGHT-%")).all()
        for p in leftovers:
            from app.services import cleanup as cleanup_svc
            cleanup_svc.delete_project(p)    # 级联清子表，避免留孤儿
        return len(leftovers)
def cleanup(pid):
    """删掉这套自检项目（级联清掉图斑下面的子表）。"""
    with app.app_context():
        from app.services import cleanup as cleanup_svc
        if db.session.get(MinePatch, pid):
            cleanup_svc.delete_patch(pid)
        p = Project.query.filter_by(code="WEIGHT-%s" %
    TAG).first()
        if p:
            cleanup_svc.delete_project(p)
def run_eval(pid):
    c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ) \
            .order_by(EvaluationResult.id.desc()).first().fi
    nal_score
print(LINE)
print("一、默认应使用系统内置权重")
print(LINE)
print("  （开跑前清掉 %d 个上一次残留的自检项目）" % purge_fixtures())
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
```

```
    w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "系统默认权重" in label,
   "未配置时权重来源为系统默认权重", label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自系统内置权重",
   "土壤指标 %.2f" % w["l1"]["土壤指标"])
pid = build_case()
base_score = run_eval(pid)
print("  默认权重下的评价值 F = %.4f" % base_score)
print()
print(LINE)
print("二、保存 AHP 算出的权重后，评价应改用新权重")
print(LINE)
with app.app_context():
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",
    0.20),
                                ("terrain", 0.10)])
    n_items = WeightItem.query.count()
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)
with app.app_context():
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",
    label2)
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,
   "未覆盖的二级权重仍保留系统内置取值",
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])
new_score = run_eval(pid)
print("  新权重下的评价值 F = %.4f" % new_score)
ok(abs(new_score - base_score) > 1e-4,
   "权重变化确实改变了评价结果",
   "%.4f → %.4f" % (base_score, new_score))
print()
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
        .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
   "评价记录中标注了权重来源",
   stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到系统默认权重")
print(LINE)
```

```
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "系统默认权重" in label3,
   "已恢复为系统默认权重", label3)
back_score = run_eval(pid)
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
    base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
```

```
# ==== app/services/cleanup.py ====
# -*- coding: utf-8 -*-
"""级联清理：删图斑、删项目时，把挂在它下面的数据一并清掉。
为什么单独抽出来：`Project.patches` 与 `MinePatch.units` 上有 ORM 级联，
但图斑下面的**指标数据、限制性判定、评价结果、得分明细、参照生态系统**
都没有级联关系。原先只有「删除图斑」那条路径里手写了清理，
于是「删除项目」会留下一批孤儿行——自检实测：删一个项目后，
上面五张表各残留 1 行（项目与图斑本身都删掉了，数据却还在库里）。
两条删除路径都调用这里，才不会漏。
"""
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail,
    ConstraintCheck,
                        ReferenceEcosystem)
from sqlalchemy import text
# 孤儿判定：子表的某个外键指向了已经不存在的父记录。
# （表名， 孤儿条件， 说明）
# 这套规则同时被三处使用：工具/清理孤儿数据.py（人工清理）、
# 演示数据重建（seed_demo）、最终验收（verify_all 跑完自检后核对）。
ORPHAN_RULES = [
    ("indicator_data", "unit_id NOT IN (SELECT id FROM
    micro_units)",
     "指标数据挂在已删除的微地质单元下"),
    ("constraint_checks", "patch_id NOT IN (SELECT id FROM
    mine_patches)",
     "限制性判定挂在已删除的图斑下"),
    ("reference_ecosystems", "patch_id NOT IN (SELECT id
    FROM mine_patches)",
     "参照生态系统挂在已删除的图斑下"),
    ("evaluation_results", "patch_id NOT IN (SELECT id FROM
    mine_patches)",
     "评价结果挂在已删除的图斑下"),
    ("score_details", "result_id NOT IN (SELECT id FROM
    evaluation_results)",
     "得分明细挂在已删除的评价结果下"),
    ("micro_units", "patch_id NOT IN (SELECT id FROM
    mine_patches)",
     "微地质单元挂在已删除的图斑下"),
    ("standard_rules", "standard_id NOT IN (SELECT id FROM
    standards)",
     "标准规则挂在已删除的标准下"),
    ("expert_matrices", "scheme_id NOT IN (SELECT id FROM
    weight_schemes)",
     "判断矩阵挂在已删除的权重方案下"),
    ("weight_items", "scheme_id NOT IN (SELECT id FROM
    weight_schemes)",
     "权重项挂在已删除的权重方案下"),
]
def purge_orphans(apply=False):
```

```
    """统计（并可选删除）库里残留的孤儿行。
    为什么需要：删图斑、删项目如果绕过 delete_patch_children，
    图斑下面那几张表就会留下孤儿；自检夹具若被中断也会留下。
    返回 [（表名， 条数， 说明）]，调用方据此打印或断言。
    """
    report = []
    for table, cond, note in ORPHAN_RULES:
        try:
            n = db.session.execute(
                text("SELECT COUNT(*) FROM %s WHERE %s" %
    (table, cond))).scalar()
        except Exception:                                  #
    表还不存在（旧库）时跳过
            db.session.rollback()
            continue
        report.append((table, n, note))
        if apply and n:
            db.session.execute(text("DELETE FROM %s WHERE
    %s" % (table, cond)))
    if apply:
        db.session.commit()
    return report
def delete_patch_children(patch_id):
    """清掉挂在某个图斑下的全部附属数据（不含图斑本身）。
    返回各表清理条数，便于调用方打印或断言。
    """
    counts = {}
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patc
    h_id=patch_id).all()]
    if unit_ids:
        counts["indicator_data"] = IndicatorData.query.filte
    r(
            IndicatorData.unit_id.in_(unit_ids)).delete(synch
    ronize_session=False)
    result_ids = [r.id for r in
                  EvaluationResult.query.filter_by(patch_id=
    patch_id).all()]
    if result_ids:
        counts["score_details"] = ScoreDetail.query.filter(
            ScoreDetail.result_id.in_(result_ids)).delete(sy
    nchronize_session=False)
    counts["evaluation_results"] = EvaluationResult.query.fi
    lter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["constraint_checks"] = ConstraintCheck.query.filt
    er_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["micro_units"] = MicroUnit.query.filter_by(
        patch_id=patch_id).delete(synchronize_session=False)
    counts["reference_ecosystems"] = ReferenceEcosystem.quer
```

```
    y.filter_by(                                                    13251
        patch_id=patch_id).delete(synchronize_session=False)        13252
    return counts                                                   13253
def delete_patch(patch_id):                                         13254
    """删除一个图斑及其全部附属数据。图斑不存在时返回 (None, {})。"""          13255
    patch = db.session.get(MinePatch, patch_id)                     13256
    if patch is None:                                               13257
        return None, {}                                             13258
    counts = delete_patch_children(patch_id)                        13259
    db.session.delete(patch)                                        13260
    db.session.commit()                                             13261
    return patch, counts                                            13262
def delete_project(project):                                        13263
    """删除一个项目：先清掉每个图斑的附属数据，再删项目本身。                    13264
    项目下的图斑由 `Project.patches` 的 ORM 级联删除，但图斑下面那五张表        13265
    必须先手工清，否则会留孤儿。                                          13266
    """                                                             13267
    cleaned = 0                                                     13268
    for patch in project.patches.all():                             13269
        counts = delete_patch_children(patch.id)                    13270
        cleaned += sum(counts.values())                             13271
    name = project.name                                             13272
    db.session.delete(project)                                      13273
    db.session.commit()                                             13274
    return name, cleaned                                            13275
#  ==== app/services/standard_check.py ====                         13276
#  -*- coding: utf-8 -*-                                           13277
"""评价结论与地方标准 / 政策文件的对照。                                    13278
标准知识库只作文件储备与比对，**不参与评价因子 F 的计算**：评价仍按内置的          13279
指标体系与权重算，算完之后再拿结论去对照文件里的判定项，回答                      13280
「这个结论是否符合当地文件的要求」。                                       13281
对照结果有三种，映射不上的项如实标为「待人工判断」，不由系统替文件下结论：          13282
    符合          —— 系统里有对应数据，且满足文件里的判定条件                 13283
    不符合        —— 系统里有对应数据，但不满足                           13284
    待人工判断    —— 文件里的定性要求，系统内没有对应数据（如「地形地貌与周边协调」）   13285
对照用到的数据全部来自系统已有记录：限制性条件判定结果、各微地质单元实测值、          13286
参照生态系统基准、最近一次评价记录。                                      13287
"""                                                                 13288
import re                                                           13289
from app.models import (db, MinePatch, MicroUnit,                   13290
    IndicatorData, ConstraintCheck,                                 13291
                          Standard, StandardRule,                   13292
    ReferenceEcosystem, EvaluationResult)                           13293
OK = "符合"                                                         13294
BAD = "不符合"                                                      13295
MANUAL = "待人工判断"                                               13296
#  结论 → 页面上的标签样式（沿用既有的 tag 样式，不新增样式）                 13297
TAG_CLASS = {OK: "tag--ok", BAD: "tag--warn", MANUAL:               13298
    "tag--muted"}                                                   13299
def province_of(region):                                            13300
```

```
    """从行政区里认出已收录的省份：「河北省唐山市玉田县」→「河北」。
    库里没有这个省份的标准时返回 None，由调用方给出「未收录」的提示。
    """
    if not region:
        return None
    provinces = {p for (p,) in db.session.query(Standard.province).distinct()
                 if p}
    for p in sorted(provinces, key=len, reverse=True):
        if p in region:
            return p
    return None
def _num(text):
    if text is None:
        return None
    m = re.search(r"-?\d+(?:\.\d+)?", str(text))
    return float(m.group(0)) if m else None
def _cmp(value, op, threshold):
    """按比较符判断；阈值不是数字或取值缺失时返回 None（无法判断）。"""
    t = _num(threshold)
    if value is None or t is None:
        return None
    try:
        v = float(value)
    except (TypeError, ValueError):
        return None
    if op == ">=":
        return v >= t
    if op == ">":
        return v > t
    if op == "<=":
        return v <= t
    if op == "<":
        return v < t
    if op == "==":
        return abs(v - t) < 1e-9
    return None
def _coverage_stats(patch_id):
    """各单元植被覆盖度的面积加权值与最低值（只统计录了实测值的单元）。"""
    rows = (db.session.query(MicroUnit, IndicatorData)
            .outerjoin(IndicatorData, IndicatorData.unit_id
    == MicroUnit.id)
            .filter(MicroUnit.patch_id == patch_id).all())
    values, num, den = [], 0.0, 0.0
    for unit, data in rows:
        cov = getattr(data, "coverage", None) if data is
    not None else None
        if cov is None:
            continue
        values.append(cov)
```

```
            w = unit.ratio or 0.0                                                13351
            num += cov * w                                                       13352
            den += w                                                             13353
        if not values:                                                           13354
            return None, None                                                    13355
        return (num / den if den else None), min(values)                         13356
def _context(patch):                                                             13357
    """把对照要用到的数据一次性取出来。"""                                                  13358
    check = (ConstraintCheck.query.filter_by(patch_id=patch.id)                  13359
                .order_by(ConstraintCheck.id.desc()).first())                    13360
    ref = ReferenceEcosystem.query.filter_by(patch_id=patch.id).first()          13361
    result = (EvaluationResult.query.filter_by(patch_id=patch.id)                13362
                .order_by(EvaluationResult.id.desc()).first())                   13363
    weighted, lowest = _coverage_stats(patch.id)                                 13364
    return {                                                                     13365
        "patch": patch,                                                          13366
        "constraint": check,                                                     13367
        "benchmark": getattr(ref, "coverage_benchmark", None) if ref else None,  13368
        "has_reference": ref is not None,                                        13369
        "result": result,                                                        13370
        "grade": getattr(result, "grade", None) if result else None,             13371
        "score": getattr(result, "final_score", None) if result else None,       13372
        "coverage": weighted,                                                    13373
        "coverage_min": lowest,                                                  13374
    }                                                                            13375
def _judge(item, op, threshold, ctx):                                            13376
    """把一条文件规则映射到系统数据，返回 (结论, 依据说明)。"""                                    13377
    name = item or ""                                                            13378
    cov, cov_min = ctx["coverage"], ctx["coverage_min"]                          13379
    check = ctx["constraint"]                                                    13380
    def cov_text():                                                              13381
        if cov is None:                                                          13382
            return "尚未录入植被覆盖度"                                                 13383
        return "面积加权覆盖度 %.2f%%（最低单元 %.2f%%）" % (cov, cov_min)                 13384
    # ---- 植被覆盖度（含趋势）----                                                   13385
    if "趋势" in name:                                                           13386
        return MANUAL, "需连续多年的监测数据，系统内没有时间序列"                                13387
    if "植被覆盖" in name:                                                         13388
        if cov is None:                                                          13389
            return MANUAL, cov_text()                                            13390
        r = _cmp(cov, op, threshold)                                             13391
        if r is None:                                                            13392
            return MANUAL, cov_text() + "；文件给出的条件为「%s %s」"                    13393
```

```
    % (op, threshold)
        return (OK if r else BAD), cov_text()
    # ---- 参照生态系统 ----
    if "参照生态系统调查" in name or "参照系调查" in name:
        return ((OK, "已录入参照生态系统的植被覆盖度与生物多样性基准")
                if ctx["has_reference"] else
                (MANUAL, "尚未录入参照生态系统基准"))
    if "参照系" in name:
        if cov is None or ctx["benchmark"] is None:
            return MANUAL, "缺参照生态系统基准或覆盖度实测值"
        gap = (ctx["benchmark"] - cov)
        gap = 0.0 if gap < 0 else gap
        r = _cmp(gap, op, threshold)
        if r is None:
            return MANUAL, "与参照系差距 %.2f 个百分点" % gap
        return (OK if r else BAD), "与参照系差距 %.2f 个百分点（基准
    %.2f%%）" % (
            gap, ctx["benchmark"])
    # ---- 污染与地质灾害 ----
    if "污染" in name:
        if check is None or check.pollution_pass is None:
            return MANUAL, "尚未完成限制性条件判定"
        return ((OK, "限制性判定：土壤污染一项通过") if
    check.pollution_pass else
                (BAD, "限制性判定：土壤污染一项未通过"))
    if "地质灾害" in name:
        if check is None or check.geo_pass is None:
            return MANUAL, "尚未完成限制性条件判定"
        if not check.geo_pass:
            return BAD, "限制性判定：存在地质灾害"
        if "发育程度" in name:
            return MANUAL, "系统只记录「有无地质灾害」，未记录发育程度分级"
        return OK, "限制性判定：地质灾害一项通过"
    # ---- 责任主体 ----
    if "责任主体" in name:
        missing = ctx["patch"].responsible_entity_missing
        if missing is None:
            return MANUAL, "图斑档案未填写责任主体是否缺失"
        if missing:
            return OK, "图斑档案记录责任主体缺失，按文件属政府主导治理"
        if "政府" in name:
            return MANUAL, "责任主体未缺失，是否符合「政府」要求需人工核实"
        return OK, "图斑档案记录责任主体明确"
    # ---- 适宜性分级 / 指数 ----
    if "适宜性分级" in name or "适宜性评价" in name:
        # 文件只要求「做过自然恢复适宜性分级」，系统已经给出等级
        if ctx["grade"] is None:
            return MANUAL, "尚未执行定量评价"
        return OK, "系统已给出适宜性等级：%s（F=%.4f）" % (ctx["grade"],
    ctx["score"])
```

```
    if "适宜性指数" in name:
        # 指数是要跟阈值比的：文件没给具体阈值时不能替它下结论
        if ctx["grade"] is None:
            return MANUAL, "尚未执行定量评价"
        t = _num(threshold)
        if t is None:
            return MANUAL, ("文件未给出具体阈值，系统评价因子
F=%.4f（%s），需人工比对"
                            % (ctx["score"], ctx["grade"]))
        r = _cmp(ctx["score"], op, t)
        if r is None:
            return MANUAL, "系统评价因子 F=%.4f" % ctx["score"]
        return (OK if r else BAD), ("系统评价因子 F=%.4f，文件要求 %s
%s"
                                    % (ctx["score"], op,
threshold))
    # ---- 其它定性要求 ----
    return MANUAL, "系统内无对应数据，需人工核实"
def compare(patch_id):
    """生成某图斑的评价结论与当地标准 / 政策文件的对照结果。"""
    patch = db.session.get(MinePatch, patch_id)
    if patch is None:
        return None
    province = province_of(patch.admin_region)
    out = {
        "patch": patch,
        "region": patch.admin_region or "",
        "province": province,
        "covered": False,
        "message": "",
        "standards": [],
        "totals": {OK: 0, BAD: 0, MANUAL: 0},
        "necessary": {"符合": 0, "共计": 0},
    }
    if province is None:
        out["message"] = ("未收录该省份的地方标准与政策文件"
                          "（图斑所在行政区：%s）" %
(patch.admin_region or "未填写"))
        return out
    standards = (Standard.query.filter_by(province=province)
                 .order_by(Standard.category.desc(),
Standard.name).all())
    if not standards:
        out["message"] = "未收录该省份的地方标准与政策文件（%s）" % province
        return out
    ctx = _context(patch)
    out["covered"] = True
    for s in standards:
        rules = (StandardRule.query.filter_by(standard_id=s.
id)
```

```
                    .order_by(StandardRule.id).all())
        rows = []
        for r in rules:
            verdict, evidence = _judge(r.item, r.operator,
    r.threshold, ctx)
            rows.append({
                "item": r.item, "operator": r.operator,
    "threshold": r.threshold,
                "condition_type": r.condition_type or "",
    "region": r.region,
                "note": r.note, "verdict": verdict, "tag":
    TAG_CLASS[verdict],
                "evidence": evidence,
            })
            out["totals"][verdict] = out["totals"].get(verdi
    ct, 0) + 1
            if (r.condition_type or "") == "必要条件":
                out["necessary"]["共计"] += 1
                if verdict == OK:
                    out["necessary"]["符合"] += 1
        out["standards"].append({
            "name": s.name, "doc_number": s.doc_number,
    "category": s.category,
            "summary": s.summary, "rules": rows,
            "counts": {k: sum(1 for x in rows if
    x["verdict"] == k)
                       for k in (OK, BAD, MANUAL)},
        })
    return out
# ==== app/services/unit_config.py ====
# -*- coding: utf-8 -*-
"""标准化参数与必填项：让界面上能改的东西真正参与计算。
以前有个说不通的地方：图斑的「参照生态系统基准」、指标定义里的「理想取值」
和「启用单元」在界面上都能改、也能落库，**但计算链只读代码常量**
（`app/utils/constants.py`），从不查这两张表。于是报告里写着「基准 60%」，
实际按常量 75% 算，同一份报告两个口径。
现在统一从这里取：
- 基准值（植被覆盖度、Shannon）来自图斑的 `reference_ecosystems`；
- 各指标的理想取值来自 `indicators.ideal_value`（能解析成数字的才覆盖，
  像「5~9」「25° 分界」这种区间/规则型仍按代码里的算法处理）；
- 启用单元来自 `indicators.enabled_units`（岩质边坡无土壤、平台无地形）。
库里没有行、或值不是数字时一律回落到代码常量 —— 所以历史数据与
`验证数据/` 里的 8 个算例结果不会因为这次改动而变化。
"""
from app.models import Indicator, ReferenceEcosystem
from app.utils.constants import (
    UNIT_TYPES, IDEAL_SOIL_RATIO, IDEAL_THICKNESS,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METALS,
```

```
)
#  二级指标编码 → 标准化参数名（同时也是 IndicatorData 的字段名）
NUMERIC_IDEALS = {
    "soil_ratio": ("soil_ratio", IDEAL_SOIL_RATIO),
    "thickness": ("thickness", IDEAL_THICKNESS),
    "organic_matter": ("organic_matter", IDEAL_ORGANIC),
    "coverage": ("coverage", IDEAL_COVERAGE),
    "shannon": ("shannon", IDEAL_SHANNON),
}
#  重金属编码 → 金属名
METAL_CODES = {"hm_" + m: m for m in HEAVY_METALS}
#  必填的实测字段（污染指标按方法约定允许「未检测」，故不在必填之列）
REQUIRED_CODES = ["soil_ratio", "thickness", "ph",
    "organic_matter",
                  "coverage", "shannon", "slope", "aspect"]
#  字段 → 界面上的中文名，用于提示「缺什么」
FIELD_LABELS = {
    "soil_ratio": "土石比", "thickness": "土层厚度", "ph": "pH",
    "organic_matter": "有机质", "coverage": "植被覆盖度",
    "shannon": "生物多样性（Shannon）", "slope": "坡度", "aspect":
    "坡向",
}
def _numeric(text):
    """把理想取值的文本解析成数字；区间、规则型返回 None。"""
    if text is None:
        return None
    try:
        return float(str(text).strip())
    except (TypeError, ValueError):
        return None
class UnitSettings:
    """一个图斑上下文的标准化参数。空对象表示「全部按代码常量」。"""
    def __init__(self, params=None, enabled=None):
        self.params = params or {}              #  覆盖用的标准值
        self.enabled = enabled or {}            #  单元类型 →
    启用的二级指标编码集合
    def ideal(self, code, fallback):
        """取某指标的理想取值：库里有数字就用库里的，否则用常量。"""
        value = self.params.get(code)
        return fallback if value is None else value
    def group_enabled(self, unit_type, l1_name, l1_code):
        """该单元类型下某个一级指标是否启用（库里没配就按代码常量）。"""
        codes = self.enabled.get(unit_type)
        if codes is None:
            return None                         #  交给调用方按常量判断
        return l1_code in codes
    def enabled_codes(self, unit_type):
        """该单元类型启用的全部二级指标编码；库里没配返回 None。"""
        return self.enabled.get(unit_type)
    def required_codes(self, unit_type):
```

```
        """该单元类型必填的实测字段。
        按「启用的二级指标」推导：岩质边坡不启用土壤，平台不启用地形。
        库里没配该项时用代码常量里的结构性规则。
        """
        from app.utils.constants import
    UNIT_ENABLED_INDICATORS
        codes = self.enabled.get(unit_type)
        if codes is None:
            groups = UNIT_ENABLED_INDICATORS.get(unit_type,
    ())
            l2_by_group = {
                "土壤指标": ["soil_ratio", "thickness", "ph",
    "organic_matter"],
                "植被指标": ["coverage", "shannon"],
                "地形指标": ["slope", "aspect"],
            }
            codes = {c for g in groups for c in
    l2_by_group.get(g, [])}
        return [c for c in REQUIRED_CODES if c in codes]
    def missing_fields(self, unit_type, data):
        """返回该单元缺哪些必填项（中文名列表）。"""
        missing = []
        for code in self.required_codes(unit_type):
            value = data.get(code)
            if value is None or (isinstance(value, str) and
    value.strip() == ""):
                missing.append(FIELD_LABELS.get(code, code))
        return missing
    def undetected_metals(self, data):
        """按方法约定，未检测的重金属按「未受污染」计入；这里列出是哪些。"""
        measured = data.get("heavy_metals") or {}
        return [m for m in HEAVY_METALS if measured.get(m)
    is None]
def for_patch(patch_id):
    """读某个图斑的标准化参数；库里有什么就用什么，没有就用常量。
    优先级：图斑自己的参照生态系统基准  **高于**  指标定义里的全局理想取值  ——
    前者说的是「这个图斑的参照生态系统是什么样」，后者是「一般情况下的理想值」，
    越具体的越应该说了算。两者都优先于代码常量。
    """
    params = {}
    enabled = {}
    for ind in Indicator.query.all():
        code = ind.code
        if code in NUMERIC_IDEALS:
            value = _numeric(ind.ideal_value)
            if value is not None:
                params[code] = value
        elif code in METAL_CODES:
            #  重金属的「理想值」存的是限值，作为越小越优的极值使用
            value = _numeric(ind.ideal_value)
```

```
                if value is not None:                                  13651
                    params[code] = value                               13652
            units = (ind.enabled_units or "").strip()                  13653
            if units:                                                  13654
                for unit in UNIT_TYPES:                                13655
                    if unit in units:                                  13656
                        enabled.setdefault(unit,                       13657
        set()).add(code)                                               13658
        #  再用该图斑的参照生态系统基准覆盖：它是「这个图斑的参照生态系统」，                13659
        #  比指标定义里的全局理想取值更具体，应该它说了算。                                13660
        ref = ReferenceEcosystem.query.filter_by(patch_id=patch_       13661
        id).first()                                                    13662
        if ref is not None:                                            13663
            if ref.coverage_benchmark is not None:                     13664
                params["coverage"] = float(ref.coverage_benchmar       13665
        k)                                                             13666
            if ref.shannon_benchmark is not None:                      13667
                params["shannon"] = float(ref.shannon_benchmark)       13668
        return UnitSettings(params=params, enabled=enabled)            13669
def metal_limits(settings):                                            13670
        """重金属的极值表：库里有配置的用库里的，其余用常量。"""                          13671
        limits = dict(HEAVY_METAL_LIMIT)                               13672
        for code, metal in METAL_CODES.items():                        13673
            value = settings.params.get(code) if settings else         13674
        None                                                           13675
            if value is not None:                                      13676
                limits[metal] = value                                  13677
        return limits                                                  13678
#  ==== app/templates/standard/_check_block.html ====                  13679
{#  结论对照内容块：标准知识库对照页与评价结果页共用同一份渲染  #}                          13680
{% if check %}                                                         13681
<h3>与地方标准 / 政策文件的对照</h3>                                                13682
{% if not check.covered %}                                             13683
<p class="muted">{{ check.message }}。标准知识库只作文件储备与比对，不参与评价计算；          13684
  如需对照，请先在「标准知识库」中收录该省份的标准或政策文件。</p>                                13685
{% else %}                                                             13686
<p class="muted">                                                      13687
  图斑所在行政区：{{ check.region or "未填写" }}（对照 {{ check.province          13688
    }} 的标准与政策文件）。                                                      13689
  对照只是把文件里的判定项逐条映射到系统已有数据：共 {{ check.totals["符合"] +          13690
    check.totals["不符合"] + check.totals["待人工判断"] }} 条，                      13691
  符合 <strong>{{ check.totals["符合"] }}</strong> 条、不符合                    13692
    <strong>{{ check.totals["不符合"] }}</strong> 条、                           13693
  待人工判断 <strong>{{ check.totals["待人工判断"] }}</strong> 条。                 13694
  文件里的定性要求（如「地形地貌与周边协调」）系统没有对应数据，一律标为待人工判断，不代为结论。                13695
</p>                                                                   13696
{% for s in check.standards %}                                         13697
<h4 style="margin:14px 0 6px;">{{ s.name }}{% if                       13698
    s.doc_number %} <span class="muted mono">{{                        13699
    s.doc_number }}</span>{% endif %}</h4>                             13700
```

```
<table>
  <caption class="muted">{{ s.category }}｜符合 {{ s.counts["符合"] }} 条、不符合 {{ s.counts["不符合"] }} 条、待人工判断 {{ s.counts["待人工判断"] }} 条</caption>
  <thead>
    <tr>
      <th scope="col">文件判定项</th>
      <th scope="col">条件</th>
      <th scope="col">类型</th>
      <th scope="col">对照结果</th>
      <th scope="col">依据（系统内的对应数据）</th>
    </tr>
  </thead>
  <tbody>
    {% for r in s.rules %}
    <tr>
      <td style="text-align:left;">{{ r.item }}</td>
      <td class="mono nowrap">{{ r.operator }} {{ r.threshold or "" }}</td>
      <td>{{ r.condition_type or "—" }}</td>
      <td><span class="tag {{ r.tag }}">{{ r.verdict }}</span></td>
      <td style="text-align:left;">{{ r.evidence }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endfor %}
<p class="muted">
  对照结论仅供评价与报送时参考：符合全部必要条件的，说明评价结论与文件要求一致；
  出现「不符合」或「待人工判断」的，需由评价人员结合现场情况核定后再报送。
</p>
{% endif %}
{% endif %}
# ==== app/templates/standard/check.html ====
{% extends "base.html" %}
{% block title %}结论对照 · {{ check.patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>结论对照 · {{ check.patch.name }}</h2>
  {% include "standard/_check_block.html" %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/check">返回图斑列表</a>
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/standard/check_index.html ====
{% extends "base.html" %}
```

```
{% block title %}结论对照{% endblock %}
{% block content %}
<div class="card">
  <h2>评价结论与地方标准 / 政策文件的对照</h2>
  <p class="muted">
    标准知识库不参与评价计算，只在评价完成后回答一个问题：
    本次评价结论是否符合图斑所在省份的文件要求。选择一个图斑查看逐条对照。
  </p>
  {% if not rows %}
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑并录入数据。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">矿山图斑</th>
        <th scope="col">所在行政区</th>
        <th scope="col">已收录该省份文件</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for row in rows %}
      <tr>
        <td style="text-align:left;">{{ row.patch.name }}</td>
        <td>{{ row.patch.admin_region or "—" }}</td>
        <td>
          {% if row.province %}<span class="tag tag--ok">{{ row.province }}</span>
          {% else %}<span class="tag tag--muted">未收录</span>{% endif %}
        </td>
        <td><a class="link-btn" href="/standard/check/{{ row.patch.id }}">查看对照</a></td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
{% endblock %}
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/parsing.py ====
# -*- coding: utf-8 -*-
"""表单数值的安全解析。
```

背景：页面上的数字输入框原来写成 `float(request.form.get("area"))`，
用户填「约 3．5」或全角数字就会撞上 ValueError —— 页面上是默认英文 500 页。
真人演示时这是最容易踩到的一类。统一走这里：
没填返回 None，填了但不是数字抛 FieldError，让视图 flash 一句中文再回退。
"""
import math
class FieldError(ValueError):
    """表单里某个字段填得不对，携带给用户看的中文提示。"""
def parse_float(raw, field, *, minimum=None, maximum=None):
    """把表单值解析成 float。
    raw   表单原始值（可能是 None / "" / "abc" / "1e400"）
    field   字段名，用于拼提示
    minimum / maximum   可选的取值范围，越界同样抛 FieldError
    """
    if raw is None:
        return None
    text = str(raw).strip()
    if text == "":
        return None
    try:
        value = float(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填数字" % field)
    # nan / inf 能通过 float()，但存进库里会把标准化算成满分，必须挡住
    if not math.isfinite(value):
        raise FieldError("「%s」请填正常的数字" % field)
    if minimum is not None and value < minimum:
        raise FieldError("「%s」不能小于 %g" % (field, minimum))
    if maximum is not None and value > maximum:
        raise FieldError("「%s」不能大于 %g" % (field, maximum))
    return value
def parse_int(raw, field, *, default=None, minimum=None,
              maximum=None):
    """解析整数，用于查询参数与计数类字段。"""
    if raw is None:
        return default
    text = str(raw).strip()
    if text == "":
        return default
    try:
        value = int(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填整数" % field)
    if minimum is not None and value < minimum:
        value = minimum
    if maximum is not None and value > maximum:
        value = maximum
    return value
def parse_text(raw, field, *, max_len=None, required=False,
               default=None):

```
    """文本字段的统一处理：去空白、查必填、查长度。
    SQLite 不强制 String(128) 的长度，超长名称会原样写进库（自检实测：
    300 字的项目名照样落库），换到 MySQL/PostgreSQL 会直接报错；
    界面版式也会被撑坏。所以在入口处挡一次。
    """
    text = "" if raw is None else str(raw).strip()
    if not text:
        if required:
            raise FieldError("「%s」不能为空" % field)
        return default
    if max_len is not None and len(text) > max_len:
        raise FieldError("「%s」不能超过 %d 个字符（当前 %d 个）"
                         % (field, max_len, len(text)))
    return text
# ==== app/utils/timefmt.py ====
# -*- coding: utf-8 -*-
"""时间显示：库里一律存 UTC，展示时换成服务器本地时间。
为什么不在写库时直接用本地时间：一个系统里时间基准只能有一个。
`app/models.py` 里所有 `created_at` / `checked_at` 都用
    `datetime.utcnow`，
存的是 UTC；如果写库改成本地时间，已经存进库的几千条记录会新旧混着差 8 小时。
所以统一在「展示」这一层换算，老数据不用动。
用法（Jinja）：
    {{ row.created_at | localdt }}                    →
    2026-09-26 23:33
    {{ row.created_at | localdt("%Y-%m-%d %H:%M:%S") }} →
    2026-09-26 23:33:28
"""
from datetime import datetime, timezone
DEFAULT_FORMAT = "%Y-%m-%d %H:%M"
def to_local(value, fmt=DEFAULT_FORMAT, empty="—"):
    """把 UTC 时间换算成服务器本地时间并格式化。
    传进来的值按「无时区即 UTC」处理；带时区的原样换算，便于以后改用
    带时区的时间戳。空值返回占位符，免得页面上出现 None。
    """
    if value is None:
        return empty
    if value.tzinfo is None:
        value = value.replace(tzinfo=timezone.utc)
    return value.astimezone().strftime(fmt)
def now_utc():
    """取当前 UTC 时间（带时区），用于只作展示、不落库的时间戳。"""
    return datetime.now(timezone.utc)
def utcnow():
    """取当前 UTC 时间（无时区），用于数据库默认值。
    `datetime.utcnow()` 在 Python 3.12 已标记弃用，导入时会刷告警；
    这里用带时区的 now() 再摘掉时区，得到的仍是 naive-UTC，行为一致。
    """
    return datetime.now(timezone.utc).replace(tzinfo=None)
# ==== app/views/__init__.py ====
```

```
# -*- coding: utf-8 -*-                                                    13901
```